



**GREEN
CLIMATE
FUND**

Meeting of the Board

4 – 7 October 2021

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Provisional agenda item 15

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14 September 2021

Consideration of funding proposals - Addendum IX

Funding proposal package for FP177

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Cooling Facility";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Cooling Facility
Country(ies):	Bangladesh, El Salvador, Kenya, Malawi, North Macedonia, Panama, Sao Tome and Principe, Somalia, Sri Lanka
Accredited Entity:	The World Bank
Date of first submission:	2020/04/29
Date of current submission	2021/09/12
Version number	V. 9



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

LIST OF ACRONYMS

AE	Accredited Entity
AC	Air Conditioning
BFAR	Bureau of Fisheries and Aquatic Resources
BAU	Business As Usual
CDM	Clean Development Mechanism
CLASP	Collaborative Labelling and Appliance Standards Programs
CO ₂ eq	Carbon Dioxide Equivalent
DRR	Disaster Risk Reduction
ECCP	Efficient, Clean Cooling Program
EDGE	Excellence in Design for Greater Efficiencies
EE	Executing Entity
EEff	Energy Efficiency
EERF	Energy Efficiency Revolving Facility
ESA	Energy Service Agreement
ESMAP	Energy Sector Management Assistance Program
ESF	Environment and Social Framework
ESCO	Energy Service Company
ESCP	Environmental and Social Commitment Plan
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
ESMS	Environmental and Social Management System
ESPC	Energy Savings Performance Contract
ESRC	Environmental and Social Risk Classification
ESS	Environmental and Social Standards
FAO	Food and Agriculture Organization of the United Nations
FI	Financial Institution
FM	Financial management
GHG	Greenhouse gas
GWh	Gigawatt-hours
GWP	Global Warming Potential
GoP	Government of Panama
GoU	Government of Uganda
GCF	Green Climate Fund
GHG	Greenhouse Gas
GRM	Grievance Redress Mechanism
HCFC	Hydrochlorofluorocarbons
HFC	Hydrofluorocarbon
IFC	International Finance Corporation
ICR	Implementation Completion Report
IFR	Interim financial report
ISR	Implementation Status and Results Report
IEG	Independent Evaluation Group
IDP	Information Disclosure Policy
IPCC	Intergovernmental Panel on Climate Change
IBRD	International Bank for Reconstruction and Development

IDA	International Development Agency
IEA	International Energy Agency
LMP	Labor Management Plan
MSME	Micro small and medium enterprise
MEPS	Minimum Energy Performance Standards
MP	Montreal Protocol
NAMA	Nationally Appropriate Mitigation Action
NAP	National Adaptation Plan
NBS	Nature-based solution
NDA	National Designated Authority
NDC	Nationally Determined Contribution
NGO	Non-Governmental Organization
PFI	Participating Financial Institution
PIE	Project Implementing Entity
PMU	Project Management Unit
PPSD	Project Procurement Strategy Document
PS	Performance Standard
SSA	Sub-Saharan Africa
SDG	Sustainable Development Goal
SEforALL	Sustainable Energy for All
TA	Technical Assistance
TFs	Trust Funds
USAID	United States Agency for International Development
WBG	World Bank Group
WHO	World Health Organization

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A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Programme	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)	Check the applicable GCF result area(s) that the overall proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of GCF budget devoted to it. The total of the percentages when summed should be 100%.		
	Mitigation: Reduced emissions from: ☐ Energy access and power generation: ☐ Low-emission transport: ☒ Buildings, cities, industries and appliances: ☐ Forestry and land use: Adaptation: Increased resilience of: ☐ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☒ Infrastructure and built environment: ☐ Ecosystem and ecosystem services:		GCF contribution: : <u>0</u>% <u>0</u>% <u>77</u>% <u>0</u>% <u>0</u>% <u>0</u>% <u>23</u>% <u>0</u>%
A.5. Expected mitigation impact	16,240,201 tCO ₂ eq	A.6. Expected adaptation impact	4.2 million direct and 16.8 million indirect beneficiaries of which 50% of women
			10.3% of total population
A.7. Total financing (GCF + co-finance)	<u>879.8 million</u> USD	A.9. Project size	<u>Large (Over USD 250 million)</u>
A.8. Total GCF funding requested	<u>157 million</u> USD		
A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.		
	☒ Grant 32 million USD ☒ Loan 125 million USD ☐ Guarantee 0	☐ Reimbursable grants 0 ☐ Results-based payment <u>0</u>	
A.11. Implementation period	10 years	A.12. Total lifespan	27 years
A.13. Expected date of AE internal approval	Tbc	A.14. ESS category	B I-2

A.15. Has this FP been submitted as a CN before?	Yes ☐ No ☒	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☐ No ☒
A.19. Complementarity and coherence	Yes ☒ No ☐		
A.20. Executing Entity information¹	Democratic Republic of Sao Tome and Principe, represented by the Ministry of Finance, Planning and Blue Economy and acting through the Ministry of Health Democratic Socialist Republic of Sri Lanka, represented by the Ministry of Finance, and acting through the Ministry of Power Federal Republic of Somalia, represented by the Ministry of Finance, and acting through the Ministry of Health People's Republic of Bangladesh, represented by the Ministry of Finance, and acting through the Ministry of Power, Energy and Mineral Resources Republic of El Salvador, represented by the Ministry of Finance, and acting through the Ministry of Health Republic of Kenya, represented by the National Treasury of the Republic of Kenya and acting through the Ministry of Energy and Petroleum Republic of Malawi, represented by Ministry of Finance, Economic Planning and Development, and acting through Ministry of Industry, Trade and Tourism Republic of North Macedonia, represented by the Ministry of Finance and acting through the Ministry of Economy Republic of Panama, represented by Ministry of Economy and Finance and acting through the Ministry of the Presidency		

¹ For the purposes of the loans and grants, the Executing Entities would be the Host Countries, acting through the relevant Ministry(ies). A Project Implementing Entity (PIE) may be designated by the Host Country, to implement Components 1-3, as identified and confirmed during Project preparation.

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

The Cooling Facility is proposed as an innovative, multi-sector and multi-country programmatic financing mechanism to help address the key sustainable development challenge of providing access to cooling while minimizing negative climate impacts.

In a warming world, access to cooling – from cold chains and refrigeration to ensure the safety of foods, medicine and vaccines, to space cooling to ensure comfortable, healthy and productive homes, institutions and workplaces – is not a luxury, as over 1 billion people globally are deemed at risk from the lack of it. The need to adapt and build resilience in a rapidly warming climate – together with growing populations, urbanization, and rising disposable income levels in developing countries – are driving the exponentially growing demand for cooling services in many developing countries, which is associated with a projected rapid increase in cooling-related energy demand and greenhouse gas (GHG) emissions. There is a window of opportunity to support access to cooling services while avoiding maladaptation and limiting GHG emissions if we can shift this growing demand toward climate-friendly, passive and low-carbon cooling solutions, thereby creating an opportunity for driving scale, reducing costs and ultimately creating sustainable markets for sustainable cooling. Such actions would contribute to responding urgently to the challenge of climate change, meeting Sustainable Development Goals (SDGs), planning for disaster risk reduction (i.e. from heatwaves) and supporting the goals of the Kigali Amendment.

Developing countries have many competing demands for capital -- it is estimated that only 0.04 percent of total Overseas Development Assistance is directed to cooling solutions².-- and the COVID-19 pandemic has further compounded these constraints. The onset of the global pandemic due to COVID-19 and its impact across economies, further heightens the relevance of sustainable cooling solutions, which have a role to play in support of recovery efforts. This is reflected, for example, in the need for proper cooling and ventilation in health (as well as other) facilities, as well as the need for efficient cold chains to safely store vaccines, and to support food value chains from harvest to markets.

In this context, a dedicated Facility for cooling would target addressing the barriers that have so far limited the deployment of clean cooling technologies and also put into place the support for a broader paradigm shift by working across key cooling target areas and with a range of countries from different parts of the developing world to amplify the potential to leverage the experiences and learnings. The Facility seeks to scale and mainstream sustainable cooling applications in a number of key sectors (agriculture, health, buildings), thereby accelerating the deployment of climate-friendly solutions. It will help countries facing risks of extreme heat hazard enhance access to cooling by putting place climate resilient sustainable infrastructure and equipment, thereby contribute to adaptation efforts. At the same time, the Facility also aims to help countries build enabling environments and support investments to scale up access to sustainable cooling with a lower carbon footprint. The Facility is designed to promote passive and low-carbon cooling in a manner that will lead to the improved energy efficiency of buildings, cooling equipment and appliances; enhance access to climate-friendly and heat-resilient cooling solution, as well help strengthen enabling environments for the scale-up of climate friendly cooling solutions. It is expected that the Cooling Facility will also lead to broader development impacts, such as helping lower pressures on already strained energy systems, lowering local air pollution, as well as helping lower losses of food and medicine.

In so doing, the Facility, taking into account the cross-sectoral nature of cooling, the different cooling needs and priorities of countries and the necessity to adapt to different national contexts, hopes to develop and demonstrate implementation and business models and approaches that can be fit-for-purpose in more countries with similar circumstances, and thus provide a foundation to further scale-up access to cooling

² Kigali Cooling Efficiency Program (<https://www.k-cep.org/why-cooling/>)

services and the shift to sustainable cooling investments, supporting mitigation objectives as well as goals of building resilience to increased temperatures, even beyond the scope of this Facility.

The Cooling Facility will support projects³ that meet the Facility's criteria in the following cooling target areas (in either grid or off grid contexts): (i) space cooling and cool/green surfaces (including passive cooling strategies in the built environment, cooling equipment; building automation and controls, as well as solar and vegetative roofs and walls); and (iii) refrigeration, cold chains and logistics (including refrigeration, storage, and distribution activities, along with associated equipment and logistics).

Components

The Cooling Facility consists of three main components which support activities to be adapted to the needs and specific context of each country that is part of the Facility:

Component 1 - Policy, regulatory and enabling environment support (US\$15.7 million of GCF grant).

This component will support activities that aim to strengthen institutional, policy and regulatory frameworks, raise awareness and stimulate behavior changes, as well as build the capacity of key stakeholders to plan, finance and implement sustainable cooling investments as well as to plan for responses to increased heatwaves and other extreme events. Enabling activities will include cooling-specific inputs (related to both climate mitigation and adaptation) into the country's Nationally Determined Contributions planning process under the Paris Agreement such as Cooling Action Plans, as well as inputs on national policies and regulations that encourage sustainable cooling. It will also support activities linked to the Facility's Gender Action Plan.

Component 2 - Financing for cooling investments (US\$141.3 million of GCF funds, of which \$125 million in loan and \$16.3 million in grant). The Facility will provide financing to eligible projects for investments that support and help lock-in more resilient, climate-friendly as well as affordable development paths for cooling. Concessional financing will foster the adoption or scale-up the deployment of sustainable cooling technologies, appliances and business models. The deployment of these funds will be tailored to the needs of the projects, through suitable financing modalities for each project's context, for example, credit lines to financial intermediaries, or loans to households, municipalities and small, medium and large enterprises, or subsidies.

Component 3 – Project management (US\$0 million of GCF funds). This component will support the Executing Entities (EEs) and Project Implementing Entities (PIEs) in their management, coordination and implementation of their project activities, in compliance with the EEs' and PIEs' contractual obligations included or referred to in the legal agreements entered into with the Accredited Entities (AE).

Facility's impact

The Facility will provide critical support to the transition to sustainable cooling through the mobilization of approximately US\$879.8 million in financing (including both GCF and IBRD/IDA). It is expected to avoid or reduce over 9.1 MtCO₂e over the 10-year implementation period of the Facility (reduction that would reach 16.2 MtCO₂e over the 27 year lifespan of the Facility's underlying assets), strengthen local capacities, policy and regulatory environments to mainstream and support sustainable cooling, facilitate countries' efforts to meet NDCs, and importantly reach approximately 4.2 million direct beneficiaries across the 4 countries most vulnerable to extreme heat risk hazards.

³ The Cooling Facility is referred to as a "Program", which supports a number (9) of "projects"

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

I. Rising global temperatures, cooling needs and GHG emissions

a. Global perspective

The world is off-track to meet the 1.5°C objective of the Paris agreement. The global demand for cooling resulting from this temperature rise is generating a vicious cycle: as temperatures increase demand for cooling increases. The consequent increases in GHG emissions as a result of increased energy use and refrigerants will further increase temperatures. Population growth, increased urbanization and increased incomes in developing countries – especially in the world’s hotter regions that are increasingly susceptible to extreme heatwaves and related weather shocks - further contribute to the projected exponential increase in the demand for cooling and its associated energy consumption and greenhouse gas emissions (GHGs). Worldwide, there are currently an estimated 3.6 billion cooling appliances in use; a figure projected to increase to 9.5 billion by 2050 – and this would not even provide cooling for all who will need it in a warming world (as opposed to only to those who can afford it), which could require in the order of 14 billion cooling appliances (Dreyfus et al 2020).

Greater and rapid access to affordable cooling has become a development necessity. In 2019, the Intergovernmental Panel on Climate Change (IPCC) reported that at 1.5°C of warming, 2.3 billion people could be both exposed and vulnerable to heatwave events – a threshold that could be reached as early as 2030. SEforALL (2020) estimates over 1 billion people in developing countries are already facing immediate risk from lack of access to cooling, which includes 699 million slum dwellers living in hotter-climate urban areas where electricity services do not exist, are intermittent or are too expensive; and about 318 million people living in poor rural areas without access to cold chains for storage and preservation of food and medicine. This situation has implications for the health and well-being of people, food security, productivity and economic development and livelihoods. In terms of health, extreme heat can cause illnesses, including heat cramps, fainting, heat exhaustion heat stroke and even death. In terms of productivity and livelihoods, the International Labour Organization (ILO 2019) estimates the productivity loss due to heat by 2030 to amount to 80 million full time jobs, representing GDP losses averaging 1.5 percent in low income countries and 4 percent in lower-middle income countries – figures that are likely under-estimates as projections assume that the increase in global temperatures by the end of the century will not exceed 1.5°C. Productivity losses may reach 5 percent by 2030 in the worse affected regions of South Asia and West Africa. The worse affected sectors will be construction and agriculture – which in many countries will disproportionately affect women given their proportionately larger participation in that sector.

The global demand for cooling resulting from this temperature rise – which is trending towards higher than the 1.5°C target set under the Paris agreement according to multiple reports - is generating a vicious cycle: as temperatures increase, demand for electricity increases to meet the growing cooling needs. The consequent increases in GHG emissions because of increased fossil-fuel driven energy use and high pollutant refrigerants will further increase temperatures. Population growth, increased urbanization and increased incomes in developing countries – especially in the world’s hotter regions that are increasingly susceptible to extreme heatwaves and related weather shocks - further contribute to the projected exponential increase in the demand for cooling and its associated energy consumption and GHGs.

b. Sector context

The scale of the cooling challenge is significant across many sectors as illustrated below.

Agriculture and rural development. Higher temperatures contribute to droughts can increase undernourishment through impact on crop yields resulting in malnutrition and affecting food security in many developing countries, where it is estimated that about 45 percent of food spoils mainly due to inadequate cold storage and cold chains. The GHG emissions associated with food loss and waste contribute to about 4.4 GTCO₂e per year, representing 8-10 percent of total GHG emissions from 2010 to 2016 (Mbow C., et al 2019). Supply chain disruptions further threaten food security and nutrition of millions of people, are lead to urban centers struggling to access fresh fruits, vegetables, dairy, meat and fish, making the existing gaps felt even more. The FAO (2018) warns that “addressing food loss and waste through cold chain extension without any technology improvement would have serious adverse effects on climate change as a result of increased GHGs”. As governments around the world plan to boost local employment, the agriculture and fisheries sectors stand-out; and are also areas where integrating efficient, clean cooling (from efficient building constructions to efficient cooling and refrigeration systems powered with renewable energy) could support low-carbon development.

Health. The health sector is a priority. Hospitals and many health facilities are often open 24 hours a day and require heating, ventilation and air conditioning (HVAC) systems to control room temperature, pressurization, air change rates and humidity in order to mitigate airborne contaminants and provide a healing space for patients while also providing a high level of protection for healthcare personnel. The increased electricity load from hospitals should be met as efficiently as possible to avoid burdening electricity systems and lowering costs for health facilities so they can dedicate scarce resource for treatments. Rural clinics may need to be built and should be done integrating energy efficiency into the design and construction; their increased energy needs for cooling (and other uses) can be met with distributed renewable energy. The availability of sustainable cold chains to safely store and preserve medicine and vaccines – was already inadequate⁴ – and becomes more urgent. The global healthcare sector, with its need for temperature control facilities and storage for medicine and vaccines, is estimated to account for about 4.4 percent of net GHG emissions globally (E3G, 2020). A well-functioning cold chain and health system infrastructure is essential; it is also an opportunity to build institutional capacity and sustainability to put countries’ health systems on a more sustainable and climate-friendly path with sustainable cooling.

The COVID-19 pandemic has shed light on – and compounded - the pre-pandemic inequities in the health cold chain and the challenges in achieving universal immunization coverage, especially in Africa. It has been estimated that the cost of vaccine lost due to exposure to temperature fluctuations amounts to approximately USD 34 billion annually – a figure that does not include the physical and financial cost associated with avoidable illnesses⁵. The COVID-19 pandemic will have both short- and long- term effects on populations around the world, causing hardship practically everywhere, but with particularly severe risks for the poorest countries and under-resourced communities. COVID-19 will likely have long-lasting impacts on healthcare systems globally, especially in developing countries where health facilities are already lacking in capacity and reach, but also in other sectors and across the economy. It is clear that priorities of countries are shifting. A multi-pronged approach will be needed to respond to the crisis and help address the strains put on health, food, energy, housing, and other systems and plan a recovery. The World Bank Group has been responding aggressively to address these needs, making available up to \$160 billion in financing capacity through June 2021, including an envelope of \$12 billion for vaccine purchase and deployment⁶, which could help cover vaccination for up to 1 billion people, and support strong immunization and health systems, including cold chain infrastructure. This overall \$160 billion support is tailored to the health, economic and social shocks countries are facing and includes over \$50 billion of IDA resources on grant and highly concessional terms. A

⁴ Hinnant and Mednick (2020) estimate that nearly 3 billion live where temperature-controlled vaccine storage is insufficient for an effective COVID-19 immunization campaign.

⁵ Peters, T. and B. Hartley (12 November 2020), <https://www.seforall.org/news/sustainable-cold-chains-needed-for-equitable-covid-19-vaccine-distribution>

⁶ World Bank (October 13, 2020), “IBRD and IDA Project Paper on a Proposed Additional Financing to the COVID-19 Strategic Preparedness and Response Program Using the Multiphase Programmatic Approach (Global COVID-19 MPA)” (P173789).

portion of the cooling solutions supported by this Facility will contribute to those efforts, and help “green” the response and recovery.

Refrigerants. About 20 percent of the climate impact comes from the direct emissions of hydrofluorocarbons (HFC) refrigerants.⁷ Further, the leakage of refrigerants used are responsible for 15-20 percent of the global warming effect of refrigeration. (Carbon Trust, 2020). This highlights the importance of tackling the energy dimension of cooling across sectors, as efforts to only address the climate impact of refrigerants (under the Montreal Protocol’s 2016 Kigali Amendment, countries committed to cut use of HFCs, a potent greenhouse gas widely used as refrigerants, by more than 80 percent over 30 years), would leave out the largest share of the cooling mitigation challenge. The HFC phase down schedule of the Kigali Amendment could avoid up to 0.5°C of warming by 2100. If combined with efforts to maximize energy efficiency in cooling equipment, close to 1°C of global warming can be avoided by the end of the century (G. Dreyfuss et al, 2020).

Buildings. Cooling is the fastest growing end-use in buildings. Forty percent of the world’s population resides within the tropics, with dramatically lower access to space cooling in relation to their need – and this figure is projected to grow to 50 percent by 2050⁸. The use of air conditioners and electric fans is becoming increasingly common and already accounts for about a fifth of the total electricity in buildings around the world – or 10 percent of all global electricity consumption (IEA, 2018). Globally, the energy consumption of buildings represents 36 percent of the global final energy demand and almost 40 percent of GHG emissions⁹. The global energy use for space cooling is projected to grow 300% between 2016 and 2050, and by then space cooling could account for 30-50 percent of peak electricity loads, requiring an increase of 2,500 GW to meet that additional demand. Improvements are needed for the existing building stock to become less energy and GHG intensive, and new buildings (keeping in mind that half of the buildings that will be standing in 2060 have not yet been built and the majority of these will be in developing countries) will need to be built with a lower energy and GHG footprint.

Energy. About 80 percent of the climate impact of cooling equipment is associated with energy use (i.e. the indirect CO₂ emissions from fossil-fuel based electricity generation). Much of the use of energy for cooling is driving peak demand which is typically the least efficient and carbon intensive and driving the need for more power plants. In hot countries like India the share of air conditioners (ACs) in peak electricity load could reach 45 percent in 2050, up from 10 percent today. Globally, space cooling-associated emissions from electricity use are set to double by 2050 (even as the grid gets cleaner). In fact, an estimated 2.3 billion people in lower-middle class in developing countries are on the verge of purchasing the most affordable – and also least efficient – ACs. Worldwide, there are currently an estimated 3.6 billion cooling appliances in use; a figure projected to increase to 9.5 billion by 2050 – and this would not even provide cooling for all who will need it in a warming world (as opposed to only to those who can afford it), which could require in the order of 14 billion cooling appliances (Dreyfus et al 2020). Moving to best available technologies would reduce cumulative emissions by 38-60 GtCO₂eq by 2030 and up to 130-260 GtCO₂eq by 2050.

c. Facility country-specific context

Each of the countries covered by the Facility face the need to adapt to a warming world as they are exposed to risks of extreme heat hazard¹⁰, albeit at different intensity levels, which translates into a growing demand for cooling. For example, Bangladesh¹¹, Malawi, Somalia, and El Salvador are exposed to a high risk, while

⁷ Refrigerants are critical to vapor compression-based equipment used in refrigeration and air conditioning.

⁸ ESMAP (2020), Primer for Space Cooling.

⁹ Global Alliance for Buildings and Construction (www.globalabc.org).

¹⁰ See the World Bank Global Facility for Disaster Reduction and Recovery (GFDRR)’s *ThinkHazard!* web-based tool (<https://thinkhazard.org/en/>)

¹¹ According to SEforALL 2020, Bangladesh ranks among the top 10 countries with urban poor populations at risk from lack of cooling.

the others are facing a medium risk. Table 1 provides a summary of the climate trends and drivers for each of these four high risk countries.

Table 1: Summary of climate trends and drivers for each country at high risk of extreme heat hazard

Country	Historical extreme temperature	Extreme temperature projections (base period 1986-2005)	Cooling degree day (CDD) - linked to temperature projections
Bangladesh	Observed data between 1985 and 1998 indicate an increasing trend in temperatures of about 1°C in May and 0.5°C in November. The seasonal frequency of days with Tmax≥36°C has been showing an increasing trend since 1986 in most parts of Bangladesh, with the highest rate of +1.018 day/year at Mongla. Average temperatures have been increasing more rapidly during the monsoon season, from June to August, as is evident from historical data for the last 30 years (1991-2020). Maximum and minimum temperatures during monsoon season increased by 0.05°C and 0.03°C respectively.	Mean annual temperatures are predicted to increase by about 1.2°C by mid-century, and by 1.9°C by the end of the century compared to the 1986-2005 base period. Compared to the 1986-2005 base period, the total number of hot days in a year where the daily maximum temperature is above 35°C will rise by 13.23 by 2039, and by 41.11 days by 2099. The probability of heat waves, defined as a period of three or more days where average temperature is above 95th percentile, will grow by 5% in 2040-2059, and by 9% between 2080-2099 compared to 1986-2005 base period.	Projected change in annual cooling degree days is projected to rise from 482 CDD between 2020-2039 to 778 CDD between 2040 and 2059 and 1210 CDD by the end of the century.
El Salvador	One of the most climate affected countries in the world. El Salvador experienced a sustained increase in monthly temperature over the last 30 years. The summer months of April to August have witnessed consistent increase in temperature. The average temperature recorded in April 2020 was 27.9 °C which exceeded the average April temperature of 1986-2005 (26.6 °C) by 1.3°C Since, 1950s, the temperature increased on average by 1.3°C compared with the global average of 0.8°C.	Mean annual temperature will rise by 1.24°C in 2050. The projected change in the annual probability of heatwaves in El Salvador will grow by 19% by 2040-59, 25% by 2060-79 (compared to 8% between 2020-2039). The projected change in hot days (Tmax>35°C) will increase by 14.49 days by 2059, and 36.12 days by the end of the century.	Projected change in annual cooling degrees is projected to rise to 946.81 CDD by 2050 and 1370.68CDD by the end of the century (compared to 578.98 CDD in 2020-2039 period).

Malawi	Between 1960 and 2006, mean annual temperature increased by 0.9°C, with an average increase of 0.21°C per decade. Climate trends over this period show the total number of hot days in a year have increased by 30 days.	Mean annual temperature will rise by 1.58°C in 2040-2059 and 2.16°C by the end of the century. The number of days where the daily maximum temperature exceeds 35°C will increase by about 21 days by 2059 relative to the period 1986-2005, and by 32 days by the end of the 21st century.	Projected change in annual cooling degree days is projected to rise by 1,154 CDD between 2040 and 2059, and 1317 CDD by the end of the century.
Somalia	Between 1960 and 2020, mean annual temperature increased by about 0.5°C in the hottest months (April and May) and 0.3°C in the coldest months (January and December).	Mean annual temperature is projected to increase by 1.3°C by mid-century and 1.7°C by the end of the century, with the number of hot days (Tmax>35°C) in a year expected to increase by 70 days.	Annual cooling degree days are projected to rise by 836 CDD between 2040 to 2059, and 1182 CDD by end of the century.

The rationale for climate change adaptation of the countries at high risk extreme heat hazard is presented below.

Bangladesh. Bangladesh is highly exposed to extreme heat, drought, and wildfires. Extreme heat hazard is classified as high throughout the country, which indicates that prolonged exposure to extreme heat, resulting in heat stress, is expected to occur at least once in the next five years¹². The seasonal frequency of days with Tmax ≥ 36°C has been on an increase since 1986 in most parts of Bangladesh, as highlighted in Table 1. Average temperatures have been increasing more rapidly during the monsoon season from June to August, according to historical data for the last 30 years (1991-2020). The increment in temperature has been consistent in the last 5 years alone. Looking into the future, extreme temperatures are expected to occur with high frequency, duration, or severity. Mean annual temperatures are predicted to increase by about 1.2°C by mid-century, and by 1.9°C by the end of the century, with warming more likely pronounced during winter months from December to February. Most notably, the total number of hot days in a year (where daily maximum temperature is above 35°C) is projected to increase by 13.23 days by 2039, 22.27 by 2040-59, and by 41.11 days by 2099. By mid-century, annual cooling degree days (CDD) are projected to rise from 482 CDD (202-2039) to about 778 CDD by 2059, and 1210 by the end of the century. Overall, this will impact the energy sector as any time the temperature rises above 18.3°C, there is an expected increase in power demand for cooling¹³. Without decarbonization efforts, this could result in significant increases in GHG given high share of fossil fuel in the power generation mix.

On average, Bangladesh is affected by a major country-wide drought every five years. There have been 19 major droughts between 1960 and 1991 that have affected around 53% of the population and 47% of the country¹⁴. Seasonal drought events commonly occur throughout the country, mostly in the relatively drier northwestern region. Climate change is projected to exacerbate drought conditions in Bangladesh, especially

¹² ThinkHazard! – Bangladesh <https://thinkhazard.org/en/report/23-bangladesh/EH>

¹³ Climate Change Knowledge Portal: Bangladesh – Climate Data: Historical and Projections <https://climateknowledgeportal.worldbank.org/country/bangladesh>

¹⁴ Climate Policy Institute: https://www.climateemergencyinstitute.com/uploads/M__Drought_in_Bangladesh_and_its_adaptive_measures.pdf

in the northwestern parts. Areas of the country that currently experience moderate droughts will likely turn into severely drought-prone areas in the next 20-30 years¹⁵. This can put significant stress on the energy system as it is likely to increase energy demand while decreasing energy supply reliability. With higher aridity, wildfires are also expected to be a prevalent risk in Bangladesh. Wildfire hazard is classified as high in the country. This means that there is greater than 50% chance of encountering weather that could support a significant wildfire that is likely to result in both life and property loss in any given year. Climate projections identify a likely increase in the frequency of fire weather occurrence, including an increase in temperature and greater variance in rainfall. In particular, the fire season is likely to increase in duration with greater number of days with weather that supports fire spread and longer periods without rain during fire seasons. Projections also indicate that there could be an increase in the severity of fire, and even areas that are not currently at risk of wildfire hazard could see an increase due to an expansion of the wildfire hazard zone.

These changes in climate will only lead to increase in energy demand for both space cooling and cold storage. This increase will be on top of energy demand growth driven by increasing urbanization and rise in income levels. Thermally efficient design and construction and passive measures that limit heat entering into the building, efficient air conditioning and ventilation will be needed, especially to protect vulnerable segments of the population, such as children, the elderly and the sick during hot days and heatwaves. Furthermore, various economic and social sectors will rely more on cooling solutions as adaptation mechanisms as temperatures increase across regions. Therefore, the Facility will support investments in sustainable cooling solutions as essential adaptation measures delivered through low-carbon efficient solutions.

El Salvador. El Salvador is highly exposed to extreme heat, hurricanes, droughts, and wildfires. Extreme heat hazard is classified as high throughout the country, which indicates that prolonged exposure to extreme heat, resulting in heat stress, is expected to occur at least once in the next five years¹⁶. Historical trends suggest that the temperature increase in El Salvador has been more pronounced than in other parts of the world, with the temperature increase since 1950s of 1.3°C compared to 0.8°C globally¹⁷. El Salvador experienced a sustained increase in monthly temperature over the last 30 years (1991-2020). In particular, the summer months (i.e., April to August) have witnessed consistent increase in temperature. The average temperature recorded in April 2020 was 27.9 °C which exceeded the average April temperature of 1986-2005 (26.6 °C) by 1.3°C. Looking into future, mean annual temperature is projected to increase by 1.24°C by 2059. Most notably, the number of days where the daily maximum temperature exceeds 35°C will increase by 36.12 days by the end of the 21st century, and the probability of heat waves will increase by 30% by 2100. The annual cooling degree days are projected to rise by 46.81 CDD by 2050, and 1370.68 CDD by the end of the century¹⁸.

El Salvador is situated in the dry corridor of Central America, and prolonged dry periods often result in significant economic losses. More than 10,000km² of land area in El Salvador is likely to be regularly affected by severe, moderate, or weak droughts¹⁹. The probability of severe drought will grow by 36% by 2059, compared to 1986-2005²⁰. Droughts can put significant stress on the energy system as they increase energy demand while decreasing energy supply reliability (which have a detrimental impact on key health services and vaccine safety). Additionally, the annual precipitation is predicted to decrease by -65.33mm by 2059.

¹⁵ GFDRR: https://climateknowledgeportal.worldbank.org/sites/default/files/2018-10/wb_gfdr气候_change_country_profile_for_BGD.pdf

¹⁶ Think Hazard – El Salvador: <https://thinkhazard.org/en/report/75-el-salvador>

¹⁷ Climate & Development Knowledge Network: https://cdkn.org/project/planning-around-temperature-increases-san-salvador/?loclang=en_gb

¹⁸ Climate Change Knowledge Portal: El Salvador – Climate Data, Historical and Projections: <https://climateknowledgeportal.worldbank.org/country/el-salvador/climate-data-projections>

¹⁹ The CGIAR Research Program on Climate Change, Agriculture and Food Security: <https://ccafs.cgiar.org/regions/latin-america/el-salvador>

²⁰ Climate Change Knowledge Portal: El Salvador – Climate Data: Historical and Projections

Similar to other countries in Central America, El Salvador has seen a steady increase in the intensity and frequency of extreme weather events and storms in the last 30 years. Due to its social and economic vulnerability, economic assets and infrastructure in El Salvador are at a high risk of damage from hurricanes. Annual average loss from hurricanes is estimated at US\$2.9million (0.01% of GDP) and the probable maximum loss at US\$ 374million (1.5% of GDP)²¹. Adding to that, the frequency and intensity of hurricanes and extreme weather events is likely to increase in the future. Wildfire hazard is classified as high in the country. This means that there is greater than 50% chance of encountering weather that could support a significant wildfire that is likely to result in both life and property loss in any given year. Warming climate, rising sea level rise, combined with the increasing unpredictability of the EL Niño Southern Oscillation's patterns, will likely put additional stress on the country's energy system, especially considering that ~35% of El Salvador's energy supply comes from hydropower energy²² that might be impacted by droughts and decrease in annual precipitation.

These changes in climate will only lead to a significant increase in demand for both space cooling and cold storage. This increase will be on top of energy demand growth driven by increasing urbanization and rise in income levels. Efficient air conditioning and ventilation will be needed, in particular to protect vulnerable segments of the population, such as children, the elderly and the sick during hot days and heatwaves. Further, in health facilities, for example, adaptation mechanisms such as adequate cooling and ventilation will be needed to create the ambient temperature ranges for optimal functionality and to safely preserve medications and vaccines. This will be another factor contributing to increasing energy demand. The impacts of increased demand will be exacerbated by the supply variability from hydropower as a result of droughts and decrease in annual precipitation thus putting additional stress on the power system.

Therefore, the investments in sustainable cooling solutions supported by the Facility will not only help adapt to extreme temperature hazard on the demand side (for space cooling and cold chains), but also help reduce system stress due to hydropower variability, which will help safeguard the efficacy of vaccines and medicines, as well as essential medical services necessary for the health and well being of the population.

Malawi. Malawi is prone to adverse climate hazards including extreme temperatures, dry spells, seasonal droughts, intense rainfall, and flash floods. Extreme heat hazard is classified as medium to high risk throughout the country, which indicates that prolonged exposure to extreme heat that results in heat stress is likely to occur at least once in the next five years²³. The southern region is particularly highly exposed to extreme temperatures and heat stress. Between 1960 and 2006, mean annual temperature increased by 0.9°C, with an average increase of 0.21°C per decade. Climate trends over this period show that average temperatures are increasing more rapidly during the rainy summer season (i.e., December to January), and the total number of hot days in a year has increased by 30 days. As climate change intensifies, **mean annual temperature will rise by 1.58°C (1.38°C to 1.96°C) in 2040-2059**. By mid-century, annual cooling degree days are projected to rise by 953 CDD, with the number of hot days in a year expected to increase by about 21days²⁴. This can significantly impact the energy sector as any time the temperature rises above 18°C, there is an expected increase in power demand for cooling.

²¹ GFDRR: El Salvador, Earthquake and Hurricane Profile
<https://www.gfdr.org/sites/default/files/EI%20Salvador1.pdf>

²² IRENA: Renewables Readiness Assessment, El Salvador https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2020/Dec/IRENA_RRA_EI_Salvador_2020.pdf

²³ ThinkHazard! – Malawi <https://thinkhazard.org/en/report/23-bangladesh/EH>

²⁴ Climate Change Knowledge Portal: Malawi – Climate Data: Historical and Projections
<https://climateknowledgeportal.worldbank.org/country/malawi/climate-data-projections>

Beyond extreme temperatures, droughts and floods are two of the most severe of climate hazards in Malawi, which have only increased in frequency and severity over the past twenty years. On average, 1.5 million people are affected by water scarcity every year, mainly in the central and southern parts of the country²⁵, and close to 6 million people were affected by drought between 1985 and 2018. This is especially important for Malawi, where about half of the population lives below the national poverty line, as severe droughts (1-in-25-year drought) can increase the poverty rate by 17 percentage points (roughly an additional 2.1 million people falling below the poverty line)²⁶. Drought conditions are a significant risk to the country's energy supply as 70% of generation capacity comes from hydro, as well as to agriculture production – a key economic sector in the country. Such conditions are worsened by extreme temperatures and wildfires. Even when looking at the next few decades, wildfire hazard is projected to continue to be a high risk throughout the country given that in any given year, there is greater than a 50% chance of encountering weather that could support a significant wildfire²⁷. Extreme heat events will likely put additional stress on the country's energy system, especially considering that majority of Malawi's energy supply comes from hydropower that might be impacted by droughts and decrease in annual precipitation. At the same time, the agriculture sector will need to adapt to a warming and changing climate which is reducing yields and impacting livelihoods.

These changes in climate will have a negative impact on many economic sectors, and agriculture in particular – affecting food security and livelihoods. The changes will also lead to a significant increase in demand for both space cooling and cold storage. This increase will be on top of energy demand growth driven by efforts to achieve higher access rates. Specifically, in the agriculture sector – which employs a very large share of the working population and is vulnerable to external shocks, particularly climatic shocks²⁸, adaptation mechanisms such as proper cold storage for optimal functionality and to preserve perishable agricultural products will be critical for farmers and for avoiding food security problems. This will be another factor contributing to increasing energy demand with rising temperatures and increased extreme heat events. The impacts of increased demand will be exacerbated by the impact of potential supply decrease from hydropower as a result of droughts and decrease in annual precipitation thus putting additional stress on the power system, in a country where there are significant energy shortages.

As the agriculture sector is both one of the most vulnerable to climate change and one of the largest contributors of GHG emissions in the country, facilitating the deployment of climate-friendly (including energy efficient, low GWP refrigerants, use of solar energy) cooling technologies would contribute to achieving a low-carbon and climate-resilient economy. The support provided through the Cooling Facility will not only help adapt to extreme temperature hazard on the cold chains for the agriculture sector but also help increase agricultural productivity, reduce power system stress due to hydropower variability and avoid the use of diesel.

Somalia. Somalia is generally arid and semi-arid with two seasonal rainfall seasons. The country's climate is influenced by the Inter-Tropical Convergence Zone (ITCZ), monsoonal winds and ocean currents, jet-streams, and conditions of the neighboring Indian Ocean and Red Sea. Mean annual temperatures was about 27°C in the period between 1901-2016, with June to September being the hottest months in the north and December to March hottest in the south. Between 1960 and 2020, mean annual temperature increased by about 0.5°C in the hottest months (April and May) and 0.3°C in the coldest months (January and December). By mid-century, mean annual temperature is projected to increase by 1.3°C and by around 1.7°C by the end of the

²⁵ GFDRR: https://www.gfdr.org/sites/default/files/publication/malawi_low.pdf

²⁶ Climate Change Knowledge Portal: Malawi – Vulnerability
<https://climateknowledgeportal.worldbank.org/country/malawi/vulnerability>

²⁷ ThinkHazard! – Malawi <https://thinkhazard.org/en/report/23-bangladesh/EH>

²⁸ See <https://www.worldbank.org/en/country/malawi/overview>

century across all areas of Somalia²⁹. In addition, the El Niño Southern Oscillation (ENSO) influences Somalia's climate variability in several ways, bringing more rainfall and flooding during El Niño and droughts in La Niña years³⁰. Majority of Somalia is at risk of several natural hazards, including extreme heat, drought, floods, and cyclones due to these factors that affect the level of temperature and precipitation in the land. For example, extreme heat hazard risk is considered high, with prolonged exposure to extreme heat that results in heat stress expected to occur at least once in the next five years³¹. The number of days where the daily maximum temperature exceeds 35°C will increase by about 70 days by 2059 relative to the period 1986-2005, and by 93 days by the end of the 21st century. Annual cooling degree days are expected to rise by more than 836 CDD between 2040 to 2059, and by 1182 CDD by the end of the century. This significant increase in temperatures will impact the energy sector as there will be higher power demand for cooling.

Beyond extreme temperatures, droughts and floods pose the most severe hazards to the country. According to Somalia's National Adaptation Program of Action, the country has endured multiple, severe drought episodes since 1965. Somalia experienced a severe drought event in 2011, which resulted in 258,000 deaths in the country and affected 13 million people throughout the Horn of Africa between October 2010 and April 2012. Continued increases in temperature could have vast impacts on the Somali population, livestock, and crop yields. Heat stress could also alter livestock's feed intake, health, mortality rates, agricultural production, and thus the country's overall economic growth.

These changes in climate will only lead to a significant increase in demand for both space cooling and cold storage. This increase will be on top of energy demand growth driven by increasing urbanization and rise in income levels. Efficient air conditioning and ventilation will be needed, especially to protect vulnerable segments of the population, such as children, the elderly and the sick during hot days and heatwaves. Further, in health facilities, for example, adaptation mechanisms such as proper cooling and ventilation will be needed to create the ambient temperature ranges for optimal functionality, ensure safe and healthy indoor environments, and to preserve medications and vaccines. This will be another factor contributing to increasing energy demand with rising temperatures and increased extreme heat events.

In summary, there is an opportunity to set the course for affordable and sustainable cooling pathways now. The proposed cooling-related investments in the Facility's Somalia project will help improve the quality of health infrastructure and services, while also limiting operational costs which is key in the context of the health ministry's depleted budget. The investments in sustainable cooling systems supported by the Cooling Facility will not only help enhance the health system's resilience and adaptation to extreme temperature hazard on the demand side (for space cooling and cold chains), but also make it more sustainable and help improve health outcomes in the country's population by making available medicine and vaccines longer to more people.

II. Cooling Facility strategic approach

The wide range of cooling needs and priorities implies an equally diverse set of challenges and solutions. Many issues are also country and context specific. There is no one-size-fits-all solution or approach. Given the nascent and still early stage focus on cooling (globally) to address for both development and climate reasons, the Cooling Facility seeks to cater to different contexts – geographic, socio-economic and including within countries' efforts and priorities to respond to the COVID crisis. In fact, it is hoped that the diversity of the selected countries in the Facility will provide pioneering examples for mitigation and adaptation activities related to cooling in different contexts. The Facility aims to support different approaches and generate lessons that can then be adapted and scaled more broadly to enhance access to affordable cooling while at same time

²⁹ Climate Change Knowledge Portal: Somalia – Climate Data: Historical and Projections: <https://climateknowledgeportal.worldbank.org/country/somalia/climate-data-projections>

³⁰ Somalia National Adaptation Programme of Action to Climate Change: <https://unfccc.int/resource/docs/napa/som01.pdf>

³¹ Think Hazard – Somalia <https://thinkhazard.org/en/report/226-somalia/EH>

kick start a transformation towards more sustainable cooling, offering examples to be emulated and adapted in countries beyond the scope of the Facility. Operationally, the nature of cooling needs is such that often interventions will consist of projects within larger country programs; this is particularly likely for measures to address cold chain issues in health, food security, and labor productivity, for example (e.g., the sustainable cold chain investments will typically be a part of a broader projects targeting support to a country's health sector).

Where growing cooling demand is creating pressures on country energy systems and upward pressures on GHG emissions, the priority is to support space cooling, both in terms of passive measures (such as building design and high efficiency thermal envelopes) and energy efficient cooling equipment and systems (this is the case for Bangladesh³², Sri Lanka, Panama, North Macedonia). In agriculture-dominant economies with needs to expand access cooling to rural and off-grid areas, the priority is to support efficient and climate friendly cold chain (as through the Facility's support to Malawi and Kenya).

In hot climate countries, ensuring health infrastructure is adequate for patients and storage of pharmaceuticals is a priority, as pharmaceuticals expire and become ineffective more quickly in hot climatic conditions. Efficient and solar-powered cold chain and health facilities can expand quality health services and ensure safe storage of vaccines while making them more resilient to warmer temperatures, in addition to keeping their carbon footprint low (e.g. Somalia, Sao Tome and Principe and El Salvador).

Overall, the Cooling Facility is aligned and supports countries' climate change goals and ambitions, as reflected in their respective climate change Nationally Determined Contributions (NDCs).

III. Related interventions / programs

The Cooling Facility draws on and coordinates with other initiatives and programs³³ related to cooling, which are funded through other financiers but are not part of the Funded Activities.

The ESMAP³⁴ Efficient, Clean Cooling program (ECCP) is a multi-sectoral technical assistance (TA) program dedicated to cooling implemented in collaboration with the World Bank's Montreal Protocol unit (housed within the Bank's Environment, Natural Resources and Blue Economy Global Practice). It was established in 2019 with seed funding from the philanthropic Kigali Cooling Efficiency Program (K-CEP), with the aim of kick-starting a new business line within the WBG by supporting the design and inclusion of efficient, clean cooling components or characteristics in WBG country engagements, lending and investments, as well as activities to raise awareness and disseminate knowledge. The ECCP also seeks to develop strategic partnerships within and outside the WBG and help mobilize funding. Under ESMAP's new business plan for FY21-24, the Efficient Clean Cooling TA Program is proposed to be sustained and scaled up, including to support project preparation activities and develop a pipeline of projects across the World Bank's Global Practices. The ESMAP Efficient Clean Cooling TA program will also support knowledge exchange events and the development of knowledge products (e.g. case studies) to share experiences and lessons from World Bank projects, including from those supported by the Facility – both among the beneficiaries of the projects and other key national and international

³² Bangladesh also ranks among the top 10 countries with lower-middle income populations on the brink of purchasing the most affordable air conditioner or refrigerator – and least energy efficient – on the market, causing increased energy demand (along with GHG emissions) that will further strain the country's energy system.

³³ The listed programs and initiatives, while having the potential to strengthen project's impact and producing analytical and knowledge products which will inform project design, are not part of the Cooling Facility nor the projects supported by the Facility.

³⁴ For more information on the World Bank Energy Sector Management Assistance Program (ESMAP), a global technical assistance and knowledge program administered by the World Bank (housed within the Bank's Energy and Extractives Global Practice), see www.esmap.org.

stakeholders, as well as with other countries– to help raise awareness and inform further scaling-up of sustainable cooling more broadly.

The Climate Investment Funds (CIF). The Cooling Facility builds on synergies with the CIF, a multi-donor fund administered by the World Bank. For example, CIF funding could address the barriers to the deployment of standalone-solar technologies in general, while the Cooling Facility would focus on cooling solutions for productive uses. The Cooling Facility will continue to identify and leverage partnerships with the CIF to enhance its development impact.

The Bank's Energy Leap proposal for Africa³⁵ seeks to help reach universal energy access by 2030. It specifically aims to promote conditions for increased use of productive efficient refrigeration and cooling electric appliances for households, micro, small and medium enterprises (MSME), and agriculture sector to help boost both household income and energy demand – and in doing so help strengthen the business case for electrification and provide households access to welfare-enhancing appliances.

The ESMAP Gender and Energy Program is helping to strengthen women's roles as consumers, employees, and entrepreneurs in the energy sector. Aligned with the WBG Gender Strategy (FY16-23), the program works with countries to generate crucial knowledge of actions and design interventions needed to close gender gaps in the sector and improve development outcomes. The ESMAP Gender and Energy program will provide support to identify, assess and suggest options to respond to specific gender issues in the context of cooling-informed projects, including in new areas, such as women in energy leadership and productive uses of energy.

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

The theory of change diagram (Figure 1) presents the logical underpinning of the Facility.

There are multiple barriers and risks to the scale-up deployment of climate-friendly cooling solutions. The first barrier, i.e. lack of regulatory/policy framework for sustainable cooling system along with limited institutional capacity to design, implement, monitor, and enforce cooling standards, is a fundamental hurdle to address to foster a transformational change in the cooling sector. However, the extent to which market transformation can be sustained relies particularly on changes to people's behavior whether that means changes in consumption patterns or in approaches to meet their cooling needs. Low demand for sustainable cooling usually hampers market development. Addressing the barriers related to the lack of public awareness of the benefits of climate-friendly cooling, and the absence of suitable business models and market mechanisms becomes critical to stimulate the demand.

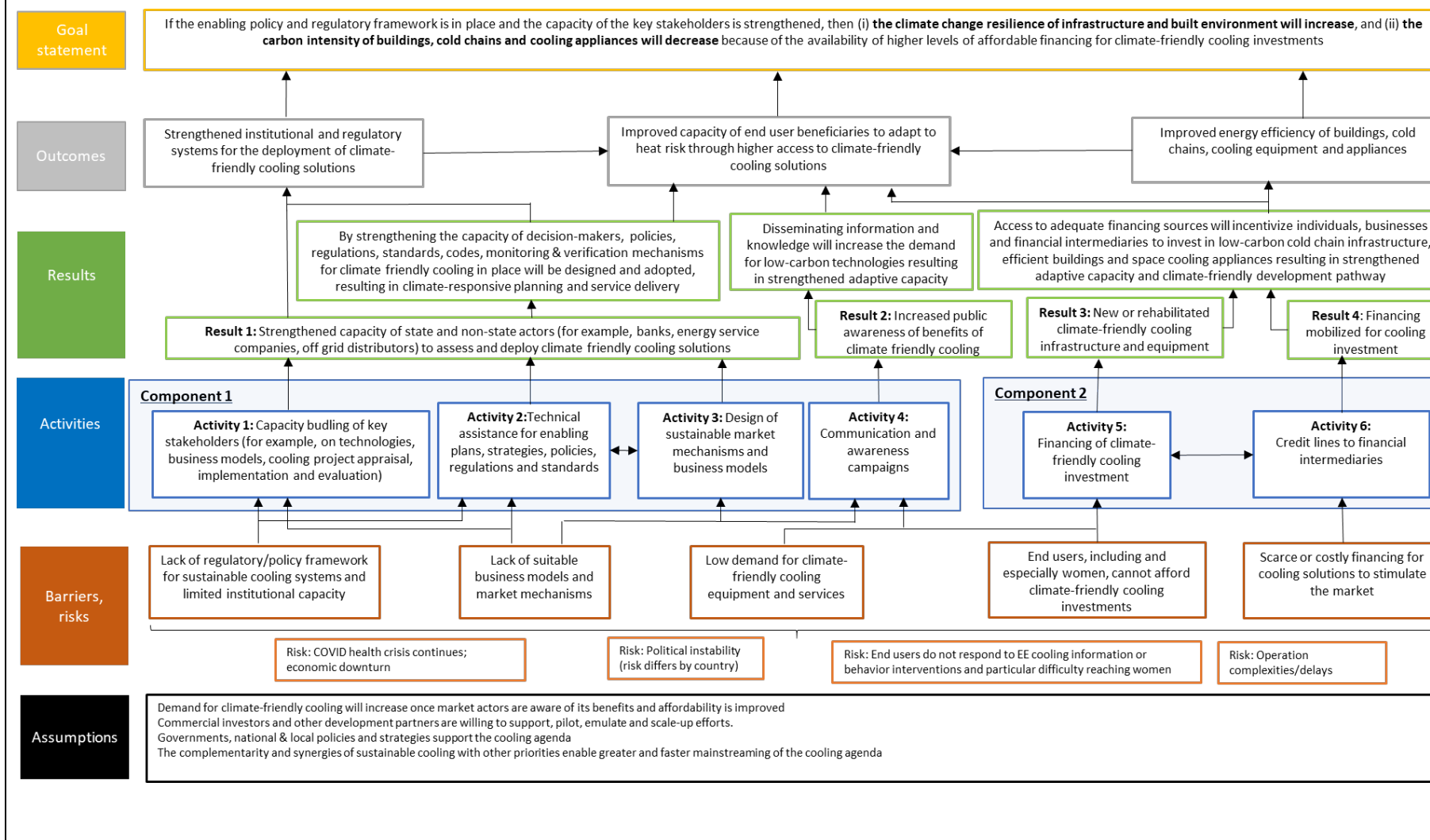
Meeting the demand for climate-friendly cooling solutions requires access to sufficient and affordable sources of financing. Gaps in financing for investments in energy efficient buildings and efficient and climate friendly equipment have been highlighted by the IEA, SEforALL and others. According to the NGO CLASP (Collaborative Labelling and Appliance Standards), consumer affordability is the most significant constraint for growth of demand for off-grid appliances. On the demand side, consumers cannot afford the retail price of an off-grid appliance without upfront financing. On the supply side, producers and retailers struggle to lower the retail price due to high distribution costs. Access to finance for consumers, product innovators, and distributors will be critical to achieving scale

In this context, the Facility will support activities that address and help remove regulatory/policy-related, institutional, market and behavioral barriers to the development of sustainable cooling investments by (i) providing technical assistance and building the capacity of state and non-state actors (such as commercial banks, private investors and technical companies, end user beneficiaries), (ii) supporting the design of sustainable implementation mechanisms and business models; (iii) increasing end user awareness of the benefits of low-carbon cooling solutions, (iv) facilitating access to affordable sources of financing; and v)

³⁵ The Africa Energy Access Leap is a program proposing a WBG global effort to help reach universal access to modern energy, aiming to achieve SDG7 ("Access to affordable, reliable, sustainable and modern energy for all") by 2030. The proposal is currently under development.

provide credit lines to financial intermediaries. In turn, these activities will strengthen the capacity of key stakeholders, stimulate the demand for climate-friendly cooling solutions, foster the penetration of climate-friendly equipment and infrastructure and mobilize capital from public and private sources. These interventions will strengthen institutional and regulatory systems, improve access to cooling, improve the adaptive capacity of end user beneficiaries to respond to rising temperatures, as well as the energy efficiency of cooling buildings and cooling infrastructure. Ultimately, the climate change resilience of built infrastructure will increase, while the carbon intensity of buildings, cold chains and cooling appliances decreases. The Facility seeks to provide pioneering examples of countries, cooling sectors, approaches and technologies and measures that can be scaled up including in other countries, beyond the Facility, that are facing similar situations and opportunities. It is indeed hoped that other countries may emulate or adapt the Facility's experience and lessons to their own circumstances to enhance access to cooling services which is critical to building resilience in a warming world while breaking the vicious circle of rapidly increasing emissions that is projected to be associated with meeting cooling needs. At a time when there are many pressing demands on developing country governments' priorities and budgets, including responding to the global COVID-19 pandemic that has left no country untouched, the Facility opts for a pragmatic approach, seeking to take advantage of available entry points that build on synergies between sustainable cooling and other sector and development priorities to trigger the changes that need to be put in place and lead, over time, to broader mainstreaming of sustainable cooling among the priorities of target countries. Building on these synergies between priorities, identifying the counterpart(s) in target countries that can become champions of sustainable cooling and the ability to stimulate interest and generate results to build confidence in sustainable cooling solutions will be key factors to ensure success. The Program is designed in full awareness of the barriers and particular risks that may affect the successful implementation of the Facility's project, including (i) hard to predict disruptions due to the COVID-19 pandemic and its impact; (ii) as well as political instabilities, and (iii) risks associated with potential operational delays, and (iv) end-users not responding as anticipated to information and interventions highlighting the benefits of energy efficient and climate friendly cooling.

Figure 1: Cooling Facility Theory of Change



B.3. Programme description (max. 2000 words, approximately 4 pages)

I. Objectives and goals

The World Bank's proposed Cooling Facility is a multi-sector, multi-country financing mechanism as part of its efforts to support countries to mainstream and bring to scale sustainable cooling solutions across key sectors. It aims to support the eligible countries to:

- Mobilize additional financing to support transition and market transformation in support of the most climate friendly means of meeting cooling needs;
- Enhance access to energy efficient and climate friendly cooling services;
- Strengthen enabling environments to catalyze a demand for sustainable cooling solutions;
- Raise public awareness of the benefits, financial viabilities and positive impacts that result from investments and public programs to promote sustainable cooling;
- Build and strengthen skills and capacity among the range of stakeholders involved in the deployment and scale-up of sustainable cooling solutions; and
- Integrate and mainstream cooling into both long-term climate planning for mitigation and adaptation and Sendai framework disaster risk reduction (DRR)³⁶ planning, and support the achievement of the Kigali Amendment Goals.

In doing so, the Facility will (i) increase access to cooling, contributing to multiple Sustainable Development Goals; (ii) mitigate the climate impacts of meeting the projected increased cooling demand; (iii) promote interest and support for innovative means of providing cooling services; and also (iv) contribute to improving climate change resilience, contributing to countries' Nationally Determined Contributions (NDCs), as well as National Adaptation Plans and Disaster Risk Reduction Plans, while also delivering a wide range of co-benefits including those related to health, economic impacts, and development of nature-based solutions (NBS). Recognizing the magnitude of financial needs, the proposed Facility is the first phase, which will expand to be the basis for substantial scaling and replication to follow.

To achieve this goal, the Facility will channel concessional climate finance from the GCF to co-finance IBRD- and IDA-financed operations that:

- support **investments** in efficient, affordable and sustainable cooling;
- catalyze technology, financial and business model **innovations** across cooling value chains and sectors to reduce barriers for private sector investment;
- help build an **enabling environment** for sustained, long-term systemic changes.
- Support the integration of sustainable cooling into long-term plans to address climate change, DRR and SDGs.

At country level, the Facility development objectives will be adapted to reflect the specific needs and priorities of countries and thereby to align the objectives and impacts of the investments.

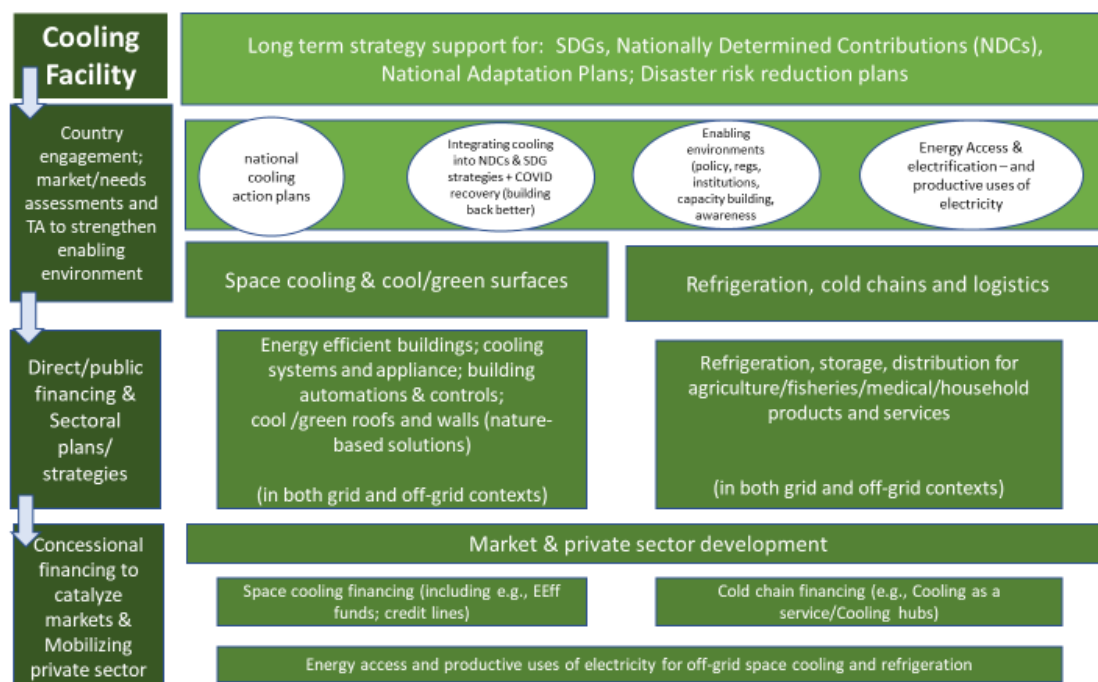
Figure 2 provides an overview of the Cooling Facility's overall framework, including its goals to contribute to the transformation of the cooling sector through policy, regulatory and enabling environment support that would help mainstream sustainable cooling into countries' overall development and climate change strategies, its key cooling target areas, (i.e., (i) space cooling³⁷ and cool/green surfaces; as well as (ii) refrigeration, cold

³⁶ The Sendai Framework for Disaster Risk Reduction 2015-2030 provides Member States with concrete actions to protect development gains from the risk of disaster, including heatwaves.

³⁷ "Space cooling", also referred to as "comfort cooling", refers to the means by which people are provided thermal comfort from heat by maintaining the optimum temperature, humidity, and ventilation within the built environment (See ESMAP 2020, *Primer on Space Cooling*). See below for more details.

chains and logistics, along with its proposed approach enabled by GCF concessional climate finance. It is important to note that the Facility's two cooling target areas could be implemented in on-grid as well as off-grid contexts.³⁸

Figure 2: Overview of the Cooling Facility



Space Cooling and cool/green solutions. “Space cooling³⁹” refers to the means by which people are provided thermal comfort from heat by maintaining the optimum temperature, humidity, and ventilation within the built environment. Space Cooling includes (i) passive cooling strategies and measures to reduce or avoid building cooling loads (energy efficient building construction and design, including building thermal insulation and ventilation); (ii) equipment used to achieved required amount of cooling (e.g. fans, air conditioning, and centralized cooling systems); and (iii) building automation, controls and good operation and maintenance practices. Cool surfaces⁴⁰ are solar reflective roofs and walls; while green surfaces refer to vegetative roofs and walls (also referred to as nature-based solutions).

Space cooling projects may (i) improve the thermal efficiency of buildings (to reduce or avoid cooling loads), (ii) improve the energy efficiency of cooling equipment and appliance, (iii) use low GWP refrigerants available in the local market, in line with the goals and requirements of the Kigali Amendment; and may also (iv) further lower the emissions associated with cooling with the installation of clean energy (mainly solar) supply.

Space cooling projects may target greenfield investments or refurbishments in different sectors, i.e., public (including health facilities, education, and administrative buildings), residential- (multi-family and single family), commercial or industrial.

³⁸ For example, off grid cooling appliances could be deployed in the context of broader energy access programs, e.g. the deployment of efficient off-grid cooling technologies (refrigeration and cooling, including fans) in homes, farms, rural schools and health facilities. Such deployment may be in the context of productive uses of electricity, which are typically diverse applications across agricultural, commercial, industrial and public sectors as a direct input to the production of goods or provision of services that enable or enhance income generation.

³⁹ See ESMAP 2020 *Space Cooling Primer* (<https://www.esmap.org/primer-for-space-cooling-report>)

⁴⁰ See ESMAP and Global Platform for Sustainable Cities 2020 *Primer for Cool Cities: Reducing Excessive Urban Heat* (<https://www.esmap.org/primer-for-cool-citiesreducing-excessive-urban-heat>)

Refrigeration, cold chain and logistics

Refrigeration and cold chain refer to managing the temperature (at specified low-temperature range) of perishable products in order to maintain quality and safety from the point of origin through the distribution chain to the final consumer⁴¹. Refrigeration has many applications, including household refrigerators, industrial freezers. A cold chain includes refrigeration, storage and distribution activities, along with associated equipment and logistics. The projects may target energy efficient refrigeration equipment and facilities in residential, public, commercial and industrial sectors.

For the health cold chain, projects may target the different components of cold chains, i.e., national and regional facilities and cold stores, as well as district-level stores and local clinics and vaccination outreach. The projects may include efficient, clean built infrastructure (e.g., walk-in refrigerators and freezers) and cold storage equipment (e.g., refrigerators and freezers, as well as passive cooling solutions).

Similarly, for food cold chains (e.g., in agriculture and fisheries sectors), projects may include efficient, clean cold chain equipment linked efficient, clean build infrastructures (e.g. refrigerated warehouses, hubs, and packhouses) and cold storage equipment (e.g., refrigerators and freezers).

Projects may include the solarization of the cold chain (built infrastructure and cold storage equipment). Project may also include digital devices and to monitor and track temperature and performance of the equipment, as well as logistics planning systems to enhance overall efficiency of the cold chain. An eligible project under the Cooling Facility may involve the replacement of existing equipment or refurbishment of existing facilities or target new equipment and facilities.

II. Program components

The Cooling Facility is composed of three complementary components that will be adapted to each eligible country's context and focus cooling area, and will be calibrated according to its needs and in accordance to the Facility's criteria. The Facility's components are structured to help countries address the barriers to sustainable cooling by (i) helping address the upfront cost and financing barriers, along with (ii) supporting measures that help build an enabling foundation to enable and sustain sustainable cooling through activities such as informing policy and regulatory frameworks, reaching out to stakeholders through effective communication and supporting training and capacity building. (the last component – not funded with GCF resources is for project management – essential for effective implementation). Subject to receiving World Bank corporate approval in due time, the projects supported by the Facility will be co-financed by IBRD/IDA. The indicative total co-financing amounts to \$722.8 million.

Component 1 - Policy, regulatory and enabling environment support⁴² (US\$15.7 million in GCF grant).

This component will support activities that aim to strengthen institutional, policy and regulatory frameworks, support program design and roll-out, raise awareness and stimulate behavior changes, as well as build the capacity of key stakeholders. These are critical to help address some of the key barriers and market failures to a systematic shift towards sustainable cooling. Indeed, support provided through Component 2 seeks to support countries address fundamental market and non-market barriers to sustainable cooling, including lack of awareness, lack of transparency (about the cost savings and benefits of energy efficient buildings, cooling and refrigeration appliances and equipment, as well as nature-based solutions and energy efficient cold chains), the first-cost bias, mis-aligned policies and incentives and the capacity gaps. A key feature will be support for developing an enabling environment that can support the effective implementation of sustainable cooling investments, mainstream sustainable cooling policies and actions, and sustain the development of regional, national or local industry and services in the relevant sectors, while ensuring alignment with the Kigali Amendment.

⁴¹ Global Cold Chain Alliance website: <https://www.gcca.org/about/about-cold-chain>

⁴² ESMAP Efficient Clean Cooling technical assistance program would finance upstream activities and ultimately help identify potential investment opportunities, including for the Facility.

Activities relevant to the eligible cooling areas supported under Component 1 may include, for instance, (i) capacity building of key stakeholders (for example, on technologies; business models; cooling project appraisal, implementation and evaluation); (ii) technical assistance for enabling plans, strategies, policies, regulations, standards; (iii) designing sustainable market mechanisms and business models; and (iv) communications and awareness campaigns.

Component 2 - Financing for cooling investments (US\$141.3 million of GCF funds, of which \$125 million in loan and \$16.3 million in grant). In order to promote and accelerate the uptake of investments in innovative and climate friendly cooling technologies and systems, including in markets with low penetration and unmet cooling demand, the Cooling Facility aims to support increased access and deployment of sustainable cooling solutions performing better than current market practice and at the same time, ensure that the level of ambition is appropriate and relevant to each market (i.e. affordable) to expand access to – and deployment of – more climate-friendly cooling at scale (and avoid promoting a prohibitively expensive market niche, which would hinder the goals of scaling-up access and deployment). The Facility will provide concessional funds to foster the adoption or scale-up the deployment of sustainable cooling technologies, appliances through (i) financing of climate friendly cooling investment and (ii) credit lines to financial intermediaries. The financial instruments will be calibrated to what is needed to make the cooling project viable and to overcome barriers to implementation, including addressing the incremental cost associated with the adoption of efficient cooling solutions or serve remote off-grid areas, which include higher upfront capital costs compared to baseline options, and market entry barriers arising from efficient, clean cooling solutions underrepresentation in many developing country regions and sectors. The deployment of these funds will be tailored to the needs of the projects, through various financing modalities, for example, public financing to sovereign or sub-sovereign entities credit lines to commercial banks, loans or subsidies to households, municipalities and small, medium and large enterprises.

High concessional loans and grant from the GCF will largely be targeted toward investment projects where upfront affordability is a major obstacle and the quantity or concessionality of the World Bank's IDA/IBRD resources⁴³ are insufficient to ensure affordability and/or to make the efficient, clean cooling project viable.

Different financing and implementation mechanisms and models – suitable for different cooling sectors and different countries may be deployed by different projects of the Facility based on specific country and sector conditions. Examples of these financing and implementation mechanisms and models are provided below (Section B.4).

Component 3 – Project management (US\$ 0 in GCF funds). This component will support the Executing Entities (EEs)' and Project Implementing Entities (PIEs)' management, coordination and implementation of their project activities, in compliance with the EEs' and PIEs' contractual obligations included or referred to in the legal agreements entered into with the Accredited Entities (AE).

III. General approach for activities supported by the Cooling Facility

Integrated and holistic approach: The scale of the cooling challenge and the window of opportunity to get now on a path to affordable, efficient, clean cooling demand an approach that integrates the development and strengthening of an enabling environment. Doing this involves technical assistance, capacity building, and concessional financing to mainstream sustainable cooling and establish sustainable implementation and business models. Moreover, in line with the Facility's goals of supporting access to affordable, efficient, clean cooling in a warming world, the Facility will integrate efficiency (starting with passive measures) and low carbon

⁴³ The World Bank offers IBRD and concessional IDA lending. The lending terms can be found here (<https://treasury.worldbank.org/en/about/unit/treasury/ibrd-financial-products/lending-rates-and-fees>) for IBRD and here (<https://treasury.worldbank.org/en/about/unit/treasury/ida-financial-products/lending-rates-and-fees>) for IDA.

considerations and adopt a holistic approach to sustainable cooling consisting of one or more of the following elements⁴⁴:

- **Minimize** - Reduce and avoid cooling loads, for example, with passive building design, increase of thermal efficiency of buildings and cold storage warehouses/facilities, and nature-based solutions such as green roofs.
- **Shift to cooling solutions with lowest GHG emissions**, for example solar cooling, climate-friendly (low GWP) refrigerants-based cooling systems, or centralized cooling solutions where the right enabling conditions are in place.
- **Improve efficiency** – use energy efficient technologies to achieve cooling with less energy (and less GHG emissions), for example, super-efficient appliances and equipment and demand management to reduce peak demand and the need for new generating power capacity.
- **Optimize** – control cooling loads, for example, smart sensing and behavior change approaches.
- **Protect the most vulnerable** in urban and rural areas from extreme heat and consequences of unreliable or insufficient medical and agriculture cold chains.

National and sectoral contexts: The Cooling Facility recognizes that the technologies, opportunities and modalities to enhance access to efficient, clean cooling depend on the sector of application (e.g. space cooling vs agriculture cold chains), as well as specific national – and sub-national (e.g. urban or rural) circumstances, and capabilities. As such, assessments of potential cooling-informed activities should consider, as relevant, country-specific context and technology benchmarks (including those derived from regional benchmarks).

Link with climate change and cooling policy goals: Projects supported by the Cooling Facility should be aligned with, and contribute to the respective country's NDC revisions under the Paris Climate Change Agreement, and Cooling Action Plan (where available), as well as support and/or strengthen as appropriate, NAMAs, National Adaptation Plans and Disaster Risk Reduction Plans and support the goals and requirements of the Kigali Amendment to the Montreal Protocol.

Link with gender equality objectives: The Cooling Facility will seek to contribute to enhancing the knowledge base on gender issues associated with access to cooling and will seek to contribute to reducing gender gaps in cooling-related sectors (including agriculture, buildings, health and energy) by (i) identifying gaps relevant to the four pillars of the World Bank Group (WBG) Gender Strategy⁴⁵, and (ii) aiming to address these gaps through specific actions supported by the project.

Link with countries' COVID-19 response and 'build better' efforts - The Cooling Facility may also support efficient cooling-related interventions in the context of the Bank's overall COVID-19 response and related operations. As an example, the Cooling Facility may support the deployment of efficient, climate friendly cold chains for COVID-19 and routine immunization vaccines as a component of projects seeking to strengthening the countries' overall health services. This could include:

- strengthening the health sector cold chain by financing the gap at service delivery points at national, regional, district and sub-district levels;
- energy efficient and climate friendly built infrastructure, including vaccine stores, walk-in cold rooms, walk-in freezers, etc;
- efficient and climate friendly cooling equipment (including refrigerators and freezers) with lower GWP refrigerants;
- preventive maintenance and as-needed repairs of walk-in cold/freezer rooms at national and sub-national levels;

⁴⁴ This approach is consistent with the approach promoted by the Cool Coalition (<https://coolcoalition.org/about/overview/>).

⁴⁵ The four pillars: (i) improving human endowments, (ii) removing constraints for more and better jobs, (iii) finding solutions that help increase women's ownership of and control over assets, and (iv) enhancing women's voice and agency and engaging men and boys.

- strengthening vaccine logistics systems and energy efficiency upgrades of ventilation and refrigeration systems in cold storage hubs for vaccines and medicines;
- Solarization of health and agricultural cold storage facilities.

Nine countries are currently considering potential investments to be supported from the Facility. The World Bank will work with each Executing Entity covered under this Facility to finalize the scope and co-financing of these investment opportunities, taking into account (i) the needs of the clients as they may evolve, (ii) the results of the ongoing market assessments, and (iii) stakeholder consultations.

IV. Eligibility criteria for Facility-supported projects

The Cooling Facility will allocate a maximum amount equivalent to an aggregate of 30% of the total GCF proceeds to any given Host Country, regardless of the financial instrument(s) deployed. Supported projects will be assessed by the Bank as meeting the following eligibility criteria. A market assessment may be conducted prior to appraisal to inform the design of the project and demonstrate how it meets the criteria to access funds from the Cooling Facility.

- **Cooling investments.** Projects selected under the Cooling Facility will support investments in the following cooling target areas: (i) space cooling and cool/green solutions; and (ii) refrigeration, cold chain and logistics, as defined in section B3.I. above.

In the case of (i) space cooling and cool/green solutions, projects should demonstrate a pathway towards financially sustainable models that reduce the need for public resources over time and eventually, leverage commercial and private sector funding.

In the case of Facility-supported projects financing investments in off-grid cold storage capacity additions (under (ii) refrigeration, cold storage), Executing Entities will need to demonstrate that the main source of energy to power these additional cold storage facilities is renewable energy (e.g. solar).

- **World Bank operation linkage.** A Cooling Facility-supported project may: (i) co-finance a new IDA or IBRD⁴⁶ operation; or (ii) be an additional financing to an existing IDA or IBRD operation. The ratio of GCF funds to IDA/IBRD funds for each supported project will be at least 1:2.
- **Performance threshold.** To receive support from the Facility the proposed project will meet the following performance thresholds:
 - At least 20% energy efficiency improvement; and/or
 - At least 20% avoided/reduced GHG.

To that effect, for each project, an estimate of GHG impact should be developed based on publicly available GHG methodologies and calculation tools and consistent with guidelines outlined in Annex 24⁴⁷.
- **Alignment with the transition to low GWP refrigerants** - Facility supported projects will be aligned with the goals and requirements of the Kigali Amendment to the Montreal Protocol, under which HFCs are being phased down and replaced with climate friendly refrigerants that do less damage to the climate. As such, investments supported by the facility will promote the use of climate friendly cooling equipment/appliances commercially available for the cooling applications

⁴⁶ Together IBRD and IDA form the World Bank. IDA focuses on the world's poorest countries, while IBRD assists middle-income and credit-worthy poorer countries.

⁴⁷ Annex 24 provides calculations of estimated GHG emissions for each Facility project.

considered in the respective project that either (i) provide cooling services without the use of refrigerants (e.g. fans and passive cooling solutions); and/or (ii) use refrigerants that are aligned with the Kigali amendment and meet relevant safety standards, as commercially available in respective local markets⁴⁸.

- **Adaptation to climate change.** The Host Country will be vulnerable to climate change, facing extreme heat risk hazards, and in need of cooling services. The Cooling Facility-supported projects will bring cooling-related benefits associated with services provided by the buildings, cold chains, health facilities and other related cooling investments.
- **Paradigm shift potential.** Supported projects will propose activities/ technology/applications that aim to achieve market transformation, create new markets and business activities at local, national or international levels, and can potentially be replicated in other sectors or geographic areas.
- **Country ownership & alignment with country priorities.** Facility-supported projects will be aligned with country priorities and climate change/cooling action plans as presented in national development plans or other policy documents.
- **GCF Concessionality Level.** The concessionality levels would be assessed based on the pricing of the WB co-financing, risk of external debt distress of the Host Country as published by the IMF under joint WB-IMF debt sustainability framework, and the financial analysis of the projects. In doing so, it would apply the following principles:
 - (a) *Grant only.* Countries in external debt distress and for which the WB co-financing is exclusively grant-based, would receive 100% grant from the Facility.
 - (b) *Combination of high concessional loans and grants.* Projects would meet any of the three criteria below:
 - Host country in high risk of debt distress;
 - Non-revenue generating project or where the reflows from the ultimate beneficiaries are not cost-reflective;
 - Projects that target vulnerable communities or groups that are more vulnerable than the overall population (for example, refugees, people in rural areas with low economic income).
 - (c) *Financing terms:* 40-year maturity, 10-year grace period with an all-in fee of 0.25 percent (i.e., 0.00 percent interest rate, 0.00 percent commitment fee, 0.25 percent service fee). *Low concessional loan.* Projects not meeting the conditions set out in (a) or (b) above would be eligible for low concessional loans for investment.

Financing terms: 20-year maturity, 5-year grace period, 0.75 percent interest rate, 0.5 percent service fee, 0.75 percent commitment fee.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

The Cooling Facility is essentially designed as an umbrella facility (a Program) that enables aggregating, through a programmatic approach, different types and sizes of sustainable cooling interventions. This will enable to lower transaction costs, reach a broader range of countries (of different sizes and income levels), and generate the necessary experience and lessons to underpin further scaling-up of sustainable cooling more broadly and the needed transformation of the cooling sectors. In fact, without the possibility of aggregating

⁴⁸ This criteria takes into account that very low GWP refrigerant alternatives may not be readily available in all local markets for all the applications considered in the project

through the Facility's programmatic approach to sustainable cooling, it would not be possible to reach the diverse set of countries covered by the Facility, especially smaller and more fragile countries. Moreover, the programmatic approach is essential to take advantage of opportunities to introduce and support climate friendly cooling solutions in the context of different sectoral operations - that may otherwise be overlooked.

The Cooling Facility currently proposes to support nine projects (one project per country) subject to the results of their appraisal. This Funding Proposal includes indicative information for each of the currently proposed projects. Under the programmatic approach, the GCF would rely on the AE's due diligence on the proposed projects included in the Program. Subject to the outcome of such due diligence, the World Bank will approve funding for such projects that meet the eligibility criteria as presented in Section B.3.

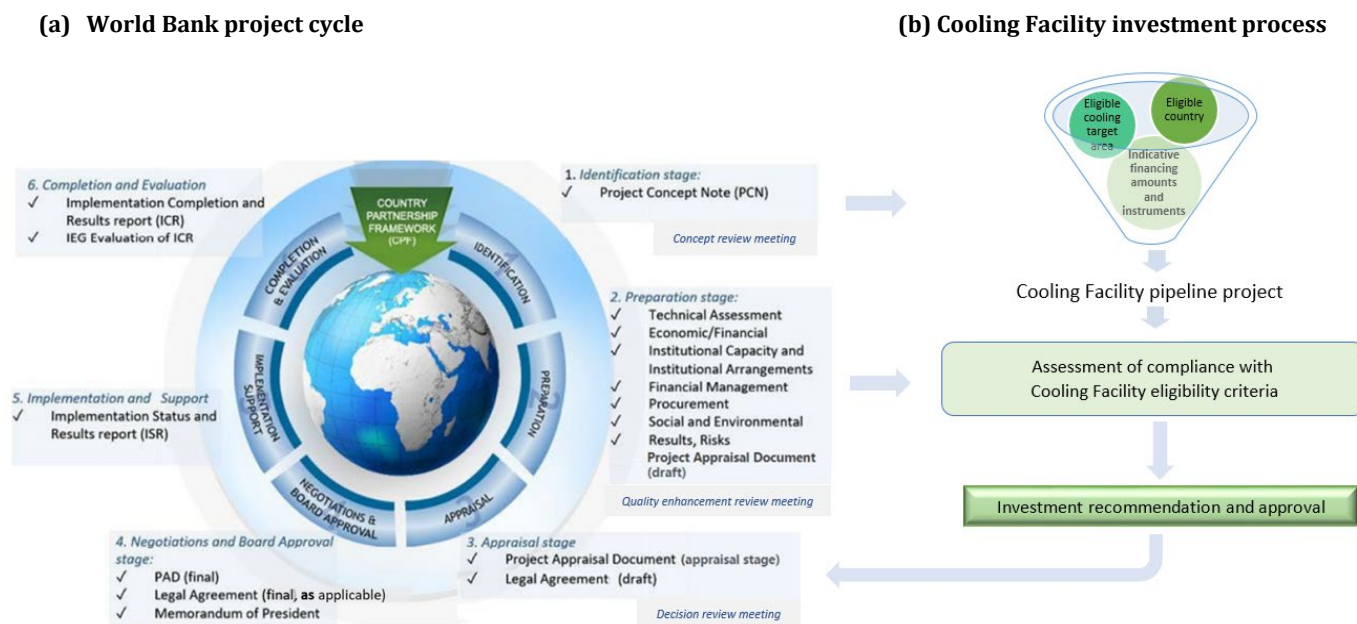
I. Facility-level arrangements

As part of its role as GCF AE and co-financier of the projects to be supported under the Facility, the World Bank will assess and confirm the eligibility of each such Project to be included in the Facility's pipeline for Bank and GCF Financing. In order to do so, for each project proposed to be included in the Facility, the World Bank team will comprise energy and climate finance experts to manage and be responsible for the Facility's overall objectives and ensure that the projects selected meet all the eligibility criteria as presented in Section B3. The team will aggregate findings from each project team's implementation progress reports and facilitate the dissemination of lessons learned across Facility projects and beyond the Facility.

Following the GCF Board approval, the World Bank and the GCF will, based on the Accreditation Master Agreement (AMA), enter into one Funded Activity Agreement (FAA) for the Facility for the provision of the funds under all GCF instruments supporting the Program. The World Bank will be solely responsible for the management and administration of GCF resources and will carry out such management and administration in accordance with its policies, procedures and practices, and the relevant provisions of the FAA and AMA. The World Bank will apply its own fiduciary principles and standards relating to any integrity checks, anti-corruption, countering of financing of terrorism, fraud, financial sanctions, embargoes and anti-money laundering.

The process for reviewing and approving GCF funding for the Facility-supported projects is embedded in the WB's project cycle (Figure 3) and based on its standard policies and procedures.

Figure 3: Project review and investment approval process



During the identification stage (stage 1), candidate-projects are screened for formal inclusion in the Facility pipeline. This preliminary review focuses on the proposed use of the GCF funds, the scope of activities, the paradigm shift potential and the financing instruments envisaged. The WB project teams then carry out a technical, economic, financial, environmental and social, fiduciary, procurement appraisal of the project, in line with WB's operational policies and procedures, and prepare a draft Project Appraisal Document (PAD) or equivalent for a technical review meeting (stage 2). This due diligence is separate from the responsibility of the recipient countries to prepare and implement the projects according to WB's policies and procedures. The scope of the technical review will cover the assessment of the GCF financing against the eligibility criteria for the Facility-funded projects. Upon determination by the Bank that the projects meet the eligibility criteria, the WB project team will prepare the appraisal-stage project documents and the draft legal agreements (stage 3). The next stages (negotiations, WB institutional approval, project implementation through completion and evaluation) will follow the WB's applicable rules and procedures. Following approval of projects, the World Bank will sign Subsidiary Agreements with the Executing Entities to make available GCF investment financing in line with the provisions of the FAA.

Relevant experience and track record of the World Bank. The proposed facility builds on the WBG's decades of experience in supporting client countries across the globe implement clean energy projects. Over the past dozen years (FY10-21), the Bank has built a strong track record in financing clean energy, providing support to developing countries across the globe with a clean energy portfolio of 520 operations amounting to US\$33.9 billion (approximately US\$3 billion per year), helping countries achieve their energy and climate (NDCs) goals as well as SDG goals. The renewable energy share of the portfolio amounts to about US\$1.6 billion per year, and the share of solar PV - the most likely source of RE to be associated with sustainable cooling activities – is growing fast. The Bank's EEff portfolio of US\$ 16.5 billion (about US\$1.5 billion per year). The Bank's EEff portfolio alone has saved more than 1,300 TWh to date, associated with 575 Mt of avoided CO2 emissions. Demand-side EEff accounted for about a third of the total FY-10-21 World Bank EEff portfolio and covers different sectors (public, industrial, and residential), implemented primarily through the Bank's Energy and Extractive Global Practice, but with an increasing share with other WB Global Practices, including Urban, Water, Transport and Agriculture. Demand side energy efficiency interventions have also

been combined with other energy sector focus, such as energy access, and have been combined with renewable energy investments. This EEff experience has included interventions related to cooling, such as building energy efficiency projects and energy efficient appliance programs.

The WBG has an extensive history of EEff projects aimed at overcoming the market failures and barriers to scaling-up energy efficiency which include weak policy/regulatory frameworks, higher first cost, lack of access to affordable financing, new technology risk, lack of credit worthiness, transaction/aggregation costs (for financiers) and risk of financing low-income consumers. The World Bank has helped deploy a menu of effective clean energy financing and delivery mechanisms across sectors and across the developing world, including dedicated credit lines, financing facilities, results-based schemes and others. Projects have encompassed both public and private financial institutions. The design of these mechanisms has evolved, building on lessons learned and in response to local needs and circumstances; and they have achieved considerable success as documented in multiple evaluation reports.

The Cooling Facility builds on this deep operational foundation, combined with the Bank's traditional energy – and sectoral - dialogues with client countries. It also builds on the Bank's experience as one of four Implementing Agencies of the Multilateral Fund of the Montreal Protocol, supporting the transition towards climate friendly refrigerants. The Facility will also benefit from the Bank's extensive track record in providing energy efficiency and clean energy technical assistance – principally through ESMAP -, to support the upstream design of lending operation, setting up the regulatory, policy and institutional foundations to help mobilize financing and catalyze private sector investments. In particular, ESMAP's Efficient, Clean Cooling technical assistance program (established under ESMAP) which is assisting the Bank in developing a new business line for sustainable cooling –will be leveraged by the Facility. Already, the Bank has gained experience with the inclusion of sustainable cooling activities in the urban, agriculture, transportation, and health sectors.

II. Project Management Arrangement

For each project, a project management unit will be established by the relevant Executing Entity with a team expected to include a team lead, technical expert, procurement specialist, financial management specialist, environmental specialist and social/gender specialist, and a lawyer. Each project will have its own monitoring and evaluation (M&E) and reporting process.

Figure 4: Typical project management unit key personnel



Project-level implementation arrangements are defined during the appraisal phase and vary according to the design of the project and are adapted to the country-specific context.

Cooling cuts across different sectors (public, residential, commercial and industrial), and targets different areas (space cooling; off-grid appliances; and refrigeration/cold chains). In this context and given the different circumstances in individual countries, there is no single, one-size-fits-all implementation model. In addition, the Bank's experience with demand-side energy efficiency and off-grid energy access projects in particular provide valuable insights that have been considered for implementation. For these reasons, the Cooling Facility will provide individual projects the opportunity to adopt the implementation model that suits their own

situation best among those described below, and in doing so enhance the prospects for successful implementation – and inform a sustainable cooling transformation in other countries.

III. Indicative countries' implementation arrangements

The Facility will support one project per country and thus, a total of nine projects. Each project comprises activities from all three components of the Facility and may include one or more implementing models as shown in Table 2 below. Several models may be considered for projects at their current early stage of development. The actual implementation models/arrangements will be finalized during the preparation stage of each project depending on the specific situation at project preparation and negotiations with the government. It is expected that all Projects shall be approved by the World Bank within five years of the GCF Board approval of the Funding Proposal.

Table 2: Indicative implementing model by country

Country	Target cooling area	Implementation models				
		Component 1: Policy, regulation and enabling environment support	Component 2: Financing for cooling investments support			Component 3: Project management
		PP&C	PP&C	CL	EEFF	PP&C
Bangladesh	S	✓	✓	✓	✓	✓
El Salvador	S,R	✓	✓			✓
Kenya	R	✓			✓	✓
Malawi	R	✓		✓	✓	✓
North Macedonia	S	✓	✓		✓	✓
Panama	S	✓		✓		✓
Sao Tome and Principe	S,R	✓	✓			✓
Sri Lanka	S	✓	✓	✓	✓	✓
Somalia	S,R	✓	✓			✓

Notes

S: Space cooling & cool/green surfaces
R: Refrigeration, cold chains and logistics

PP&C: Public procurement and contracting
CL: Credit lines
EEFF: Energy efficiency financing facility including the public sector ESCO and the revolving facility variation

IV. Implementation models description

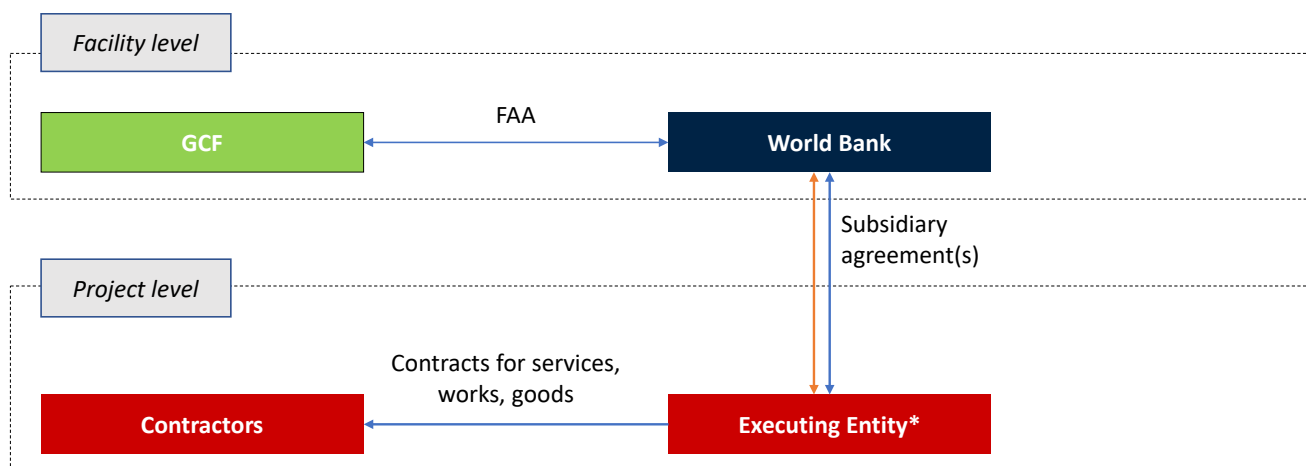
This section describes the implementation models and arrangements that would be followed by the projects supported by the Cooling Facility.

(i) Public procurement and contracting for cooling infrastructure service delivery and technical assistance⁴⁹

GCF funds would be used to support investments through a traditional public procurement and contracting model. Under this model, the EE (or the Project Implementing Entity (PIE) as designated by the EE) would need to possess the necessary capacities to conduct the required activities consistent with relevant Bank procedures (see below). This may be in the form, for example, of a dedicated unit within a government ministry or designated national agency such as a rural electrification agency.

The EE (or PIE as applicable) would: (i) prepare the technical specifications and terms of reference of the cooling infrastructure or equipment it wishes to procure; (ii) subject to no-objection of the World Bank, launch the procurement process to invite bids from suitable contractors; (iii) award the procurement contract; (iv) sign contractual agreements with the winning bidders; (iv) monitor the delivery of the goods and services to ensure quality control using its own resources or through an owner's engineer procured by the project, and (v) remunerate the contractor according to the agreed terms. Depending on its capacity and operating model, the EE (or PIE as applicable), may subsequently enter into an operations and maintenance (O&M) contract to ensure the sustainability of the infrastructure. (Figure 5). Otherwise, the EE would ensure that the beneficiaries receive adequate training to conduct O&M to ensure continuous operation of the equipment and infrastructure and that the impact is achieved as per estimates.

Figure 5: Implementation arrangement for public procurement and contracting model



*If a project has a Project Implementing Entity (PIE), there may be additional arrangements including a project agreement between the World Bank and such PIE

This implementation model is well-suited where the private sector does not have the capacity or the appetite to implement the cooling activities and thus, public intervention is critical for service-delivery. Such cases include:

- Interventions that strengthen the enabling environment by establishing the conducive policy, legal, regulatory and economic frameworks, reduce barriers to investment, improve access to knowledge, and strengthen institutional capacities.

⁴⁹ This model is primarily relevant for both Component 1 and Component 3.

- Despite their economic viability, interventions that are not commercially viable to adequately remunerate⁵⁰ private investors (for example, dissemination of off-grid appliances in poor and marginalized communities with limited economic opportunities and low capacity to pay).
- Interventions that promote the deployment of a public good (for example, refrigeration and cold chain for vaccines and medicines such as in El Salvador, Sao Tome and Principe and Somalia).
- Pilot interventions that aim to demonstrate the benefits of energy efficient cooling options to improve energy services and energy access and generate lessons for a replication at scale.

GCF concessional funds provided through loan or grant agreements with the Government would help reduce the systemic risk of undertaking cooling investment and reduce transaction cost by establishing conducive investment, legal and policy frameworks. Further, it would promote social equity and protection by ensuring that the projects benefit poor and marginalized communities. To this end, GCF funding will be priced lower than and blended with WB funds, to reduce the overall financing cost and thereby absorbing the incremental cost to deploy efficient cooling solutions. Final beneficiaries could include households, sole traders, micro-, small- and medium- enterprises, communities. Through these interventions, GCF concessional funds would play a catalytic role in creating a market for and increasing access to clean, efficient and sustainable cooling.

For the purposes of the FAA, this model may be used for the activities under Component 1, financing of climate-friendly cooling investments under Component 2 and project management activities under Component 3.

(ii) Credit lines

A key barrier to investing in energy efficiency in general – and sustainable cooling in particular – is the limited availability of commercial financing for these projects. The Facility will support the provision of credit lines,⁵¹ for example in Malawi, to help financial institutions establish business lines for sustainable cooling investments. Credit lines can be well-suited in less mature markets; they help mitigate the perceived high financial risk of energy efficient and clean cooling, which may be carried out by energy service companies (ESCOs). An ESCO is an organization that provides a range of services to energy users to design and implement cooling and energy efficiency measures. The services may also include providing or arranging financing. Such services are provided by the ESCO using a performance-based contract under which the payments from the energy user to the ESCO are contingent upon achieving certain pre-specified performance levels. Also, the ESCO generally allows energy users to pay for its services using the energy cost savings that result from the cooling/energy efficiency project. This approach allows energy users to transfer much of a project's technical, construction, and performance risks to the ESCO. The concept of energy savings performance contracts (ESPCs) implemented by ESCOs has been recognized as a promising approach to overcome some of the most intractable market barriers. Under an ESPC, an ESCO identifies and selects, develops, implements, and finances a cooling and energy efficiency project (at the customer level), and uses the stream of income from the cost savings to repay the costs of the project, including the costs of the investment).

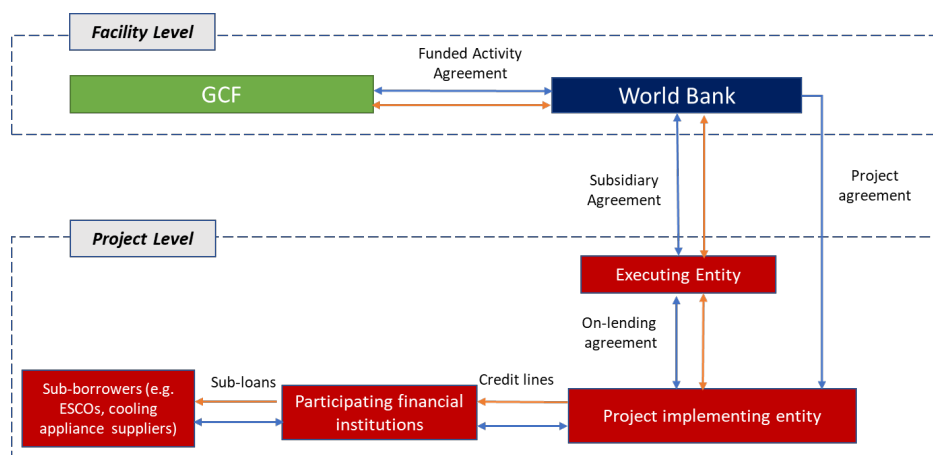
The WB would channel GCF finance along with World Bank co-financing to the EE (the Host Country as the Borrower/Recipient), through a Subsidiary Agreement. The EE would extend the credit line to one or more participating financial institutions (PFIs), which would typically be banks or micro-lending institutions, for on-

⁵⁰ At the minimum, investment rates of return should cover the cost of debt incurred by private actors, and the cost of equity. While the cost of debt can be easily determined as the cost at which commercial banks would lend to cooling projects, the cost of equity depends on the financial robustness of the prospective investor and its risk appetite (investors with an international footprint and a well-diversified portfolio would tend to be less risk averse than local investors). As a general rule, it would be higher than the rate that can be earned through a passive investment (for example, a AAA-rate sovereign bond) plus premia for the additional risks incurred (for example, investments with longer payback periods will command and higher maturity premium).

⁵¹ See for example, World Bank-ESMAP livewire (2014) "Designing Credit Lines for Energy Efficiency", by Ashok Sarkar, Jonathan Sinton and Joeri de Wit. (<https://openknowledge.worldbank.org/handle/10986/18410>)

lending to eligible sub-borrowers (for example, industry and other private sector entities). In doing so, the PFIs will select and approve sub-loan applications based on the agreed investment criteria. Subsidiary Agreements would require the EE to ensure that the PFIs comply with the sub-lending and other terms, and monitor the performance of the PFIs. As such, the EE has contractual control and oversight responsibility over the implementation of the Funded Activity. Repayment occurs through the same channels followed to disburse the credit line. Where the EE designates a Project Implementing Entity (PIE) to provide credit lines to the PFIs, the World Bank would require the EE to enter into an on-lending agreement with the PIE. In addition, it may enter into a project agreement with the PIE, outlining the obligations of the EE that are passed down to such PIE with respect to the implementation of the funded activities (as shown in Figure 6).

Figure 6: Implementation arrangement for credit lines



GCF concessional funds would be critical to help create markets for clean cooling. For illustrative purpose, the example of Malawi is provided below.

An agriculture-driven economy, Malawi is adversely impacted by climate change impacts such as floods and droughts. Energy is one of the priorities of the third Malawi Growth and Development Strategy (MGDS III), which has identified promoting the use of energy efficient technologies and systems as an enabling strategy. As the agriculture sector is one of the largest contributors of GHG emissions in the country, facilitating the deployment of low carbon cooling technologies would contribute to achieving a climate-resilient economy. However, commercial banks in Malawi seldom provide loans for agricultural activities. There is overdependence on firewood and charcoal for heating and domestic use and knowledge and availability of cooling services is limited. Limited awareness and access to financing ultimately affect the deployment of much needed investments and income generating activities. Malawi is considering the provision of credit lines to participating financial institutions to stimulate the market adoption of cooling solutions.

GCF and WB funds will be used to extend credit lines to commercial banks, micro-finance institutions, and licensed Savings and Credit Cooperatives (collectively referred to as "participating financial institutions" or "PFIs"), for on-lending to micro-, medium and small-scale enterprises ("MSMEs"). The GCF funds would supplement IDA funds mobilized through a recently approved project⁵² to promote access to climate friendly refrigeration and cold storage by rural farmers.

Flow of GCF funds. The World Bank would enter into a financing agreement with the Republic of Malawi through the Ministry of Finance, Economic Planning and Development (the Executing Entity), which will in turn sign an on-lending agreement with the Reserve Bank of Malawi (the RBM) for the administration of the funds. In such a case, the World Bank would also sign a project agreement with the RBM setting forth the obligations

⁵² The project, approved in August 2020, aims to increase access to financial services, entrepreneurship and capabilities for micro small and medium enterprises. Additional financing from the Facility is critical to introduce sustainable cooling among the target development areas.

of the Republic of Malawi that are transferred to RBM with respect to the implementation of the funded activities. The RBM has a track record of managing World Bank-funded projects and has the legal mandate to extend wholesale financing to PFIs for on-lending to MSMEs. Moreover, the RBM's banking and micro-finance institutions supervision functions give it an added advantage in terms of its ability to manage the implementation of the credit lines.

Selection of participating financial institutions (PFIs). The selection of the participating financial institutions is an important aspect of the credit lines model, and is made according to criteria acceptable to the World Bank. For example, in the case of Malawi, criteria to assess the participating financial institutions include, *inter alia*, (i) proof of possessing a license to operate in the country; (ii) compliance with all national prudential and other applicable laws and regulations; (iii) compliance with total capital to risk-weighted assets ratio prescribed by national prudential regulations; (v) adequate liquidity; (vi) adequate internal audit and control systems; (vii) adequate management information systems; as well as (viii) commitment and previous experience lending out to MSMEs, preferably with a focus on businesses owned by women.

For the purposes of the FAA, this model may be used for credit lines to participating financial institutions under Component 2.

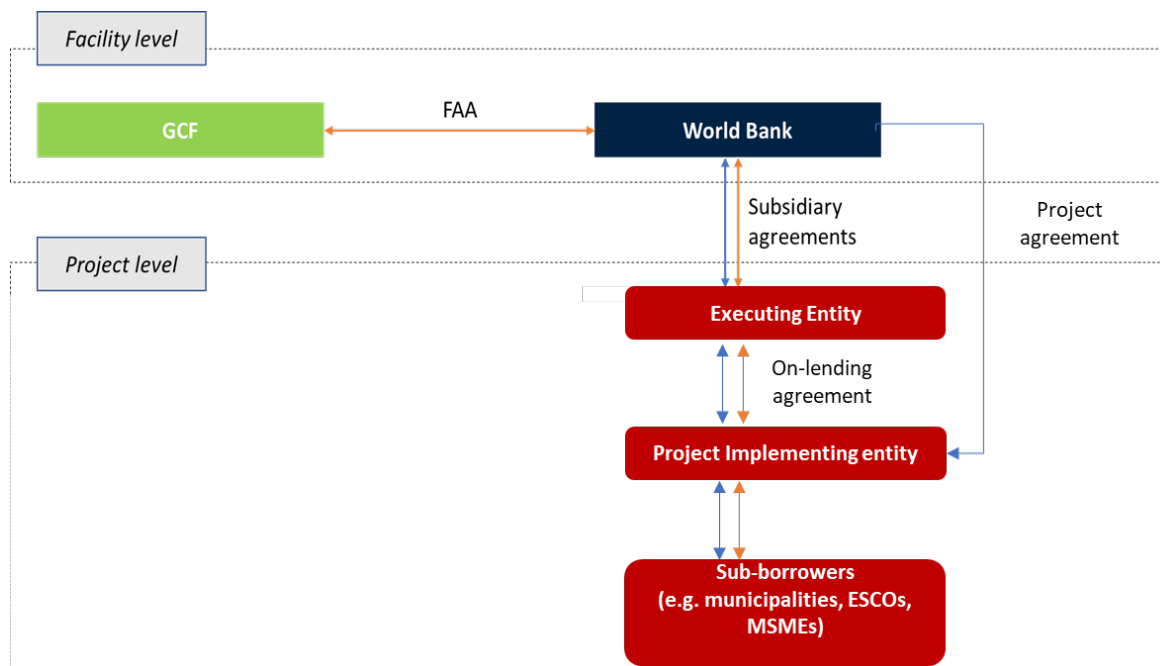
(iii) Energy Efficiency Financing Facility

Under this model, the GCF proceeds are channeled through the Executing Entity (the Host Country as the Borrower) according to the terms of the Subsidiary Agreement signed with the World Bank. The EE either directly, or through an entity that it designates (the Project Implementing Entity or PIE), on-lends the funds to sub-borrowers (for example, private energy service companies, cooling as a service operators⁵³, municipalities, small- and medium-enterprises) to undertake eligible cooling investments. Where there is a PIE, the World Bank would require the Host Country to enter into an on-lending agreement with the PIE which would include provisions that require the PIE to execute the funded activities in accordance with the requirements of the Subsidiary Agreement. In this manner, the EE has contractual control and oversight responsibility over the PIE. In addition, the World Bank may enter into a project agreement with the PIE setting forth the obligations of the Host Country that are passed down to the PIE with respect to the implementation of the funded activities. The PIE would extend sub-loans, partial guarantees, or grants, as applicable, to sub-borrowers (e.g., energy service companies or cooling-as-a-service companies), as shown in

Figure 7. Where there is no PIE involved, the Executing Entity directly provides financing to the sub-borrowers and there is no project agreement or on-lending agreement.

⁵³ For example, Community Cooling Hubs may adopt cooling as a service model.

Figure 7: Energy efficiency financing facility implementation arrangement

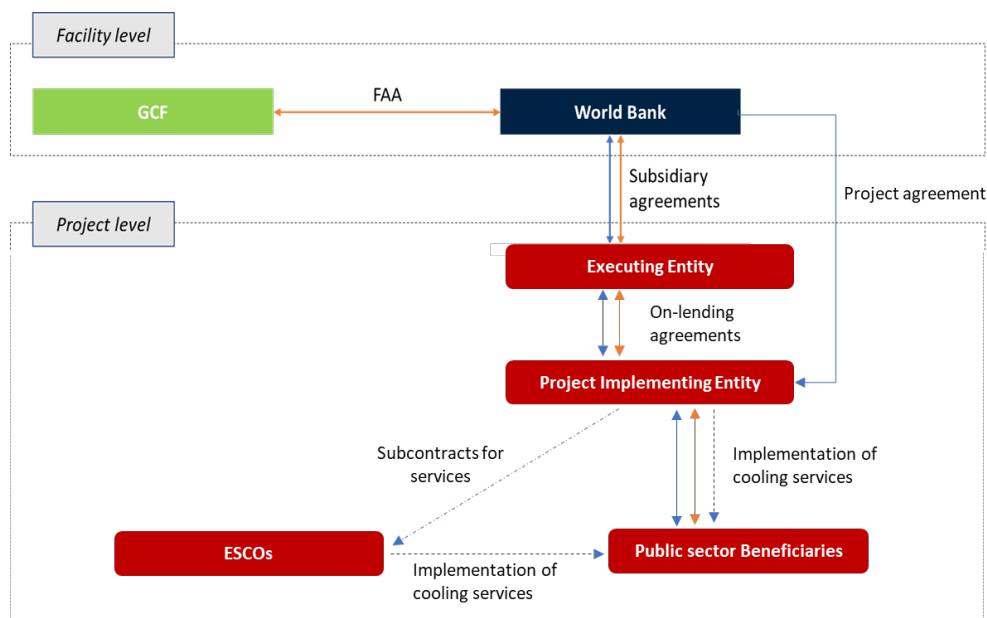


For the purposes of the FAA, this model may be used for financing of climate-friendly cooling investment under Component 2.

The energy efficiency financing facility implementation arrangement involves two main variations, as presented below.

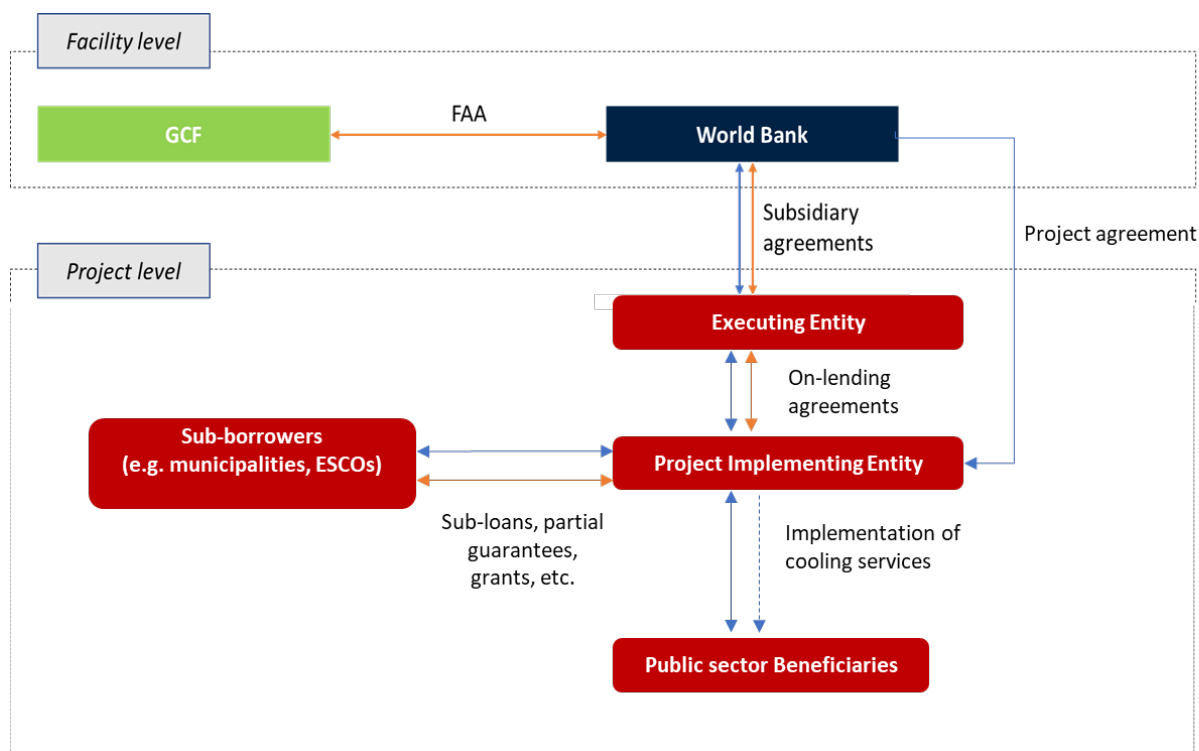
(iii-a) Public super ESCO model. In this variation of the Eff financing facility model, the Host Country designates a public energy service company (the public super ESCO) as the PIE. The public super ESCO would typically be a public entity established by the government to (a) support cooling and energy efficiency investments in the public sector (public buildings and sustainable cooling projects) and (b) strengthen private sector ESCOs through demand aggregation, bulk procurement of equipment (therefore reducing the cost of equipment for ESCOs), standardization of ESPCs, M&V templates and capacity building, project development, and facilitation. Figure 8 illustrates a potential structure for a public super ESCO implementation model. As shown in Figure 8, such PIE may even sub-contract with private ESCOs to implement energy efficiency projects through ESPCs in public buildings and facilities. The functions of public sector ESCO acting as a PIE are multifaceted and can potentially cover all the building blocks of an energy efficiency ecosystem, from energy audits to project design to performance contracting, procurement of cooling and energy efficiency measures or services (including through private ESCOs), installation, and M&V of energy savings to operations and maintenance. Thus, the PIE would not only moderate stakeholders' risks but also would help build trust, including between private ESCOs and public sector end users. This variation of the energy efficiency financing facility arrangement is envisaged in Bangladesh.

Figure 8. Public super ESCO implementation model



(iii-b) Energy efficiency revolving facilities. This modality combines the features of the plain-vanilla financing facility and the super ESCO model in that, in addition to providing financing to eligible sub-borrowers, the PIE would invest in energy efficiency activities using its own capital (for example, retrofitting of existing buildings) as shown in Figure 9

Figure 9: Energy efficiency revolving facility (EERF) model



Under the EERF model, the PIE uses revolving mechanisms to recover its investment and operating costs based on contracts such as Energy Service Agreements (ESAs)⁵⁴ or also uses more traditional revolving financing mechanisms such as debt financing and budget capture.⁵⁵ Under an ESA, the facility finances and undertakes the investment on behalf of the beneficiary. In return, the beneficiary makes regular payments based on the energy savings generated by the investments to the EERF over an agreed period so that the EERF can recover the investment and operating costs. For the beneficiaries, the recurring payment (consisting of the reduced energy bill and the payment to the EE or PIE, as applicable) is equal or lower to their pre-investment baseline energy bill. The ESA covers an agreed period, typically up to 15 years depending on criteria including, but not limited to, the size of the investment, the amount of energy saved and the price of energy. As ESAs represent long-term contractual commitments and are typically not classified as debt, they provide major advantages to the beneficiary since they are relatively simple to carry out, do not require debt financing, and pose little risk, thereby addressing key barriers to public investments in efficient space cooling and energy efficiency.

The World Bank does not bear the risk associated with the performance of these investments and the associated recovery mechanism. That is, in any case, the Host Country is obligated to make repayments, as per the terms of the Subsidiary Agreements, in line with the provisions of the FAA between the GCF and the World Bank. Energy efficiency revolving facilities (EERFs) can be effective financing and implementation mechanisms to incentivize efficient, clean cooling where there are market failures, where commercial financing for energy efficiency/sustainable cooling is under-developed, and where local banks perceive energy efficiency – and sustainable cooling – as too risky. It helps stimulate the market for sustainable cooling – mostly for space cooling (e.g., improve the thermal efficiency of buildings and the efficiency of cooling equipment/appliances, stimulate investments in new energy and cooling efficient buildings) in the target country by building capacity, establishing confidence in the energy savings, introducing new financing approaches, and encouraging the establishment of supply chain contractors, suppliers, and financial and legal experts in this area. The World Bank has experience in various countries with the establishment of EERFs that have been in operation for many years.⁵⁶

North Macedonia envisages to deploy GCF funds through the EERF model. The North Macedonia Ministry of Finance will establish the EERF within the Development Bank of North Macedonia (DBNM), a development and export bank wholly owned by the Republic of North Macedonia. GCF funds along with WB funds will be provided to the Host Country (the Borrower). The Host Country, through the Ministry of Finance, will make part of the GCF proceeds available to the DBNM to support the initial capitalization of the EERF. In so doing, the Host Country will (i) ensure that the on-lending agreement is on the terms and conditions acceptable to the World Bank and the GCF, particularly ensuring that the GCF concessionality of the loan financed out of the GCF Reimbursable Funds is substantially passed on for the financing of the GCF funded activities, and (ii) cause the DBNM to carry out the funded activities in accordance with the provisions of the Subsidiary agreement between the World Bank and the Republic of North Macedonia.

The DBNM would use the GCF proceeds to finance and implement efficient cooling and other energy efficiency investments and offer the full package of services required to (i) identify and select subprojects presented by the participating municipalities in response to calls for proposals; (ii) finance eligible subprojects through various mechanisms (including ESAs, loans, budget capture, partial guarantees and grants); (iii) procure and manage all required consultants to conduct energy audits and prepare technical designs, and contractors to

⁵⁴ Details about EERFs using ESAs can be found at <https://openknowledge.worldbank.org/handle/10986/30388>

⁵⁵ Budget capture: After the EERF completes an investment for a public agency, the government reduces its budgetary transfers to the public client by an amount equivalent to the energy cost savings (thereby capturing the savings gained through energy efficiency) and transferring this amount to the EERF.

⁵⁶ The World Bank will incorporate lessons-learned from its experience in supporting such mechanisms in, e.g., Kosovo, Armenia, Bulgaria, and India. See World Bank-ESMAP Livewire (2018) "Financing Energy Efficiency, Part 1: Revolving Funds" (<https://www.esmap.org/node/170491>).

carry out the building retrofits and installation or upgrade of equipment of central government buildings⁵⁷ (i.e. primary healthcare facilities identified in consultation with the Ministry of Health); and (iv) carry out and monitoring and evaluation activities. Investments financed by the DBNM, through the EERF, would need to meet the eligibility criteria for investments under the Cooling Facility and eligible beneficiaries would include government owned buildings (e.g., schools, universities, government administration buildings, health facilities).

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Sustainable cooling will remain a small, boutique niche unless targeted efforts are made to mainstream it and catalyze a paradigm shift. As documented by the IEA, SEForAll, and others, current trends are neither sustainable or adequate to support development goals. If current trends continue, the rapid growth in purchases of residential and industrial air conditioners using inefficient technologies, often with high GWP refrigerants, will make the Paris climate goals unattainable. Current trends are also far from sufficient to support the cooling services needed if health, food security, and other SDGs are to be achieved. Opportunities for triggering a sustainability shift in traditional cooling approaches are often found within different sectors (e.g. agriculture, fisheries, health, cold chains, buildings, etc) and may be missed, as countries face competing priorities - especially as they respond and navigate the health and economic challenges exacerbated and caused by the emergence of the COVID-19 pandemic.

The GCF loan and grant are required for the development of the anticipated implementation models in the selected countries under the Facility, which involves adapting the models to the specific country context; testing, improving, and increasing the scale of the models; capacity building of government and private sector stakeholders; awareness raising among beneficiaries; development of investment pipelines; and dissemination of results; etc. This phase of developing the models and the associated policy and regulatory support to enhance institutional and technical capacity require concessional financing before the models can be scaled-up relying on non-concessional financing and private sector financing. Without the availability of long-term, concessional finance from the GCF, the initial costs to develop these models would be too high for the governments and result in either no, insufficient, or delayed adaptation in the critical area of cooling. Without appropriate adaptation mechanisms, rising temperatures would translate into higher electricity demand to meet cooling needs and consequently, higher cooling-related GHG emissions. GCF funding is critical to remove barriers and address market failures identified and demonstrate ways in which countries can adapt to rising temperatures in a climate-friendly manner.

GCF funding will be allocated to eligible Facility projects in accordance with the WBG's Guiding Framework on Strategic Use of Climate Finance to Maximize Climate Action⁵⁸ which advises that to maximize the impact of climate finance, the degree of concessionality allocated to a program or project should be equal to what is needed to overcome the identified barriers to implementation and make the program/project viable.

Different financing instruments will be assembled to form a cascade of concessionality or risk tolerance, which entails leveraging the private sector in ways that optimize the use of scarce public resources. Consistent with GCF criteria, projects will be selected and designed so as to provide the least concessionality needed to make the proposal viable without crowding out other funding sources.

⁵⁷ These investments would be done on a grant basis, i.e., there would not be any sub-loan agreement signed with the beneficiaries.

⁵⁸ World Bank Group, *Strategic Use of Climate Finance to Maximize Climate Action: A Guiding Framework*. 2018

The following factors justify the use of concessionality:

- Misaligned national and global incentives to reduce GHG emissions. This explains the lack of incentives for governments to invest adequately in climate action; more specifically, the fact that national and global incentives to invest in mitigation do not necessarily overlap accounts for much of the underinvestment. As decarbonization is considered a GPG, concessionality should support mitigation projects and help align national and global plans for climate action.
- Developing countries lack the resources to implement the projects required to achieve deep decarbonization. Even though many climate policies bring significant domestic co-benefits, they often do incur significant short-term costs, e.g. investments required for deep decarbonization and the social costs linked to any economic transition (e.g., retraining of workers in declining industries).
- In countries with limited access to capital markets, finance with a concessional component helps leverage private sector finance and helps reduce the funding gap that hamper mitigation and adaptation efforts.
- Temporary interventions financed on grants or concessional terms can ease policy transitions and create new markets for low-carbon technologies. Without concessional support, nascent technologies will not break through fast enough and certainly not at the scale needed to avoid a higher than 2°C global temperature rise.

Several factors specific to cooling technologies add to the rationale for concessionality described in the Guiding Framework:

- First, is the reality that there are market failures leading consumers to favor less climate-friendly solutions, even if the lifecycle cost, and impact on consumers and households' pockets, is greater. As highlighted in ESMAP's Primer for Space Cooling (2020), expenditures to achieve space cooling already account for an estimated 5-15 percent of the median household income in many parts of the world. With rising temperatures and increased demand for comfort, the operation of air conditioning units to provide cooling will only increase. A less efficient entry-level room AC can cost one-and-a-half to two times more over its lifetime compared with commercially available high efficiency room AC units..
- Second, the lack of capital access and large transaction costs that many developing countries face and, in particular, hinders affordability for poorer communities, see below. (This, and other barriers are further highlighted in the feasibility study)
- Third, waiting for markets to evolve without support, risks missing cost-effective opportunities and can lead to substantial avoidable increases in GHG emissions; action is urgently needed to mobilize concessional climate finance, as well as to better coordinate and create synergies between climate finance for cooling with funding from the Multilateral Ozone Fund supporting refrigeration replacement projects. Proving performance and expanding the size of markets for efficient equipment, regulatory and fiscal incentives, and consumer awareness efforts can accelerate the time-frame for economies of scale that are necessary to lower costs and achieve sustainability.
- Another factor is the need to bridge the affordability gap, a major barrier in large urban slums, a category that according to the most recent SEForAll estimate includes 680 million people. Another 2.2 billion people are at risk because income limitations only allow purchase of the cheapest and typically least efficient air conditioners. (SEForAll 2019) Finally, concessional financing is necessary to enable a programmatic approach with opportunities for synergies, knowledge sharing, and dissemination of lessons learned, thereby accelerating the transition to sustainable cooling across regions and countries.
- There is also the need to accompany physical cooling-related investments with non-revenue generating financing to build and/or strengthen the policy, regulatory, institutional and general enabling framework needed to ensure the sustainability and scaling up of the physical cooling investment and the climate and other co-benefits.
- Priority for grants will be given to cooling solutions whose beneficiaries are the poorer marginalized communities.
- **The learning generated by efforts to test and put in place new models and create a new market demand for climate friendly cooling solutions.** When a climate initiative involves new approaches, tools, or solutions, it entails a *risk* for the implementing country, and *benefits* for all others—a key

characteristic of global public goods. The lack of (historical) data, uncertainties regarding future climate conditions, and the need for new strategies and technologies imply a need for capacity and financial resources.

- **International transfer of knowledge about factors accounting for success/failure in climate action related to cooling could reduce the cost of a low carbon cooling pathway globally.** However, it also creates perverse incentives. Being able to learn from the mistakes of others creates an incentive to delay action and stick to proven solutions rather than experiment. Knowledge transfer does not happen automatically and is facilitated by financial support.
- There is a real risk of **negative global spillovers from low access to sustainable cooling which will hinder populations' resilience and livelihoods as a result of climate change.** Climate change impacts often spill over internationally and do not remain within country borders. Examples include health-related impacts, with the possible spread of communicable diseases on a global scale, and food security impacts caused by lower agricultural yields in one region leading to higher food prices worldwide. Other, more uncertain but potentially very large sources of spillovers are:
 - *Migration:* If some places become unable to provide food security and jobs to the population; significant internal displacement and migrations could result, with large negative impacts on regional or even global stability;
 - *Conflict:* Some evidence suggests that climate change may increase local and subnational conflicts, again with impacts that could extend throughout a region and globally;
 - *Humanitarian crisis cost.*

In summary, the proposed level of concessionality is justified by the significant role of public goods; the presence of both substantial market and non-market barriers; the expected transition to sustainability over the life of the project; and the large leverage of investment expected from GCF funds.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

The Facility will help the recipient countries to develop adequate systems to sustain the results and benefits beyond project closure, where possible. As provided in the Guiding Framework for the Use of Climate Finance, in its use of concessional funds, the WB is prepared to stay the course to support a change in trajectory up to the tipping point, where the targeted barriers to implementation are overcome and commercial finance can take over. Projects will signal that concessionality is timebound, delimited by a clear exit strategy, and intended to encourage and support policy changes that promote the adoption of nascent technologies and solutions.

Financial. The financing package (including the GCF portion of the financing) will be designed to ensure that it does not distort the market or displace other sources of funds. Depending on the nature of the project, the design will incorporate actions that aim to stimulate the participation of the private sector and provide economic opportunities for end-user beneficiaries.

Knowledge exchanges will be organized to foster cross-pollination and the dissemination of best practices at local, national and regional levels and help leverage the lessons learned. The experience shows that events that involve experience sharing among different communities, institutions or countries sharing their experience with cooling is a proven and rich instrument to support continuity, dissemination of best practices and replication at a larger scale.

Technical and implementation arrangements. The World Bank will help EEs develop and put in place implementation arrangements with features (e.g., structure, budget and capacity requirements) that can be sustained beyond the life of the respective projects. In order to ensure the sustainability of the investments, the EEs will ensure that the beneficiaries receive adequate training to conduct O&M to ensure continuous operation of the equipment and infrastructure and/or may subsequently enter into an O&M contract. A key area of focus will be contributing to the development of local technical capacity, taking into account gaps, needs and context relevant to each project. Further, and importantly, the Facility will provide support to strengthen EE's enabling environments, including, for example, the development and adoption technical standards, policies and practices for sustainable cooling that can be sustained, as well as expanded over time.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount		Currency		
		157		million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	29	40 years	10 years	No interest 0.75% interest rate		
(ii) Senior loans	96	20 years	5 years			
(iii) Equity						
(iv) Guarantees	-	-				
(v) Reimbursable grants	-	-				
(vi) Grants	32					
(vii) Results-based payments						
(b) Co-financing information ⁵⁹		Total amount		Currency		
		722.84		million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
World Bank (IBRD)	Senior Loans	130	million USD (\$)	tbd	tbd	Options
World Bank (IBRD)	Senior Loans	27.4	million USD (\$)	10 years 3 years	Floating interest rate 0.25% front end fee 0.25% commitment fee	Options
World Bank (IBRD)	Senior Loans	20	million USD (\$)	25 years 5 years	Floating interest rate 0.25% front end fee 0.25% commitment fee	Options
World Bank (IDA)	Senior Loans	86	million USD (\$)	38 years 7.5 years	no interest 0.75% service fee 0.5% commitment fee	Options
World Bank (IDA)	Senior Loans	150	million USD (\$)	30 years 5 years	0.71% interest rate 0.75% service fee 0.5% commitment fee	Options
World Bank (IDA)	Senior Loans	150	Million USD (\$)	tbd	tbd	
World Bank (IBRD)	Guarantee (TBD)	50	Million USD (\$)	tbd	tbd	
World Bank (IDA)	Grants	80.5	million USD (\$)			
Global Financing Facility	Grants	25	million USD (\$)			
Health Emergency Preparedness and Response Trust Fund	Grants	3	million USD (\$)			
Pandemic Emergency Facility	Grants	0.94	million USD (\$)			
Amount			Currency			

⁵⁹ The World Bank (IBRD/IDA/TF) co-financing amounts are indicative and subject to approval by the World Bank Board and agreement with the EEs.

(c) Total financing (c) = (a)+(b)	879.84	million USD (\$)
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Not applicable.	

C.2. Financing by component

Table - Financing by Component

Component	Indicative cost million USD (\$)	GCF financing		Co-financing ⁶⁰		
		Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Component 1 – Policy, regulatory and enabling environment support	91.6			46.1	Senior loans	World Bank
		15.7	Grants	21.6	Grants	World Bank
				0.25	Grants	Health Emergency Preparedness and Response Trust Fund
				8	Grants	Global Financing Facility
Component 2- Financing for cooling investments	769.04	125	Senior loans	509.8	Senior loans	World Bank
		16.3	Grants	47.5	Grants	World Bank
				50	Guarantee (TBD)	World Bank
				2.5	Grants	Health Emergency Preparedness and Response Trust Fund
				0.94	Grants	Pandemic Emergency Facility
				17	Grants	Global Financing Facility

⁶⁰ The World Bank (IBRD/IDA/WB-administered TFs) co-financing amounts are indicative and subject to approval by the World Bank Board and agreement with the EEs.

Component 3 – Project management	19.2			7.5	Senior loans	World Bank
				11.45	Grants	World Bank
				0.25	Grants	Health Emergency Preparedness and Response Trust Fund
Indicative total cost (million USD)	879.84	157		722.84		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

GCF funds will be used to support activities that aim to strengthen policy, regulatory and enabling environments that support scaling up and sustaining cooling-informed investments and building the capacity of key stakeholders. This includes inputs and learnings relevant for inclusion in cooling action plans as well as in the NDC revision process. Important co-benefits such as increasing resilience and adaptive capacity would also be highly relevant to inform national adaption and disaster risk strategies and plans. Activities linked to capacity building will be important and can take various forms, including, for example for the testing, labelling and standards for cooling technologies (e.g., different appliances and equipment for space cooling and/or cold chains), the labelling of energy efficient buildings. A key feature will be supporting the removal of barriers in order to help increase the demand for efficient cooling technologies and investments and stimulate self-sustaining markets in the relevant sectors. Experiences and lessons will be documented to contribute to building capacity even beyond the context of countries supported by the Facility.

The Cooling Facility will be technology agnostic and support continued technological innovations, as long as the proposed technologies lead to more energy-efficient and climate-friendly cooling services and are in line with the criteria established for the Cooling Facility. The performance of proposed cooling solutions should have been demonstrated through laboratory testing and/or field-based data.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

- Programme level

The proposed Facility is designed as an innovative, multi-sector and multi-country programmatic financing mechanism with the goal of transforming the cooling sector in the target countries. With a cross cutting-adaptation and mitigation- approach, the Facility will bring about access to cooling which is essential in a warming world and especially critical to help build resilience in countries facing risks of extreme heat hazards, as well as help chart a sustainable path to low carbon development (impact potential).

- Adaptation

In the Facility countries at high risk of facing extreme heat hazards- Bangladesh, El Salvador, Malawi, and Somalia - the impact of the Facility-supported sustainable cooling investment are expected to reach over 8 million direct (and over 16 million indirect) beneficiaries⁶¹. It is estimated that the expected direct beneficiaries will amount to approximately 0.44 million in Bangladesh (and more than 4.5 million indirect beneficiaries), 1.3 million in El Salvador (and almost 4 million indirect beneficiaries), 251 thousand in Malawi (and 1.7 million indirect beneficiaries) and 2.2 million in Somalia (and 6.6 million indirect beneficiaries)⁶². The Facility's projects will increase resilience to climate impacts of buildings in Bangladesh, of agriculture cold chains in Malawi, and of health cold chains and facilities in El Salvador and Somalia. These climate-resilient cooling assets (more than 16,000 cooling assets) will also contribute to the Facility's mitigation impact (along with the cooling investments in the Facility's 5 other countries).

- Mitigation

Overall, it is conservatively estimated that the Cooling Facility will lead to approximately 1 million ton of avoided or reduced GHG emissions (compared to BAU) per year across the sustainable cooling investments made in the 9 countries that are part of the Cooling Facility. This amounts to over 10.7 MtCO₂e over the 10-year implementation lifetime of the Facility, and over 18 Mt CO₂e over the lifespan of the cooling assets (which represents about 2% of today's global cooling-related emissions). This estimated GHG mitigation potential for the Facility is calculated based on an ex-ante assessment of each of the identified project in the different cooling target areas utilizing best-practice and publicly available tools and methodologies and conservative assumptions. The program's impact potential stems from the combination of interventions of different types of projects that engage different business, financing and delivery models based on the barriers faced in different markets. The program encompasses projects that, for example,

- promote passive measures to provide comfort while avoiding the need for mechanical cooling (for example, advanced thermal protection in building envelope, natural ventilation and cool roofs) – *space cooling*;
- increase energy efficiency by deploying energy-saving cooling technologies in buildings (for example, super-efficient air conditioners, energy-efficient chillers) – *space cooling*;
- promote sustainable refrigeration and cold chain infrastructure – *refrigeration and cold chains*; or
- promote the deployment of renewable energy to power cooling such as solar refrigerators and agriculture chillers– *refrigeration and cold chains*.

Given the diversity in the types of projects/sectors covered under the two target cooling areas, different calculation models and resources are required to estimate GHG emissions reductions, complemented by empirical data to derive assumptions. The main tools used to calculate GHG emissions are consistent with the International Financial Institution Technical Working Group (IFI-TWG) Guidelines for Harmonized Approaches to GHG Accounting. These tools are:

- 1) IFC EDGE "Excellence in Design for Greater Efficiencies" software application to estimate (EEff buildings)⁶³
- 2) HOMER Powering Health (vaccines cold chains and health facilities)⁶⁴

Annex 23 of the funding proposal outlines in more detail the GHG calculations for each of the projects covered by the Facility and Annex 24 describes the methodology and assumptions considered to perform the calculations.

⁶¹ Initially the supported climate-friendly and heat resilient health cold chains and facilities will support health system strengthening for the countries' COVID-19 response, but later in the life of the assets potentially for vaccines and treatments to combat other existing and new diseases, including possible climate induced diseases

⁶² See Annex 17 as well as Annex 23 for an overview of the assumptions and methodology to estimate the projects' beneficiaries.

⁶³ See <https://app.edgebuildings.com>

⁶⁴ See <https://poweringhealth.homerenergy.com>

The Facility's experience and approach in assessing and achieving significant GHG impact through its 2 cooling target areas in different countries and sectors will contribute to building the information and know-how basis to facilitate and stimulate the expansion of sustainable cooling more systematically in climate change plans and strategies more broadly.

- *Space cooling*

Through the Facility's support, space cooling projects will be promoting both higher efficiency and lower GWP refrigerants uses, thereby shifting the BAU trajectory of cooling GHG impact towards sustainable high efficiency/low GWP cooling. It is expected that the largest share (more than three quarter, i.e. 12.6 MtCO₂e from energy efficiency in space cooling) of the program's GHG mitigation impact over its lifespan⁶⁵ will be achieved from EEff space cooling – in line with the large contribution of this cooling target area to global emissions (amounting to 1,135 MtCO₂e/yr worldwide and growing significantly to about 2,070 Mt Co₂e in 2050 – representing 18% of the increase in the total CO₂ emissions associated with global electricity consumption to 2050⁶⁶) and the significant potential to shift downward that global space cooling emissions curve⁶⁷. The GHG impact is further enhanced when adding the lower and lower GWP refrigerant use, considering that refrigerants are powerful GHGs in their own right.

The space cooling projects under the program assume a **minimum energy efficiency improvement of 20%** compared to BAU, with ex-ante calculations for EEff buildings projects using the internationally recognized IFC "EDGE" green buildings software application⁶⁸ to calculate energy savings for different types of buildings. The EDGE software and methodology has been customized to the specific local circumstances providing base cases for typical buildings by using empirical data from actual buildings reflecting current practice and taking into account climatic conditions of the location, building type and occupant use, with assumptions updated as the market evolves⁶⁹.

- *Refrigeration and cold chains*

The Facility will help catalyze a greening of the cold chain by improving the efficiency of refrigeration, storage and distribution infrastructure for agriculture and health using low GWP refrigerants.

The Facility aims to also support climate friendly cooling in the context of health facilities and cold chains to safely store vaccines and medicine. Some of the cold chain investments will involve solarization to meet cooling needs of the health facilities and cold chains. To estimate the energy requirements (to be provided by renewable energy), the energy savings, the lower GWP refrigerants and the overall GHG impact, the following well known tools will be used: the IFC EDGE tool; the HOMER (Hybrid Optimization of Multiple Energy Resources) Powering Health tool⁷⁰ (for the assessment of distributed solar power).

⁶⁵ The lifespan of the program – the estimated period during which the impact of the program will be felt) is 27 years. This is different from the program's implementation period (10 years) which is the basis for calculating the Facility's indicators.

⁶⁶ IEA (2018) *The Future of Cooling*.

⁶⁷ In fact, it is estimated that the current global emissions from space cooling could have been less than half with more efficient buildings and efficient cooling appliances (ESMAP 2020). The IEA estimates that EEff space cooling – as represented in its "Efficient Cooling Scenario", would lead to less than half of the energy used for space cooling under the Baseline Scenario.

⁶⁸ See <https://edgebuildings.com/>. The EDGE software can be used for free to help design an energy and resource efficient building in more than 160 countries. The EDGE software calculates utility savings and reduced carbon footprint of a buildings. (A building achieving a minimum projected reduction of 20% in energy, as well as water use and embodied energy in materials compared to a standard BAU building in that same market may be eligible for EDGE certification.)

⁶⁹ More information on the EDGE methodology can be found here:

<https://edgebuildings.com/learning/methodology/>

⁷⁰ Information on Homer Powering Health Tool is publicly available on their website:

<https://poweringhealth.homerenergy.com/>. The tool is free online and can be used to create initial designs of electric power systems for health care facilities that have no other power supply or have grid electricity available for a predictable period of hours each day.

The GHG emissions reductions associated with the promotion of more climate-friendly refrigeration and cold chain through the Facility's projects are estimated to reach about 2.7 MtCO₂e over the lifespan of the program, accompanied by important co-benefits (see below).

- *Off-grid cooling powered by renewable energy*

Enabling the expansion of affordable cooling in off-grid contexts will help reach some of the poorest communities and to contribute to enhancing their resilience to the impact of climate change. Indigenous and non-GHG emitting renewable energy (solar) will mostly power (126 MW) these critical cooling services.

- *Kigali Amendment and Refrigerants.*

The Facility will also support the goals and requirements of the Kigali Amendment to the Montreal Protocol by helping countries in the transition to more climate-friendly refrigerants by catalysing more energy efficient cooling equipment (air conditioners and refrigeration) in parallel with HFC phase down. The more climate-friendly refrigerant for specific projects would be determined based on the specific market and application of the more efficient cooling application and guided by the May 2019 Report of the Technology and Economic Assessment Panel (TEAP) to the Montreal Protocol Parties Report on *Cost and Availability of Low-GWP Technologies and Equipment that Maintain/Enhance Energy Efficiency*⁷¹. The GHG benefits from this refrigerant transition supported by the Facility have been estimated ex-ante, in line with global estimates indicating that GHG emissions from the refrigerants represent about 20 percent of cooling equipment's overall GHG emissions. These GHG reductions achieved under the Facility's projects are estimated to amount to about 1.1Mt CO₂e over the lifespan of the Facility.

- *Other impacts*

The Cooling Facility is expected to generate a number of co-benefits, notably in terms of health. The Facility's support for expanding and retrofitting the climate friendly vaccine cold chains in 3 countries (El Salvador, Sao Tome and Principe and Somalia) will help them respond to the current COVID-19 pandemic, increasing the capacity to store vaccines and lower the incidence of wasted vaccines and help reach larger shares of their populations in different parts of the country – which is critical for ensuring equity in their vaccination efforts. Health co-benefits – albeit of a different nature are also expected from the Facility's support for more sustainable space cooling (in Bangladesh, Panama, and Sri Lanka). These investments will help lower the incidence of heat-related illnesses⁷², as well as mental stress⁷³.

Other expected co-benefits are linked to improved local air quality through the displacement/avoidance of diesel generators, especially in the context of support going towards the solarization of cold chains.

The Cooling Facility's support in sustainable cooling in the agriculture sector in Malawi as well as Kenya, will also contribute to efforts to lower current high rates of food loss and indirectly to help boost efforts to address malnutrition.

As countries' face economic downturns associated from the COVID-19 pandemic, investments in sustainable cooling supported by the Cooling Facility will contribute to countries' efforts to address unemployment and efforts to build back better. In fact, space cooling investments in new buildings and retrofits in particular have been identified in several countries to help boost local employment opportunities⁷⁴.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

⁷¹ Volume 4: Decision XXX/3 Task Force Report on Cost and Availability of Low-GWP Technologies and Equipment that Maintain/Enhance Energy Efficiency (<https://ozone.unep.org/science/assessment/teap>)

⁷² Heat stress starts to occur when the human body reaches a temperature above 38°C (i.e., 1°C above its normal temperature of 37°C). (Steffan, Hughes and Perkins 2014, as reported in ESMAP 2020).

⁷³ Experts have found links between heat and aggressive or violent behavior, which also has implications for domestic violence, with women and children bearing the brunt of such situations (Cooper 2019; Anderson and DeLisi 2011. Discussion also reported in ESMAP 2020)

⁷⁴ For example, see the IEA December 2020 report on Energy Efficiency 2020 (<https://www.iea.org/reports/energy-efficiency-2020/energy-efficiency-jobs-and-the-recovery>) and IRENA (<https://www.irena.org/publications/2020/Apr/Global-Renewables-Outlook-2020>)

At a time when the needs for cooling are intrinsically linked with efforts to meet SDGs and to build resilience to impacts from a dangerously warming planet, and when the global cooling demand is increasing exponentially, but on a trajectory that is costly, inefficient and high in GHGs, there is an opportunity now to set the stage for changing course and support a paradigm shift. The potential is huge: maximizing energy efficiency in the cooling sector (refrigeration and space cooling) while simultaneously phasing down HFCs contained in refrigerants, the world can potentially avoid close to 1°C by 2100.

The Cooling Facility proposes to catalyze a paradigm shift by working across key cooling target areas and with a range of countries from different parts of the developing world to amplify the potential to leverage its experiences and learnings. In so doing, the Facility seeks to provide pioneering examples that can be emulated through development and demonstration of models and approaches that can be scaled up and be fit-for-purpose in more countries with similar circumstances, and thus further stimulate scaling-up the shift to sustainable cooling investments. Through a combination of targeted technical assistance and policy dialogue offered in conjunction with investment support for country-identified national priority sectors, the Facility seeks to build evidence and generate a track record, working with local leaders and champions to elevate sustainable cooling as a multi-focal investment and regulatory, administrative and policy action area at national level that can contribute to – and has synergies with – key climate and development goals (including efforts to respond to the COVID-19 pandemic and to build back better). The investment component will allow governments to see how the policy activities are delivering at sector level, put in place implementation structures and financial mechanisms that can be scaled up and/or see their scope broadened, and ensure that valuable lessons are learned while technology to enable achievement of ambitious objectives at country level is being transferred. In order to shift from one-off investments to financially sustainable and scaled-up investments, the Facility includes projects adopting different implementing models that aim at (i) revolving-financing that allow financing to continue after the end of the projects and (ii) enabling private sector entities to engage and invest in project activities in the longer term, as well as (iii) sending strong market signals for shifts towards sustainable cooling through public procurement (in the case of the health sector in particular) and incentives.

For example, credit lines help local banks develop a business line for cooling-investments and build their capacity to identify, assess, and originate loans for cooling-investments so that local banks can continue supporting these investments beyond the project duration. Developing ESCOs that provide financing as well as services to design and implement cooling and energy efficiency measures would address both the financing and capacity barriers that prevent cooling-investments from being implemented. EERFs collect energy cost savings to recover the capital expenditure for cooling-investments so that the funds can be re-invested and the EERF can become a sustainable entity. In addition, the EERF sustains its staff over the long term and builds its capacity – in contrast to project management units that are usually dissolved after the project end. The development, implementation, and demonstration of such models will allow investments to continue and scale-up and create experience that can be shared with and replicated in other countries after tailoring the solutions to the specific context.

Moreover, the Facility will help countries mainstream sustainable cooling into development and climate plans, procurement strategies and policy/regulatory frameworks and also help train and build the capacity of relevant stakeholders, while also helping raise awareness of the availability and benefits of sustainable cooling measures and technologies. These efforts are aimed at building an enabling environment for further scale-up of sustainable cooling beyond the Facility's investments.

The Facility's support to El Salvador, Sao Tome and Principe, and Somalia is targeting efficient and climate friendly cooling in the health sector, seeking to internalize sustainable cooling in efforts to strengthen health systems to leave countries better prepared for current and any future public health emergencies with greater sustainability.

In the buildings and cooling appliances sectors (e.g., in Bangladesh, North Macedonia, Panama, Sri Lanka), the Facility also seeks to facilitate and mainstream better access to cooling by supporting enabling environments and sustainable implementation mechanisms adapted to local institutional and policy frameworks, working with strong local partners and mobilizing private sector capacity and financing. The pioneering examples are designed to be taken to scale, and with positive results, the expectation is that these examples would inspire others to emulate them.

In the agriculture and off-grid sectors (Kenya, Malawi), the Facility will support new financing mechanisms and help build capacity and awareness and greater affordability highlighting contribution to productivity. Positive results with these projects are expected to provide powerful ammunition for scaling up with the countries and beyond. As many African countries' economies have high reliance on the agriculture sectors and also vulnerable to climate change, the ground is literally fertile for a push for sustainable cooling if we can demonstrate practical mechanisms that can bring greater affordability.

At the same time the Facility's projects are supportive and aligned with each country's climate change priorities and can support further ambition.

D.3. Sustainable development (max. 500 words, approximately 1 page)

The delivery of cooling efficiently is essential for both **Affordable and Clean Energy (SDG 7)** and **Climate Action (SDG 13)**. If cooling efficiency remains unchanged, additional power generation capacity of 2,500 GW will be needed to meet electricity demand for air conditioning by 2050 – an estimated cost of ~US\$1.7 trillion. Increased access to cold chains is critical to ensuring food safety and reducing food loss and waste. Climate friendly technologies are necessary to reduce associated GHG emission (estimated at 4.4 GT CO₂e, i.e., 8% of global GHG emissions). By maximizing energy efficiency in the cooling sector (refrigeration and space cooling) while simultaneously phasing down HFCs contained in refrigerants, the world can potentially avoid close to 1°C by 2100. In addition to climate mitigation, increasing access to sustainable cooling in developing countries, especially those located in the tropics and sub-tropics, will also contribute to SDG 13 by supporting greater resilience and improved health and livelihood of vulnerable people, communities, and regions.

Protecting workers and communities from extreme heat is critical in order to achieve **Decent Work and Economic Growth (SDG 8)** and **Sustainable Cities and Communities (SDG 11)**. Higher temperatures and lack of cooling negatively impact performance labor productivity. It is estimated that by 2050, work hours lost due to heat may be as high as 12 percent in the worst affected regions of South Asia and West Africa – representing 6 percent of annual GDP. In urban environments, average temperatures have been increasing at twice the rate of global change, largely due to the Urban Heat Island (UHI) effect. Annual mean air temperatures of cities can be 1-3°C higher than their surroundings (at night, the temperature difference can be as high as 12°C). The need to reduce excess urban heat is one of the key resilience and sustainability challenges of the 21st century.

Cooling and thermal comfort are also key to **Zero Hunger (SDG 2)**, **Good health and well-being (SDG 3)**, and **Gender Equality (SDG 5)**.

As noted in Section B, in developing countries, many with undernourished populations, around 40 percent of food produced (amounting to more than 250 million tons), including 1 in 4 fish caught, is lost between harvest and market.

- Heat is now the deadliest global natural disaster in an average year, posing the greatest risk for vulnerable populations. High temperatures increase the risk of heatstroke, breathing issues, heart attacks, asthma and kidney problems. Unreliable medical cold chains put at risk the effectiveness of vaccines and other medical supplies

- While more data and analysis are needed to better understand the gender aspect of access to sustainable cooling, gender risks are evident (SEforALL 2019). Women and girls are often disproportionately impacted by a lack of access to sustainable cooling and enhanced access could lead to lower vulnerability to mortality

risk and poverty. Furthermore, through the Facility, has a unique opportunity to pro-actively address gender gaps across entrepreneurs, employees and at the consumer level.

There are numerous examples of potential contributions to support of the SDGs from possible projects. For example, the support to cooling appliances in Panama would lead to multiple benefits for consumers including cost-savings for households, particularly women, who typically spend more time homes, reduced food waste, and lower electricity bills. There are also significant health benefits associated with access to refrigeration in health care from the safe-storage of vaccines and other temperature-sensitive medication. Access to refrigeration will also reduce food waste for agricultural, dairy and livestock farmers, whilst agricultural and commercial refrigeration will boost revenues and incomes, and agricultural cold chains will reduce hunger and malnutrition.

The proposed North Macedonia project will also have significant gender benefits through increased women's representation in the engineering professions with a focus on EEff, by providing targeted training program for women and increasing awareness about employment opportunities under the Project investments. The Project includes an indicator—the percentage of female staff hired under service contracts for technical and design work under the Project—to measure these impacts. The Malawi project seeks to specifically target women-owned enterprises.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Recipient needs to adapt to rising temperatures vary with the country context. For example, from the energy perspective, countries such as Bangladesh, North Macedonia, Sao Tome and Principe are fossil-fuel dominant. Others face rapidly growing peak energy demand, including for cooling, which strain already unreliable electricity systems and increase the use of high carbon-emitting sources of supply (e.g., Bangladesh, Sri Lanka, Panama, El Salvador)⁷⁵. Finally, countries such as Somalia, Malawi⁷⁶, El Salvador, Sao Tome and Principe and Kenya are likely to default to carbon-intensive cooling solutions, as they focus on providing universal access to electricity services.

The Facility-supported interventions would enable these countries to meet their cooling needs in a more climate-friendly way and contribute to lower GHG emissions by: (i) strengthening their institutions and implementation capacity to adapt the rising temperatures; and (ii) addressing the barriers (as outlined in Section B.1⁷⁷), including the absence of adequate sources of financing for climate-friendly cooling investments. Instead of proposing a single solution to all, the Facility takes a bottom-up approach to sustainable cooling solutions, favoring measures, mechanisms and strategies that are grounded on each country's institutional and policy framework and building on countries' priorities, the respective sector context and best suited to address the challenges and needs of the local recipients. The section below summarizes the Facility's alignment with the individual countries' needs.

Bangladesh. Bangladesh is among the World's most vulnerable countries to the effects of climate change. The country is also facing both serious energy security challenges and severe heat stress, equally across its high-income residents, low-income neighborhoods and informal slum settlements. Challenges to mainstreaming sustainable cooling include making solutions affordable among poor populations, overcoming consumer resistance to the higher first cost of more efficient equipment, and enabling greater consideration of cooling in new building design and approval. The share of energy consumption for space cooling, in buildings in particular, is rising rapidly due to increased urbanization and growing income and increased reliance on air conditioning in public, commercial and high-end residential buildings.

Pressures to meet growing demand for sustainable cooling services, could result in significant increases in coal-fired generation by 2030 if nothing changes. Greater use of renewable energy and faster progress to scale-up energy efficiency will be needed to help steer development in a more climate-friendly way. Buildings and appliances EEff are an important part of the EEff efforts under the Environment Conservation Act and

⁷⁵ See, for example, World Bank Open Data (data.worldbank.org).

⁷⁶ Malawi ranks among the top-10 countries with rural poor at cooling risk (SEforALL, 2020).

⁷⁷ Section B.1 also outlines the key challenges from a sector perspective.

the Bangladesh Energy Efficient & Conservation Master Plan. Integrating EEff and passive building design features and energy efficient cooling at scale now is key for building a more sustainable infrastructure at lower costs and lower pressures on the electricity grid.

El Salvador. El Salvador is the smallest country in Central America with a poverty rate (based on a US\$5.5 per person per day poverty line) of 29% and extreme poverty (US\$3.2 per person per day) of 8.5% as of 2017. Given its tropical geographic location, biodiversity, and topography, El Salvador is particularly vulnerable to increasing temperatures (which are predicted to increase between 1.4 and 2 degrees by 2050) and rising sea levels (a rise of up to 8cm by 2050 is predicted).

The challenges to El Salvador's health system and the Government's ability to implement measures as a response to COVID-19 are exacerbated by the observed and anticipated impacts of climate change. These impacts are expected to lead to more frequent natural disasters, such as floods, droughts, and tropical storms. These disasters in turn could lead to increasing damages to health care facilities, possibly disabling them completely at times when their services are most required. A large proportion of El Salvador's vulnerable population are the elderly (11.7%) and the adult population with underlying risk factors such as obesity and non-communicable diseases (40%). These two groups are those also most at risk to COVID-19.

To ensure an effective COVID-19 response and generate long-lasting resilience and be prepared to handle future health crisis, the Government of El Salvador seeks to strengthen the overall national immunization and related health delivery systems. This includes a variety of issues, including safe and appropriate cold chain and related services. The key challenges are timing and financial means to enable the Government of El Salvador to take advantage of the window of opportunity of revamping its cold chain to ensure it is safe, as well as climate friendly.

Kenya. Demand for cooling is projected to grow significantly in the Nairobi and Mombasa markets -- 1.8 million cumulative unit sales between 2020 and 2030 for off-grid households alone. In addition, there are an estimated 3.4 million off-grid households in Kenya that could realize a direct net economic benefits from owning a refrigerator. Energy efficiency will help lower the projected energy consumption to meet Kenya's cooling demand, and in doing so reduce utility bills for households and businesses, decrease the need for additional power supply, and increase energy access while also reducing GHG emissions in line with the country's climate mitigation goals. In the off-grid context, furthering access to cooling solutions will contribute to the development of productive uses of electricity, which is necessary for the viability of energy access efforts (with renewable energy). In the context of nearly 40% of food perishing before it reaches a consumer in Sub-Saharan Africa, enhancing access to cooling will contribute to increasing overall climate resilience of the population by improving food security through reduced post-harvest losses, create employment opportunities, enhance resilience to shocks, and stimulate economic growth. Challenges include the need for innovative business models to make cold chains available to small farmers as well as financing for the higher first cost of more efficient AC systems for residential and business applications.

Malawi. Malawi is a small, low-income landlocked country in south-eastern Africa, with most of its people living below the poverty line. The country is subject to frequent macroeconomic and climatic shocks. Malawi's private sector is mostly informal with a few largescale companies capturing most economic opportunities. Micro, small and medium enterprises (MSMEs) often operate in the informal sector, have low capacity, and are primarily concentrated in the service sector, such as wholesale, retail and trade. A key constraint indicated by MSMEs is the lack of access to financing and perceptions of risk for new technologies and innovative business models within the financial sector. MSMEs can potentially drive employment; yet, they face numerous challenges that constrain their ability to contribute to economic growth. Lack of access to cooling, in particular more sustainable cooling solutions with lower operational costs, is hindering

economic opportunities for MSMEs involved in a number of important sectors, such as dairy and horticulture (fruits and vegetables).

Many firms, especially smaller ones, face capacity and capability gaps, going from attitude/mindset to skills with respect to financial literacy, management practices, business language, market assessment, and behavioral characteristics of successful entrepreneurs. The majority of MSMEs (89 percent) are informal, 36 percent do not keep records, and 45 percent lack innovation and diversification (FinScope survey 2019). Among several challenges, most women and youth-led entrepreneurship are said to be unaware of the process or the importance of formalizing their activity. Promoting investment opportunities in sustainable cooling investments, with a focus on women and the youth, will contribute to providing economic opportunities and strengthening the resilience to climate change of vulnerable and marginalized population groups, while also avoiding expensive and polluting energy consumption. In this context and given barriers mentioned earlier, Malawi is looking at credit lines as a suitable implementation model that can kick start the transformation towards more sustainable cold chains in its agriculture sector targeted at MSMEs by facilitating access to financing among MSMEs.

North Macedonia. North Macedonia is a landlocked country at the heart of the Balkans, characterized by its mountainous terrain that is intersected by valleys and lowlands. Many public buildings (including schools, health care centers/clinics, universities, etc.) are 50+ years old and classified as underheated, unsafe (due to structural instability and seismic risks) and dilapidated. In fact, buildings, which consume about 39 percent of the energy, have been identified in the country as a major priority, with estimates of savings from 20-40 percent. The public sector has the greatest potential, with about 35-40 percent savings, mostly in the health and education sectors. While the focus in the past has been on heating, projected increases in summer temperatures in Europe are expected to substantially increase the demand and need for cooling, especially in schools and hospitals. Improving thermal renovations of public buildings will substantially reduce their future cooling loads, which will contribute to the country's adaptation to the changing climate.

Several challenges need to be addressed for the public sector to invest in sustainable cooling. Central government buildings typically have very limited budgetary provisions for capital improvements, if any, and usually are not allowed to borrow. Municipalities must undergo the fiscal decentralization process and, even then, have stringent debt limitations which many have already reached. Few have formal credit ratings. Thus, while energy efficiency investments can pay for themselves through cost savings, traditional debt financing has been difficult to mobilize for the public sector, effectively forming a barrier to EEff renovations in public buildings. A combination of capacity building, institutional reforms, and demonstration, along with a financing mechanism designed specifically for the public – including municipal – context, will be needed to overcome these barriers. In the context of the COVID-19 situation, rehabilitation of health care facilities, with adequate air conditioning systems, are expected to become higher priorities. Given the specific conditions and barriers mentioned above, North Macedonia is seeking to tackle space cooling in the public sector – and lower its energy consumption and associated GHG emissions - through the establishment of an energy efficiency fund, a sustainable financing mechanism that is well suited for its circumstances and targeted sector.

Panama. In the past ten years, Panama's cooling appliances market has been growing rapidly for both ACs and refrigerators technologies. AC penetration in the residential sector is relatively low but is expected to rapidly increase in line with the country's dynamic economic development (e.g., residential household AC ownership stood at 8.9 percent in 2000, compared to 15.6 percent in 2010 and is expected to grow to an estimated 66 percent in 2050). According to the National Energy Plan of Panama (Plan Energético Nacional 2015-2050), cooling systems' consumption currently represent about 40 percent of the overall electricity consumption in Panama. The electricity sector is facing serious challenges in keeping up with rising electricity demand, which is growing at an average annual rate of 8.5 percent per year mainly due to an increased use of cooling equipment in the residential and commercial sectors. In order to align with Panama's climate change objectives and commitments (NDC), the pace of energy demand growth needs to be reduced from an average 6 percent in the business-as-usual scenario to around 4 percent, which would reduce the need of capacity additions by 38 percent.

Several barriers must be addressed to respond to this rapidly growing problem. Though Panama has a sophisticated banking sector, the absence of i) dedicated EEff credit lines priced to incentivize investments in EEff and ii) local long term capital market for mortgages (access of local banks to capital markets is limited to up to 5-years bonds), and iii) limited technical capacity in local banks to assess EEff products and investments, represent a major barrier for EEff in Panama. The higher initial cost of EEff equipment and lack of access to affordable financing are hindering the adoption of more sustainable cooling solutions and could lock-in large quantities of inefficient, polluting but cheaper (on a first cost basis) cooling equipment. Law 69-12 mandates the establishment of an energy efficiency fund. The national Fondo de Eficiencia Energética (UREE Fund) aims to be transformational by catalyzing private investment, reducing the high-risk perception associated with green buildings and energy efficient cooling appliances, and improve market liquidity for energy efficiency initiatives. Credit lines facilities will help transform the residential and commercial sectors and leverage private sector capital.

Sao Tome and Principe. The Republic of Sao Tome and Principe is a small low-middle-income country comprised of two main islands. It has a total population of approximately 200,000 people, 42.6% of whom are 14 years of age or younger. In addition to having a small population and a remote location, there is a high fixed cost of public goods— all factors that affect the country's trade, fiscal accounts and human development outcomes.

With 1,054 COVID-19 confirmed cases as of January 8, 2020, Sao Tome and Principe showed one of the highest contamination rates (441/100,000 people) in the region and still has limited capacity to handle the COVID-19 outbreak. The Government was unable to quickly establish laboratory capacity to diagnose COVID-19 due to the interruption of commercial flights to the island in the begging of the pandemic. After several months without having laboratory capacity to diagnose COVID-19, early efforts to contain local transmission (first case diagnosed in early April) were hindered by incapacity to confirm diagnosis and isolate cases, perform contact tracing and slow transmission. This capacity was finally established in July, alongside with improved hospital capacity to manage cases and trained surveillance teams to perform contact tracing. Notwithstanding the fact that the country is now better prepared to diagnose, isolate, treat and track COVID-19 contacts, a significant and more comprehensive strategy to strengthen the public health systems will be key in ensuring that COVID-19 and future epidemics will be properly managed. As highlighted by the WHO, the investments and expenditures for COVID-19 should also lead to longer-term, wider benefits, in line with national needs for sustainable capacities. International collaboration and support to the Government will be crucial to continue mitigations measures to contain the spread of the pandemic in the country and to design and implement its National COVID-19 Immunization Plan.

Somalia. Somalia has experienced prolonged conflict beginning in 1991 and is among the poorest countries in the world with a per capita GDP of US\$819 in 2019. An estimated 69% of the Somali population live in poverty (below US\$1.9 per day, 2018 estimate) with many more living just above the poverty line. Somalia's struggling health outcomes reflect the country's insecurity, vulnerability and deep-rooted poverty, limiting opportunities for people to access basic social services, including education and health. Health service delivery data indicate weak health systems. The most recent reliable health service delivery data are from the recently released 2019 Somalia Health and Demographic Survey (SHDS). While geographically disaggregated SHDS are only available for Puntland and Somaliland, national-level and the limited disaggregated data show tremendous service delivery gaps. Health service utilization is low, particularly in the public sector, estimated at 0.23 outpatient visits per person per year and 0.81 hospital discharges per one hundred people per year (SARA, 2016). Clan structures are believed to have major impacts on service utilization, dictating which facilities people visit.

The global COVID-19 pandemic is a pressing problem in Somalia, exacerbated by an extremely weak primary health system, inadequate surveillance and laboratory capacity, and limited hospital services. As of January 7, 2021, Somalia had 4,726 confirmed coronavirus cases and 130 deaths, with anecdotal reports of additional, unconfirmed cases and quickly rising deaths. The government and development partners have taken important steps to improve Somalia's COVID-19 response, including establishing testing capacity at the National Public Health and Reference Laboratory in Mogadishu, constituting rapid response teams, and

increasing intensive care unit bed capacity from 15 early in the outbreak to 47. However, very weak health system capacities in Somalia severely constrain the response. Moreover, pharmaceuticals and medicines expire and become ineffective more quickly in the very hot climatic conditions experienced in Somalia. There are major gaps in cold-chain capacity, which prevents expanding the currently extremely low vaccine coverage and will make distribution of COVID-19 vaccines difficult. Moreover, without support, it is expected that kerosene and petrol / diesel generator fueled equipment will be moved to increase cold storage capacity

Sri Lanka. Commercial (including government buildings) and industrial buildings contribute 60% of the electricity consumption in the country. Driven by economic growth, increasing income, rapid urbanization, and an expanding middle class, buildings also represent the fastest growing source of energy consumption in Sri Lanka. This rapid growth in the building sector also has significant implications for the power sector. Despite commendable progress in the last decade, development of the energy sector faces major challenges that, if not addressed, could bring into question its sustained long-term viability.⁷⁸ With inefficient energy consumption typical of existing buildings, there is a large untapped potential for energy efficiency improvement, estimated to be 30-50% across the public, commercial, industrial, and residential sectors. Air conditioning and ventilation are main energy uses. A large share of the commercial building stock is old and uses chillers that can be replaced with much more efficient and more climate friendly equipment.

In view of the above developments, the Sustainable Energy Authority (SEA) has identified the commercial buildings sector – where a large stock of existing buildings has old, inefficient chillers for space cooling – as one of the key targets for achieving its EEff improvement goals. SEA also has defined an ambitious Demand Side Management (DSM) program that includes chillers, fans, refrigerators, air conditioning and green buildings (including the potential of nature-based solutions such as green roofs and green walls). The key investment barriers are: lack of suitable financing, resistance to the higher initial cost of efficient equipment, and insufficient public awareness. Dedicated EEff credit lines, capacity building and enabling environment support would help address these concerns. Sri Lanka's selection of suitable implementation model needs to be applicable to commercial, residential as well as public buildings, enable greater energy efficiency space cooling at scale and foster the development and strengthening of the national ESCO (Energy Sector Companies) market in the country, and thereby establish the foundation for the transformation of space cooling.

D.5. Country ownership (max. 500 words, approximately 1 page)

Each of the proposed projects in the Facility has been discussed with relevant local authorities (usually line ministries) and has been presented to the relevant GCF National Designated Authorities (NDAs) who have issued no-objection letters. Engagements with several local stakeholders and partners to discuss issues related to the project have also occurred (as outlined Annex 7), although the extent and format of these engagements have had to take into account the disruption and restrictions linked to the COVID-19 pandemic.

The alignment of the Facility with the policy and strategy context of the beneficiary countries is summarized below:

- **Bangladesh:** The Government has prepared the Energy Efficiency Conservation Master Plan 2015-2030, which describes the adoption and implementation of EEff and conservation regulatory measures. In addition, the project will inform the Government's efforts in developing a national Cooling Action Plan. These efforts are aligned with the Facility's support of space cooling and EEff measures in buildings.
- **Kenya:** Under the Paris Agreement, Kenya has committed to reducing GHG emissions by 30%, or 142 MTCO_{2e}, by 2030. The National Electrification Strategy presents a roadmap for achieving universal access to electricity by 2022. Supporting off-grid cooling appliances and access under the

⁷⁸ World Bank and International Finance Corporation 2019. Sri Lanka Energy InfraSAP © World Bank, <http://documents.worldbank.org/curated/en/843901561438840086/pdf/Sri-Lanka-Energy-Infrastructure-Sector-Assessment-Program-Executive-Summary.pdf>

Facility will contribute to both of these goals. Kenya is also in the process of developing a National Energy Efficiency and Conservation Strategy, which includes targets for energy savings in households, buildings and industry.

- **Malawi:** Promoting the use of energy efficient technologies and designs is one of the strategies advocated for in the Government's priority areas under the Malawi Growth and Development Strategy (MDGS III). Since agriculture is one of the major contributors to GHG emission, the Facility's support of clean and sustainable cold chains in agriculture will contribute to the achievement of the MDGS III goal.
- **North Macedonia:** As a contracting party of the Energy Community Treaty, the Government has committed to reduce energy use by 12% (about 200 ktoe) by 2018 in its 2020 Energy Efficiency Strategy (based on a 2010 baseline), more than the 9% required under the ECT's National Energy Efficiency Action Plans. Buildings, which consume about 39% of the energy, have been identified in the country as a major priority, with estimates of energy savings from 20-40%. The Facility will support achieving the Government's energy saving commitments through establishing a financing mechanism and investing into efficient space cooling of buildings.
- **Panama:** In 2012, the Government made a national commitment to EEff by approving the Law 69-12 "Rational and Efficient Use of Energy". In 2015, the National Secretariat of Energy published the National Energy Plan 2015-2050, which contemplates the design and implementation of the Rational and Efficient Use of Energy Program, as one of the key actions to meet the sector objectives. The country also ratified the Kigali Amendment in 2017, which aims to phase-down hydrochlorofluorocarbons (HFC). Panama is currently implementing the Stage II of its HFC Phase-out Management Plan (HPMP) under the Montreal Protocol. The Facility will support achieving the Government's commitments through its support for efficient space cooling.
- **El Salvador:** The Government is integrating climate friendly investments in its national programs and its NDC includes health as a sector affected by climate change and the promotion of a larger use renewable energy. This is aligned with the Facility's support for climate friendly health facilities and cold chain, which will help develop a more sustainable and resilient health system in the country to be able to respond to the COVID-19 pandemic as well as future public health crisis and challenges.
- **Sao Tome and Principe:** The country's NDC include promoting the greater use of renewable energy and list health as one of the sectors negatively impacted by climate change. The Facility will contribute to the Government's NDC through supporting space cooling and cold chains in the health sector, including through the increased use of solar power.
- **Somalia:** Somalia's NDC related to scaling up the use of solar energy is consistent with the Facility's to support for more climate friendly health facilities and cold chain, including through greater use of solar power.
- **Sri Lanka:** Demand Side Management is identified as one of the seven mitigation actions of the energy sector NDC of Sri Lanka to be implemented during 2020-2030. Regarding the regulatory framework, the Energy Efficiency Building Code was first introduced in 2002 and revised in 2008. The code sets standards for EEff in design and retrofits in buildings, at the same time providing methods for determining compliance. These efforts are aligned with the Facility's support of space cooling and EEff measures in buildings.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

The financial structure of the Facility adequately balances the need to support low-carbon cooling investments, leverage additional resources from private and commercial investors, and enhance the enabling environment for cooling. The facility's requested level of concessionality from GCF is in line with providing the minimum level required to successfully address market failures. A more detailed justification of GCF funding is provided in section B5.

I. Co-financing and leverage

The US\$157 million from GCF will be complemented by an expected US\$ 722.8 million in World Bank⁷⁹ co-financing.

Cost-effectiveness

According to the International Energy Agency's Energy Technology Perspectives 2008 Report, a marginal cost of reducing a ton of carbon dioxide-equivalent of \$200 is the lower-end estimate of the incentive needed to achieve the objectives of the "BLUE Map Scenario."⁸⁰ The Facility provides a cost-effective way to reduce GHG emissions with an overall cost-effectiveness ratio of \$54.2 per ton of carbon dioxide equivalent avoided or reduced (accounting for the total investment, i.e. GCF plus World Bank co-financing).

II. Economic viability

An economic analysis was prepared for the cooling facility to assess its economic net value. The analysis considered the following costs and benefits: (i) investment outlays to the supported projects; (ii) reflows from the funds provided through the facility; (iii) avoided or reduced greenhouse gas emissions; and (iv) recycling and/or disposal of cooling appliances. Other economic outcomes such as improvement in health conditions of people living in underserved areas, or reduced food loss and waste due to efficient cold chains, have not been monetized. However, they are expected to significantly drive the economic viability of the proposed program. The results of the analysis show that the program yields a financial and an economic internal rate of return of -0.8 and 16.3 percent, respectively. The economic value add of the Facility is largely due to its contribution to GHG emission reduction potential of the Funded Activities. Further, the economic value of is very sensitive to the estimated cost of recycling (otherwise conducting environmental remedial actions) for appliances that have exceeded their useful lives. This finding highlights the importance of disposing of appliances in a climate-friendly and economic manner. The following table presents a summary of the results of the analysis.

Table 3: Economic Model Spreadsheet Summary⁸¹

Facility total financing			Core indicator targets			Financial value add		
Grant	\$M	32	20.4%			Financial discount rate		3.0%
Loan	\$M	125	80%	Total project financing	\$M	879.8	PV of cash outlays	\$M (151.5)
Guarantee	\$M	0	0%	Total public financing leveraged	\$M	722.8	PV of cash inflows	\$M 90.6
Total	\$M	157	100%	Total private financing leveraged	\$M	-	Financial net present value	\$M (61.0)
IBRD/IDA/TrustFund co-financing							Financial internal rate of return	-0.8%
Grant	\$M	109.4	15.1%	Expected lifetime emmis. reduction	MtCO2eq	16.2		
Loan	\$M	563.4	78%	Est. total cost per tCO2eq	\$	54.2		
Guarantee	\$M	50	7%	Est. GCF cost per tCO2eq	\$	9.7		
Total	\$M	722.8	100%	Total leverage ratio	times	4.6	Sensitivities	
Private sector funds lev	\$M	0		Public co-financing ratio	times	4.6	Unit cost of recycling/disposal	\$ 375.0
				Private co-financing ratio	times	0.0	Economic net present value	\$M 504.2
Facility cash inflows							Economic discount rate	5%
Loan repayment	\$M	125.00		Economic value add			Economic net present value	\$M 248.8
Interest payment	\$M	9.00		Economic discount rate		1.0%		
Service charge	\$M	4.13		PV of economic benefits	\$M	792.4		
Reflows from guarantee	\$M	-		PV of economic costs	\$M	407.4		
Total	\$M	138.13		Net present economic value	\$M	385.0	Economic discount rate	15%
				Economic internal rate of return		16.7%	Economic net present value	\$M 19.6

The detailed calculations, underlying inputs and assumptions are presented in Annex 3.

⁷⁹ The World Bank (IBRD/IDA) co-financing amounts are indicative and subject to approval by the World Bank Board and agreement with the EEs."

⁸⁰ IEA BLUE Map scenario explores a reduction of global GHG to 50 percent of current levels by 2050.

⁸¹ The model considers the economic benefits of the assets over their economic lives. Such economic lifespan may exceed the implementation period of the Facility.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's [Performance Measurement Frameworks](#) under the [Results Management Framework](#) to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

- ☒ Shift to low-emission sustainable development pathways
- ☒ Increased climate resilient sustainable development

E.2. Core indicator targets

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	912,658 t CO ₂ eq
	Lifetime ⁸²	16,240,201 t CO ₂ eq
E.2.2. Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	<u>879,840,000</u> USD
	(b) Requested GCF amount	<u>157,000,000</u> USD
	(c) Expected lifetime emission reductions	<u>16,240,201</u> t CO ₂ eq
	(d) Estimated cost per t CO₂eq (d = a / c)	54.18 USD
	(e) Estimated GCF cost per t CO₂eq removed (e = b / c)	9.67 USD
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	<u>922,840,000</u> USD
	(g) Public source co-financed	<u>722,840,000</u> USD
	(h) Private source finance leveraged	=
	(i) Total Leverage ratio (i = f / b)	<u>5.9</u>
	(j) Public source co-financing ratio (j = g / b)	<u>4.6</u>
	(k) Private source leverage ratio (k = h / b)	=
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	4.22 million (of which 2.15 million are female) ⁸³
	Indirect	16.85 million (of which 8.51 million are female) ⁸⁴
<i>For a multi-country proposal, indicate the aggregate amount here and provide the data per country in annex 17.</i>		
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	
	Indirect	

⁸² The lifespan of the Facility is 27 years based on the lifetime of the sustainable cooling assets (15 years for appliances and cold chains and 20 years for buildings) and the distribution of funds per year (See Annex 4 on Budget) .

⁸³ Direct beneficiaries estimated only for the projects in 4 countries at high risk of extreme heat hazard. See Annex 23 for calculations and Annex 24 for explanation of methodology

⁸⁴ Indirect beneficiaries for the projects in 4 countries at high risk of extreme heat hazard

For a multi-country proposal, leave blank and provide the data per country in annex 17.

E.3. Fund-level impacts

Select the appropriate impact(s) to be reported for the project/programme. Select key result areas and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected impact result. The result areas indicated in this section should match those selected in section A.4 above. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
M3.0 Reduced emissions from buildings, cities, industries and appliances	M3.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced or avoided - buildings, cities, industries, and appliances	Ex-ante and ex-post analyses conducted by an independent contractor – calculations of annual CO ₂ e reduced or avoided associated with energy savings; procurement documents ⁸⁵ ; energy audit/assessments reports	0	3.65 Mt (assuming 40% of final target to be achieved at mid-term)	9.13 Mt CO ₂ e	Target defined as the ex-ante GHG impact expected from Facility projects targeting EEff in space cooling, and cold chains compared to BAU over a 10 year period (i.e., the Facility's implementation period)⁸⁶. See D.1 and Annex 23 for more details Main assumptions: Methodology⁸⁷: IFI Guidelines for a Harmonized Approach to Greenhouse Gas Accounting, IFC EDGE and HOMER Powering Health Tools are applied. Lifespan of the Cooling Facility: 27 years Lifetime of assets: 15 years for appliances and 20 years for buildings Annual emission reductions⁸⁸: 912,589 tCO₂eq

⁸⁵ Procurement documents will be used to verify, for example, energy efficiency investments in buildings, as well as number and types of space cooling and refrigeration/cold chain appliances procured.

⁸⁶ Estimates calculated based on the implementation period of the Cooling Facility. However, over the lifespan of the cooling appliances and equipment (15 years) and energy efficient buildings (20 years), the investments are estimated to reduce emissions by 18.6 MtCO₂e

⁸⁷ See Annex "A24. Methodology" for more details

⁸⁸ This is the annual average number. The distribution of GHG emissions savings per year is included in Annex 23 (GHG calculations)

						Lifespan emission reductions: 16,240,201 tCO ₂ eq
A3.0 Increased resilience of infrastructure and the built environment to climate change	A3.1 Number and value of physical assets made more resilient to climate variability and change considering human benefits	Ex-ante and ex-post analyses conducted by an independent contractor Procurement documents ⁸⁹ ;	0 Physical assets 0 USD million	7,223 physical assets (USD 121.7 million) of which: 681 are resilient buildings (USD98 million), 5,669 are agriculture cold chains (USD19.1 million), and 872 are health cold chains (USD4.8 million)	18,057 physical assets (USD304.4 million) of which: 1,703 are resilient buildings (USD244.7 million), 14,173 are agriculture cold chains (USD47.7 million), and 2,181 are health cold chains (USD11.9 million)	Value target based on (i) estimate cost of equipment and appliances and (ii) estimated cost of building retrofits. Target defined as the number (and estimated value) of the more heat resilient cooling assets in the 4 countries in the Facility facing high risk of extreme heat hazard (Bangladesh, Malawi, Somalia and El Salvador). ⁹⁰ Number and value of assets derived from GHG calculation model (Annex 23).

E.4 Fund-level outcomes

Select the appropriate outcome(s) to be reported for the project/programme. Select key expected outcomes and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected outcome. Add rows as needed.

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term)	Final	
M7.0 Lower energy intensity of buildings, cities, industries and appliances	M7.1 Energy intensity/improved efficiency of buildings, cities, industries and appliances as a result of Fund support	Ex-ante and ex-post analyses conducted by an independent contractor; energy audits and energy/market assessments reports - Annual	0	5,936 GWh energy saved Share of energy savings from space cooling =	14,841 GWh energy saved Share of energy savings from space cooling	Facility-funded projects supporting improvements in energy efficiency of space cooling (buildings and cooling systems) and cold chains by at least 20% compared to BAU. The target value is

⁸⁹ Procurement documents will be used to verify, for example, energy efficiency investments in buildings, as well as number and types of space cooling and refrigeration/cold chain appliances procured, along with the costs.

⁹⁰ This actually under-estimates the value of the entire building, which is dependent on local market conditions.

		calculations of energy savings resulting in lower energy intensity of project-supported EEff cooling activities		5,249 GWh Share from cold chains = 687 GWh	=13,122 GWh Share from cold chains = 1,718 GWh.	cumulative, assuming Cooling Facility's implementation period of 10 years ⁹¹ See D1 and Annex 23 for more details.
M5.0 Strengthened institutional and regulatory systems	M5.1 Institutional and regulatory systems that improve incentives for low-emission planning and development and their effective implementation	planning and policy instruments developed in each country; surveys, consultant reports	0 (no incentive mechanism in place)	6 countries adopt policy or decision to introduce incentive mechanism or regulation promoting sustainable cooling	6 policies or incentive mechanisms to incentivize sustainable cooling are operationalized ⁹² [target to be re-assessed at the Interim Evaluation stage]	By mid-term, with TA support provided through the Cooling Facility for necessary assessments, consultations and design work, assume 6 countries adopt policy or decision to introduce policy or incentive mechanisms for more climate friendly cooling-related investments/purchases. By end of project, expect 6 policies or incentive mechanisms to be in place and operational, but this target will be re-assessed at the Interim Evaluation Stage and updated accordingly
	Number of technologies and innovative solutions	Ex-ante and ex-post analyses;	0	1 (out of the 3 types outlined	3 (out of the 3 types outlined in	Assessments confirm suitability of technologies.

⁹¹ Estimates calculated based on the duration of the Cooling Facility. However, over the lifespan of the cooling appliances and equipment (15 years) and energy efficient buildings (20 years), the investments are estimated to save a total of 31,676 GWh.

⁹² E.g., applications for sustainable cooling-related financing are received/accepted; energy efficient/green buildings certifications/labelling.

	transferred or licensed to support low-emission development as a result of GCF	procurement documents		in the assumption column)	the assumption column)	Technologies and innovative solutions are the following: (i) solar based efficient cooling solutions for health, (ii) solar based efficient cooling solutions for agriculture, and (iii) innovative solutions for cool roofs.
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	A7.1 Use by vulnerable households, business and public sector services of GCF-supported tool, instruments, strategies and activities to respond to climate change and variability.	Ex-ante and ex-post analyses conducted by an independent contractor; surveys (surveys will be designed and adapted to each specific project and target beneficiaries, including participants in trainings and workshops)	0 people	8,020 entities 6,000 farmers, 2,000 households and business in the buildings sector, and 20 hospitals and clinics have received information through the GCF tool, have implemented at least 1 sustainable cooling measure (installation of EEff equipment and/or practice) and have reported benefits (e.g., enhance	20,050 entities 15,000 farmers, 5,000 households and business in the buildings sector, and 50 hospitals and clinics have received information through the GCF tool, have implemented at least 1 sustainable cooling measure (installation of EEff equipment and/or practice) and have reported benefits (e.g., enhanced comfort, reduced operational costs,	The indicator will be counted in countries at high risk of extreme heat hazard. Targets based on farmers, households, commercial buildings, hospitals and clinics benefitting from the Facility. The communications tool/activities will inform of: (i) agriculture sustainable practices and efficient equipment to improve agriculture production, (ii) cooling sustainable practices and efficient equipment to improve comfort levels in buildings, and (iii) cooling sustainable practices and efficient equipment to

				d comfort, reduced operation al costs, reduced wastage of vaccines/ medicine or food, and/or increased income)	reduced wastage of vaccines/ medicine or food, and/or increased income)	optimize vaccines conservation
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E.5. Project/programme performance indicators

The performance indicators for progress reporting during implementation should seek to measure pre-existing conditions, progress and results at the most relevant level for ease of GCF monitoring and AE reporting. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Result 1: Strengthened capacity of state and non-state actors (for example, banks, energy service companies, off-grid-distributors) to assess and deploy climate-friendly cooling solutions	1.1 Number health, financial and energy institutions integrating climate-friendly cooling solutions in procurement/lending/bidding documents	Ex-ante and ex-post analyses conducted by an independent contractor; surveys; procurement documents; supervision mission reports	0	9 countries conduct capacity building and training activities on sustainable cooling	9 institutions/agencies integrate provisions for reliable and climate friendly cooling in procurement/bidding/lending documents. At least four also develop provisions/guidelines for O&M of cooling equipment.	Capacity building and training is provided to officials and stakeholders in health sectors of 3 countries and to financial institutions and energy agencies in 6. TA to support development of procurement documents and O&M guidelines for well functioning, sustainable cooling.
	1.2 Number of legal/regulatory/policy instruments drafted/amended ⁹³	Planning and policy instruments developed in each country	0	1	3	Legal/regulatory instruments (e.g., code/standard/ green or energy efficient procurement) drafted

⁹³ In some cases, entirely new legal/regulatory/policy instruments may be drafted; in others, where there is already a basis, we may see amendment (e.g., we may see an update to a building code/ procurement strategy or Minimum Energy Performance Standard).

						(or amended/updated) in at least 3 countries
Result 2: Increased public awareness of the benefits of climate-friendly cooling	2.1 Number of people adopting/ switching to climate friendly cooling facilities compared to baseline at home or at the workplace.	Ex-ante and ex-post analyses conducted by an independent contractor; surveys	0	1.26 million of which: 0.7 million are people living in improved homes, and 0.56 million are people working in improved workspaces	3.15 million, of which: 1.74 million are people living in improved homes, and 1.41 million are people working in improved workspaces	Target based on estimated (i) people living/working in green/retrofitted buildings in space cooling projects, (ii) farmers and households purchasing climate friendly cooling equipment/appliances
Result 3: New or rehabilitated climate-friendly cooling infrastructure and equipment	3.1 Number of health facilities (clinics, hospitals) with improved cooling and refrigeration operating as a result of the project.	Ex-ante and ex-post analyses conducted by an independent contractor; reports from supervision missions	0	69	172	Target based on estimated health facilities (hospitals and clinics) in the 3 countries over the project lifetime having more reliable and climate friendly equipment that is functioning
	3.2 MW of solar power installed through the solarization of cooling as a result of Cooling Facility support	Ex-ante and ex-post analyses conducted by an independent contractor; Bidding documents	0	50MW	126 MW	Solarization of cooling services in the Facility's cooling investment in agriculture and health sectors (in 5 countries).
	3.3 Number of energy efficient buildings financed through the Cooling Facility	Ex-ante and ex-post analyses conducted by an independent contractor; Bidding documents	0	941 buildings	2,354 buildings	Target based on number of energy efficient buildings in 4 countries targeting space cooling (i.e., Sri Lanka, Bangladesh, Panama and North Macedonia). (Multi-family buildings are considered as one building)
	3.4 Number of energy efficient space cooling	Ex-ante and ex-post analyses	0	0.21 million appliances	0.53 million appliances	Target based on number of space cooling appliances (air

	appliances deployed	conducted by an independent contractor; procurement documents				conditioners) in four countries targeting space cooling. (Note: where appliances are included in the new EEff buildings above, they are <i>not</i> counted here).
	3.5 Energy expenditure savings by users (\$)	Ex-ante and ex-post analyses conducted by an independent contractor ; energy audits/assessment reports	0	\$712 million	\$1,781 million	Target refers to all Facility countries and is based on: Estimated energy savings per country (kWh) * Energy source price per country (\$/kWh) ⁹⁴ over the program implementation period (10 years).
	3.6 Percentage of priority population vaccinated	Ex-ante and ex-post analysis conducted by an independent contractor; ministry of health reports;	[to be determined for each project during project preparation]	[x% - to be determined during project preparation for each country]	[50% - to be confirmed during project preparation and refined at Interim Evaluation stage]	This indicator is relevant for El Salvador, Sao Tome and Principe, and Somalia. Priority population will be determined according to each country's national plans during project preparation, along with targets. Targets will be updated in annual performance reports and finalized by midterm.
	3.7 Improved thermal comfort	Building occupant surveys and/or building measurements	Occupant survey and/or Ex-ante indoor air temperature and indoor humidity measurements	Majority of building occupants (disaggregated by gender) expressing improvement in thermal comfort; [XX]% of total renovated buildings with improved	Majority of building occupants (disaggregated by gender) expressing improvement in thermal comfort; [YY]% of renovated buildings with improved indoor air	For Bangladesh (country at high risk of extreme heat hazard with Cooling Facility intervention targeting space cooling), surveys will be prepared for building occupants (gender disaggregated). This can be complemented with ex-ante and ex-post measurement of ambient temperature;

⁹⁴ See Annex 23 (GHG Calculations) for estimates for individual countries and data sources.

			To be determined during project preparation	indoor air temperature and indoor humidity measurement	temperature and indoor humidity measurement	ex-ante and ex-post measurement of indoor air temperature (dry-bulb temperature (DBT) and indoor air humidity (wet-bulb temperature) The targets will be defined during project preparation and updated in annual performance reports (starting when completion of first EE building renovation works) and finalized at mid-term (Interim evaluation stage)
	3.8 Proportion of agriculture production stored in cold chain equipment/facilities	Ex-ante and ex-post surveys with farmers (disaggregated by gender) conducted by an independent contractor.	Proportion of agriculture production stored in cold chain equipment/facilities [baseline to be determined during preparation]	[x]% of agriculture production stored in cold chain equipment/facilities	[x]% of agriculture production stored in cold chain equipment/facilities	For Malawi (country at high risk of extreme heat hazard, the baseline will be determined with country statistics and surveys during project preparation and targets will be defined during project preparation and updated in annual performance reports and finalized at mid-term (Interim evaluation stage)
Result 4: Financing mobilized for cooling investment	4.1 Additional public and private capital for climate-friendly cooling investments mobilized by the GCF through programmatic Cooling Facility	Ex-ante and ex-post analyses conducted by an independent contractor; disbursement reports	\$0	\$0.38 billion (24% private sector)	\$0.95 billion (24% private sector)	Target based on expected financing to be mobilized for climate friendly cooling (all target areas and sectors) in the countries included in the Facility. This includes government own funds; IBRD/IDA; other WB administered trust fund; and private capital

E.6. Activities			
<i>All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in the implementation timetable. Add rows as needed.</i>			
Activities	Description	Sub-activities	Deliverables ⁹⁵
Component 1: Policy, regulatory and enabling environment support			
Activity 1: Capacity building of key stakeholders (for example, on technologies, business models, cooling project appraisal, implementation and evaluation)	Activities that help strengthen the capacity of state and non-state actors (such as commercial banks, energy service companies and off-grid distributors)	- Technical and economic assessments - Review of procurement documents and technical specifications - Survey development and administration - Development of training materials	- At least 15 training events/workshops - Training material linked to assessment, financing, procurement, design and/or operation and maintenance of sustainable cooling solutions developed/used in at least 8 countries
Activity 2: Technical assistance for enabling plans, strategies, policies, regulations, and standards	Activities that support the preparation of relevant plans, strategies, policy and regulatory instruments and that promote gender equality in cooling-related sectors	- Advisory services - Market assessments - Policy analyses - Regulatory review - Measures to address sustainable cooling included in NDCs and other national policy documents - Assessment of options to advance gender equality (data, policy reform, finance and skills development/employment)	- Regulatory/legal/institutional assessments (including, e.g., labeling and certification of energy efficient buildings/equipment) and recommendations for development and/or implementation of sustainable cooling in at least 4 countries - Recommendation and input towards development of National or local climate/adaptation/cooling plans and/or strategies in at least 4 countries - Gender assessments in 9 countries. - Guidelines for procurement and/or design/implementation of programs for sustainable cooling in at least 3 countries - Monitoring and evaluation mechanisms developed in at least 4 countries.
Activity 3: Designing sustainable market mechanisms and business models	Activities that support the planning and development of context-specific business models and financing/implementation	- Advisory services (for example, for auction design, to establish dedicated financing windows, or to design sustainable financing schemes) - Technical, legal, financial due diligence for cooling investments	- Assessments and designs of cooling-related financing/implementation mechanisms/business models in at least 4 countries - Market assessments (which may include baseline context and willingness to pay) in at least 2 countries.

⁹⁵ Deliverables are indicative.

	mechanisms climate-friendly cooling deployment	- Consultations with stakeholders - Cooperation with other global initiatives, financiers, think tanks, universities, etc.	
Activity 4: Communication and awareness campaigns	Activities that increase public awareness and promote behavior change	- Survey development and administration - Opinion research - Public consultations - Focus group discussion - Message design and testing - Market assessments - Gender gap analyses	- At least 5 countries will develop and implement at least 1 communications/awareness campaigns

Component 2: Financing for cooling investments

Activity 5: Financing of climate-friendly cooling investment	Financing of public and private stakeholders of investment cooling infrastructure equipment	- Engineering, procurement and construction of cooling infrastructure - Procurement, installation and maintenance of cooling equipment including operations and maintenance arrangements - Provision of loans and grants to public and private stakeholders for cooling investment - Coordination with other public and private financiers, technical partners and investors	- Financing mechanisms that engage private sector designed in at least 6 countries. - Public procurement documents of cold chains integrate climate-friendly features in 3 countries.
Activity 6: Credit lines to financial intermediaries	Design and implementation of credit lines for cooling investments and equipment	- Advisory services - Analytical support - Provision of credit lines - Coordination with other public and private financiers, technical partners and investors	- Credit lines designed in at least 3 countries.

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Besides the arrangements laid out in the AMA between the World Bank and the GCF, the implementation of the Cooling Facility involves (a) monitoring of program performance indicators (as per Section E5); (b) periodic progress reports; and (c) midterm review of implementation progress. It is noted that the end targets of the Program are best estimates as it is not known upfront exactly the results from each project that will be financed and the targets may be revised during AE's due diligence, project preparation and the mid-term review as necessary. The AE will report to GCF the implementation status at both Facility and country level by aggregating specific country-level information and data.

▪ Monitoring

The implementation of each project under the Cooling Facility will be managed, monitored and reported on at project level by the Executing Entities and the Project Management Units (which may include staff and consultants). The World Bank has dedicated staff in its sectoral teams, operation departments, Environment and Social departments and country offices that will conduct due diligence, monitor compliance and

performance risks, implementation of the Environmental and Social Commitment Plans and the stakeholder engagement plans.

The World Bank will follow its standard monitoring and reporting policies and procedures, which will include semi-annual reports on the implementation status and performance of each project. By the end of each project supported by the Cooling Facility, the Bank will conduct jointly with each respective Executing Entity an implementation completion assessment and prepare an implementation completion report (ICR) which will review the performance of the project, assess effectiveness and efficiency of project implementation, the achievement of the project development objective, and provide relevant lessons learned. Financial reporting is further discussed in Section G3.

▪ Reporting

Reporting of Executing Entities to World Bank. As specified in the subsidiary agreements between the Executing Entities and the World Bank, the Executing Entities are obliged to report on the implementation of the Project and the use of the financing proceeds, as applicable. This reporting includes the environmental and social performance of the project and the achievement of the indicators agreed between the WB and the Executing Entity. For each project supported by the Cooling Facility, the respective Executing Entity will be responsible for overall M&E of the project, including collection of project performance information and reporting on project impacts and results.

Reporting of World Bank to GCF. The World Bank will provide to the GCF (i) annual activity performance reports (APRs) on the status of the Facility throughout the relevant reporting period, (ii) mid-term evaluation report at the midpoint of the implementation period of the Facility and (iii) final evaluation report at the end of the implementation period of the Facility as per the FAA provisions.

▪ Evaluation

The evaluation methods for mid-term and final evaluations will include:

- Key informant interviews with relevant beneficiary staff and clients, as well as other key stakeholders, based on stakeholder mapping; and.
- Desktop review of relevant Facility documentation, including Project Management Unit reports, based on an agreed list of evidence to be provided by the client.

Please refer to Annex 11 for further details.

As described earlier, each country's cooling operation, a PMU will be established with a team expected to include at least a team lead, technical expert, procurement specialist, financial management specialist, environmental specialist and social/gender specialist, and a lawyer. Each project will have its own monitoring and evaluation (M&E) and reporting process, and the EE will be accountable for its implementation.

The Bank's evaluation of projects included in the Cooling Facility is done in the context of the Bank's standard Implementation Status and Results (ISR) report⁹⁶, which is the key tool used for internal reporting on the implementation performance and prospective outcomes of projects. It is through this tool that the Bank's project team leader conducts the evaluation of the project's implementation status and performance, progress toward reaching its development objective, and evolution of risks. Safeguards, financial management, and procurement specialists advise the team leader on evaluation of project's performance from a fiduciary and safeguards perspective.

⁹⁶ See The Bank Guidance on Investment Project Financing: Implementation Support (2017) - <https://worldbankgroup.sharepoint.com/sites/ppfonline/PPFDdocuments/ee96edefb68040df8af93027280ab4e1.pdf>

The Bank's operational task team will document progress toward meeting project objectives by tracking and updating the project's cooling or cooling-informed indicators as outlined above through the Bank's periodic Implementation Status and Results reporting (ISR) process⁹⁷. The Bank will draw on these inputs to report aggregate data annually across the portfolio as part of the AE's agreed obligations to the GCF. It is noted that the end targets of the Facility are best estimates as it is not known upfront exactly the results of each investment that will be financed, and the targets may be revised during AE's due diligence, project preparation and the mid-term review as necessary.

At the end of the project, the Bank will conduct jointly with each PMU, an implementation completion assessment and prepare an Implementation Completion and Results (ICR) report which will review the performance of the project, assess effectiveness and efficiency of project implementation, the achievement of the development objective, and provide relevant lessons learned. The individual project ICR reports will form the basis of the Facility terminal evaluation report.

⁹⁷ The ISR is the Bank's key tool used for internal reporting on the implementation performance and prospective outcomes of projects. It is through this tool that the Bank's project team leader conducts the evaluation of the project's implementation status and performance, progress toward reaching its development objective, and evolution of risks.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1: Project readiness risk

Category	Probability	Impact
Operational	Medium	Medium

Description

There is a risk that some of the early stage projects do not mature and thus, are not able to mobilize funding from the Facility. Developing the relatively new concepts of cooling for some projects, may take significant time during the start-up phase, especially for activities that involve cold chain or mobile cooling where beneficiaries have limited experience and/or data and information on technology and related cost savings are missing.

Mitigation Measure(s)

To mitigate this risk, the Facility is designed to leverage the ESMAP multi-sectoral Efficient, Clean Cooling Technical Assistance Program. The TA is intended to help beneficiaries and Bank operational teams in the design of cooling informed projects including activities such as cooling market studies, sector gap analysis and strategies, knowledge sharing, stakeholder consultations, business models identification, etc. The TA will provide dedicated grants and tailored technical experts to ensure a solid pipeline of well-prepared projects, designed to meet the eligibility criteria of the Facility. Continued dialog with national stakeholders will support the preparation of cooling projects aligned with the goals and criteria of the Facility.

Selected Risk Factor 2: Political Risk

Category	Probability	Impact
Political	Medium	Medium

Description

If project investments result in direct or indirect increases in tariffs, fees, consumer product prices, or other cost increases where constituents might bear the cost burden (in order to ensure the financial sustainability of project investments), this may prove politically unattractive for client governments to adopt, especially with respect to reelection.

Mitigation Measure(s)

WB mitigates this risk by ensuring that the design of the projects consider the affordability constraints of the end user beneficiaries. For example, the WB will ensure that low income groups are provided with financial support if affordability thresholds are surpassed, and, whenever possible, combine tariff/product increases with measures to improve the payback efficiency of the utility/supplier in order to ensure a positive impact on costs and reduce the cost passed on to consumers. These efforts will be sustained to lower risk level to Low as much as possible for all projects under the Facility.

Selected Risk Factor 3: COVID-19 risk

Category	Probability	Impact
Operational	High	Medium

Description

Given the size and scope of the economic impact of the pandemic, there is a growing concern that developing countries will be hit the most which will cause disruptions in various economic and social sectors. There is a risk that client countries will shift their priorities to focus on immediate emergency response and recovery interventions to address the COVID crisis and investment in cooling could be deterred.

Mitigation Measure(s)

The World Bank teams will closely monitor the current COVID-19 crisis and its potential impacts on the progress of the projects supported by the Facility. In addition, the WB operational teams take part of the ongoing dialogue with the governments on the C-19 and will ensure to position the cooling agenda as part of the solutions to help client countries in the immediate response as well as in their economic recovery from the current difficult situation. The need for cooling will be even more critical in the current situation for instance in hospitals, warehouses and testing facilities, cold chains supply for food and/or pharmaceutical/medical goods storage, etc. The technical assistance activities provided through ESMAP will ensure continued engagement with the various countries as listed in this proposal.

Selected Risk Factor 4: Limited uptake of EEff Clean Cooling technologies

Category	Probability	Impact
Operational	Medium	Medium

Description

The penetration of energy efficient and clean cooling solutions may be very low particularly in sectors such as residential and commercial as end-users often undervalue energy savings due to lack of information and awareness about the energy savings, reluctance to pay for a higher initial cost, etc. Therefore, the implementation of some of the projects could be exposed to the risk of a lack of market demand for efficient clean cooling solutions.

Mitigation Measure(s)

This risk can be mitigated through dedicated capacity building activities targeting relevant stakeholders, awareness raising, and communication combined with market studies that would identify and recommend policy options and business models adapted to the country circumstances to facilitate the rapid uptake of the EEff clean cooling technologies. The World Bank has a longstanding experience working with developing countries on EEff, and the teams will build on the lessons learnt from these experiences as well.

Selected Risk Factor 5: Lack of technical capacity

Category	Probability	Impact
Operational	Medium	Medium

Description

The lack of technical capacity, in some countries, could negatively affect the implementation of the projects, continuous operation of the equipment and infrastructure, and thus the expected outcomes. A proper installation, operating, and maintenance is key to ensure the projects are not underperforming, hence the results are not compromised.

Mitigation Measure(s)

The risk will be mitigated through the M&E framework that will be developed during the preparation phase and is an integral part of the project documentation and specific to each country. M&E arrangements will be used to assess periodically the project performance and impact as expected. In order to ensure the sustainability of the investments, the EEs will ensure that the beneficiaries receive adequate training to conduct O&M to ensure continuous operation of the equipment and infrastructure and/or may subsequently enter into an O&M contract

Selected Risk Factor 6: Credit risk

Category	Probability	Impact
Credit	Low	Medium

Description		
GCF loans are provided to sovereign entities and will not bear any credit risk beyond that the sovereign risk of the Borrowers. In those projects where GCF loans are deployed, and where qualified PFIs on-lend, they, not the GCF, will be exposed to the credit risk of their customers and will employ their risk management processes.		
Mitigation Measure(s)		
The GCF credit exposure is limited to EE listed in Section A.20. As such, the GCF would only bear the sovereign risk of the countries in which the loans will be deployed.		
Selected Risk Factor 7: Risk of co-financing not materializing		
Category	Probability	Impact
Financial	Low	Medium
Description		
Risk of IBR/IDA co-financing not materializing or being lower than specified in this Funding Proposal and its impact on the proposed target for adaptation and mitigation. .		
Mitigation Measure(s)		
About 54 percent (or \$392.84 million of \$722.84 million) of the co-financing amount has already materialized, with WB and WB-managed trust funds approved and the corresponding legal agreements signed with the recipient countries for six of the nine projects envisaged. Should the balance the remaining projects materialize albeit with a ratio of GCF funds to IDA/IBRD funds not greater than 1:1 (the minimum level set in the eligibility criteria), the Facility would achieve 71% of its mitigation target (as measured by GHG emissions reduced or avoided) and reach 96% of the target direct beneficiaries. In the very unlikely case that none of the remaining projects take place (and this GCF counterpart financing is not provided) the facility will only achieve 33% of its mitigation impact but reach 90% of the direct adaptation beneficiaries. The risk is mitigated by (i) the strong alignment of the objectives supported by the Cooling Facility, with the country needs as expressed in national strategies, policies and plans; (ii) the country ownership reflected through the non-objection letters received from each National Designated Entity; and (iii) the robust sector-level dialog led to the World Bank in the supported countries.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

The main aspects considered from an E&S perspective are summarized below. However, each project under the Cooling Facility will have its own assessment completed at concept/appraisal stage. The ESMF will include a list of expected environmental and social impacts and suggest mitigation and monitoring measures for each potential impact due to the project activities.

- **ESS1 (Assessment and Management of Environmental and Social Risks and Impacts)**

The Facility's cooling-informed projects are expected to generate substantial direct social and environmental benefits by increasing energy efficiency in buildings and cold chain infrastructure and deploying climate-friendly cooling equipment including off-grid cooling solutions that use renewable energy. Environmental benefits include, inter alia: (i) reduced greenhouse gas emissions due to energy savings and reduction or avoidance of food losses/waste which will ultimately contribute to climate change mitigation; (ii) reduced local air pollutants (fine particulates, SOX, NOX); and (vi) reduced emissions of ozone depleting substances and/or high GWP chemicals used as refrigerants in cooling equipment.

The projects also generate significant positive social impacts through job creation and market development for energy efficiency products and services; reduction in consumer electricity bills thus, improving affordability and increasing competitiveness and income of small businesses/farmers; increased comfort and well-being including a better overall working environment in buildings. By improving and expanding access to cooling solutions including cold chain, the projects will reduce the vulnerability to extreme heat events, increase access to vaccines and medicines as well as to nutritional food; hence generating positive impacts on health and food security.

Some of the projects may cover geographically remote, politically sensitive and socially marginalized areas and communities. The projects therefore benefit not only mainstream communities, but also those that are often marginalized or maybe subject to discrimination. Improved cooling services and access to affordable energy efficient cooling products and increased employment opportunities are likely to also provide benefits to women and disadvantaged or vulnerable individuals or groups. The E&S risks and impacts assessment of the projects will include impacts on gender and disadvantaged or vulnerable individuals or groups. Differentiated measures will be adopted so that adverse impacts do not fall disproportionately on the disadvantaged or vulnerable, and that they are not disadvantaged in sharing development benefits and opportunities resulting from the project, which would enhance project development outcomes by promoting inclusion, equality and non-discrimination. The Facility will work with countries to collect data and generate crucial knowledge of actions and design interventions needed to avoid or minimize gender impacts and impacts on the poor and vulnerable. Particularly, the actions will be guided by the WB Good Practice Notes (GPNs) on: (i) Gender; (ii) Non-Discrimination and Disability; (iii) Non-Discrimination: Sexual Orientation and Gender Identity (SOGI)⁹⁸.

If not properly managed, construction of new buildings and cold chain facilities may result in potential adverse environmental and social impacts associated with land acquisition or land conversion. During the construction or retrofitting of buildings, common risks are safety and health hazards on workers and the community, which are due to noise, dust, handling and disposal of old parts and equipment which may contain hazardous waste and materials.

⁹⁸ <http://pubdocs.worldbank.org/en/158041571230608289/Good-Practice-Note-Gender.pdf>
<http://documents1.worldbank.org/curated/en/573841530208492785/Environment-and-Social-Framework-ESF-Good-Practice-Note-on-Disability-English.pdf>
<http://pubdocs.worldbank.org/en/590671570796800429/Good-Practice-Note-SOGI.pdf>

However, these potential impacts can vary in terms of intensity/significance and are localized in nature. For each project, an assessment of environmental and social risks and impacts will be carried out and measures to address them will be identified in accordance with the mitigation hierarchy as described in the previous section. For example, some of the impacts can be prevented by aligning site selection for buildings with ESS requirements and mitigated by sound operational practice and consistency with ESS3 principles during operation.

- **ESS2 (Labor and Working Conditions)**

Project workers are expected to be mobilized under most projects. This will include direct workers and contracted workers. The Government will have to comply with the ESS2 requirements as specified in Labor Management Procedures (LMP) to be prepared for each project. Occupational health and safety (OHS) plans will need to be prepared as per the World Bank Group EHS guidelines. The labor influx would have a series of impacts, including labor conditions, health, safety, GBV, conflict etc., not only among the project workers but also upon local communities. These risks need to be assessed as part of the E&S assessment and managed in accordance with the LMP and other E&S documents. All relevant measures will be reflected in the bidding and contractual documents. A dedicated grievance redress mechanism for project workers needs to be established under each project. This mechanism is separate from the overall project Grievance Redress Mechanism.

- **ESS3 (Resource Efficiency and Pollution Prevention and Management)**

Construction of new buildings or retrofitting existing buildings are expected to generate typical construction impacts such as, dust; construction waste generation, transportation, transfer and disposal due to demolition; and handling, and disposal of asbestos waste, resulting in negative impacts on air quality and occupational and community health and safety. These risks will be assessed for each project. However, most of these potential impacts are low to moderate in intensity/significance and are reversible and localized in nature, and therefore can be mitigated.

Other hazardous waste generation is specifically associated with the dismantling and disposal of old cooling equipment and refrigerant gases. These gases could be ozone depleting substances or have a high GWP and if not properly handled, they risk to be released into the atmosphere. These impacts require appropriate mitigation assessments and management measures and include adequate and safe management of refrigerants (e.g. recovery, recycling or storage) in line with national legislation and countries' obligations under the Protocol of Montreal and relevant Good International Industry Practice (GIIP).

- **ESS4 (Community Health and Safety)**

ESS4 requirements will be considered in the implementation of project's activities. During the construction works of the buildings (including cold chain buildings), potential risks may arise from noise, dust, labor safety, and disposal of old parts and equipment which may contain hazardous waste. In the case where the construction works are carried out in existing facilities used by employees or neighboring communities, there may be a temporary transitional impact for these users. These impacts require appropriate mitigation assessment and management measures to contain them.

Additional risks are associated with the type of refrigerants used in cooling equipment. Some can be toxic and/or flammable and may result in accidents during their handling and installation. These safety risks can be minimized through proper mitigation assessments and management measures to contain them.

- **ESS5 (Land Acquisitions, Restrictions on Land Use and Involuntary Settlements)**

Land acquisition for construction of new buildings or cold storage facilities will need to be done in a consistent manner with ESS5 requirements - to avoid, minimize, reduce/mitigate and compensate at replacement cost land acquisition based on due diligence and plans prepared in accordance with ESS5. An indicative outline of a resettlement policy framework is presented in Annex 5 (in ESSF Annex). The Executing Entities will be responsible for preparing, in a manner and substance satisfactory to the Bank, the project-specific resettlement policy frameworks and/or resettlement action plans in line with ESS5 requirements

- **ESS6 (Biodiversity Conservation and Sustainable Management of Living Natural Resources)**

Impacts, arising out of the siting and selection of land for the future cold storage facilities, may include loss of habitats and other adverse encroachment impacts, depending on the scale of the projects. Care should be taken by these future projects to ensure that their impacts on natural habitats, terrestrial and aquatic ecosystems and species are well assessed and managed.

- **ESS7 (Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local communities)**

The presence of indigenous people in the areas identified for the projects will be screened and confirmed by the World Bank in consultation with the national authorities. Once it is confirmed, consistency with ESS7 objectives and requirements will be integrated into the design of the project and E&S documents, including the development of an indigenous peoples' plan or other types of plans depending on the country and project context, and Free, Prior and Informed Consent in the circumstances specified in ESS7.

- **ESS8 (Cultural Heritage)**

The project will screen the project area and assess potential impacts on cultural heritage, both tangible and intangible. The site selection for each project will be mindful of cultural heritage and the application of ESS8 will be further assessed under each project. In case of no apparent cultural heritage in civil works sites, a chance find procedure will be a part of the E&S instrument.

- **ESS9 (Financial Intermediaries)**

The deployment of funds under the Cooling Facility will be tailored to the needs of the projects. Some projects are expected to channel financing to Financial Intermediaries (FI). For such projects, ESS9 will be applied to the FIs going to receive financial support under the Cooling Facility. In this respect the FI is required to establish and maintain an Environment and Social Management System (ESMS) to identify, assess, manage, and monitor the environmental and social risks and impacts of FI subprojects on an ongoing basis. The ESMS will be commensurate with the nature and magnitude of environmental and social risks and impacts of FI subprojects, the types of financing, and the overall risk aggregated at the portfolio level. Where the FI can demonstrate that it already has an ESMS in place, it will provide adequate documented evidence of such an ESMS, indicating which elements (if any) will be enhanced or modified. The FI's ESMS will include the following elements: (i) environmental and social policy; (ii) clearly defined procedures for the identification, assessment and management of the environmental and social risks and impacts of subprojects; (iii) organizational capacity and competency; (iv) monitoring and review of environmental and social risks of subprojects and the portfolio; and (v) external communications mechanism. An FI may be required to adopt and implement additional or alternative environmental and social requirements, depending on the nature of the FI, its activities, the sector or countries of operation, and the environmental and social risks and impacts of the potential FI subprojects.⁹⁹

- **ESS10 (Stakeholder Engagement and Information Disclosure)**

The Project affected parties include implementing agencies at national level, regional level, local authorities of participating provinces, private investors and concerned communities. In addition, other interested parties include media, local/international NGO and development partners working in the same region.

A Stakeholder Engagement Plan (SEP) will be developed for each project, to ensure transparency and meaningful consultation with the affected and interested parties. Stakeholder engagement and consultations will be conducted throughout the project cycle. This will include discussions of project design and impacts as well as multi-stakeholder discussions on these issues during the preparation phase. The SEP, along with other social and environmental instruments, will be subject to public consultation and disclosure per requirements of ESS10 and will be treated as a live document to be regularly updated along the pace of project implementation.

- **WB Performance Standards for Private Sector Activities**

WB Performance Standards (PSs)¹⁰⁰ apply to activities supported by the Private Entities. For this purpose, a Private Entity is a natural or legal person, whether publicly or privately owned, which is established for a business purpose and operates on a commercial basis, is financially and managerially autonomous, and whose day-to-day operations are not controlled by the government. Private Sector Activities refer to activities (a) designed, constructed, operated and/or owned by Private a Entity (or consists of TA in preparation for such activity), and (b) for which the Private Entity is fully responsible for identifying, assessing and managing the environmental and social risks associated with the activity it carries out or with the subprojects it finances in the case of FIs.

PS1 (Assessment and Management of Environmental and Social Risks and Impacts) applies to all projects that have environmental and social risks and impacts. Depending on project circumstances, other Performance Standards may apply as well. The Private Entity is responsible for developing an ESMS. FIs are required to develop an ESMS, acceptable to the Bank, which the intermediary applies in identifying, assessing and managing environmental and social risks and impacts under its Bank-supported portfolio of subprojects. The Bank reviews the Private Sector Activity, the ESMS, and environmental and social assessment and management plans prepared for activities under the ESMS for consistency with the WB Performance Standards. The Bank categorizes the activity based on the nature of the potential environmental and social risks and impacts of the Private Sector Activity. Finally, the Bank monitors the social and environmental aspects of the Private Sector Activity.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

The Facility aims to strengthen women's roles as employees, entrepreneurs and consumers in cooling-related sectors in project countries, including enhanced female economic participation in cooling-related sectors and increased women's access to services (e.g., energy, cooling, health, financial) and enhanced productivity and income generating opportunities for female-led businesses and women. Proposed interventions under the Gender Action Plan are both aligned with the World Bank Group Gender Strategy (2016-2021): (i) advancing equality between men and women through climate change mitigation and adaptation actions, and (ii) minimizing gender-related risks in climate change actions.

While more data and analysis are needed to better understand the gender aspect of access to sustainable cooling, gender risks are evident (SEforALL 2019). Women and girls are often disproportionately impacted by limited access to sustainable cooling and enhanced access could ensure gains in terms of entrepreneurship opportunities, and enhanced livelihoods.

Taking into account the diversity of projects supported by the Facility, in terms of sectors (e.g., agriculture, buildings, energy etc) and geographies (both in terms of countries and in terms of urban versus rural contexts), **the Facility's gender approach will be adapted to each projects' specific design, local barriers and circumstances**, but issues related to closing gender gaps in terms access to cooling-related services, economic participation, access to finance, as well as opportunities to promote gender equality and inclusion will be effectively identified and embedded throughout each project's cycle.

⁹⁹ These will be incorporated into the ESMS, the environmental and social procedures and/or set out in the legal agreement.

¹⁰⁰ These are: Assessment and Management of Environmental and Social Risks and Impacts (PS1), Labor and Working Conditions (PS2), Resource Efficiency and Pollution Prevention (PS3), Community Health, Safety and Security (PS4), Land Acquisition and Involuntary Resettlement (PS5), Biodiversity Conservation and Sustainable Management of Living Natural Resources (PS6), Indigenous People (PS7), and Cultural Heritage (PS8).

These issues will be tackled in a sustainable manner, raising awareness, improving gender policies, building capacities and generating knowledge on the program's differentiated impact on men and women. Preliminary gender assessments for each of the identified countries under the Facility highlighted the following gender gaps and their potential mitigation actions:

- **Entrepreneurship:** Women entrepreneurs often lack information about access to finance options in the country. In addition, development of enterprises requires technical assistance support tailored to women's needs. Partnering with business networks, such as female exporters or associations under chambers of commerce, is key to creating awareness about credit lines among women entrepreneurs. From a consumer perspective they will also have different needs and wants in terms of cooling technology products and services and the sectors they operate in (e.g. agriculture or transport).
- **Financial services**—not only accounts, but payments, savings, and credit—give women some basic tools they need to enhance their livelihoods and economic status. Under the current ecosystem, bank accounts are key to ensure payments for cooling technologies, and strategies need to be developed to ensure women's participation to the financial sector and cooling market. The use of digital payments can further reduce access for women. Affordability is a key part of this focus.
- **Jobs and Skills Development:** Job creation and livelihood opportunities will be mapped out as part as part job creation and skills development aspects of the Facility. For instance, having women sales agents and local suppliers will create opportunities in terms of jobs or looking at labor force participation gaps between women and men in the e.g. cooling sector and implementing entities.
- **Consumer Engagement:** Growing evidence points to significant differences in the needs and wants of women versus men as e.g. consumers and entrepreneurs, as well as caregivers. For example, differences can be observed in how they use energy on a daily basis, how much they are willing to adopt cooling technology and measures e.g. in agriculture sector), and how they respond to campaigns marketing products and services. Differences have also been observed in the context of caregiving responsibilities in communities and families. It is therefore essential to tailor any efforts toward consumers specific behaviours of men and women. This points to a need for communications strategies in these countries to look beyond written materials, communication mode (e.g. radio, television, internet) as they transmit information related to benefits of cooling-related services, consumption patterns, and so forth. It could also be advisable to leverage women's groups.
- **COVID-19 response and health measures:** Women often lack empowerment and autonomy to access services, sometimes related to health and decision making. Women must not be excluded from accessing services, as they can act as agents of change. Families and communities can benefit from informed actions taken by women to counter the negative effects of the global pandemic.
- **Data and information:** collecting sex-disaggregated data and gender-specific information help to assess the projects' results and impacts on gender equality and also help to scale-up refine activities for future interventions

The Facility will follow a systematic approach and provide support to projects interventions with a focus on data collection, policy engagement, project level actions and capacity-building activities.

The Facility will leverage support for the gender-related activities from the ESMAP Gender and Energy Program, the ESMAP Efficient Clean Cooling Program, as well as from the Bank's ecosystem of gender focal points for relevant WB Global Practices. The Facility will encourage the recruitment of a balanced female-to-male employment ratio recruitment of women within the project management unit (PMU) teams.

Obtaining stakeholders buy-in will be key in ensuring that interventions to promote women's employment and enhance their livelihood are effectively implemented with measurable outcomes. Institutional and behavioral change are a long-time process that will require sustained support of all relevant

stakeholders (public and private) throughout the program cycle. Countries adherence to international agreements and national normative frameworks on gender equality, will be a good starting point to strengthen government's support for gender interventions.

At the project and program level, the following approach will be followed:

- **Data collection to document the baseline situation and identify gaps.** Baseline data will be collected to set targets and measure progress during implementation, it will also support existing efforts where baseline sex-disaggregated data are weak. Research will build upon existing country gender diagnostics and action plan when available, to ensure national priorities are understood and considered. To support project design and implementation, stakeholders' consultations will involve female participants, to understand men and women differentiated needs. In addition, initial assessments will include an analysis of the differentiated impact on men and women, through the involvement of gender and energy experts (Environmental and Social Impact Assessment, Resettlement Action Plans etc.). Gaps could be identified but not limited to female labor issue or inadequate access to finance for the cooling-related sectors, and design mitigation measures that minimize gender gaps such as enhanced productive uses of energy, enhanced knowledge and skills of small and micro-business, households and farmers.
- **Identifying and implementing relevant actions.** Based on the initial diagnosis and inclusive stakeholders' consultations, targeted activities will be identified to address key gender gaps. These may include (i) actions to promote female employment (including capacity building and programs for women to gain hands-on experience in the sector such as support for school-to-work transition), (ii) financing local communities projects empowering women in the area of implementation of the projects (such as activities generating revenues), (iii) implementation of prevention measures to mitigate associated incidents or risks in the area of implementation of the projects and (iv) facilitating energy access for female-headed households (including capacity-building workshops to maintain and develop energy services).
- **M&E and Knowledge management.** In 2017, the WBG started implementing the "gender-tag" mechanisms to identify projects that effectively promote equality between men and women and are aligned with the WBG gender strategy. The Cooling Facility will adopt a gender perspective and aim to close key gaps in its supported projects. Results will be monitored throughout the project and be used for real-time adjustments during implementation. To expand the knowledge base, projects will build on lessons learnt from prior projects as well as regional and country base knowledge production and exchange learning opportunities.
- **Engaging stakeholders** with specific needs and risks of gender inequality in meaningful consultation, information disclosure and responsive grievance redress throughout the project life (as set out in the Stakeholder Engagement Plan). Focus will be given to activities to strengthen women's participation and voice during consultations. Consultations and public meetings with stakeholders will be conducted throughout the project cycle in a participatory manner and with a special focus on engaging women. This will help gaining knowledge about community concerns during project preparation and help monitor impact and undertake real-time correction during implementation. Special attention will be paid to using local dialect, cultural concerns when relevant and including women from minority ethnic groups and other socially vulnerable groups.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Each project financed through the Cooling Facility will be required to conduct financial management and procurement according to the World Bank policies and procedures.

- **Procurement**

Each project supported by the Cooling Facility will follow the World Bank's Procurement Framework and Regulations (<https://www.worldbank.org/en/projects-operations/products-and-services/brief/procurement-new-framework#framework>). World Bank procurement staff support borrowers throughout the procurement process. They work with governments to achieve the highest bidding and contract management standards to get the best development results. Each project is developed, designed and monitored by the Executing Entity (or PIE as applicable) with support and supervision by a World Bank task team that includes a procurement specialist.

For each project, a PMU will be established by the relevant Executing Entity (host countries or governments) – or PIE as applicable - with a team expected to include a procurement specialist.

Each project, with the exception of the guarantee operation (see below), is also be subject to the 'World Bank's Anti-Corruption Guidelines, dated July 1, 2016'.

- **Financial Management**

Each project supported by the Facility is required to maintain financial management arrangements (including planning and budgeting, accounting, internal control, funds flow, financial reporting, and auditing arrangements of the borrower and entities responsible for project implementation) acceptable to the World Bank. A financial management specialist is expected to be part of the team forming the PMU established by each relevant Executing Entity (or PIE as applicable) for its respective project.

Moreover, for each project, the World Bank task team includes an accredited Financial Management (FM) specialist who provides support during the entire project cycle (preparation, supervision and implementation) in accordance with the World Bank's operational policies and procedures (to work with partner countries to design financial management and disbursement arrangements and to supervise and support the performance of World Bank-finance operations). The team's FM specialist closely monitors the performance of the Executing Entity's (or PIE as applicable) FM arrangements and their continued capacity to provide reasonable assurance that loan proceeds are being used for the purposes intended.

Financial reporting is also required for each project which is specified in each project's Disbursement and Financial Information Letter (DFIL). Unless otherwise agreed by the World Bank, the Executing Entity must submit annual audited project financial statements six months after the close of the financial year and unaudited interim financial (IFR) reports periodically. Audits are carried out by auditors with independence and capacity acceptable to the World Bank, under terms of reference acceptable to the World Bank.

G.4. Disclosure of funding proposal

The Access to Information Policy (2015) governs the public accessibility of information of operational documents prepared by the World Bank. The policy can be accessed through the following link: <http://pubdocs.worldbank.org/en/393051435850102801/World-Bank-Policy-on-Access-to-Information-V2.pdf>

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence.

Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☒ Environmental and Social Sustainability Framework (ESSF)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement framework
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Projects information (confidential)
- ☒ Annex 23 Greenhouse gas calculations
- ☒ Annex 24 Methodology and assumptions for greenhouse gas and beneficiaries calculations

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter issued by the national designated authority(ies) or focal point(s)



Secretary
Economic Relations Division
Ministry of Finance
Government of Bangladesh

D.O. No-09.00.0000.105.05.011.17-06

17 January 2021

Re: Funding proposal for the GCF by The World Bank regarding the programme "Cooling Facility"

Dear Sir,

We refer to the programme "Cooling Facility" in Bangladesh as included in the funding proposal submitted by The World Bank to us on 15 January 2021.

The undersigned Fatima Yasmin is the duly authorized representative of the National Designated Authority of Bangladesh.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme "Cooling Facility" as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Bangladesh has no-objection to the programme "Cooling Facility" as included in the funding proposal;
- (b) The programme "Cooling Facility" as included in the funding proposal is in conformity with Bangladesh's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme "Cooling Facility" as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme "Cooling Facility" as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Fatima Yasmin

Executive Director
Green Climate Fund
Songdo Business District, 175 Art Center-daero
Yeonsu-gu, Incheon 22004, Republic of Korea



MINISTERIO DE
MEDIO AMBIENTE
Y RECURSOS
NATURALES

MARN-DM-001/2021

San Salvador, January 5th, 2021

Re: Funding Proposal for the programme "The Cooling Facility in El Salvador" submitted by The World Bank.

Mr. Javier Manzanares
Deputy Executive Director
Green Climate Fund Secretariat
Republic of Korea

Dear Mr. Manzanares:

We refer to the programme, "The Cooling Facility in El Salvador", as included in the funding proposal submitted by The World Bank to us January 5, 2021.

The undersigned is the duly authorized representative of the Ministry of Environment and Natural Resources (MARN), as the National Designated Authority of El Salvador.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no objection, to the programme as included in the funding proposal

By communicating our no-objection, it is implied that:

- (a) The government of El Salvador has no-objection to the programme as included in the Funding Proposal;
- (b) The programme as included in the funding proposal, is in conformity with El Salvador's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process ascertaining no objection to the project as included in the proposal has been duly followed. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Fernando Andrés López Larreynaga
Minister



Kilómetro 5 ½ Carretera a Santa Tecla, colonia y calle Las Mercedes, edificios MARN
San Salvador, El Salvador, Teléfono (503) 2132-9418. Correo electrónico: despacho@marn.gob.sv
www.marn.gob.sv



**REPUBLIC OF KENYA
THE NATIONAL TREASURY AND PLANNING**

Telegraphic Address: 22921
FAX NO. 310833
Telephone: 2252299

**THE NATIONAL TREASURY
P O BOX 30007 – 00100
NAIROBI**

When Replying Please Quote

Ref: TNT/CONF/36/021/C/ (74)

Date: January 6, 2021

Mr. Yannick Glemarec
Executive Director
Green Climate Fund
Songdo Business District
175 Art center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Dear *Yannick*

**RE: FUNDING PROPOSAL FOR THE GCF BY THE WORLD BANK REGARDING THE
'COOLING FACILITY' PROGRAMME**

We refer to the programme **"Cooling Facility"** as included in the funding proposal submitted by The World Bank as the Accredited Entity to us in October 2020.

The undersigned is the duly authorized representative of the National Treasury, the National Designated Authority of Kenya.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

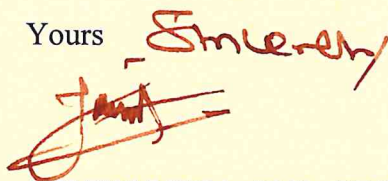
By communicating our no-objection, it is implied that:

- (a) The Government of Kenya has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Kenya's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed. Further, we confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours



JULIUS MUIA, PhD, CBS
PRINCIPAL SECRETARY/NATIONAL TREASURY

Copy to: **Hon. (Amb.) Ukur Yatani EGH**
Cabinet Secretary
The National Treasury and Planning
NAIROBI

Dr. Eng. Joseph K. Njoroge, CBS
Principal Secretary
Ministry of Energy
NAIROBI

Mr. Keith E. Hansen
Country Director
The World Bank in Kenya
NAIROBI

Telephone: 01 771 111
Telefax No.: 01 773 379
Our Reference No.: EAD/99/7/5
Your Reference
No.....



ENVIRONMENTAL AFFAIRS DEPARTMENT
LINGADZI HOUSE
CITY CENTRE
PRIVATE BAG 394
LILONGWE 3
MALAWI

Communications should be addressed to:
The Director of Environmental Affairs

The Secretariat
Green Climate Fund (GCF)
175 Art Center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Lilongwe, 29th April 2020

Re: Funding proposal for the GCF by The World Bank regarding the Cooling Facility

Dear Madam/ Sir,

We refer to the programme, the Cooling Facility, in Malawi as included in the funding proposal submitted by The World Bank to us on 28th April 2020.

The undersigned is the duly authorized representative of Environmental Affairs Department, the National Designated Authority of Malawi. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of Malawi has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Malawi's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

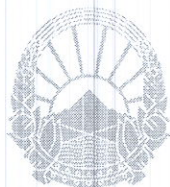
We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

T.G Mbale-Luka
DIRECTOR OF ENVIRONMENTAL AFFAIRS & GCF FOCAL POINT



Government of the Republic of North Macedonia

- DEPUTY PRESIDENT OF THE GOVERNMENT, IN CHARGE OF ECONOMIC AFFAIRS, COORDINATION OF ECONOMIC SECTORS AND INVESTMENTS -

09-4297/1
Skopje, 08-04-2021 2021

To: The Green Climate Fund ("GCF")

Subject: No objection Letter to the Funding proposal for the GCF by the World Bank Group regarding the Cooling Facility

Dear Madam, Sir,

We refer to the programme titled Cooling Facility in the Republic of North Macedonia as included in the funding proposal submitted by the World Bank to us on 12.05.2020 nr.26-3089/1 and 08.03.2021 nr. 26-2045/1.

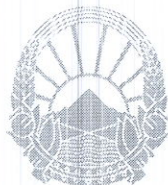
The undersigned is the duly authorized representative of the North Macedonia NDA-the Cabinet of Deputy President of the Government of Republic of North Macedonia in charge of Economic Affairs, Coordination of Economic Sectors and Investments, Mr. Fatmir Bytyqi, of the Republic of North Macedonia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Republic of North Macedonia has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Republic of North Macedonia;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.



Government of the Republic of North Macedonia

- DEPUTY PRESIDENT OF THE GOVERNMENT, IN CHARGE OF ECONOMIC AFFAIRS, COORDINATION OF ECONOMIC SECTORS AND INVESTMENTS -

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Sincerely,

**DEPUTY PRESIDENT OF THE GOVERNMENT
OF THE REPUBLIC OF NORTH MACEDONIA IN
CHARGE OF ECONOMIC AFFAIRS, COORDINATION OF
ECONOMIC SECTORS AND INVESTMENTS**

Fatmir Bytyqi





Влада на Република Северна Македонија
КАБИНЕТ НА ЗАМЕНИКОТ НА ПРЕТСЕДАТЕЛОТ НА ВЛАДАТА ЗАДОЛЖЕН ЗА ЕКОНОМСКИ
ПРАШАЊА, КООРДИНАЦИЈА СО ЕКОНОМСКИТЕ РЕСОРИ И ИНВЕСТИЦИИ

Архивски број:

09-42 97/1

08-04-2021

До: Зелен климатски фонд

Предмет: Писмо без приговор за Предлог-проект за Зелениот климатски фонд поднесен од Светска банка за енергетска ефикасност во јавниот сектор во Република Северна Македонија

Почитувани,

Повикувајќи се на Програмата за поддршка за системи за ладење на Зелениот климатски фонд, односно Проектот за енергетска ефикасност во јавниот сектор во Република Северна Македонија како што е наведено во предлог-проект поднесен од Светската банка до Национално назначениот орган за Зелениот климатски фонд на 12.05.2020 арх.бр. 26-3089/1 и 08.03.2021 арх. бр. 26-2045/1 ве известуваме за следното:

Долопотпишаниот претставник е овластен претставник на Национално назначениот орган на Република Северна Македонија, национално лице за контакт (Focal Point) за Република Северна Македонија, г-дин. Фатмир Битиќи, Заменик на претседателот на Владата на Република Северна Македонија задолжен за економски прашања, координација со економски ресурси и инвестиции.

Во согласност со одлуката на Зелениот климатски фонд бр. В.08/10, со чија содржина сме запознаени, го доставуваме ова Писмо без приговор за програмата како што е назначено во предлог-проектот.

Со доставувањето на ова Писмо без приговор, ве известуваме дека:

- Владата на Република Северна Македонија нема забелешки по однос на програмата како што е наведено во предлог-проектот.
- Програмата како што е наведена во предлог-проектот, е во согласност со националните приоритети, стратегии и планови.
- Во согласност со насоките за заштита на природата и социјалните ризици на Зелениот климатски фонд, програмата како што е наведено во предлог-проектот е во согласност со соодветните национални закони и регулатива.



Влада на Република Северна Македонија
КАБИНЕТ НА ЗАМЕНИКОТ НА ПРЕТСЕДАТЕЛОТ НА ВЛАДАТА ЗАДОЛЖЕН ЗА ЕКОНОМСКИ
ПРАШАЊА, КООРДИНАЦИЈА СО ЕКОНОМСКИТЕ РЕСОРИ И ИНВЕСТИЦИИ

Исто така, потврдуваме дека националниот процес за издавање на писмо без приговор за програмата како што е наведено во предлог-проектот е спроведена.

Воедно, ве известуваме дека писмото за приговор се однесува на сите активности кои ќе бидат спроведени во рамките на предлог-проектот.

Се согласуваме ова писмо да биде сподделено преку веб страната на Зелениот климатски фонд.

Ви благодариме за соработката.

Со почит,

ЗАМЕНИК НА ПРЕТСЕДАТЕЛОТ НА ВЛАДАТА НА
РЕПУБЛИКА СЕВЕРНА МАКЕДОНИЈА ЗАДОЛЖЕН
ЗА ЕКОНОМСКИ ПРАШАЊА, КООРДИНАЦИЈА НА
ЕКОНОМСКИТЕ РЕСОРИ И ИНВЕСТИЦИИ

Фатмир Битиџи



Panamá, 16 de septiembre de 2020.
DM-1024-2020

Señor
YANNICK GLEMAREC
Secretario Ejecutivo,
Fondo Verde del Clima (FVC)
E.S.D

Estimado Señor,

Nos referimos al programa, Cooling Facility/Fondo UREE , en la República de Panamá tal como se incluye en la propuesta de financiamiento que nos presentó el Banco Mundial el 19 de marzo de 2020.

El abajo firmante, es el representante debidamente autorizado del Ministerio de Ambiente de Panamá, la Autoridad Nacional Designada / punto focal de Panamá.

De conformidad con la decisión B.08 / 10 del GCF, cuyo contenido reconocemos haber revisado, por la presente comunicamos nuestra no objeción al programa tal como se incluye en la propuesta de financiamiento.

El comunicar nuestra no objeción, implica que:

- (a) El gobierno de Panamá, no tiene ninguna objeción a este proyecto, como se incluye en la propuesta de financiamiento;
- (b) Este proyecto que se incluye en la propuesta de financiamiento, está en conformidad con las prioridades, estrategias y planes nacionales de Panamá;
- (c) De acuerdo con las salvaguardas ambientales y sociales del FVC, el proyecto, tal como se incluye en la propuesta de financiamiento, está en conformidad con las leyes y regulaciones nacionales pertinentes.

Confirmamos, que nuestro proceso nacional para determinar la no objeción al proyecto, como se incluye en la propuesta de financiamiento, ha sido debidamente implementado.

Confirmamos, que nuestra no objeción, aplica a todas las actividades que se implementarán dentro del programa del proyecto. Finalmente, reconocemos que esta carta se pondrá a disposición del público en el sitio web de FVC.

Atentamente,



MILCIADES CONCEPCIÓN
Ministro de Ambiente



MC/JUQ/LC/jmg

13.9.20

Albrook, Calle Broberg, Edificio 804
República de Panamá
Tel.: (507) 500-0855

www.miambiente.gob.pa

Panama, September 16th, 2020
DM-1024-2020

Mr.

YANNICK GLEMAREC

Executive Director

Green Climate Fund

Re: Funding proposal for the GCF by The World Bank regarding the Cooling Facility

Dear Sir,

We refer to the programme, the Cooling Facility, in the Republic of Panama as included in the funding proposal submitted by The World Bank to us on March 19, 2020.

The undersigned is the duly authorized representative of the Ministry of Environment of Panama, the National Designated Authority/focal point of Panama.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- A. The government of the Republic of Panama has no-objection to the programme as included in the funding proposal;
- B. The programme as included in the funding proposal is in conformity with Panama's national priorities, strategies and plans;
- C. In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


MILCIADES CONCEPCIÓN

Minister of Environment



MC/JUQ/LC/jmg



Albrook, Calle Broberg, Edificio 804
República de Panamá
Tel.: (507) 500-0855

www.miambiente.gob.pa

REPÚBLICA DEMOCRÁTICA DE S.TOMÉ E PRÍNCIPE
(Unidade - Disciplina - Trabalho)
Ministério do Planeamento, Finanças e Economia Azul
GABINETE DO MINISTRO

To: The Green Climate Fund ("GCF")

N./Ref. 0038/MPFEA-GM/2021

Re: Funding proposal for the GCF by The World Bank regarding the Cooling Facility

Dears,

We refer to the programme, the Cooling Facility, in Sao Tome and Principe as included in the funding proposal submitted by The World Bank to us on 01/06/2021. The undersigned is the duly authorized representative of Ministry of Planning, Finance and Blue Economy, the National Designated Authority/focal point of Sao Tome and Principe.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Sao Tome and Principe has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Sao Tome and Principe's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

São Tome and Príncipe, January 07, 2021

Kind regards,

Osvaldo Tavares dos Santos Vaz
Minister of Planning, Finance and Blue Economy





The Federal Republic of Somalia
Office of the Prime Minister

Ref: OPM/ 0012 /01/2021

Date: 06/01/2021

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by The World Bank regarding the Cooling Facility

Dear Madam/Sir,

We refer to the program, the Cooling Facility, in Somalia as included in the funding proposal submitted by The World Bank to us on 28th December 2020.

The undersigned is the duly authorized representative of the National Designated Authority/focal point of the Federal Government of Somalia Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Somalia has no-objection to the program as included in the funding proposal;
- (b) The program as included in the funding proposal is in conformity with Somalia's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programmed as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

H.E Mahdi Mohammed Gulaid
Deputy Prime Minister
The Federal Government of Somalia





පරිසර අමාත්‍යාංශය
சுற்றுடல் அமைச்சு
Ministry of Environment

"සොබාදාම පියස", අංක 416/සී/1, රොබට් ගුනවර්ධන මාවත, බත්තරමුල්ල, ශ්‍රී ලංකාව.

"சோபாதம் பியச", இல. 416/சீ/1, ரொபர்ட் குணவர்தன மாவத்தை, பத்தரமுல்லை, இலங்கை.
"Sobadam Piyasa", No. 416/C/1, Robert Gunawardana Mawatha, Battaramulla, Sri Lanka.

Gen. Tel. +94-11-2034100

ලේකම්

செயலாளர்

Secretary

+94-11-2034121

ෆැක්ස්

தொலை நகல்

Fax

+94-11-2879944

මගේ අංකය
எனது இல
My No

04/04/10/350/-GCF IV

ඔබේ අංකය
உமது இல
Your No

දිනය
திகதி
Date

03.09.2021

The Secretariat
Green Climate Fund
Songdo-dong, Incheon,
South Korea

Re: Funding proposal for the GCF by World Bank regarding Cooling Facility Programme

Dear Sir,

We refer to the multi country programme "Cooling Facility" also executing in Sri Lanka as included in the funding proposal submitted by the World Bank to us on the 29th of April 2020.

The undersigned is the duly authorized representative of the Ministry of Environment, the National Designated Authority of Sri Lanka.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- The government of Sri Lanka has no-objection to the programme as included in the funding proposal;
- The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Sri Lanka;
- In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Dr. Anil Jasinghe
Secretary

"මේ මහාපොළව සහ ගහනොළ මිනිසාට මෙන්ම අනෙක් පියාසරන සියොතුවන්ට ද මිනිසා සරන සිවුපාවන්ට ද සියලු සතුන්ට ද එකසේ අයිතිය"
"இப் பூமியும், மரங் செடி கொடிகளும் மனிதனுக்கும் மோன்றோ வானில் பறந்து திரியும் பறவைகளுக்கும், பூமியில் வாழும் உயிர்விலங்குகளுக்கும், அனைத்து விலங்குகளுக்கும் ஒருமித்துச் சொந்தமானது"
"This great earth and the flora on it equally belong to the man and the birds flying in the sky, the quadrupeds and all creatures living on earth"

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Cooling Facility
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	World Bank
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	The target countries are: Bangladesh, El Salvador, Kenya, Malawi, North Macedonia, Panama, Sao Tome and Principe, Somalia, and Sri Lanka.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, September 1, 2021
Language(s) of disclosure	English, Bengali, Macedonian, Portuguese, Sinhala, Somali, and Spanish.
Explanation on language	English is an official language of Kenya, and the official language of Malawi; Bengali is the official language of Bangladesh; Macedonian is the official language of North Macedonia; Portuguese is the official language of Sao Tome and Principe; Sinhala is an official language of Sri Lanka; Somali is an official language of Somalia; Spanish is the official language of El Salvador, and Panama.
Link to disclosure	English: https://esmap.org/sites/default/files/esmap-files/A6_Cooling%20Facility_ESSF_26%20August%202021_English.pdf Bengali: https://esmap.org/sites/default/files/publication-home/A6_Cooling%20Facility_ESSF_26August2021_Bengali.pdf Macedonian: https://esmap.org/sites/default/files/publication-home/A6_Cooling%20Facility_ESSF_31August2021_Macedonian.pdf Portuguese: https://esmap.org/sites/default/files/publication-home/A6_Cooling%20Facility_ESSF_26August2021_Portuguese.pdf Sinhala: https://esmap.org/sites/default/files/publication-home/A6_Cooling%20Facility_ESSF_27August2021_Sinhalese.pdf

	Somali: https://esmap.org/sites/default/files/publication-home/A6_Cooling%20Facility_ESSF_31August2021_Somali.pdf Spanish: https://esmap.org/sites/default/files/publication-home/A6_Cooling%20Facility_ESSF_27%20August%202021_Spanish.pdf
Other link(s)	N/A
Remarks	An ESIA template consistent with the requirements for a Category B project is contained in the "Environmental and Social Sustainability Framework (ESSF)".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, September 1, 2021
Language(s) of disclosure	English, Bengali, Macedonian, Portuguese, Sinhala, Somali, and Spanish.
Explanation on language	English is an official language of Kenya, and the official language of Malawi; Bengali is the official language of Bangladesh; Macedonian is the official language of North Macedonia; Portuguese is the official language of Sao Tome and Principe; Sinhala is an official language of Sri Lanka; Somali is an official language of Somalia; Spanish is the official language of El Salvador, and Panama.
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	Spanish: https://esmap.org/sites/default/files/publication-home/A6 Cooling%20Facility ESSF 27%20August%202021 Spanish.pdf
Other link(s)	N/A
Remarks	An ESMP template consistent with the requirements for a Category B project is contained in the “Environmental and Social Sustainability Framework (ESSF)”.
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity’s website	The “Environmental and Social Sustainability Framework” includes the templates the following ESS reports: Resettlement Policy Framework (Appendix 5), Resettlement Action Plan (Appendix 6), Indigenous People Planning Framework (Appendix 7) and Indigenous Peoples Plan (Appendix 8/ August 2021
Language(s) of disclosure	English, Bengali, Macedonian, Portuguese, Sinhalese, Somali, and Spanish.
Explanation on language	English is an official language of Kenya, and the official language of Malawi; Bengali is the official language of Bangladesh; Macedonian is the official language of North Macedonia; Portuguese is the official language of Sao Tome and Principe; Sinhala is an official language of Sri Lanka; Somali is an official language of Somalia; Spanish is the official language of El Salvador, and Panama.
Link to disclosure	English: https://esmap.org/sites/default/files/esmap-files/A6 Cooling%20Facility ESSF 26%20August%202021 English.pdf Bengali: https://esmap.org/sites/default/files/publication-home/A6 Cooling%20Facility ESSF 26August2021 Bengali.pdf Macedonian: https://esmap.org/sites/default/files/publication-home/A6 Cooling%20Facility ESSF 31August2021 Macedonian.pdf Portuguese: https://esmap.org/sites/default/files/publication-home/A6 Cooling%20Facility ESSF 26August2021 Portuguese.pdf

	Sinhala: https://esmap.org/sites/default/files/publication-home/A6 Cooling%20Facility ESSF 27August2021 Sinhalese.pdf Somali: https://esmap.org/sites/default/files/publication-home/A6 Cooling%20Facility ESSF 31August2021 Somali.pdf Spanish: https://esmap.org/sites/default/files/publication-home/A6 Cooling%20Facility ESSF 27%20August%202021 Spanish.pdf
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	N/A
Place	N/A
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 4, 2021

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP177

Proposal name:	Cooling Facility
Accredited entity:	World Bank
Country/(ies):	Bangladesh, El Salvador, Kenya, Malawi, North Macedonia, Panama, Sao Tome and Principe, Somalia, and Sri Lanka
Project/programme size:	Large

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
Cooling Facility is a multi-sector, multi-country, cross-cutting proposal that is designed to address key barriers to scale by providing a focused and tailor made technical assistance to create enabling environment and develop multiple financial model for best adoption, scaling and replication of cooling solutions. This integrated approach can make a difference, particularly when countries are prioritizing the scarce resources available for an emergency response to COVID-19, focusing on healthcare.	The Cooling Facility and the projects for investment are at an early development stage. Proper assurances, including implementation arrangements, need to be put in place to ensure the achievement of the targeted impacts that the Facility will deliver as indicated in the funding proposal. It is important that the Accredited Entity (AE) strictly complies with the agreed set of parameters in the term sheet throughout the process.
The nine (9) countries are representative of different types of cooling context and varied needs and challenges for cooling. The data generated, application of the models and lessons learned from these countries will inform the larger cooling community of the innovative and integrated approach.	
The programme has a high level of impact both on mitigation and adaptation, while meeting Sustainable Development Goals (SDGs), planning for disaster risk reduction (e.g. from heatwaves) and supporting the goals of the Kigali Amendment.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVIII, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Programme background

2.1.1. Programme context

3. Cooling is critical for health and well-being, livelihoods and the environment. Cooling is an essential part of vaccines and food systems to keep vaccines safe and food fresh, and is central to industrial and transport infrastructure. Cooling has also become a mainstay for maintaining “comfort” in buildings for both work and leisure to ensure people have comfortable buildings to live and work in. Population growth, including a growing middle-class population, and changes in lifestyles, rapid urbanization and rising ambient temperatures are leading to an increase in cooling demand year on year. Therefore, an estimated 3.6 billion cooling appliances currently in use is projected to increase to 9.5 billion by 2050, according to Dreyfus et al. 2020, and the demand would be at around 14 billion if all people in need of cooling were to be catered for, as opposed to only those who could afford it.

4. According to SEforALL (2020), over 1 billion people in developing countries are already facing an immediate risk from lack of access to cooling, of whom 699 million are slum dwellers living in hotter-climate urban areas where electricity services do not exist, are intermittent or are too expensive; and about 318 million are people living in poor rural areas without access to cold chains for storage and preservation of food and medicine. Therefore, cooling is not a luxury but a necessity for sustainable development.

5. This new warming world poses numerous climate challenges related to both mitigation and adaptation. Firstly, the increase in cooling demand leads to high levels of greenhouse gas (GHG) emissions, as the primary energy comes from carbon-intensive sources. Secondly, most of the refrigerants in use today for cooling are fluorinated gases (F-gases) – hydrochlorofluorocarbons (HCFCs) or hydrofluorocarbons (HFCs) – these F-gases have a very high global warming potential (GWP) ranging from 92 to 14,000, and if leaked or vented will cause substantial emissions. Thirdly, the Intergovernmental Panel on Climate Change in 2019 reported that at 1.5°C of warming – a threshold that could be reached as early as 2030 – 2.3 billion people could be both exposed and vulnerable to heatwave events, which can cause illnesses including heat cramps, fainting, heat exhaustion, heat stroke and even death. Heat stress will also impact productivity. The International Labour Organization (2019) estimates the productivity loss due to heat by 2030 to amount to the equivalent of 80 million full-time jobs.

2.1.2. Programme description

6. As mentioned above, cooling has become a critical part of daily lives, be it in comfort, supply chain management, transport or industrial process. However, cooling has always been a “part” of the overall sector and never a stand-alone initiative. Therefore, clear focus on cooling has not been given. For example, cold chains are part of the cooling action, but it is embodied within the overall agriculture value chain.

7. The Cooling Facility will be one of the world’s first cooling-focused facilities with the aim of providing key solutions to cooling from multiple angles, including regulation and policy, technical assistance, and most importantly financing to address multiple barriers inhibiting the broader adoption of clean cooling solutions, which is energy efficient and with low-GWP refrigerant.

8. Key barriers to be addressed by the Cooling Facility are the following:

- (a) Lack of regulatory/policy framework for sustainable cooling systems along with limited institutional capacity to design, implement, monitor, and enforce cooling standards;
- (b) Lack of enforceable standards like Minimum Energy Performance Standards (MEPS) and Standards and Labels (S&L), which target a specific sector – in this instance cooling.

- MEPS and S&L can be further operationalized into government fiscal and monetary policy as indirect incentives and lead to better enforcement;
- (c) Lack of awareness among users of cooling equipment. Few users are fully aware of its emissions impact, energy consumption, maintenance requirements and operating costs over time. The driving incentive for users is price and not efficiency; this locks in people into inefficient equipment for the life duration of that inefficient equipment;
 - (d) Lack of skills: in many countries, there are not enough sufficiently trained engineers/technicians to install and maintain cooling equipment. Even the most efficient equipment, if not installed and maintained correctly, will not perform in the manner it was designed to;
 - (e) Lack of data and credible forecasts for the future scaling up of cooling needs including costs, benefits and climate impact. Quality baseline data about equipment stocks, equipment sales and refrigerant inventories in many countries are either not present or not organized for ease of access; and
 - (f) In addition, market barriers include the following:
 - (i) Lack of access to finance for cooling investment (greenfield and retrofit);
 - (ii) Lack of innovation in business models; most countries still have high collateral requirement and do not offer other services;
 - (iii) Lack of technology choice; generally most markets are dominated by few suppliers;
 - (iv) Lack of bargaining power, as most projects are small in size and thus do not have the leverage to bargain with financiers for better terms; and
 - (v) Lack of pipeline; consumption patterns and consumer behaviour have to demand efficient services as opposed to accepting the lowest price principle.

2.1.3. Cooling Facility design

9. The wide range of cooling needs and priorities implies an equally diverse set of challenges and solutions as outlined above. Therefore, the Cooling Facility is uniquely designed based on the principle of “no one-size-fits-all solution or approach”. Given the nascent and still early stage focus on cooling, globally, to address both development and climate change purposes, the Cooling Facility seeks to cater to different contexts – countries’ geographic and socio-economic contexts, and efforts and priorities to respond to the crisis caused by the COVID-19 pandemic crisis (e.g. vaccination supply chain).
10. In this context, the Cooling Facility will support activities that address and help remove regulatory/policy-related, institutional, market and behavioural barriers to the development of sustainable cooling investments by:
 - (a) Providing technical assistance and building the capacity of state and non-state actors (such as commercial banks, private investors and technical companies, and end user beneficiaries);
 - (b) Supporting the design of sustainable implementation mechanisms and business models;
 - (c) Increasing end-user awareness of the benefits of low-carbon cooling solutions;
 - (d) Facilitating access to affordable sources of financing; and
 - (e) Providing credit lines to financial intermediaries.
11. In turn, these activities will: (1) strengthen the capacity of key stakeholders; (2) stimulate the demand for climate-friendly cooling solutions; (3) foster the penetration of

climate-friendly equipment and infrastructure; and (4) mobilize capital from public and private sources.

12. These interventions will strengthen institutional and regulatory systems, improve access to cooling, improve the adaptive capacity of end-user beneficiaries to respond to rising temperatures, as well as increase energy efficiency of cooling buildings and cooling infrastructure. Ultimately, the climate change resilience of built infrastructure will be enhanced, while the carbon intensity of buildings, cold chains and cooling appliances will decrease.

13. In fact, the diversity of the selected countries targeted by the Facility is expected to provide pioneering examples for mitigation and adaptation activities related to cooling in different contexts. Aside from different cooling sub-sectors, many issues are country-specific, and thus the Cooling Facility aims to support different approaches and generate lessons that can then be adapted and scaled up more broadly to enhance access to affordable cooling, while at the same time kick start a transformation towards more sustainable cooling, particularly by offering examples to be emulated and adapted in countries beyond the scope of the Facility.

14. Where growing cooling demand is creating pressures on a country's energy systems and upward pressures on GHG emissions, the priority is to support space cooling, in terms of both passive measures (e.g. building design and high-efficiency thermal envelopes) and energy-efficient cooling equipment and systems (this is the case for Bangladesh, North Macedonia, Panama and Sri Lanka). In agriculture-dominant economies where there is a need to expand access to cooling into rural and off-grid areas, the priority is to support an efficient and climate-friendly cold chain (e.g. through the Facility's proposed support to Malawi and Kenya).

15. In hot climate countries, ensuring health infrastructure is adequate for patients and storage of pharmaceuticals is a priority, because pharmaceuticals expire and become ineffective more quickly in hot climatic conditions. Efficient and solar-powered cold chain and health facilities can expand quality health services and ensure safe storage of vaccines while making them more resilient to warmer temperatures, as well as keeping their carbon footprint low (e.g. El Salvador, Sao Tome and Principe, and Somalia).

16. There is an urgent need to cut emissions and meet increasing demand for future cooling sustainably. The Kigali Amendment to the Montreal Protocol, which has already been ratified by over 100 countries, and regional frameworks such as the European Union F-Gas Regulation, demonstrate how refrigerant emissions can be reduced, yet further refrigerant phase-downs and phase-outs are needed. Meeting future cooling needs sustainably can reduce the costs of renewable energy build-out by up to USD 3.5 trillion by 2030 and accelerate the net-zero transition by up to eight years. Hence, the Cooling Facility will pioneer the way forward to meeting these targets.

2.1.4. Integrated and holistic approach

17. The scale of the cooling challenge and the window of opportunity to get on a path to affordable, efficient and clean cooling require an approach that integrates the development and strengthening of an enabling environment. Doing this involves technical assistance, capacity-building, and concessional financing to mainstream sustainable cooling and establish sustainable implementation and business models. Moreover, in line with the Facility's goals to support access to affordable, efficient and clean cooling in a warming world, the Facility will integrate efficiency, starting with passive measures, and low-carbon considerations and adopt a holistic approach to sustainable cooling consisting of one or more of the following elements:

- (a) Minimize: reduce and avoid cooling loads, for example, with passive building design, increase of thermal efficiency of buildings and cold storage warehouses/facilities, and nature-based solutions such as green roofs;

- (b) Shift to cooling solutions with the lowest GHG emissions, for example solar cooling, climate-friendly (low GWP) refrigerant-based cooling systems, or centralized cooling solutions where the right enabling conditions are in place;
 - (c) Improve efficiency: use energy-efficient technologies to achieve cooling with less energy (and less GHG emissions), for example, super-efficient appliances and equipment and demand management to reduce peak demand and the need for new generating power capacity;
 - (d) Optimize: control cooling loads, for example, with smart sensing and behavioural change approaches; and
 - (e) Protect the most vulnerable in urban and rural areas from extreme heat and the consequences of unreliable or insufficient medical and agriculture cold chains.
18. The total project budget is USD 879.84 million, of which USD 157 million are requested from GCF, of which USD 125 million is the form of loans, and USD 32 million is in the form of grants. The environmental and social safeguards categorization of this proposal is assessed to be I-2.

2.2 Component-by-component analysis

Component 1: Policy, regulatory and enabling environment support (total cost: USD 91.6 million; GCF cost: USD 15.7 million)

19. This component will support activities that aim to strengthen institutional, policy and regulatory frameworks, raise awareness and stimulate behavioural changes, as well as build the capacity of key stakeholders to plan, finance and implement sustainable cooling investments, and to plan for responses to increased heatwaves and other extreme events. Enabling activities will include cooling-specific inputs (related to both climate mitigation and adaptation) into the country's nationally determined contributions planning process under the Paris Agreement such as Cooling Action Plans, as well as inputs on national policies and regulations that encourage sustainable cooling. It will also support activities linked to the Facility's Gender Action Plan.
20. Activities relevant to the eligible cooling areas supported under Component 1 may include the following:
- (a) Capacity-building of key stakeholders, for example, on technologies, business models, cooling project appraisal, implementation and evaluation;
 - (b) Technical assistance for enabling plans, strategies, policies, regulations and standards;
 - (c) Designing sustainable market mechanisms and business models; and
 - (d) Communications and awareness campaigns.

Component 2: Financing for cooling investments (total cost: USD 769 million; GCF cost: USD 191.3 million)

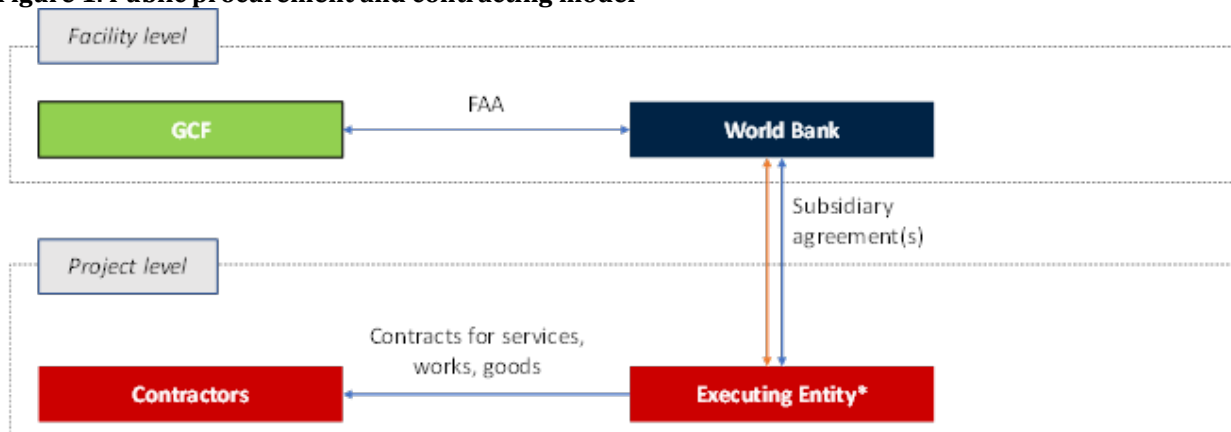
21. In order to promote and accelerate the uptake of investments in innovative and climate-friendly cooling technologies and systems, including in markets with low penetration and unmet cooling demand, the Cooling Facility aims to support increased access and deployment of sustainable cooling solutions performing better than current market practice and at the same time ensure that the level of ambition is appropriate and relevant to each market (i.e. affordable) to expand access to – and deployment of – more climate-friendly cooling at scale and avoid promoting a prohibitively expensive market niche, which would hinder the goals of scaling up access and deployment.

22. The Facility will provide concessional funds to foster the adoption or scale up the sustainable cooling technologies, appliances' deployment through financing of climate-friendly cooling investment and credit lines to financial intermediaries.
23. The financial instruments will be calibrated to what is needed to make the cooling project viable and to overcome barriers to implementation, including addressing the incremental cost associated with the adoption of efficient cooling solutions or serve remote off-grid areas, which include higher upfront capital costs compared to baseline options, and market entry barriers arising from the underrepresentation of efficient and clean cooling solutions in many developing country regions and sectors. The deployment of these funds will be tailored to the needs of the projects, through various financing modalities, for example, public financing to sovereign or sub-sovereign entities credit lines to commercial banks, loans or subsidies to households, municipalities and small, medium and large enterprises.
24. This component will focus on different financing and implementation mechanisms and models suitable for different cooling sectors and different countries, which may be deployed by different projects of the Facility based on specific country and sector conditions. Examples of these financing and implementation mechanisms and models are as below.

2.2.1. Public procurement and contracting for cooling infrastructure service delivery and technical assistance

25. GCF funds would be used to support investments through a traditional public procurement and contracting model. Interventions that promote the deployment of a public good (for example, refrigeration and cold chain for vaccines and medicines such as in El Salvador, Sao Tome and Principe and Somalia). This model includes the following activities:
- Prepare the technical specifications and terms of reference of the cooling infrastructure or equipment to procure;
 - Launch the procurement process to invite bids from suitable contractors;
 - award the procurement contract;
 - Sign contractual agreements with the winning bidders;
 - Monitor the delivery of the goods and services to ensure quality control using own resources or through an owner's engineer procured by the project; and
 - Remunerate the contractor according to the agreed terms.

Figure 1. Public procurement and contracting model



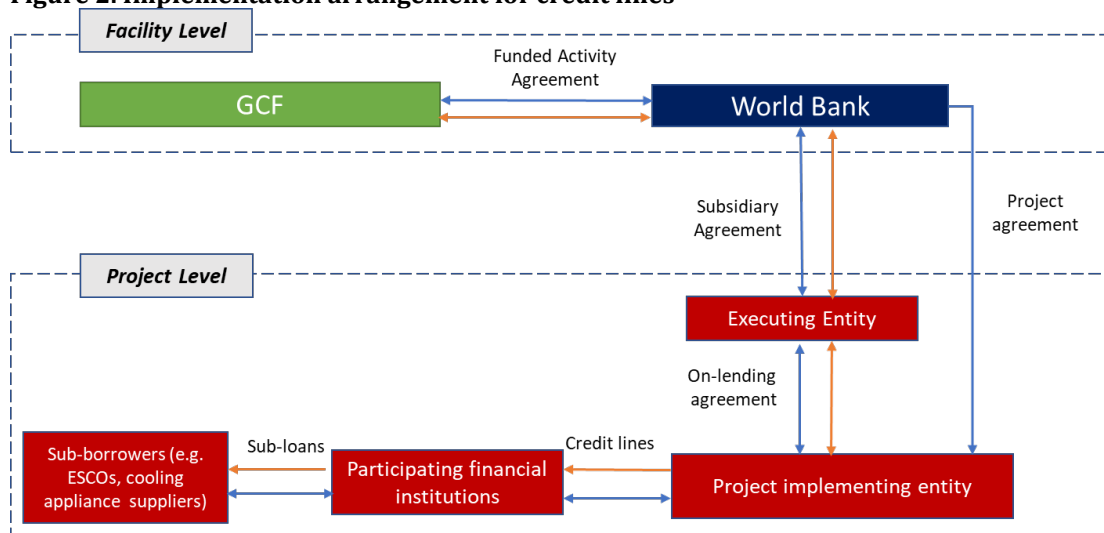
*If a project has a Project Implementing Entity (PIE), there may be additional arrangements including a project agreement between the World Bank and such PIE

2.2.2. Credit lines

26. A key barrier to investing in energy efficiency, in general, and sustainable cooling, in particular, is the limited availability of commercial financing for these projects. The Facility will support the provision of credit lines, for example in Malawi, to help financial institutions establish business lines for sustainable cooling investments. Credit lines can be well suited in less mature markets; they help mitigate the perceived high financial risk of energy efficient and clean cooling, which may be carried out by energy service companies (ESCOs).

27. An ESCO is an organization that provides a range of services to energy users to design and implement cooling and energy efficiency measures. The services may also include providing or arranging financing. Such services are provided by the ESCO using a performance-based contract under which the payments from the energy user to the ESCO are contingent upon achieving certain pre-specified performance levels. This approach allows energy users to transfer much of a project's technical, construction, and performance risks to the ESCO. The concept of energy savings performance contracts (ESPCs) implemented by ESCOs has been recognized as a promising approach to overcome some of the most intractable market barriers.

Figure 2. Implementation arrangement for credit lines



2.2.3. Energy efficiency financing facility

28. Under this model, the GCF proceeds are channelled through the executing entity (EE). The EE either directly, or through an entity that it designates (the project implementing entity (PIE)), on-lends the funds to sub-borrowers (for example, private ESCOs, cooling as a service operators, municipalities, or small- and medium-sized enterprises) to undertake eligible cooling investments.

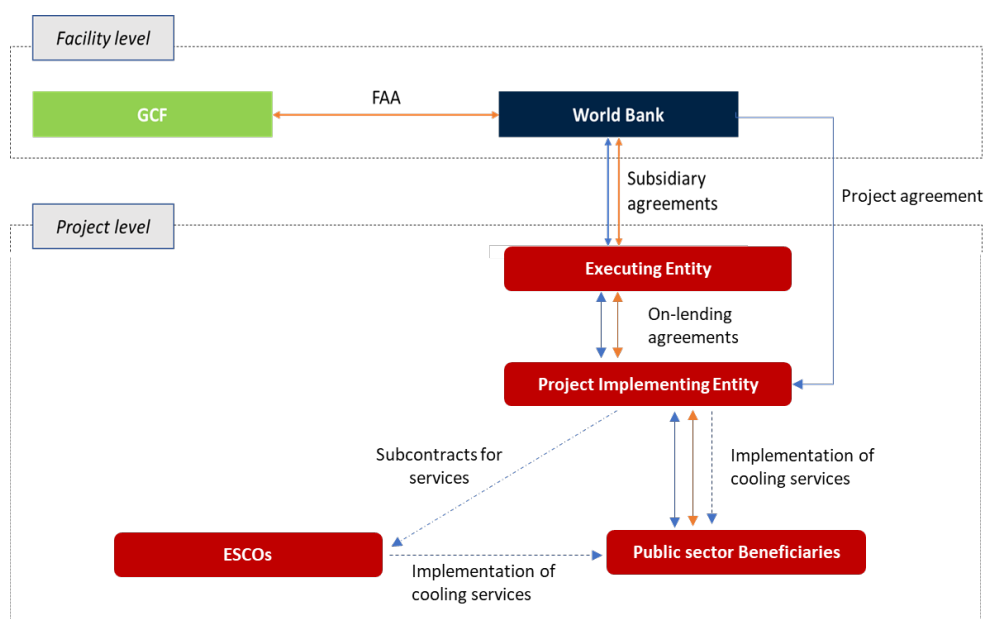
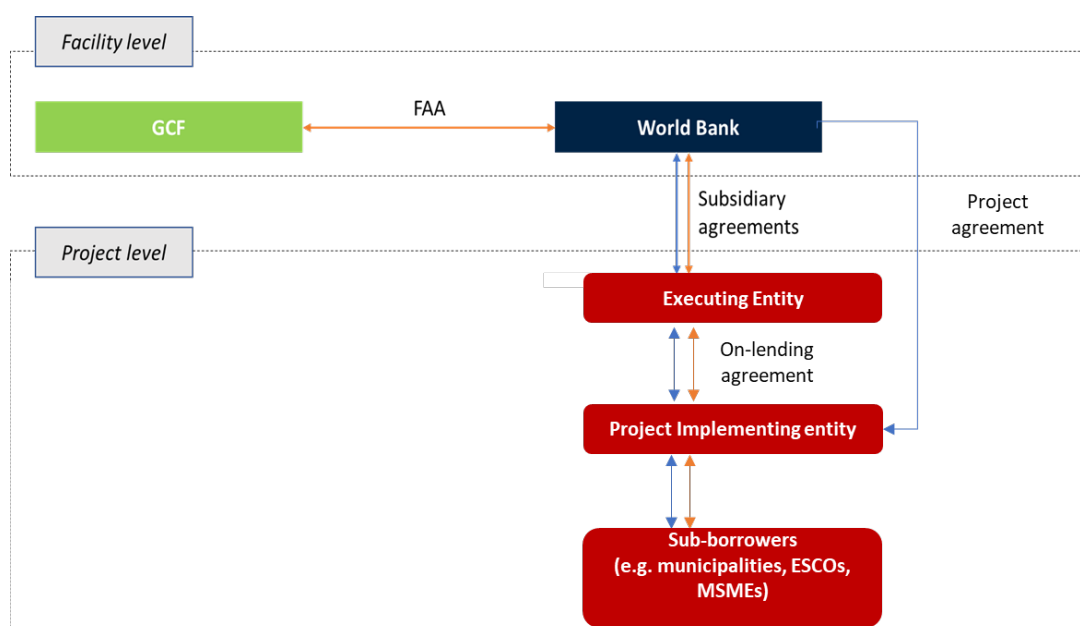


Figure 3. Energy efficiency financing facility model



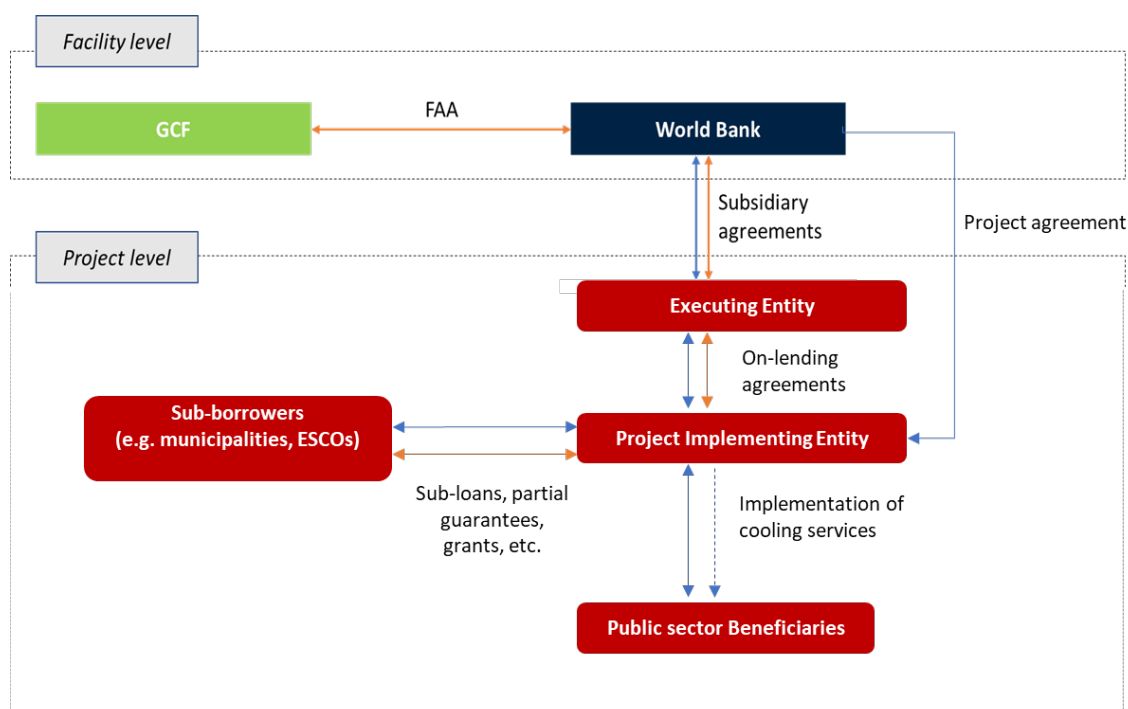
29. The energy efficiency financing facility implementation arrangement involves two main variations, as presented below.

30. (3-a) Public super ESCO model. In this variation of the energy efficiency financing facility model, the host country designates a public energy service company (the public super ESCO). The public super ESCO would typically be a public entity established by the government to (a) support cooling and energy efficiency investments in the public sector (public buildings and sustainable cooling projects) and (b) strengthen private sector ESCOs through demand aggregation, bulk procurement of equipment (therefore reducing the cost of equipment for ESCOs), standardization of ESPCs, monitoring and verification templates and capacity-building, project development, and facilitation. This variation of the energy efficiency financing facility arrangement is envisaged in Bangladesh.

Figure 3-a. Public super ESCO implementation model

31. (3-b) Energy efficiency revolving facilities (EERF). Under the EERF model, the PIE uses revolving mechanisms to recover its investment and operating costs based on contracts such as energy service agreements (ESAs) or also uses more traditional revolving financing mechanisms such as debt financing and budget capture.⁵⁵ Under an ESA, the facility finances and undertakes the investment on behalf of the beneficiary. In return, the beneficiary makes regular payments based on the energy savings generated by the investments to the EERF over an agreed period so that the EERF can recover the investment and operating costs. For the beneficiaries, the recurring payment (consisting of the reduced energy bill and the payment to the EE or PIE, as applicable) is equal to or lower than their pre-investment baseline energy bill. For North Macedonia it is envisaged that GCF funds will be deployed through the EERF model.

Figure 3-b Energy efficiency revolving facility (EERF) model



Component 3: Project management (total cost: USD 19.2 million; GCF cost: USD 0 million)

32. This component will support the EEs and project implementing entities in the management, coordination and implementation of their project activities, in compliance with the contractual obligations that are included or referred to in the legal agreements entered into with the AE. There is no funding request from GCF for project management.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: High

33. The Cooling Facility has a mitigation impact potential of 16.24 million tonnes of carbon dioxide equivalent (MtCO₂eq) avoided for the lifetime of the project. A total of 4.22 million people will directly of performance benefit from the programme, in addition to 16.86 million indirect beneficiaries. The Cooling Facility envisages enhancing the resilience of 18,057 physical assets, including buildings, agricultural and health cold chains.

34. The Cooling Facility is envisioned to increase access to cooling that is needed in a warming world and is especially critical to help build resilience in countries facing risks of extreme heat hazards, as well as help chart a sustainable path to low carbon development. As a cross-cutting facility the impact potentials are greatly heightened as described below:
- (a) **Adaptation** – The Cooling Facility will be engaged in countries facing a high risk of extreme heat hazards: Bangladesh, El Salvador, Malawi and Somalia. The impacts of the Facility-supported sustainable cooling investments are expected to reach over 4 million direct and over 16 million indirect beneficiaries. It will facilitate more than 16,000 climate-resilient cooling assets; and
 - (b) **Mitigation** – It is estimated that the Cooling Facility will lead to approximately 1 million tonnes of avoided or reduced GHG emissions per year across the sustainable cooling investments made in the nine countries, and more than 14 million tonnes of GHG reduced within the lifetime of the project. This amounts to nearly 2 per cent of the global cooling-related emissions, which is quite substantial from a sector point of view.
35. As a multi-sectoral facility, the impacts will be envisioned across these cooling sub-sectors:
- (a) **Kigali Amendment and refrigerants**: The Facility will also support the goals and requirements of the Kigali Amendment to the Montreal Protocol by helping countries in the transition to more climate-friendly refrigerants by catalysing more energy-efficient cooling equipment (air conditioners and refrigeration) in parallel with HFC phase-down. The GHG benefits from this refrigerant transition supported by the Facility have been estimated ex ante, in line with global estimates indicating that GHG emissions from the refrigerants represent about 20 per cent of cooling equipment's overall GHG emissions;
 - (b) **Space cooling**: Space cooling projects will promote a) passive measures to provide comfort while avoiding the need for mechanical cooling, and b) both higher efficiency and lower GWP refrigerants' uses, thereby shifting the business-as-usual trajectory of cooling GHG impact towards sustainable high efficiency/low GWP cooling;
 - (c) **Refrigeration and cold chains**: This will help catalyse a greening of the cold chain by improving the efficiency of refrigeration, storage and distribution infrastructure for agriculture and health using low GWP refrigerants. This will be of particular importance to health facilities and cold chains to safely store vaccines and medicine, reducing emissions by 2.7 MtCO₂eq over the lifespan of the programme;
 - (d) **Off-grid cooling powered by renewable energy**: Enabling the expansion of affordable cooling in off-grid contexts will help reach some of the poorest communities and contribute to enhancing their resilience to the impact of climate change. Indigenous and non-GHG emitting renewable energy (solar) will mostly power (126 MW) these critical cooling services; and
 - (e) **Other impacts**: (1) climate-friendly vaccine cold chains in three countries (El Salvador, Sao Tome and Principe, and Somalia) will help them respond to the current COVID-19 pandemic, increasing the capacity to store vaccines and lower the incidence of wasted vaccines; (2) investments will help lower the incidence of heat-related illnesses, as well as mental stress; and (3) improved local air quality through the displacement/avoidance of diesel generators.

3.2 Paradigm shift potential

Scale: High

36. There is huge climate potential for cooling by maximizing energy efficiency (refrigeration and space cooling) while simultaneously phasing down HFCs contained in refrigerants; the world can potentially avoid close to 1° C by 2100. The Cooling Facility aims to contribute to unlocking the climate potential.
37. The Cooling Facility envisions working across key cooling sub-sectors and with a range of countries from different parts of the developing world via a combination of targeted technical assistance and policy dialogue offered in conjunction with investment support for country-identified national priority sectors to:
- (a) Develop an enabling environment for cooling, via policy/regulations such as MEPS, S&L, National Clean Air Programs;
 - (b) Build and generate a track record in the cooling sector;
 - (c) Amplify the potential to leverage experience gained and lessons learned;
 - (d) Provide pioneering examples that can be emulated through different models and approaches that can be scaled up and be fit-for-purpose in more countries;
 - (e) Allow governments to see how the policy activities are delivered and delivering at sector level via investments; and
 - (f) Further stimulate scaling up the shift to sustainable cooling investments.
38. The Cooling Facility will work with countries to mainstream sustainable cooling into their development and climate plans, procurement strategies and policy/regulatory frameworks and also help train and build the capacity of relevant stakeholders, while also helping raise awareness of the availability and benefits of sustainable cooling measures and technologies.
39. The Cooling Facility will in parallel also focus on different implementing models: a) credit lines that help local banks develop a business line for cooling investments and build their capacity to originate cooling projects; b) ESCOs that provide financing as well as services to design and implement cooling and energy efficiency measures, to address both the financing and capacity barriers; and c) EERFs collect energy cost savings to recover the capital expenditure for cooling investments so that the funds can be re-invested and the EERF can become a sustainable entity.
40. The combinations of policy and investment via the Cooling Facility will: (a) set up revolving financing that allows financing to continue after the end of the projects; (b) enable private sector entities to engage and invest in project activities in the longer term; as well as (c) send strong market signals for shifts towards sustainable cooling through public procurement, thus exemplifying the financial sustainability and scale-up potential within the cooling sector.

3.3 Sustainable development potential

Scale: High

41. The programme has a high potential to generate environmental and health benefits, including reduction in air pollution, increase in access to resources and cold chains for vaccines and health infrastructure development.
42. The Cooling Facility will provide benefits over multiple Sustainable Development Goals (SDGs). For instance:
- (a) Decent Work and Economic Growth (SDG 8): Higher temperatures and lack of cooling negatively impact performance labour productivity. It is estimated that by 2050, work hours lost due to heat may be as high as 12 per cent in the worst affected regions of South Asia and West Africa – representing 6 per cent of annual GDP;

- (b) Sustainable Cities and Communities (SDG 11): In urban environments, average temperatures have been increasing at twice the rate of global change, largely due to the Urban Heat Island effect. Annual mean air temperatures of cities can be 1–3 °C higher than their surroundings (at night, the temperature difference can be as high as 12 °C). The need to reduce excess urban heat is one of the key resilience and sustainability challenges of the 21st century;
- (c) Zero Hunger (SDG 2): In developing countries around 40 per cent of food produced (amounting to more than 250 million tonnes), including one in four fish caught, is lost between harvest and market;
- (d) Good Health and Well-being (SDG 3): High temperatures increase the risk of heatstroke, breathing issues, heart attacks, asthma and kidney problems. Unreliable medical cold chains put at risk the effectiveness of vaccines and other medical supplies;
- (e) Gender Equality (SDG 5): According to SEforALL 2019, a gender equality risk is evident with regard to cooling but more data are needed. Women and girls are often disproportionately impacted by a lack of access to sustainable cooling. The Facility will be able address gender gaps across entrepreneurs, employees and at the consumer level; and
- (f) Affordable and Clean Energy (SDG 7) and Climate Action (SDG 13).

3.4 Needs of the recipient

Scale: High

43. The Cooling Facility will enable the target countries to meet their cooling needs in a more climate-friendly way and contribute to lower GHG emissions by: (a) strengthening their institutions and implementation capacity to adapt to rising temperatures; and (b) addressing the barriers for clean cooling. The bottom-up approach taken by the Facility will provide favourable conditions for local recipients to address the issue of sustainable cooling solutions by grounding mechanisms and strategies to each country's institutional and policy framework and building on countries' priorities. The sub-sector and models are geared towards addressing the challenges and needs of the local recipients. The context for each country is as follows:

- (a) **Bangladesh:** The country faces serious energy security challenges and severe heat stress, equally across its population across different income levels. Integrating energy efficiency and passive building design features and energy efficient cooling at scale now is key for building a more sustainable infrastructure at lower costs and lower pressures on the electricity grid and also in line with the Environment Conservation Act and the Bangladesh Energy Efficiency and Conservation Master Plan;
- (b) **El Salvador:** To ensure an effective COVID-19 response and generate long-lasting resilience and be prepared to handle future health crises, the Government of El Salvador seeks to strengthen the overall national immunization and related health delivery systems. This includes a variety of issues, including safe and appropriate cold chain and relates services;
- (c) **Kenya:** Demand for cooling is projected to grow significantly in the Nairobi and Mombasa markets, with a projected 1.8 million cumulative unit sales between 2020 and 2030 for off-grid households alone. There are an estimated 3.4 million off-grid households in Kenya that could realize direct net economic benefits from owning a refrigerator. Energy efficiency will help lower the projected energy consumption to meet Kenya's cooling demand, and in doing so reduce utility bills for households and businesses, and reduce GHG emissions in line with the country's climate mitigation goals;

- (d) **Malawi:** A lack of access to cooling, in particular more sustainable cooling solutions with lower operational costs, is hindering economic opportunities for micro, small and medium-sized enterprises involved in a number of important sectors, such as dairy and horticulture (fruits and vegetables). Promoting investment opportunities in sustainable cooling investments, with a focus on women and the youth, will contribute to providing economic opportunities and strengthening the resilience to climate change of vulnerable and marginalized population groups, while also avoiding expensive and polluting energy consumption;
- (e) **North Macedonia:** Buildings consume about 39 per cent of the energy in the country, and represent a major priority. The public sector has the greatest potential, with about 35–40 per cent savings, mostly in the health and education sectors. In the past, the focus has been on heating, but increases in summer temperatures in Europe have increased the demand and need for cooling, especially in schools and hospitals. Improving thermal renovations of public buildings will substantially reduce their future cooling loads, which will contribute to the country's adaptation to the changing climate. In the context of the COVID-19 situation, the rehabilitation of health-care facilities, with adequate air conditioning systems, is expected to become a higher priority;
- (f) **Panama:** According to the National Energy Plan of Panama, cooling systems' consumption currently represents about 40 per cent of overall electricity consumption. The electricity sector is facing serious challenges in keeping up with rising demand coupled with increased air conditioning penetration jumping from 8.9 per cent in 2000 to an estimated 66 per cent by 2050; the need for efficient cooling systems is a priority for Panama;
- (g) **Sao Tome and Principe:** The Government was initially unable to quickly establish laboratory capacity to diagnose COVID-19. Notwithstanding the fact that the country is now better prepared to diagnose, isolate, treat and track COVID-19 contacts, a significant and more comprehensive strategy to strengthen the public health systems is needed to continue with the mitigation measures to contain the spread of the disease in the country and to design and implement the national COVID-19 immunization plan, which includes a robust vaccine cold chain;
- (h) **Somalia:** The global COVID-19 pandemic is a pressing problem in Somalia, exacerbated by an extremely weak primary health-care system, inadequate surveillance and laboratory capacity, and limited hospital services. There are major gaps in cold-chain capacity, which prevents expanding the currently extremely low vaccine coverage and will make distribution of COVID-19 vaccines difficult. Likewise pharmaceuticals and medicines expire and become ineffective more quickly in the very hot climatic conditions. Moreover, without support, it is expected that kerosene, petrol and diesel fuelled generator equipment will be moved to increase cold storage capacity; and
- (i) **Sri Lanka:** Commercial buildings (including government buildings) and industrial buildings contribute 60 per cent of the electricity consumption in the country. The Sustainable Energy Authority has also has defined an ambitious Demand Side Management programme that includes chillers, fans, refrigerators, air conditioning and green buildings (including the potential of nature-based solutions such as green roofs and green walls). But the major barrier to meet the needs of the people for these technologies has been access to the right type of financing; the Cooling Facility via the EERF with ESCO model could be a solution.

3.5 Country ownership

Scale: High

44. The Cooling Facility is strategically aligned with the countries' nationally determined contributions, the Montreal Protocol and its Kigali Amendments, and countries' strategies. The relevant country engagements have been discussed with relevant local authorities and also presented to national designated authorities (NDAs), which have issued no-objection letters. Also, stakeholder engagements have been undertaken and will be further expanded while developing country-specific intervention. The following describes the government's efforts and strategic vision for the energy efficiency sector.

- (a) **Bangladesh:** The Government has prepared the Energy Efficiency Conservation Master Plan 2015-2030, which describes the adoption and implementation of energy efficiency and conservation regulatory measures. In addition, the project will inform the Government's efforts in developing a national Cooling Action Plan;
- (b) **Kenya:** Under the Paris Agreement, Kenya has committed to reducing GHG emissions by 30 per cent, or 142 MtCO₂eq, by 2030. The National Electrification Strategy presents a road map for achieving universal access to electricity by 2022. Supporting off-grid cooling appliances and access under the Facility will contribute to both of these goals;
- (c) **Malawi:** Promoting the use of energy-efficient technologies and designs is one of the strategies advocated for in the Government's priority areas under the Malawi Growth and Development Strategy (MDGS III). Since agriculture is one of the major contributors to GHG emissions, the Facility's support of clean and sustainable cold chains in agriculture will contribute to the achievement of the MDGS III goal;
- (d) **North Macedonia:** The Facility will support achieving the Government's energy saving commitments through establishing a financing mechanism and investing into efficient space cooling of buildings;
- (e) **Panama:** Panama has ratified the Kigali Amendment in 2017, which aims to phase down HFCs and is currently implementing Stage II of its HFC Phase-out Management Plan under the Montreal Protocol. The Facility will support achieving the Government's commitments through its support for efficient space cooling;
- (f) **El Salvador:** The Government is integrating climate-friendly investments in its national programmes and its NDC includes health as a sector affected by climate change and the promotion of a larger use renewable energy. This is aligned with the Facility's support for climate-friendly health facilities and cold chains;
- (g) **Sao Tome and Principe:** The country's NDC include promoting the greater use of renewable energy and lists health as one of the sectors negatively impacted by climate change. The Facility will contribute to the Government's NDC through supporting space cooling and cold chains in the health sector, including through the increased use of solar power;
- (h) **Somalia:** Somalia's NDC related to scaling up the use of solar energy is consistent with the Facility's aim to support for more climate-friendly health facilities and cold chains, including through greater use of solar power;
- (i) **Sri Lanka:** Demand Side Management is identified as one of the seven mitigation actions of the energy sector NDC of Sri Lanka. The Energy Efficiency Building Code sets standards for energy efficiency in design and retrofits in buildings, at the same time providing methods for determining compliance. These efforts are aligned with the Facility's support of space cooling and energy efficiency measures in buildings.

3.6 Efficiency and effectiveness

Scale: High

45. The project has a co-financing ratio of 1:4.6. The cost of mitigation is USD 54.2/tCO₂eq for the total project cost and USD 9.7/tCO₂eq in terms of GCF funding.
46. The financial structure of the Facility balances the need to support low-carbon cooling investments, leverage additional resources from private and commercial investors, and enhance the enabling environment for cooling. The Facility's approach of matching the model to the context of the country and sector will provide the right solution at the right place instead of a "one size fits all" approach which is never the case and generally not appreciated by countries.
47. The economic analysis shows that the facility yields a financial and economic internal rate of return of -0.8 and 16.3 per cent, respectively. The economic value added of the Facility is largely due to its contribution to GHG emission reduction potential of the funded activities. Further, the economic value is very sensitive to the estimated cost of recycling (otherwise conducting environmental remedial actions) for appliances that have exceeded their useful lives; this would need to be tackled as part of component 1.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

48. **Programme background.** The programme aims to support subprojects in the areas of space cooling and cool/green surfaces (including passive cooling strategies in the built environment, cooling equipment; building automation and controls; and solar and vegetative roofs and walls); and refrigeration, cold chains and logistics (including refrigeration, storage, and distribution activities, along with associated equipment and logistics). The programme consists of three components that focus on (1) supporting policy and the regulatory and enabling environment to strengthen institutional, policy and regulatory frameworks, among others; (2) provide financing for eligible cooling investments through financing modalities that will promote the adoption or scaling-up of sustainable cooling technologies, appliances and business models; and (3) project management to support the EEs and PIEs perform their project management functions.
49. **Environmental and social (E&S) risk category and safeguards instrument.** The AE has assessed that the activities under the Cooling Facility will have potential for limited adverse E&S risks and impacts which, individually or cumulatively, are few, generally site-specific, largely reversible, and can be readily addressed through mitigation measures. The supported activities are likely to have low/moderate to substantial risk as per the AE's risk definition. The Secretariat confirms that the classification corresponds to the GCF's E&S risk classification of category B and that it is within the risk accreditation of the AE. An environmental and social sustainability framework (ESSF) is prepared in accordance with the AE's safeguards requirements. The programme's environmental and social co-benefits include those relating to health since the support for more sustainable space cooling will help lower the incidence of heat-related illnesses and mental stress. Improved local air quality through the displacement/avoidance of diesel generators by replacing them with solar-powered cold chains is also expected.
50. **Compliance with GCF's Environmental and Social Safeguards (ESS) standards.** The following discussions describe the programmes compliance with the GCF ESS standards:
51. **ESS1: Assessment and Management of E&S Risks and Impacts.** The programme has submitted an ESSF that provides a description of how the E&S risks and impacts will be assessed, managed, and supervised at both the framework level and the operational level. The ESSF describes the necessary instruments and information related to environmental and social impact assessment or audit and management at the programme level. The ESSF also outlines the

impacts of the proposed programme as well as possible mitigation measures to address these impacts. The potential E&S risks and impacts that will be considered in the conduct of due diligence for the sub-projects include those related to community health and safety; climate change and other transboundary or global risks and impacts; and threats to the protection, conservation, maintenance and restoration of natural habitats and biodiversity and those related to ecosystem services and the use of living natural resources. Potential social risks and impacts, include threats to human security through the escalation of personal, communal or inter-state conflict, crime or violence; risks that impacts will fall disproportionately on disadvantaged or vulnerable individuals or groups; prejudice or discrimination towards individuals or groups in providing access to development resources and benefits; negative economic and social impacts relating to the involuntary land acquisition or restrictions on land use or regarding land and natural resources; impacts in relation to occupational health and safety of workers; and risks to cultural heritage. The programme is expected to channel financing through financial intermediaries, which will be required to establish and maintain an environment and social management system to identify, assess, manage, and monitor the E&S risks and impacts of the subprojects.

52. **ESS2: Labour and working conditions.** Construction and implementation of the subprojects will involve hiring workers which will include either direct-hired or contracted/sub-contracted workers. Occupational health and safety plans will be prepared as per the World Bank Group environment, health and safety guidelines and implementers will have to comply with these requirements including the preparation of a labor management procedure. Since an influx of migrant labour will also have potential impacts in relation to worker conditions, occupational health and safety, gender-based violence and potential conflict among the subproject workers as well as with the host local communities, these risks/impacts will be evaluated as part of the E&S risk assessments and will be managed in accordance with the labour management procedure and other appropriate plans and procedures. These measures will likewise be included in the bidding and legal documents with contractors/sub-contractors. A dedicated grievance redress mechanism for subproject workers will also be established for each subproject.

53. **ESS3: Resource efficiency and pollution prevention.** While the interventions under the programme are generally expected to promote efficiency and resource conservation, construction of such facilities such as new buildings or retrofitting existing buildings is expected to generate typical construction impacts such as dust and air pollution from excavations and vehicle exhaust; solid waste arising from construction and demolition including the potential need to handle and dispose of hazardous waste (e.g. asbestos and wastes from dismantling/disposal of old cooling equipment and refrigerant gases). These risks/impacts will be assessed in detail for each of the subprojects. However, given that these potential impacts are expected to be low to moderate and are reversible and localized in nature, appropriate mitigation measures can be readily implemented in accordance with good international industry practice.

54. **ESS4: Community health, safety and security.** Constructing facilities and implementing programme activities will have potential to adversely affect the health, safety and security of the host communities. Impacts may arise from noise and vibration; dust emission; and disposal of old parts and equipment which may contain hazardous waste. Traffic-related incidents as a result of vehicle transport in and out of construction areas may also affect the nearby communities. Nevertheless, these risks and impacts which will be mainly on the communities' safety, can be managed through proper implementation of adequate and appropriate mitigation measures as provided in the AE's standards.

55. **ESS5: Land acquisition and involuntary resettlement.** The programme may result to land acquisition for construction of new buildings or cold storage facilities and may have potential adverse environmental and social impacts associated with land acquisition or land

conversion which will be managed consistent with ESS5 requirements: that is, avoid, minimize, reduce/mitigate and compensate at replacement cost lands that will be acquired based on due diligence and plans prepared in accordance with ESS5. An indicative resettlement policy framework and an outline of a resettlement action plan is provided in the ESSF to guide EEs in preparing specific resettlement action plans (if required) in line with safeguard requirements.

56. **ESS6: Biodiversity conservation and sustainable management of living natural resources.** Depending on the scale and location of the facilities that will be financed under the programme, there may be potential loss of habitats and other adverse impacts to flora and fauna. Nevertheless, siting and careful selection of the area where these facilities will be constructed will dictate the extent of its impacts. To mitigate the impacts, care would be taken by the developers to ensure that the impacts on natural habitats, terrestrial and aquatic ecosystems and on sensitive species are well assessed and managed through preparation of appropriate plans, as needed.

57. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** The AE stipulates that the presence of indigenous people in the areas identified for the projects will be screened and confirmed by the AE in consultation with the national authorities. The AE outlines that, once a project is confirmed, consistency with ESS7 objectives and requirements will be integrated into the design of the project and E&S risk documents, including the development of an indigenous peoples' plan or other types of plans depending on the country and project context, and free, prior and informed consent in the circumstances specified in ESS7. For consistency with the GCF Indigenous Peoples Policy, the sub-project screening will need to be inclusive for groups such as "Afro-descendent communities of South America and the Caribbean", "local communities", "scheduled tribes", "hill tribes" and "tribal groups".

58. **ESS8: Cultural heritage.** The ESSF indicates that subprojects will be screened and assessed regarding potential impacts on tangible and intangible cultural heritage. Site selection for each subproject will consider the application of ESS8 and will be further assessed under each subproject. Should there be possibility for presence and/or discovery of items of archaeological, paleontological, historical, cultural, artistic and religious value in the civil works sites during construction, a "chance find procedure" will be a part of the E&S risk instrument.

59. **Implementation arrangements.** In accordance with the ESSF, the AE will manage the overall environmental and social performance of the Cooling Facility and that of the sub-projects as well as monitor their performance in accordance with the requirements of the legal agreements, including those in the environmental and social commitment plan that will be agreed between the AE and the borrowers. The AE will likewise provide implementation support regarding the environmental and social performance of the subprojects, which will include reviewing the borrower's monitoring reports regarding their compliance, and will share such information with the GCF through the annual performance reports. For each sub-project, a project management unit will be established by the relevant EE which will be expected to include a team of experts including environmental specialist and social/gender specialist that will be tasked with ensuring adequate implementation of the management measures as well as the monitoring, evaluation and reporting during implementation.

60. **Stakeholder engagement and information disclosure.** The programme has conducted preliminary stakeholder engagements in the targeted countries. These engagements included meetings and consultation workshops with representatives from implementing agencies at the national level and regional level, local authorities of participating provinces, private investors, farmer representatives, and concerned communities. In addition, other interested parties include the media, local/international non-governmental organizations and development partners working in the same region. A detailed stakeholder engagement plan (SEP) will be developed for each of the subprojects to ensure transparency and meaningful consultation with the affected peoples and interested parties. The SEP will include stakeholder identification, analysis, information dissemination, consultation and engagement that will be

carried out during subproject design and implementation. The stakeholder identification, analysis and engagement will be informed by the degree of potential risks and impacts of the subprojects and how these will affect the stakeholders.

61. **Grievance redress mechanism.** The SEP for each of the proposed subproject under the facility will include a grievance redress mechanism that will be prepared and implemented by each implementing agency to receive and facilitate resolution of concerns and grievances from affected communities as well as broader stakeholders who may be affected or have interest in the subproject's activities. The mechanism will be established to receive and record complaints as well as provide resolution and feedback on the status of the grievances to the complainants to ensure transparency and accountability. Communities and individuals may likewise submit complaints to the AE's Grievance Redress Service that allows submission of grievances from affected communities and individuals. In addition, the GCF Indigenous Peoples Policy states that the GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 The revised SEAH Policy

62. The Revised Policy on the Prevention and Protection from Sexual Exploitation, Sexual Abuse, and Sexual Harassment (Revised SEAH Policy) includes a guiding principle related to Fund-related activities. It states that the "GCF and its Covered Individuals shall not enter into future engagements with those who condone, encourage, participate in, or engage in SEAH in Fund-related activities". As a result, the AE was requested to include the relevant covenant in the Term Sheet. As a form of this provision could not be agreed in the Term Sheet, a secretariat condition has been added on this matter.

4.3 Gender policy

63. The AE provided a gender assessment and action plan with the funding proposal and therefore complies with the requirements of the GCF Gender Policy.

64. The gender assessment presents the enabling environment in terms of applicable laws for non-discrimination and protection of women. Some of the countries promote increased engagement of women in labour markets, including in the electricity, gas, steam and air conditioning supply sectors. Gender equality in terms of access to income, financial inclusion, access to health and education, representation in political bodies is promoted in the countries. Although there are a number of laws enacted in all the countries to promote gender equality and prevent discrimination, women do not always have the same rights as men under some laws and they have lower access to resources compared with men. The assessment has fallen short of providing information on the existing national institutional frameworks that could support and ensure gender equality matters are integrated well in the programme's beneficiary countries. The AE has not provided an in-depth analysis of the root causes for the inequalities and gender-based challenges faced by women, in particular from cultural, traditional and societal perspectives.

65. The gender assessment is limited in that it was compiled on the basis of a review of literature available to the AE. The assessment does not include inputs from relevant stakeholders or has not benefited from a stakeholder engagement exercise that should have been undertaken during the preparation of the programme. The stakeholder engagement exercise to ensure relevance and effectiveness of the programme and project impacts on gender issues will be undertaken during the programme's implementation.

66. Information on gender-related issues on cooling is limited and descriptions of the preliminary identification of gender gaps do not articulate in detail the challenges faced by

women and men in different countries or in the sectors that are likely to be supported by the Facility. With the limited information provided, understanding the probable impact on women's participation and access to benefits of the programme as well as drivers of change is difficult. Despite that, some gender issues have been identified which may indicate impact, the level of engagement and benefits to women, such as lack of access to jobs and skills development, access to finance and entrepreneurship. In addition, a list of possible interventions for women's economic empowerment, such as access to services and leadership and decision-making, which may be implemented to address gender gaps. These actions could potentially be included in individual project-level gender action plans. The funding proposal further indicates that the programme will aim to enhance the knowledge base of gender issues associated with access to cooling and reduce gender gaps in cooling-related sectors.

67. The assessment indicates that additional gender analyses will be undertaken at the project level, which will involve relevant stakeholders and which would have to address and include the many actions identified through the current gender assessment, address gaps and provide insight into the likely differentiated cooling needs and priorities based on defined responsibilities. Additionally, further work will need to be undertaken to ensure that various components of the programme mainstream gender and take gender issues as an important eligibility criterion in financing of projects.

68. The programme will establish a grievance redress mechanism for each project, which will be made available for reporting and addressing grievances associated with sexual exploitation, abuse and harassment.

69. The AE has provided a programme-level gender action plan that includes activities, indicators, some targets, responsibilities and timelines. As part of the budget for component 1, financial resources have been set aside for undertaking further gender assessments although there are no budget allocations for the implementation of the other activities in the action plan. Several targets have not been set and the setting of targets has been outlined as country dependent. The AE has proposed to collect baseline data during implementation which should be used to rationalize and establish targets across all activities.

70. Activities listed in the gender action plan are too general and set at a high level and fail to link and reflect the nature of the projects that have been identified and presented in a separate annex with the funding proposal, such as space cooling, cool surfaces, refrigeration, and cold chains in sectors such as health and agriculture. A few important gender co-benefits are outlined in the gender assessment and are reflected in the gender action plan, such as employment opportunities for women and access for women-owned businesses to cooling services, in an effort to ensure that direct benefits of the programme are realized by women. The activity on improving access for men and women to new cooling, refrigeration, cold chains, logistics and green solutions has not articulated how this will be achieved and a corresponding indicator will track the number of both male and female beneficiaries, although no targets have been set. Activities also included in the programme-level gender action plan are related to participation in stakeholder engagement, capacity-building and awareness-raising events. Other activities in the gender action plan are actions that contribute to the development of further gender analyses and action plans at the project level, which will be done by the AE through its EEs and do not warrant inclusion in the action plan as standalone activities (e.g. identifying barriers that need to be addressed to promote the participation of women and prioritize actions that contribute to closing gender gaps). The AE has failed to include activities such as skills development in the cooling sector, entrepreneurship and access to finance/addressing affordability constraints, consumer engagement, addressing different needs of women and men for cooling technology products and services in various sectors such as fisheries and agriculture, under component 2.

71. Gender specialists are included in the project management unit for each project. Each specialist will be responsible for overseeing the implementation and reporting on activities in each respective action plan.

72. The AE is advised to improve its gender mainstreaming approach to ensure that all components of the programme address the needs of women and men equitably. The AE is recommended to ensure that additional gender analyses at the project level for each of the nine countries integrate the needs and priorities of women and men, particularly closing the gender gaps in interventions that are dedicated to cooling. This should include opportunities to address them, in the priority areas of intervention of the programme as would also be identified through stakeholder engagement. Stakeholder engagement should provide space for all relevant stakeholders and at all levels, including communities, to engage and influence the design of the gender action plans. Additionally, the AE is required to ensure that an action plan is prepared for each country not for just six when there are nine participating countries. The programme-level gender action plan will need to be updated based on the findings of the aggregate, country and sector-based gender assessments and will need to reflect expected benefits and impacts. The updated programme-level gender action plan should include indicators, targets with baseline, timelines, responsibilities and allocated budgets for their implementation. The GCF expects to monitor progress based on the updated programme-level gender action plan. The criteria for the selection of women-owned businesses will need to be defined and activities to support women-owned businesses should be included in a revised gender action plan for the programme.

4.4 Risks

4.4.1. Overall programme assessment: medium risk

73. The funding proposal requests USD 125 million in sovereign loans and USD 32 million in grants. The programme targets support to the participating host countries to mainstream and bring to scale sustainable cooling solutions across key sectors. The host countries of the programme are Bangladesh, El Salvador, Kenya, Malawi, North Macedonia, Panama, Sao Tome and Principe, Somalia and Sri Lanka. The programme consists of three components: (1) policy, regulatory and enabling environment support (USD15.7 million of GCF grant); (2) financing for cooling investments (USD141.3 million of GCF funds, of which USD125 million in loan and USD 16.3 million in grant); and (3) project management (no GCF funds requested). The AE envisions co-financing of USD 722.84 million of which USD 50 million in guarantees, USD 109 million in grants and USD 563 million in loans. All co-financing amounts are indicative and co-financing tenor does not match GCF tenor. If the programme raises a lower amount of co-financing, the programme's climate impact may also get reduced while maintaining the same level of financing from GCF.

4.4.2. Accredited entity/executing entity capability to execute the current programme: medium risk

74. The World Bank is the AE for the proposed programme. Given the structure of the programme, the broad eligibility criteria, and limited visibility at this stage regarding the end projects, the success of the programme depends to a great extent on the capability of the AE. The AE has a strong institutional capacity and experience in executing similar programmes.

75. For the purposes of the loans and grants, the respective host countries will be the EEs. They will either act through the relevant ministries or designate a PIE. The AE has at this stage not assessed the capacity of all EEs involved. It is advised that the AE shall complete such assessment before disbursement of the funds to such an entity.

4.4.3. Programme-specific execution risks: medium risk

76. **Demand risk:** The impact achieved by the project will depend on the financing opportunities and early-stage projects maturing. Also, impact depends on continued utilization of GCF funds in such opportunities. Demand risk is partially mitigated by technical assistance under to programme to help beneficiaries and World Bank operational teams in the design of cooling informed projects including activities such as cooling market studies, sector gap analysis and strategies, knowledge sharing, stakeholder consultations, business models identification. It is advised that the terms and conditions of the project require that GCF low concessional loans shall be invested in sub-projects for the duration of the programme.

77. **Sovereign credit risk:** GCF will provide up to USD 125 million in sovereign loans under the programme. GCF will not benefit from the preferred credit status of the AE. The credit risk of the countries under the programme varies from lower medium grade (BBB by S&P, Panama) to substantial risks (CCC+ by S&P, Sri Lanka). Several countries in the programme do not have a rating by a major rating agency (e.g. Somalia and Sao Tome and Principe). The AE has included in the term sheet that GCF loans will not enjoy the AEs preferred creditor status.

78. **Co-financing risk:** Lower than indicative funding amounts provide by the AE and related entities could affect the impact of the programme. Co-financing presented in the funding proposal is indicative. The terms sheet contains no requirements on commitment of co-financing before funded activity agreement (FAA) execution. A minimum co-financing ratio of 1:2 is included in the term sheet on a per project basis, but this is below the co-financing ratio of 1: 4.6 in section E.2 of the funding proposal. However, the AE notes that 54 per cent of the co-financing has already materialized. The World Bank and World Bank-managed trust funds have approved and signed the corresponding legal agreements with the recipient countries for six of the nine projects envisaged. Achieving a lower amount of co-financing may result in reducing the climate impact of the project. It is recommended that a minimum co-financing ratio of 1:4 on a programme basis is included so that the efficiency of GCF funding remains as envisaged.

79. **Operational environment:** Execution of the project will depend on the technical capacity on the ground as well as political factors relating to affordability. The AE will develop a monitoring and evaluation framework during the preparation phase to address technical capability risk and will take into account impacts on affordability of the interventions.

80. **Carbon credits:** The termsheet includes no limitation on the sale of carbon credits. Sale of carbon credits could reduce the impact attributable to GCF. It is advised that limitations on the sale of carbon credits are included in the terms and conditions of the programme

4.4.4. Compliance risk: medium risk

81. Of the countries involved in this proposal, it is confirmed that Somalia is subject to United Nations Security Council (UNSC) [Resolution 751 \(1992\)](#). However, the World Bank, as the AE, has advised that it is not aware of any UNSC resolutions affecting Somalia that would impact on the proposed project being considered under the Cooling Facility. The World Bank acknowledges Resolution 751 (para. 14) insofar as it urges the international community to facilitate humanitarian assistance to the Somali population, and it does not list/exclude any particular territory.

82. In confirming that no individual or entity that is listed on any UNSC Sanctions list will be involved in any manner with the project or its activities, the World Bank referred to its obligations in this regard as stated in paragraph 4.05(a)(ii) the accreditation master agreement (AMA):

(a) In preparing each Funding Proposal, the Accredited Entity shall:

- (i) contractually require the Executing Entity in accordance with the Accredited Entity's policies and procedures in the Subsidiary Agreement
 - (1) to comply and to oblige compliance by all other persons and entities involved in the use of GCF Proceeds with the Accredited Entity's standards relating to fraud, corruption and other sanctionable practices (as defined in the Accredited Entity's policies and procedures regarding sanctions), and
 - (2) to ensure that the GCF Proceeds are not being used for any payments prohibited by a decision of the United Nations Security Council taken under Chapter VII of the Charter of the United Nations, if and when the Subsidiary Agreement is entered into with the Executing Entity.

83. The requirements for assessing capacities of counterparties are as summarized in the World Bank's Policy, Directive and Procedure on Investment Project Financing, to ensure that counterparties have the technical, financial management, procurement and environmental and social capacity to implement the project in accordance with the requirements of the policies. The assessment for a particular project will be documented in a project appraisal document, including capacity-building measures/action plans based on the assessment.

84. The World Bank confirmed that, as a multi-country programme covering several projects, project appraisal documents will be prepared at the project level rather than the Facility level. Such documents will be drafted during the project preparation phase, finalized during negotiations with host countries and publicly disclosed once the project is approved by the World Bank.

85. The World Bank confirmed that anti-money laundering and combatting the financing of terrorism (AML/CFT) are very important and are comprehensively covered in the AMA and FAA template, which stipulate that the World Bank will apply its own policies and procedures on these matters. These policies and procedures require the World Bank to assess all the fiduciary risks and develop mitigation measures, which is undertaken during project preparation.

86. The World Bank has confirmed that controls mechanisms to ensure materials or technology are not diverted or misused for prohibited practices are covered in the AMA and FAA templates, and that the World Bank will apply those standard covenants to the projects through the FAA. For example, clause 11.01 of the FAA template provides, as follows:

- (a) *The Accredited Entity's AML/CFT procedures share the objective of the Fund's AML/CFT Policy to ensure that the proceeds of any financing are used only for the purposes for which the financing was granted, it being understood that the Accredited Entity shall be solely bound to apply its own AML/CFT procedures in accordance with the AMA and this Agreement.*

87. The World Bank has confirmed that the GCF funds will not support the distribution of cash, vouchers, commodities, or items of value to end user beneficiaries, whether directly or indirectly.

88. The World Bank has advised that communities and individuals who believe that they are adversely affected by the project may submit complaints to existing project-level grievance redress mechanisms or the World Bank's Grievance Redress Service.

89. The Grievance Redress Service ensures that complaints received are promptly reviewed in order to address project-related concerns. Project-affected communities and individuals may submit their complaint to the World Bank's Independent Inspection Panel, which determines whether harm occurred, or could occur, as a result of non-compliance with World Bank policies and procedures.

90. The World Bank has confirmed that, for each of the proposed projects under the Cooling Facility, a grievance redress mechanism is to be prepared and implemented by each EE to receive and facilitate resolution of concerns and grievances of project-affected communities as well as broader stakeholders who may be affected or have interest in the project or its activities.

91. **Recommended risk rating:** The Office of Risk Management and Compliance (ORMC)/Compliance Team has conducted a review of the project in accordance with relevant GCF Board approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the ORMC/Compliance Team have determined a risk rating of 'medium' and has no objection to this request proceeding to the next steps.

92. The ORMC/Compliance Team would like to remind the World Bank, as the AE, of its continuing obligations and responsibilities with regard to monitoring and reporting any risks for money laundering, terrorist financing, or prohibited practices among the intended counterparties, EEs, beneficiaries, persons involved, or any of the proposed activities.

4.4.5. GCF portfolio concentration risk: low risk

93. In case of approval, the impact of this proposal on the GCF portfolio risk remains within the risk appetite.

94. **Project viability and concessionality:** The AE will differentiate the level of concessionality between the reimbursable instrument in each project by applying either low or high concessional loans. If the host country is in external debt distress, it will receive GCF non-reimbursable funds and will only receive co-financing exclusively on grant terms. A combination of high concessional loans and grants will be offered if (1) the host country is in high risk of debt distress; (2) the project does not generate any revenue or the reflows from the ultimate beneficiaries are not cost-reflective; and/or (3) the project targets vulnerable communities or groups that are more vulnerable than the overall population. Any other projects will receive low concessional loans.

95. Within the above parameters, GCF shall rely on the AE to appropriately deploy the concessionality. It is advised that the terms and conditions include a covenant that require the AE to apply the minimum concessionality principle and to contractually require of the AE that a substantial part of the concessionality is passed on to the end beneficiaries.

4.4.6. Recommendation

96. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall programme	Medium	- The AE has a strong institutional capacity and experience in executing similar programmes. - GCF is exposed to sovereign risk. - Co-financing information in the funding proposal is indicative. It is recommended that clauses are included to ensure co-financing materializes. - GCF shall rely on the AE, within the stated parameters to appropriately deploy the concessionality. It is advised
Accredited entity (AE)/executing entity (EE) capability	Medium	
Project-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	Medium	

		that the terms and conditions include a covenant that requires the AE to apply the minimum concessionality principle and to contractually require of the AE that a substantial part of the concessionality is passed on to the end beneficiaries
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4.5 Fiduciary

97. The World Bank will be the Accredited Entity for this programme. The AE will be responsible for ensuring that the Projects are developed and implemented by the relevant Executing Entities (EEs) and Project Implementing Entities (PIEs) and approved by the AE in accordance with its policies and procedures and the relevant provision of the AMA and FAA, and selected in accordance with the Eligibility Criteria. The AE will provide implementation support to the Projects including necessary monitoring, supervision, and quality assurance in accordance with its policies and procedures and the relevant provisions of the AMA and FAA.

98. Each Project will be carried out through different Executing Entities (EEs) and PIEs as further described below. EEs are as listed on FP. For indicative purposes, Table 1 on the TS, sets out certain Executing Entities and PIEs that are anticipated to carry out part or all of Projects in specific Host Countries. For the avoidance of doubt, the AE will not act as an EE for the Projects and the PIEs will not be included within the definition of EEs for this Programme. Although the PIEs are not included within the definition of EEs for this Programme, as they will carry out part or all of Projects in specific Host Countries, the AE will undertake due diligence on them in accordance with clause 10.02(a)(i) of the AMA as if they were included within the definition of EEs for this Programme and the outcome of such due diligence will be included in the respective Project appraisal document.

99. In respect of each Project for which a disbursement is requested, AE will deliver to the Fund a copy of the Accredited Entity's Project appraisal document (PAD), containing information about the relevant Project. PAD will include the capacity assessment of the EEs.

100. Each project financed through the Cooling Facility will be required to conduct financial management and procurement according to the World Bank policies and procedures. Each project supported by the Cooling Facility will follow the AE's Procurement Framework and Regulations.

101. Each project supported by the Facility is required to maintain financial management arrangements (including planning and budgeting, accounting, internal control, funds flow, financial reporting, and auditing arrangements of the borrower and entities responsible for project implementation) acceptable to the AE. A financial management specialist is expected to be part of the team forming the PMU established by each relevant Executing Entity (or PIE as applicable) for its respective project.

102. Moreover, for each project, the AE task team includes an accredited Financial Management (FM) specialist who provides support during the entire project cycle (preparation, supervision and implementation) in accordance with the AE's operational policies and procedures (to work with partner countries to design financial management and disbursement arrangements and to supervise and support the performance of World Bank-finance operations). The team's FM specialist closely monitors the performance of the EE's (or PIE as applicable) FM arrangements and their continued capacity to provide reasonable assurance that loan proceeds are being used for the purposes intended.

103. Financial reporting is also required for each project which is specified in each project's Disbursement and Financial Information Letter (DFIL). Unless otherwise agreed by the World Bank, the Executing Entity must submit annual audited project financial statements six months after the close of the financial year and unaudited interim financial (IFR) reports periodically. Audits are carried out by auditors with independence and capacity acceptable to the World Bank, under terms of reference acceptable to the World Bank.

4.6 Results monitoring and reporting

104. As a cross-cutting initiative, the programme aims at generating both mitigation and adaptation benefits. On the mitigation side, the intervention is expected to result in overall GHG emission reductions of 16,240,201 tCO₂eq over a lifetime period of 27 years, as per the metrics of the GCF core indicator for mitigation based on GHG accounting methodology detailed under annexes 23 and 24 of the funding proposal.

105. Regarding adaptation, the programme expects to benefit 4.22 million direct and 16.85 million indirect beneficiaries in the four countries at high risk of extreme heat hazard (Bangladesh, Malawi, El Salvador and Somalia) through the implementation of space cooling, refrigeration and cold chain projects. The AE provided a robust methodology and rationale for defining both direct and indirect beneficiaries in annex 24 of the funding proposal.

106. The theory of change adequately explains how change is understood to come about in the targeted countries. The goal statement is properly articulated, and the stated programme results and outcomes are defined in a manner that is clearly supportive of meeting the ultimate project goal.

107. The logical framework has been designed with relevant details, including the inclusion of solid means of verification and reporting on the appropriate core indicators for both mitigation and adaptation, as well as on the respective impact, outcome and programme-level indicators for the targeted results areas as per the GCF performance management framework and results measurement frameworks. At the design stage the AE has not been able to provide credible baselines and targets for the three programme-level indicators for adaptation in section E.5, so the Secretariat has requested through a dedicate term sheet covenant, that those are updated every year in each annual performance report and finalized prior to the interim independent evaluation.

108. The implementation timetable in Annex 5 of the Funding Proposal should be updated during the FAA negotiation when the exact timeline for submission of the initial APR will be negotiated and agreed.

109. The monitoring and evaluation plan (annex 11 of the funding proposal) has been found to follow the appropriate requirements because it includes the entire list of the programme logical framework indicators while ensuring alignment of the data/sources with established means of verification.

110. The Secretariat considers, however, that the programme does not provide sufficient safeguards for the GCF mitigation impact and, therefore, has agreed to discuss further with the AE during FAA negotiations the conditions on the potential issuance of carbon credits by the programme, while maintaining the integrity of the lifetime mitigation impact.

4.7 Legal assessment

111. The AMA was signed with the AE on 13 November 2017, and it became effective on 21 February 2019.

112. The AE has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and that it has the capacity and authority to implement the programme. It is recommended that, prior to submission of the funding proposal to the Board (a) the AE has obtained all its internal approvals and (b) the GCF has received a certificate or legal opinion from the AE in form and substance satisfactory to the GCF confirming that all final internal approvals by the AE have been obtained and that it has the authority and capacity to implement the programme.

113. The proposed programme will be implemented in nine countries in which GCF is not provided with privileges and immunities: Bangladesh, El Salvador, Kenya, Malawi, North Macedonia, Panama, Sao Tome and Principe, Somalia and Sri Lanka. This means that, among other things, GCF is not protected against litigation or expropriation in this/these country(ies), which risks need to be further assessed.

- (a) Bangladesh: A draft agreement on privileges and immunities is under negotiation. The Secretariat has liaised with the Ministry of Finance (Economic Relations Division) on the review of the draft agreement by the relevant involved ministries and agencies, as well as with the national designated authority (NDA). In February 2020, the Secretariat resent the draft agreement, together with a background note on privileges and immunities, to the NDA; however, no formal response has been received thus far;
- (b) El Salvador: The Secretariat sent a draft agreement together with background notes on privileges and immunities to the NDA in September 2015. The Secretariat met with the El Salvador delegation during COP 21 in December 2015, after which the draft agreement with background note was resent to the NDA. In October 2018, the Secretariat met with NDA to provide information about the draft agreement and the rationale for privileges and immunities, and clarified queries regarding the importance for the GCF of signing such agreements. However, no formal response has been received so far;
- (c) Kenya: The Secretariat sent a draft agreement on privileges and immunities, together with a background note, to the Treasury Department of Kenya on March 2016, and several exchanges have taken place to respond to comments received from the Attorney General's Office and the Ministry of Foreign Affairs of Kenya. However, no formal response or written comments on the draft agreement have been received thus far;
- (d) Malawi: The Secretariat sent a draft agreement together with background notes on privileges and immunities to the Ministry of Foreign Affairs in September 2015 and met with the delegation of Malawi at COP 22 in December 2016 to provide information about the draft agreement and GCF status. However, no formal response has been received thus far;
- (e) North Macedonia: the Secretariat sent a draft agreement together with background notes on privileges and immunities to the Ministry of Finance and the Ministry of Environment and Physical Planning in April 2017, and met with the Macedonian delegation at the GCF structured dialogue held in Tajikistan on September 2018. However, no formal response on the draft agreement has been received thus far;
- (f) Panama: The Secretariat sent a draft agreement together with background notes on privileges and immunities to the Minister of Presidency in September 2017. In October 2018, the Secretariat met with the NDA and provided information about the draft agreement and its rationale. The Secretariat has sent a follow-up email in November 2020; however, no formal response has been received so far;
- (g) Sao Tome and Principe: The Secretariat sent a draft agreement together with background on privileges and immunities to the NDA in February 2017 and met with the Sao Tome and Principe's delegation at the Global Programming Conference in August

2019. The Secretariat resent a draft agreement and background notes, both in English and French, to the Government in August 2019; however, no formal response has been received thus far;

- (h) Somalia: The Secretariat sent a draft agreement together with background notes on privileges and immunities to the NDA and a meeting was held with an NDA's representative during the Global NDA Conference held in October 2018. However, no formal response has been received thus far; and
- (i) Sri Lanka: A draft privileges and immunities agreement is under negotiation with Sri Lanka and there have been multiple exchanges on the text of the agreement. In June 2020, the Secretariat replied to inquiries from the Ministry of Finance in relation to the intent of entering into a host country agreement and the list of countries that have granted privileges and immunities to the GCF. Since then, the Secretariat has followed up with the Ministry of Development and Environment on the finalization of the text; however, no response has been received thus far.

114. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country(ies), or has been provided with appropriate privileges and immunities.

115. The definition of the term "Executing Entities" in the AMA includes entities that channel GCF proceeds and/or carry out project implementation. Under the proposed programme, for each project the AE will enter into a subsidiary agreement with the relevant host country that will carry out project implementation and/or channel the GCF proceeds to a "project implementing entity" that will implement all or part of the project. Therefore, as per the terms of the AMA, each of the host countries and the project implementing entities should be considered as Executing Entities for the purposes of complying with the obligations to be imposed on Executing Entities under the AMA and the FAA. During the course of negotiations on the proposed programme it has become clear that the AE will not accept the project implementing entities to be considered as Executing Entities and, therefore, they will not sign a subsidiary agreement with each of them. However, in terms of contractual arrangements, the AE has agreed that it will enter into a subsidiary agreement (as per the terms of the AMA) with each host country, and that it will also enter into a direct "project agreement" with each project implementing entity that will set out the relevant obligations of an Executing Entity set out in the AMA and the FAA.

4.8 List of proposed conditions (including legal)

116. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed programme;
- (b) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and

- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP177

Proposal name:	Cooling Facility
Accredited entity:	World Bank
Country/(ies):	Bangladesh, El Salvador, Kenya, Malawi, North Macedonia, Panama, Sao Tome and Principe, Somalia, and Sri Lanka
Project/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential

Scale: Medium to low

1. This is a large size funding proposal submitted by the World Bank as the accredited entity (AE). The programme covers nine countries from different regions: four countries in Africa (Kenya, Malawi, Sao Tome and Principe, Somalia); two countries in Asia (Bangladesh, Sri Lanka); one country from Eastern Europe (North Macedonia); and two countries from Latin America (El Salvador, Panama). The funding proposal is for the establishment of a cooling programme or/and facility¹ with the aim of reducing greenhouse gas (GHG) emissions from the cooling sector. Energy consumption by the cooling sector is increasing as a result of urbanization, increasing income of populations and increasing demand for cooling due to global warming-related local processes. This is a multi-sector and multi-country programme to establish a financing mechanism to address the challenges of providing access to cooling while minimizing negative climate impacts.
2. The programme aims to contribute to shifting from less-efficient cooling systems/no cooling systems (the baseline situation) to more efficient cooling systems, through a number of subprojects. The funding proposal is submitted as a cross-cutting project: 77 per cent of GCF proceeds is for GHG mitigation activities in the GCF results area “buildings, cities, industries and appliances” and the remaining 33 per cent is for GCF adaptation activities in the results area “infrastructure and built environment”.
3. Increasing energy demand has various drivers and among them is increasing cooling requirements driven by urgent adaptation needs caused by increasing trends of cooling degree days (CDD), heat waves and days with extremely high maximum temperatures. Of the nine countries covered by the funding proposal only four are considered eligible for adaptation activities due to high vulnerability: Bangladesh, El Salvador, Malawi and Somalia.
4. **GCF Financial support.** The total cost of funding proposal is USD 879.8 million including USD 157 million GCF contribution, which is 17.8 per cent of total cost. Twenty per cent (USD 32 million) of the total GCF contribution is grant for technical assistance, credit lines and concessional loans. The rest, 80 per cent (USD 125 million), is for a senior concessional loan. This loan has two different conditions: USD 29 million (23 per cent) has high concession (40 years tenor with 10 years of grace period and no interest rate) and the rest USD 96 million (77 per cent) will be issued for 20 years with 5 years’ grace period and an interest rate of 0.75 per cent.

¹ According to the definition in the funding proposal, “The Cooling Facility is essentially designed as an umbrella facility (a Program) that enables aggregating, through a programmatic approach, different types and sizes of sustainable cooling interventions”.

The total contribution from the AE is USD 722.84 million of which 15 per cent is grant comprising contributions from the International Development Agency (IDA), the Global Financing Facility, the Health Emergency Preparedness and Response Trust Fund (HEPRTF) and the Pandemic Emergency Fund, and 7 per cent is guarantee from the International Bank for Reconstruction and Development (IBRD) which is not still decided at the stage of submission and review.² The remaining 78 per cent is senior loans from IBRD and IDA with different pricing, grace periods and tenors (for details see section C.1 of the funding proposal).

1.1.1. Implementation arrangements

5. The funding proposal states that, for the purposes of the loans and grants execution, executing entities (EE) will be selected in the host countries, acting through the relevant ministry(ies), or the host country may designate a project implementing entity (PIE) to implement components 1 and 2, as identified and confirmed during project preparation. Accordingly, the funding proposal provides the list of EEs: Bangladesh (Ministry of Finance, acting through the Ministry of Power, Energy and Mineral Resources), El Salvador (Ministry of Finance, acting through the Ministry of Health), Kenya (National Treasury of the Republic of Kenya, acting through the Ministry of Energy and Petroleum), Malawi (Ministry of Finance, Economic Planning and Development, acting through the Ministry of Industry, Trade and Tourism), North Macedonia (Ministry of Finance, acting through the Ministry of Economy), Panama (Ministry of Economy and Finance, acting through the Ministry of the Presidency), Somalia (Ministry of Finance, acting through the Ministry of Health) and Sri Lanka (Ministry of Finance, acting through the Ministry of Power).

6. Three types of financial scheme are considered for implementing this programme: public procurement and contracting; credit line; and an energy efficiency financing facility.

7. **Public procurement and contracting:** for cooling infrastructure service delivery and technical assistance. This model is relevant for components 1 and 3 of the funding proposal. This model will be practically implemented by the AE and includes contracting with providers of goods and services, whereby contracts for services, works and goods could be submitted by an EE/PIE to the AE, which will assess them for relevancy and approve payments. Operation and maintenance contracts could be managed by the EE/PIE or the EE could ensure that the final beneficiary has relevant capacity to maintain the equipment or infrastructure operation. GCF funding will be priced lower than and blended with World Bank funds, to reduce the overall financing cost, thereby absorbing the incremental cost of deploying efficient cooling solutions. Final beneficiaries, in this financial model, could be households, sole traders, micro, small and medium-sized enterprises, or communities. Through these interventions, GCF concessional funds would play a catalytic role in creating a market for and increasing access to clean, efficient and sustainable cooling.

8. **Credit lines:** these are considered for less mature markets. The Cooling Facility will support the provision of credit lines for mitigation of the perceived high financial risk of energy-efficient and clean cooling, which may be provided through energy service companies (ESCOs). The AE will channel GCF finance along with its co-financing to the EE (i.e. the host country as the borrower/recipient), through a subsidiary agreement. The EE would extend the credit line to one or more partner financial institutions (PFIs), which would typically be banks or micro-lending institutions, for on-lending to eligible sub-borrowers such as micro, small and medium-sized enterprises (for example, industry and other private sector entities). In doing so, the PFIs will select and approve sub-loan applications based on the agreed investment criteria. Subsidiary agreements would require the EE to ensure that the PFIs comply with the sub-lending and other terms, and that they monitor the performance of the PFIs.

² Together IDA and IBRD form the World Bank. IDA focuses on the world's poorest countries, while IBRD assists middle-income and credit-worthy poorer countries.

9. The independent Technical Advisory Panel (TAP) notes that the implementation efficiency of this model depends significantly on the capacity of PFIs to manage climate change products locally. The AE has not yet conducted baseline studies for each participant country, including the assessment of PFI market capacity, so the independent TAP considers that this model has high risk. Moreover, barriers to improving the efficiency of the cooling sector vary from country to country and this has not been thoroughly assessed and presented in the funding proposal. The independent TAP notes that an understanding of the country-specific barriers to energy efficiency products should be the basis for developing the standard investment criteria for lending. The information provided in the funding proposal is general and does not enable the independent TAP to assess whether this model of credit line will be successful, because of the lack of well-developed investment criteria for country-specific cooling products. Capacity-building programmes for PFIs and standard tools for the assessment of product eligibility for lending could significantly reduce the risks related to the capacities of PFIs. Unfortunately, the funding proposal is not transparent on this subject. The independent TAP notes the following transparency issues:

- (a) The AE should transparently clarify whether or not the PFIs are only local financial institutions. Such a clarification would provide information to assess the potential capacity of the PFIs;
- (b) The funding proposal states that the PFIs are assessed by the EEs using eligibility criteria and eight such criteria are provided in funding proposal (p.37) using Malawi as an example of how the criteria are applied. However, the independent TAP noted that none of these eight criteria assess the capacity and/or experience of a PFI to manage climate change related or energy-efficient products; and
- (c) A second set of criteria for energy efficiency product investment is mentioned in the funding proposal but transparent details are not provided.

10. **Energy efficiency financing facility:** Under this model, the GCF proceeds are channelled through the EE (the host country as the borrower) according to the terms of the subsidiary agreement signed with World Bank. The EE either directly, or through a PIE, on-lends the funds to sub-borrowers (e.g. private energy service companies, municipalities and small and medium-sized enterprises) to undertake eligible cooling investments. The PIE would extend subloans, partial guarantees or grants, as applicable, to sub-borrowers (e.g. energy service companies or cooling-as-a-service companies).

11. The energy efficiency financing facility could have two implementation modalities:

- (a) **Public super ESCO model:** In this model, the host country designates a public energy service company (the public super ESCO) as the PIE. This super ESCO would typically be a public entity established by the government to (1) support cooling and energy efficiency investments in the public sector (public buildings and sustainable cooling projects); and (2) strengthen private sector ESCOs through demand aggregation, bulk procurement of equipment (therefore reducing the cost of equipment for ESCOs), standardization of energy-saving performance contracts, monitoring and verification templates and capacity-building, project development, and facilitation; and
- (b) **Energy efficiency revolving facilities:** This modality combines the features of a plain (vanilla) financing facility and the super ESCO model. In addition to providing financing to eligible sub-borrowers, the PIE would invest in energy efficiency activities using its own capital (e.g. for retrofitting of existing buildings).

1.1.2. Analysis of components

12. The components described in the funding proposal are structured to help countries remove the barriers to sustainable cooling by helping to address the financing barrier and build

an enabling environment through such activities as informing policy and regulatory frameworks, reaching out to stakeholders through effective communications, and supporting training and capacity-building. As such, the independent TAP is of the opinion that this funding proposal is not a typical “programme” but is more of a facility, as reflected in the name of the proposal. As described in the funding proposal, the Cooling Facility will consist of three main components, which will be calibrated according to countries’ needs and in accordance with the Facility’s criteria. Component 1 is mainly capacity-building, component 2 is the investment component and component 3 is the management component (fully financed by the AE).

13. **Component 1:** Support to policy, regulatory and enabling environment. The total cost of this component is USD 91.6 million, of which USD 15.7 million (17 per cent) is GCF grant, USD 46.1 million (50 per cent) is the senior loan and USD 21.6 million (23.6 per cent) is grant from the AE. The remaining USD 8.25 million (9.4 per cent) will be grant from the HEPRTF and the Global Financing Facility.

14. Activities supported under component 1 may include, for instance, (1) capacity-building of key stakeholders (e.g. on technologies; business models; and cooling project appraisal, implementation and evaluation); (2) technical assistance for developing plans, strategies, policies, regulations and standards; (3) designing sustainable market mechanisms and business models; and (4) communications and awareness campaigns.

15. **Component 2:** Financing for cooling investments. The total cost of this component is USD 769.04 million, comprising an USD 84.24 million (11 per cent) grant from GCF and the AE, and USD 50 million (6.5 per cent) guarantee from the AE, with USD 634.8 million (82.5 per cent) as a senior loan (19.6 per cent from GCF and 80.4 per cent from the AE). This component is the facility/financial scheme for financing of climate change eligible mitigation and adaptation projects. The deployment of these funds will be tailored to the needs of the projects, through suitable financing modalities for each project’s context; for example, credit lines to financial intermediaries, or loans to households, municipalities and small, medium and large enterprises, or subsidies. The second and third financial models are eligible for this component. Component 2 will finance the projects that comply with the following condition and project eligibility criteria:

- (a) **Condition.** The funding proposal states that the Cooling Facility will allocate a maximum amount equivalent to an aggregate of 30 per cent of the total GCF proceeds to any given host country/project³, regardless of the financial instrument(s) deployed. The independent TAP considers that this condition could be a threat for some countries among these nine wishing to participate but with relatively low capacities, if four or five leader developing countries take up over half of this cap. The funding proposal does not clarify how this 30 per cent maximum threshold per country was calculated. The draft term sheet demonstrates that this criterion balanced by the Secretariat having established an indicative amount for each GCF eligible country which has already provided a no-objection letter. The indicative amounts of the GCF proceeds are also secured by the term sheet, which establishes the cap for eligible reallocation of indicative amounts (15 per cent). According to the term sheet, a new country cannot be added to these nine countries and – crucially – the eligibility criteria (items (b)–(i) below) are no longer relevant for selection of a new country/project (as it is defined by

³ On the request for clarification from iTAP, the AE clarified that one country is eligible only for one project, hence, the Cooling Facility will support one project per country and nine projects in total. Annex 22 of the funding proposal provides example projects. The independent TAP is of the opinion that the AE’s understanding of a project format differs from the independent TAP’s understanding, because the provided project examples are combinations of different subprojects, with different baselines, different methodologies for calculating GHG reductions and different monitoring indicators. Some of the subprojects are even independent programmes, although the common target for all considered subprojects is the cooling sector. In response to a request for clarification made by the independent TAP, the AE clarified that one country is eligible only for one project.

the AE). However, these criteria could be applied by the AE to replace sub-projects within these nine countries;

- (b) **Cooling investments.** Projects selected under the Cooling Facility will support investments in the following target areas: (1) space cooling and cool/green solutions; and (2) refrigeration, cold chain storage and logistics. In the case of (1), projects should demonstrate a pathway towards financially sustainable models that reduce the need for public resources over time and eventually leverage commercial and private sector funding. In the case of investing in off-grid cold storage capacity additions (under (2) refrigeration, cold storage), the EE will need to demonstrate that the main source of energy to power these additional cold storage facilities is renewable energy (e.g. solar);
- (c) **World Bank operation linkage.** A Cooling Facility-supported project may (1) co-finance a new IDA or IBRD operation; or (2) be additional financing to an existing IDA or IBRD operation. The ratio of GCF funds to IDA/IBRD funds for each supported project will be at least 1:1. The independent TAP noted that it should be made very clear how the GCF proceeds that are spent for cooling purposes will be monitored when proceeds are added to the existing funds, even if those funds are for energy efficiency. Energy efficiency measures for heating purposes not always efficient for cooling. In some cases, it might be vice-versa and heating energy efficiency measures facilitate the increase of cooling energy consumption (due to location of building in favour of heating EE or other heating EE measures). For example, in the case of North Macedonia, the GCF proceeds are planned to be added to the existing energy efficiency fund. The independent TAP notes that cooling efficiency monitoring indicators should be transparently described and separated from other monitoring indicators;
- (d) **Performance threshold.** The threshold is stated as “at least 20 per cent energy efficiency improvement and/or at least 20 per cent avoided/reduced GHG emissions”. According to a clarification provided by the AE this 20 per cent will be results for the total project. A project, according to the definition confirmed during the review, is in most cases a group of different activities. The funding proposal clarifies that for each project an estimate of GHG reduction impact should be developed based on publicly available GHG methodologies and calculation tools, and consistent with guidelines outlined in annex 24 of the funding proposal. Regarding this criterion, the independent TAP notes that 20 per cent energy savings in some low-emission countries will not have any tangible impact on emission reduction and therefore the word “or” in the statement of performance threshold is not relevant: the 20 per cent GHG reduction should be obligatory and the 20 per cent energy savings should be regarded as desirable. As mentioned above, the monitoring and verification of the results of this programme will be complicated. Country baselines and national statistics related to the cooling process should be developed for each participant country;
- (e) **Alignment with the transition to low global warming potential (GWP) refrigerants:** (Note that this is not a general criterion and should be applied only to the subprojects considering refrigerators in the cooling process-TAP) Projects supported by the Cooling Facility will be aligned with the goals and requirements of the Kigali Amendment to the Montreal Protocol, under which hydrofluorocarbon (HFC) refrigerants are being phased down and replaced with refrigerants that do less damage to the climate. Investments will support the following activities: (1) providing cooling services without the use of refrigerants (e. g. fans and passive cooling solutions); and/or (2) using refrigerants that are aligned with the Kigali Amendment and meet relevant safety standards, as commercially available in respective local markets;⁴

⁴ This criterion takes into account that very low GWP refrigerant alternatives may not be readily available in all local markets for all the applications considered in the project.

- (f) **Adaptation to climate change.** (Note that this criterion should be applied only for adaptation projects -TAP.) The host country will be considered vulnerable to climate change if facing extreme heat risk hazards, and in severe need of cooling services. The Cooling Facility will support projects that will bring cooling-related benefits associated with services provided by the buildings, cold chain storage and logistics, health facilities and other related cooling investments. As mentioned above, four countries (Bangladesh, El Salvador, Malawi and Somalia) are considered countries eligible for adaptation countries. The independent TAP questioned the AE on the performance threshold for adaptation cases. The response from the AE clarified that there is no threshold to entry for adaptation projects as far as focus of the programme is mitigation activities. The independent TAP is of the opinion that if a funding proposal claims adaptation impact as well, this adaptation effect should be more transparent and a threshold should be established such as the number of direct beneficiaries that will have improved access to cooling. The independent TAP further noted that, for example, El Salvador is not listed among most vulnerable countries in the latest report about high-risk countries⁵ and it should be clarified why this country considered under adaptation eligible countries.
- (g) **Paradigm shift potential.** Supported projects will propose activities/technologies/applications that aim to achieve market transformation, create new markets and business activities at the local, national or international level, and can potentially be replicated in other sectors or geographic areas. The independent TAP is of the opinion that this criterion is formulated appropriately, though some of the example interventions/ measures provided in annex 22 of the funding proposal do not have such potential, but among them are activities related to cold chains storage and logistics, refrigerators and COVID-19 vaccination processes which could have transformative effect if successfully implemented;
- (h) **Country ownership and alignment with country priorities.** Facility-supported projects will be aligned with country priorities and climate change/cooling action plans as presented in national development plans or other policy documents. This criterion is in line with GCF criteria; however, cooling needs as such are not mentioned separately in countries' strategies and action plans related to climate change or energy efficiency. Of the nine countries, only Bangladesh and Panama target efficient cooling measures in their building energy efficiency programmes; and
- (i) **GCF concessionality level.** The funding proposal states that concessionality levels would be assessed based on the pricing of the World Bank co-financing, risk of external debt distress of the host country as published by the International Monetary Fund under the joint World Bank-IMF debt sustainability framework, and the financial analysis of the projects using the following principles:
- (i) *Grant only.* Countries in external debt distress and for which the World Bank co-financing is exclusively grant-based, would receive 100 per cent grant from the Facility;
 - (ii) *Combination of high concessional loans and grants.* Projects would meet any of the following three criteria: (1) host country in high risk of debt distress; (2) non-revenue generating project or where the reflows from the ultimate beneficiaries are not cost-reflective; and (3) projects that target vulnerable communities or groups that are more vulnerable than the overall population (e.g. refugees, people in rural areas with low economic income);
 - (iii) *Financing terms:* 40-year maturity, 10-year grace period with an all-in fee of 0.25 per cent (i.e. 0.00 per cent interest rate, 0.00 per cent commitment fee, 0.25 per

⁵ [Sustainable Energy for All \(2021\) Chilling Prospects: Tracking Sustainable Cooling for All 2021, available at https://www.seforall.org/chilling-prospects-2021.](https://www.seforall.org/chilling-prospects-2021)

cent service fee). Low concessional loan. Projects not meeting the conditions set out in (i) or (ii) above would be eligible for low concessional loans for investment; and

- (iv) *Financing terms:* 20-year maturity, 5-year grace period, 0.75 per cent interest rate, 0.5 per cent service fee, 0.75 per cent commitment fee.

16. The independent TAP is of the opinion that not all of these eligibility criteria are suitable for project/subproject selection because they are a mix of investment criteria, conditions, and so on. The independent TAP considers that these criteria should be regrouped and titled and revised according to the term sheet terminology.

17. **Component 3:** Project management. The total cost (100 per cent) of this component is covered by the AE and it equates to USD 19.2 million or 2 per cent of the total project cost. Some 39 per cent of this cost is the World Bank senior loan and the remainder (61 per cent) is grant from the AE and HEPRTF.

18. This component will support the EEs and PIEs in the management, coordination and implementation of project activities, in compliance with their contractual obligations with the AE.

1.1.3. Expected impact

19. The independent TAP notes that the project management unit, described in figure 4 of the funding proposal, is not appropriate for achieving the climate change related results anticipated. Instead, the climate change team responsible for implementing the subprojects selection criteria and designing the monitoring process should be added to the AE's management unit and similar representation should be required from each EE.

20. The programme could significantly contribute to the health-care sector of most vulnerable countries facing the lack of capacity to deal with COVID-19 vaccination requirements.

21. Annual GHG emission reductions estimated by the funding proposal amount to 912,659 tonnes of carbon dioxide equivalent (tCO₂eq) per year, or 16,240,202 tCO₂eq for the 27-year lifetime of the actions funded under the Cooling Facility (15 years lifespan for appliances and cold chains, storage and logistic and 20 years for buildings). The funding proposal calculates that there would be 4.22 million direct beneficiaries and 16.85 million indirect beneficiaries. However, these estimated beneficiaries relate only to the projects in four vulnerable countries (Bangladesh, El Salvador, Malawi and Somalia) at high risk of extreme heat hazards. The independent TAP notes that this approach is conservative, considering that other countries also have beneficiaries that will receive improved cooling appliances and cold chain benefits.

22. The independent TAP scores this criterion as medium to low, taking account of the fact that the impact of the funding proposal as a programme is not expected to be high. For example, some individual countries asking for support for the health sector could benefit from a relatively high impact but for the other countries, without relevant baseline information the expectation is that impact will be low.

1.2 Paradigm shift potential

Scale: High

23. According to the feasibility study provided by the AE, global energy use for space cooling is projected to grow three-fold between 2016 and 2050, with a majority of this growth occurring in developing countries. While the growing need for space cooling is in alignment with the developmental needs of countries, this growth must be addressed with effectively designed strategies and solutions to avoid severe economic, power system and environmental impacts.

24. The Kigali Amendment to the Montreal Protocol entered into force in 1 January 2019. The phase down of HFC refrigerants under the amendment has the potential to avoid up to 0.1°C of warming by 2050 and up to 0.4°C by 2100.⁶
25. The independent TAP recognizes that an increase in cooling demand could be regarded as a climate change adaptation process that is particularly important for those countries most vulnerable to the increasing trends of heat indexes (maximum daily temperature, CCDs and heat waves). Mitigation of this increasing rate of cooling demand through implementation of energy efficiency measures or by replacing conventional fuel sources with renewable sources in the cooling supply process will contribute to the low-emission sustainable development pathway, with the potential of global impact. The independent TAP shares the AE's rationale that enabling those countries most vulnerable to climate change to meet the increasing cooling requirement should be considered as adaptation. Hence, the independent TAP confirms that the concept considered within the funding proposal has paradigm shift potential in each country and globally.
26. The Cooling Facility considered in the funding proposal plans to demonstrate different options and in-country potentials for paradigm shift by proposing investments across the different but key cooling target areas and with a range of countries from different continents of the developing world having different barriers and national priorities. Diversity of financial mechanisms for implementing measures is another testing point in operation. In the opinion of the AE such a diversified approach could amplify the potential of the AE to leverage its experiences and learning. The AE's understanding is that in so doing, to provide pioneering example models and approaches, results can be scaled up and be fit-for-purpose in more countries with similar circumstances.
27. The independent TAP does not fully agree with the AE on its position that the selected approach of putting under one programme a variety of countries with different national circumstances and priorities and a variety of GHGs mitigation measures, as well as a number of complicated financial schemes for implementation will be a more efficient way to put this subsector on a transformational/sustainable pathway compared with the traditional programmatic approach where countries and measures are concentrated around the narrower concept than cooling demand in general, in different sectors and with different sources. Thus, the independent TAP concludes that the funding proposal is the pilot case for testing the effective approaches for mitigation of GHG emissions from the cooling sector in a way that does not limit but fully satisfies the climate change adaptation demands of vulnerable populations and countries.
28. Based on the above analysis the independent TAP concludes that this funding proposal as a pilot programme could not be rated as having a high paradigm shift impact; rather, more demonstration impact is expected, providing examples of success and failure and reasons for these outcomes. The concept itself has high transformational potential.

1.3 Sustainable development potential

Scale: Medium

29. Space cooling is increasingly recognized as a development priority and an issue of equity in a warming world. The increasing trend of cooling demand and the increasing gap between energy supply and cooling demand is significantly exacerbated by COVID-19 management requirements.
30. As mentioned in the feasibility study (annex 2a), meeting cooling demand in a business-as-usual way leads to increasing GHG emissions, but providing cooling is unavoidable if countries are to sustain resilience to climate change trends. The funding proposal describes how

⁶ See the Climate & Clean Air Coalition website at <https://www.ccacoalition.org/en/initiatives/efficient-cooling>.

the Cooling Facility contributes to sustainable development by demonstrating the options for slowing the increasing trend of energy consumption through supporting energy efficiency and clean energy development processes in pilot countries, transfer of clean technologies and solutions, and strengthening capacity in and awareness of cooling technologies. At the country level this programme is particularly important for countries that are highly vulnerable to increasing temperature and heat stress as well as for countries with energy supply challenges. From the global perspective the concept considered in the funding proposal on sustainable cooling supply deserves the urgent attention from global financial institutions.

31. Supply of sustainable cooling directly contributes to four of the Sustainable Development Goals (SDGs):

- (a) SDG 3 – Good health and well-being. From the sample projects attached to the funding proposal three are for mitigation of health sector stress due to the necessary COVID-19 management processes;
- (b) SDG 7 – Affordable and clean energy. Off-grid solar panels provide clean and affordable energy to households and small farmers mainly in rural areas, ensuring the cold chain for food security purposes;
- (c) SDG 11 – Sustainable cities and communities. The programme's contribution to this goal is significant, particularly for countries focusing on energy efficiency in the buildings sector; and
- (d) SDG 13 – Climate actions. Increase in cooling demand is exacerbated by climate change, and should be managed in a sustainable way in order to avoid further contributing to climate change processes. According to the information provided in the funding proposal if cooling efficiency remains unchanged, additional power generation capacity of 2,500 GW will be needed to meet electricity demand for air conditioning by 2050 – an estimated cost of USD 1.7 trillion. Increased access to cold chains is critical to ensuring food safety and reducing food loss and waste. By maximizing energy efficiency in the cooling sector (refrigeration and space cooling) while simultaneously phasing down HFCs contained in refrigerants, the world can potentially avoid close to 1°C temperature increase by 2100.

32. The programme offers different financial schemes to counties and clean technology developers for implementing/disseminating highly efficient cooling technologies; thereby strengthening the capacities of local financial institutions for managing funding for cooling.

33. Refrigerants in vapour compression-based equipment used in refrigeration and air conditioning account for about 20 per cent of the climate impact associated with cooling, owing to direct emissions of HFCs.⁷ The Kigali Amendment targets global consumption of HFCs,⁸ and could avoid up to 0.5°C of warming by 2100. The independent TAP notes that this is the first funding proposal in the GCF process targeting new industrial gases.

34. The independent TAP considers that the role of this subsector in the sustainable development process is crucial and should have high priority, although the independent TAP is not sure that this particular programme, as it is designed, could have high impact at the country level and therefore scores this criterion as medium to low.

⁷ [G20 agreement to reduce greenhouse gases vital to tackle climate change – UN agency | UN News](#)

⁸ The Kigali Amendment to the Montreal Protocol requires countries to phase down HFCs, beginning in 2019 for developed countries, and for most developing countries five years later. Its initial phase-down schedule will achieve a reduction in HFC consumption of more than 80 per cent by 2047. Most developed countries begin in 2019, whereas most developing countries will freeze consumption and production in 2024 and begin the phase down five years later. The aim is for developing countries' susceptible to high ambient temperatures will freeze at 2028 and begin to phase down in 2032.

1.4 Needs of the recipient

Scale: High

35. The rise in temperature and lack of access to cooling will impact the health, education, productivity and economic development of nations. In 2019, the Intergovernmental Panel on Climate Change reported that at 1.5°C of warming, 2.3 billion people could be both exposed and vulnerable to heat wave events.

36. The cooling needs of the nine countries involved in the programme significantly differ from each other, as described below.

37. Bangladesh, El Salvador, Malawi and Somalia are particularly vulnerable to heat hazards, increasing their adaptation needs as follows.

38. **Bangladesh** is highly exposed to extreme heat, drought and wildfires. Extreme heat hazard is classified as high throughout the country, which indicates that prolonged exposure to extreme heat, resulting in heat stress, is expected to occur at least once in the next five years.⁹ The seasonal frequency of days with $T_{\max} \geq 36^{\circ}\text{C}$ has been on the increase since 1986 in most parts of Bangladesh. Average temperatures have been increasing more rapidly during the monsoon season, from June to August, according to historical data for the past 30 years (1991–2020). Forecasts show that extreme temperatures are expected to occur with high frequency, duration, or severity. Most notably, the total number of hot days in a year (where daily maximum temperature is above 35°C) is projected to increase by 13 days by 2039, by 22 days by 2040–59, and by 41 days by 2099. By mid-century, annual CDDs are projected to rise from 482 CDD (2020–2039) to about 778 CDD by 2059, and 1210 by the end of the century. Overall, this will impact the energy sector because any time the temperature rises above 18.3°C , there is an expected increase in power demand for cooling.¹⁰

39. Bangladesh is among the world's most vulnerable countries to the effects of climate change. The country is also facing serious energy security challenges and severe heat stress, with the impact reaching its high-income residents, low-income neighbourhoods and informal slum settlements. In its GCF country programme, for the mitigation sector Bangladesh mainly focuses on capacity-building needs for developing bankable projects, which need relevant data and tools for assessing project parameters. Developing strong public–private partnerships, leveraging private sector resources, and access to competitive and inclusive value chain financial products and services are priority areas highlighted by the country.

40. The Climate Facility will support investments in sustainable cooling solutions for the building sector as essential adaptation measures delivered through low-carbon efficient solutions.

41. **El Salvador** is highly exposed to extreme heat, hurricanes, droughts and wildfires. Extreme heat hazard is classified as high throughout the country, which indicates that prolonged exposure to extreme heat, resulting in heat stress, is expected to occur at least once in the next five years.¹¹ El Salvador has experienced a sustained increase in monthly temperatures over the past 30 years (1991–2020). In particular, the summer months (i.e. April to August) have witnessed consistent increases in temperature. According to the future trends, most notably, the number of days where the daily maximum temperature exceeds 35°C will increase by 36 days by the end of the 21st century, and the probability of heat waves will increase by 30 per cent. The annual CDDs are projected to rise by 47 CDD by 2050, and 1,370.68 CDD by the end of the century.¹² El Salvador is situated in the dry corridor of Central America,

⁹ ThinkHazard! website, Bangladesh report. See <https://thinkhazard.org/en/report/23-bangladesh/EH>.

¹⁰ WB Climate Change Knowledge Portal: Bangladesh – Climate Data: Historical and Projections. See <https://climateknowledgeportal.worldbank.org/country/bangladesh>.

¹¹ ThinkHazard! website, El Salvador report. See <https://thinkhazard.org/en/report/75-el-salvador>.

¹² WB Climate Change Knowledge Portal: El Salvador – Climate Data: Historical and Projections. See <https://climateknowledgeportal.worldbank.org/country/el-salvador/climate-data-projections>.

and prolonged dry periods often result in significant economic losses. Droughts put significant stress on the energy system, because they increase energy demand while decreasing the reliability of energy supply.

42. To ensure an effective COVID-19 response and to generate long-lasting resilience and be prepared to handle future health crises, the Government of El Salvador is seeking to strengthen the overall national immunization and related health delivery systems. This includes a variety of issues, including safe and appropriate cold chain and related services. The key challenges are timing and financial means to enable the Government of El Salvador to take advantage of the window of opportunity for revamping its cold chain to ensure it is safe, as well as climate friendly.

43. El Salvador has asked for support for its health sector through a “Covid-19 Emergency Response Project”. The investments in sustainable cooling solutions supported by the Cooling Facility will help not only with demand-side adaptation to extreme temperatures (for space cooling and cold chains), but also with reducing system stress due to hydropower variability, which in turn will help to safeguard the efficacy of vaccines and medicines, as well as essential medical services necessary for the health and well-being of the population.

44. **Malawi** is prone to adverse climate hazards including extreme temperatures, dry spells, seasonal droughts, intense rainfall and flash floods. Extreme heat hazard is classified as medium to high risk throughout the country, which indicates that prolonged exposure to extreme heat resulting in heat stress is likely to occur at least once in the next five years.¹³ The southern region is particularly highly exposed to extreme temperatures and heat stress. Between 1960 and 2006, mean annual temperature increased by 0.9°C. Climate trends over this period show that the total number of hot days in a year has increased by 30 days. By mid-century, annual CDDs are projected to rise by 953, with the number of hot days in a year expected to increase by about 21 days.¹⁴

45. In August 2020, “Malawi Financial Inclusion and Entrepreneurship Scaling” project for agriculture sector, was approved by the World Bank Board. Refrigeration, cold storage and logistics for agriculture and food security are planned to be supported by the Cooling Facility within the framework of this project. Additional funds from the Cooling Facility would help strengthen climate change policy framework of the Government of Malawi to promote the use of low-carbon cooling technologies in other interventions as well as support cold chains for rural farmers.

46. **Somalia** is generally arid and semi-arid with two rainfall seasons. The country’s climate is influenced by the Inter-Tropical Convergence Zone. Between 1960 and 2020 mean annual temperatures have increased by about 0.5°C in the hottest months (April and May) and 0.3°C in the coldest months (January and December). Much of Somalia is at risk of several natural hazards, including extreme heat and droughts. Risk of extreme heat hazard is considered high, with prolonged exposure to extreme heat resulting in heat stress expected to occur at least once in the next five years.¹⁵ The number of days where the daily maximum temperature exceeds 35°C is expected to increase by about 70 days by 2059 compared with 1986–2005, and by 93 days by the end of the 21st century. Annual CDDs are expected to rise by more than 836 between 2040 and 2059, and by 1182 CDDs by the end of the century. This significant increase in temperatures will impact the energy sector because there will be higher power demand for cooling.

47. The Cooling Facility will support financing and implementing solar power systems for space cooling and cold chains in primary- and secondary-level health-care facilities to address

¹³ ThinkHazard! website, Malawi report. See <https://thinkhazard.org/en/report/152-malawi>.

¹⁴ WB Climate Change Knowledge Portal: Malawi – Climate Data: Historical and Projections. See <https://climateknowledgeportal.worldbank.org/country/malawi/climate-data-projections>.

¹⁵ ThinkHazard! website, Somalia report. See <https://thinkhazard.org/en/report/226-somalia/EH>.

the lack of access to electricity in Somalia's health-care facilities. This will also include solar direct-drive refrigerators that can stand up to Somalia's harsh climatic conditions to expand climate-friendly cold storage capacity for vaccines.

48. **Kenya** has included enhancement of energy and resource efficiency as one of the priorities among mitigation options in its nationally determined contribution (NDC). Despite the priority given to adaptation in Kenya's NDC, the country recognizes the importance of clean, efficient and sustainable energy technologies to reduce over-reliance on fossil and non-sustainable biomass fuels. Kenya intends to include non-CO₂ gases such as HFCs and perfluorocarbons in its updated NDC.

49. Off-grid solar access for underserved counties is targeted by the Cooling Facility. Such support would be used for innovative business models to make cold chains available to small farmers and to provide financing to mitigate the high up-front cost of efficient air conditioning and refrigeration systems (solar powered in off-grid contexts) for residential and business applications. In doing so, it will promote sustainable refrigeration and protect against extreme heat in underserved areas of Kenya.

50. In **North Macedonia** most of public buildings (including schools, health-care centres/clinics and universities) are over 50 years old and classified as under-heated, unsafe (due to structural instability and seismic risks) and dilapidated. In fact, buildings, which consume about 39 per cent of the energy, have been identified in the country as a major priority, with estimates of savings ranging from 20 to 40 per cent. The public sector has the greatest potential, with about 35–40 per cent savings, mostly in the health and education sectors. Several challenges need to be addressed for the public sector to invest in sustainable cooling. Central government buildings typically have very limited budgetary provisions for capital improvements, if any, and usually are not allowed to borrow. Municipalities must undergo the fiscal decentralization process and, even then, have stringent debt limitations which many have already reached. Few have formal credit ratings. In the context of the COVID-19 situation, rehabilitation of health-care facilities, with adequate air-conditioning systems, is expected to become a higher priority.

51. North Macedonia is seeking to tackle space cooling in the public sector – and lower its energy consumption and associated GHG emissions – by establishing an energy efficiency fund, a sustainable financing mechanism that is well suited for its circumstances and targeted sector.

52. **Panama's** cooling appliances market has been growing rapidly over the past decade, for both air conditioning and refrigeration technologies. Air conditioning penetration in the residential sector is relatively low but is expected to rapidly increase in line with the country's dynamic economic development (e.g. residential household ownership of air-conditioning equipment stood at 8.9 per cent in 2000, compared with 15.6 per cent in 2010 and is expected to grow to an estimated 66 per cent in 2050). According to the National Energy Plan of Panama (Plan Energético Nacional 2015–2050), energy consumption for cooling systems currently represents about 40 per cent of the overall electricity consumption in Panama. The electricity sector is facing serious challenges in keeping up with rising electricity demand, which is growing at an average annual rate of 8.5 per cent per year mainly due to the increased use of cooling equipment in the residential and commercial sectors. In order to align with Panama's NDC, the pace of energy demand growth needs to be reduced from an average of 6 per cent in the business-as-usual scenario to around 4 per cent, which would reduce the need of energy generation capacity additions by 38 per cent. Several barriers must be addressed to respond to this rapidly growing problem. Although Panama has a sophisticated banking sector, the absence of (1) dedicated credit lines priced to incentivize investments in energy efficiency, and (2) a local long-term capital market for mortgages (access of local banks to capital markets is limited to up to 5-year bonds), as well as (3) limited technical capacity in local banks to assess energy efficiency products and investments, represent a major barrier for energy efficiency in Panama. The higher initial cost of energy-efficient equipment and lack of access to affordable financing

are hindering the adoption of more sustainable cooling solutions and could lock-in large quantities of inefficient, polluting but cheaper (on a first-cost basis) cooling equipment. Law 69-12 mandates the establishment of an energy efficiency fund. The national Fondo de Eficiencia Energética (UREE Fund) aims to be transformational by catalysing private investment, reducing the high risk perception associated with green buildings and energy-efficient cooling appliances, and improving market liquidity for energy efficiency initiatives. Credit line facilities will help transform the residential and commercial sectors and leverage private sector capital.

53. The Cooling Facility will channel GCF funds through the Government of Panama to provide credit lines to the Panama Energy Efficiency Fund and support the enabling environment.

54. **Sao Tome and Principe** is a small low- to middle-income country comprising two main islands. It has a total population of approximately 200,000 people, 42.6 per cent of whom are 14 years of age or younger. In addition to having a small population and a remote location, there is a high fixed cost of public goods – all factors that affect the country's trade, fiscal accounts and human development outcomes. The public health sector is underdeveloped and this is a significant barrier to dealing with COVID-19. International collaboration and support to the Government of Sao Tome and Principe will be crucial to continue mitigation measures to contain the spread of the pandemic in the country and to design and implement its National COVID-19 Immunization Plan.

55. The Cooling Facility will support the health sector services through financing and implementing climate-friendly space cooling and cold chains in the country's health-care network, with the overall objective that every health facility receives a solar power system. Investments will include, for example, solar photovoltaics and battery systems to offset existing backup diesel generation, energy efficiency improvements in health facilities, walk-in cold rooms or cold-storage warehouses, and solar direct-drive refrigerators/freezers or other low-GWP cold chain equipment.

56. **Sri Lanka.** The introduction of demand-side management activities is one of the priority areas of Sri Lanka's NDC. Sri Lanka has recently taken strong initiatives to implement efficient and effective sustainable energy programmes. Promotion of solar power generation has been initiated through different modalities such as solar rooftops and solar parks. This will provide approximately 500 MW in the short term and has the potential to expand to a considerably higher level by 2030. In addition to the interventions concerning power generation, the Government of Sri Lanka has also identified demand-side management as one of the priority areas, and a Presidential Task Force has been appointed for implementing an island-wide demand-side management programme encompassing all the related sectors.

57. Improving energy efficiency in the building sector is the focus of a sample project. The proposed project will finance and promote the implementation of energy-efficient, climate-friendly chillers and other energy efficiency measures in Sri Lanka's commercial and industrial building sector.

58. The independent TAP considers that the needs of the recipient countries are high but are not adequately estimated for most of the participant countries. Limited baseline details are provided for few countries. The information provided above, extracted from the funding proposal and NDCs should be supported by and updated through a baseline study. The independent TAP scored this criterion as high mainly based on the information provided for three highly vulnerable participant countries identified in the assessment report, "Tracking Sustainable Cooling for All".¹⁶

¹⁶ See note 5.

1.5 Country ownership

Scale: Medium

59. All nine countries involved in the programme have provided no-objection letters. In addition:

- (a) All of these countries have submitted their updated NDC;
- (b) All of these countries have energy efficiency as one of the priorities for implementing their NDCs through their climate action plans;
- (c) Bangladesh and Panama are the only countries among these nine that have transparently claimed the cooling sector as priority for energy efficiency measures; and
- (d) Only Bangladesh has finalized and adopted a GCF country programme. Most of the other participant countries are at different stages of application and preparation.

60. **Bangladesh:** The Government has prepared the Energy Efficiency Conservation Master Plan 2015–2030, which describes the adoption and implementation of energy efficiency and conservation regulatory measures. Bangladesh is developing a national Cooling Action Plan. These efforts are aligned with the Cooling Facility’s support of space cooling and energy efficiency measures in buildings.

61. **El Salvador:** The Government is integrating climate-friendly investments in its national programmes, and its NDC includes health as a sector affected by climate change and the promotion of wider use of renewable energy. This is aligned with the Cooling Facility’s support for climate-friendly health facilities and cold chains.

62. **Kenya:** Under the Paris Agreement, Kenya has committed to reducing GHG emissions by 30 per cent, or 142 MtCO₂eq, by 2030. The country’s National Electrification Strategy presents a roadmap for achieving universal access to electricity by 2022. Supporting off-grid cooling appliances and access under the Cooling Facility will contribute to both of these goals. Kenya is also in the process of developing a National Energy Efficiency and Conservation Strategy, which includes targets for energy savings in households, buildings and industry.

63. **Malawi:** Promoting the use of energy-efficient technologies and designs is one of the strategies advocated in the Government’s priority areas under the Malawi Growth and Development Strategy.

64. **North Macedonia:** As a contracting party of the Energy Community Treaty, the Government has committed to reduce energy use by 12 per cent (about 200 kilo tonnes of oil equivalent) by 2018 in its 2020 Energy Efficiency Strategy (based on a 2010 baseline), more than the 9 per cent required under the Treaty’s National Energy Efficiency Action Plans. Buildings, which consume about 39 per cent of energy in the country, have been identified as a major priority, with estimates of energy savings ranging from 20 to 40 per cent.

65. **Panama:** In 2012, the Government made a national commitment to energy efficiency by approving Law 69-12 on “Rational and Efficient Use of Energy”. In 2015, the National Secretariat of Energy published the National Energy Plan 2015–2050, which contemplates the design and implementation of the Rational and Efficient Use of Energy Programme as one of the key actions to meet the sector objectives. The country also ratified the Kigali Amendment in 2017. Panama is currently implementing stage II of its HFC Phase-out Management Plan under the Montreal Protocol. The Cooling Facility will support achieving the Government’s commitments through its support for efficient space cooling.

66. **Sao Tome and Principe:** The country’s NDC includes promoting the greater use of renewable energy and lists health as one of the sectors negatively impacted by climate change. The Cooling Facility will contribute to the Government’s NDC through supporting space cooling and cold chains in the health sector, including through the increased use of solar power.

67. **Somalia:** Somalia's NDC on scaling up the use of solar energy is consistent with the Cooling Facility's support for more climate-friendly health facilities and cold chains, including through greater use of solar power.

68. **Sri Lanka:** Demand-side management is identified as one of the seven mitigation actions of the energy sector NDC of Sri Lanka, to be implemented during 2020–2030. Regarding the regulatory framework, the Energy Efficiency Building Code was first introduced in 2002 and revised in 2008. The code sets standards for energy efficiency in design and retrofits in buildings, while also providing methods for determining compliance.

69. According to the funding proposal, for the purposes of the loans and grants execution, energy efficiency will be selected in the host countries. This approach increases countries' ownership of the programme, although the independent TAP notes that the capacity of the EEs for implementing cooling energy efficiency projects should be regularly assessed and capacity-building support should be provided where gaps are identified.

70. Based on the information provided in the countries' NDCs the independent TAP considers that country ownership at this stage is medium. Almost all participant countries have energy efficiency as a priority, but they are not clearly focusing on cooling needs at the strategy and policy levels. Exceptions among these nine countries are Bangladesh and Panama. A comprehensive baseline study might change this conclusion.

1.6 Efficiency and effectiveness

Scale: Medium to low

71. The share of the GCF total contribution to the total co-financing from the AE and other donors is 1:4.6. For the loan, this proportion is 1:4.5 (USD 280 million is reported as TBD) and for the grant, the ratio is 1:6.3. Allocation of 100 per cent of GCF loan and 51 per cent of grant to component 2 is considered by the independent TAP as effective distribution of the investment by components. The GCF concessional loan fully supports the most risky investment for component 2.

72. As already mentioned in section 1.1, the independent TAP's concern regarding the maximum 30 per cent of GCF proceeds as a cap per country, is effectively balanced by the Secretariat in the term sheet and now all countries have some indicative budget. The independent TAP welcomes this approach, as it considers that all GCF eligible countries issuing no-objection letters should have equal opportunity to participate in the programme. Securing each country's opportunity for equal participation makes the programme more effective from the countries' perspective.

73. As mentioned above, the projects and measures to be implemented through the Cooling Facility are too diverse and therefore it is not possible, or efficient, to design unique eligibility criteria and the associated climate change impact monitoring methodology. Energy efficiency, renewable energy (solar) and GHG-free refrigerators are considered as eligible activities for this Cooling Facility; these systems are quite diverse, with each having specificities that should be taken into consideration when designing measures, financial schemes and monitoring modalities. Technical details are not provided for the attached example projects but the main focus is on financial schemes. These are important for GCF but only if balanced with technical/technological elements and the funding proposal results. Therefore, it is difficult for the independent TAP to confirm that, even in the case of success of the proposed financial models, the mitigation and/or adaptation results will be adequately successful. Such diversity of the scope and technologies of the Cooling Facility as well as of the financial schemes in one programme could have a negative impact on the programme's overall efficiency and effectiveness as well as at the individual country level. Grouping of nine projects (countries) considered in funding proposal is as follow:

- (a) Energy efficiency in the building sector, in some cases appliances are also included: Bangladesh, North Macedonia, Panama and Sri Lanka. Types of buildings are not specified for Bangladesh but appliances are included; only public buildings are the focus for North Macedonia;
- (b) Efficient cooling facilities (HFC-free refrigerators, cold chains, logistics): Kenya (off-grid solar energy and energy access to refrigeration, cold chains and logistics) for cooling purposes, and Malawi (refrigeration, cold chains and logistics in agriculture); and
- (c) Strengthening the health sector within the context of COVID-19 requirements: El Salvador, Sao Tome and Principe, and Somalia.

74. The funding proposal (on page 28) states that nine countries are currently considering potential investments to be supported from the Cooling Facility. The World Bank will work with each EE covered under this Facility to finalize the scope and co-financing of these investment opportunities, taking into account (1) the needs of the clients as they may evolve; (2) the results of the ongoing market assessments; and (3) stakeholder consultations. As mentioned above, in the absence of proper market analysis and stakeholder consultations that includes the needs of the clients and without a final decision on the scope of activities, the independent TAP is not disposed to assess and confirm the efficiency of programme to each participant country.

75. Regarding the efficiency and effectiveness of the whole programme, the independent TAP is of the opinion that, due to the broad diversity of the energy efficiency scope, the programme efficiency is not expected to be high, in comparison with some of the effects in the individual countries asking for support in the health sector or cool chain for food security. Three types of findings could be considered in the context of the programme efficiency:

- (a) Cooling energy efficiency is a part of energy efficiency, but it should be treated separately and not mixed with traditional energy efficiency measures when building sector efficiency is being considered. This is a sector where efficiency measures for heating and cooling do not always coincide; indeed, they could contradict each other depending on the orientation of buildings and types of measures designed for heating energy efficiency. Therefore, to automatically add GCF investment to an Energy Efficiency fund (as it is stated in the implementation modalities and demonstrated in the case of North Macedonia) which is dedicated to the energy efficiency of the building without relevant revisions of the fund's investment eligibility criteria for the cooling sector, the expected effect of the programme could be negative;
- (b) In countries where existing less-efficient appliances (including refrigerators) will be replaced, the old appliances should be monitored and if they stay in second-hand use the increase in energy consumption related to the programme activities should be calculated and deducted from the reduced amount; and
- (c) Although the independent TAP recognizes the urgency of such programmes for highly vulnerable countries that have limited access to cooling facilities under the pressing requirement of cooling due to COVID-19, this programme does not demonstrate any added value compared with individual country-level projects and particularly for highly vulnerable countries with increasing trends of heat indices (CCDs, heat waves, extreme daily temperatures, etc.). The independent TAP considers that in planning a programmatic approach the scope of targeted cooling technologies and geography and type of countries' needs should be narrowed on the basis of baseline studies conducted for each country, to ensure the added value of adopting a programmatic approach.

76. The independent TAP would particularly like to highlight the role of well-designed monitoring processes in reaching high efficiency and effectiveness from a programmatic approach comprising various financial schemes, components and subprojects. The monitoring arrangements provided in annex 11 to the funding proposal are not satisfactory to the independent TAP. The independent TAP would be interested to see more details of monitoring,

particularly for component 2 directly contributing to the final results. The AE should submit a very transparent monitoring plan for component 2 demonstrating how the final beneficiaries (not always borrowers) – consumers of energy – will be monitored in general and in particular when subprojects consider many buildings, many appliances and so on.

77. Based on the available information the independent TAP considers that the efficiency and effectiveness of this programme is expected to be medium to low. The independent TAP would also like to highlight that this programme is really a “financial facility” rather than a programme having added value to individual project. The independent TAP noted that it is receiving more and more facility and subfund type funding proposals based, in most cases, on good concepts and ideas but not supported by relevant feasibility/baseline studies. Such proposals are not fully fit to the programmatic approach and not always relevant to assess against the six investment criteria of the GCF.

II. Overall remarks from the independent Technical Advisory Panel

78. Despite the deficiencies identified by the independent TAP and commented on in the above assessment, the independent TAP fully recognizes the urgent need to regulate the cooling sector, which is not yet relevantly targeted as the most important subsector of the energy efficiency sector in terms of successful implementation of the Paris Agreement and urgent mitigation-related energy needs for COVID-19 vaccination-related cooling processes. The independent TAP recommends this funding proposal for consideration and approval by the GCF Board. Although recommending approval, the independent TAP also fully recognizes that the funding proposal has significant deficiencies which should be addressed and improved prior to the execution of the funded activity agreement. The independent TAP recognizes that the ratings for the six criteria are lower than usual for an endorsed funding proposal. However, the independent TAP considers that, once the condition for endorsement is met, these ratings will go up. Hence, the independent TAP recommends this funding proposal for approval under the following condition:

(a) Conditions to be met prior to the execution of the FAA:

- (i) Delivery by the AE to the GCF, in a form and substance satisfactory to the GCF Secretariat, of the following set of annexes, together with a relevantly revised funding proposal pursuant to those annexes:
 - (1) A cooling sector-related baseline study for each host country, which shall take into consideration (i) all the comments and recommendations expressed by the independent TAP in this assessment report and (ii) the methodology for cooling sector baseline assessment considered as the “guidance” in annex 02a (Pimer for Space Cooling) provided by the AE as part of the original submission;
 - (2) The baseline study for each host country referred to in item (1) above shall estimate, take into account and set out: (i) ongoing trends of cooling degree days and other heat indexes, for the last decades; (ii) correlation between the heat indexes and energy consumption trends for sectors where activities are planned, taking into consideration impact of COVID-19 on energy demand; and (iii) trends of energy consumption by cooling sector and related trends of GHGs where statistics are available;
 - (3) A comprehensive monitoring and verification plan for demonstrating the mitigation of energy consumption and reduction of GHGs from the cooling sectors of the respective host countries compared with the baseline situation. The monitoring methodology shall consider (i) monitoring of potential leakage and related mitigation measures, and (ii)

methodologies for monitoring the mitigation of power consumption at the final beneficiary level; and

- (4) Updated sample projects with descriptions of the baseline situation in the sub-sector specific to the relevant sample project and monitoring methodology specific to such sub-sector.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP177)

Proposal name:	Cooling Facility
Accredited entity:	World Bank
Country/(ies):	Bangladesh, El Salvador, Kenya, Malawi, North Macedonia, Panama, Sao Tome and Principe, Somalia, and Sri Lanka
Project/programme size:	Large

Impact potential

Scale: We consider the rating of “medium to low” (especially “low”) as under-estimating the Facility’s potential impact. The World Bank as AE is presenting an innovative, multi-sector and multi-country (covering 9 countries) programmatic financing facility to help demonstrate how to address the key sustainable development challenge of providing access to cooling in diverse contexts while minimizing negative climate impacts. We would like to highlight iTAP’s observation that the approach to estimating impact in the FP is “conservative”.

- Corrections/clarifications:

- iTAP notes that “only four countries are *considered eligible for adaptation activities*”. We would clarify that each of the 9 countries in the Facility are counted against mitigation-related indicators. While each of the countries covered by the Facility face the need to adapt to a warming world as they are exposed to risks of extreme heat hazard, albeit at different intensity levels, only the 4 countries facing “high risk” of extreme heat hazard (the other 5 countries face “medium risk”) are counted against the Facility’s adaptation indicators.
- In all projects, EE would be the Host country (“the Borrower”) acting through the relevant ministry(ies) and/or designated Project Implementing Entities (PIEs).
- The public procurement and contracting model may be relevant for components 1, 2 and 3. This model will be implemented by the EE, or the PIE as may be designated by the EE, (NOT the AE) and as such, already corresponds with ITAP’s recommendation.
- Rather than representing examples of projects, Annex 22 provides, for each country, a summary description of the actual projects that are being considered under the Cooling Facility, If the Facility is approved by the GCF Board and in line with the notion of a program supporting projects are early development stage, the AE, in consultation with each Host Country, will refine the scope the proposed activities considering the agreed parameters.
- The AE has provided GHG calculations for each country’s project. For each country, “base case” (i.e., baseline) assumptions were provided in Annex 23 (GHG Calculation) of the Funding Proposal.
- We agree that TA to support implementation of credit lines is important. TA is envisaged for every project supported by the Facility.

○ Component 2 can be implemented through all the 3 implementation models (i.e. public procurement, credit lines and energy efficiency financing facility) depending on the country context and the identified barriers. ○ Adaptation to climate change: The AE will report on beneficiaries associated with the 4 projects that count against the Facility's contribution to adaptation efforts. In reality, the impact of the Facility may be greater as the Facility will also generate resilience benefits in countries at medium risk of extreme risk hazard (even if they are not "counted"). ○ El Salvador is a country facing high risk of extreme heat (https://thinkhazard.org/en/report/75-el-salvador/EH). iTAP may be conflating this with the SEforALL's listing of countries that face high risk of lack of access to cooling. ○ Figure 4: structure of a typical project management unit established by the EE (not AE). - We do not share iTAP's assessment that there is a "threat" that a country would not be supported. Rather, we believe that there are provisions and safeguards, including amount of GCF financing required for each country (in Annex 17) and description of activities envisaged per country, that seek to give assurance that none of the proposed countries would be excluded from the program.
Paradigm shift potential
--
Sustainable development potential
--
Needs of the recipient
Clarification: - iTAP refers to "sample projects" giving the impression that the project descriptions are completely hypothetical. In fact, the FP package provide a summary description of the actual projects that are being considered under the Cooling Facility, based on consultations with the Host Countries and in coordination with the National Designated Authorities.
Country ownership



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Efficiency and effectiveness

- As noted above, there should be no doubt that the intention of the AE is that each country listed in the funding proposal (each has provided a NOL) will have access to funding support under the Cooling Facility. This is a fundamental aspect of the Cooling Facility. Annex 17 presents the amount sought for each country and annex 22 describes the proposed supported activities. Because of the varying absorption capacity and needs across countries, the cap of a maximum of 30% of GCF proceeds going to any one country represents an additional measure to ensure that all countries can be supported through the Facility.
- The Cooling Facility aims to address the barriers that have so far limited the deployment of clean cooling technologies and also to put into place the support for a broader paradigm shift by working across key cooling target areas and with a range of countries from different parts of the developing world to amplify the potential to leverage the experiences and learnings. The Facility seeks to scale and mainstream sustainable cooling applications in a number of key sectors (agriculture, health, buildings), thereby accelerating the potential for across-the-board deployment of climate-friendly solutions.
- The Facility adopts a pragmatic approach to deal with the nature of cooling and take advantage of diverse windows of opportunity appropriate in each country.
- iTAP expressed doubts regarding the “added value” of the Facility approach for cooling (compared to individual country project). We would note that were it not for the opportunity to aggregate cooling projects under one Facility, especially the smaller countries would not be in a position to access critical GCF financial support for their respective cooling projects. Moreover, we would note that the Facility approach offers the opportunity to provide pioneering examples of countries, cooling sectors, approaches, technologies and measures that can be scaled up including in other countries, beyond the Facility, that are facing similar situations and opportunities.
- iTAP comments that “Such diversity [...] in one programme could have a negative impact on the programme’s overall efficiency and effectiveness as well as at the individual country level”. We would respectfully disagree. The alternative to the support provided by the Cooling Facility, may be missed opportunities to enhance access to climate-friendly cooling. As just noted a learning-by-doing facility, it also offers up an opportunity to test a range of technological applications, business models and implementation approaches for emulation, replication and scale-up globally.

Overall remarks from the independent Technical Advisory Panel:

We thank iTAP for its assessment and its recognition of the urgent need to support developing countries’ move towards to climate-friendly cooling. We appreciate iTAP’s favourable recommendation for the consideration and approval of the Cooling Facility’s funding proposal by the GCF Board.

We hope that the clarifications provided above help better understand the Facility, its functioning, and its aims.

We have also communicated our concern with iTAP's recommended conditions and shared our suggested edits.

In addition to other points we raised and as we clarified in writing during ITAP's assessment period, the subsequent call with ITAP (8/18/2021), and our comments to the iTAP's assessment, each Project must not be assimilated to a national programme. The proposed Projects under the Cooling Facility support and contribute to the achievements of the Host Countries' national programs but the scope of these Projects is more narrowly defined. In this respect, our main concerns are related to iTAPs' requests for (i) the production of additional national/sectoral baseline studies and monitoring plans that go beyond the scope of individual Projects; (ii) the overly onerous cost (and challenge) associated with collecting yet more data, conducting the assessments, and producing additional documents to meet all the conditions – representing an even higher transaction cost associated with working with the GCF without commensurate benefits; and in particular (iii) the overly tight timing (i.e., “prior to the execution of the FAA”) to conduct the additional studies and plans for each of the 9 countries that are considered under the Cooling Facility, without giving any consideration to challenges and delays involved in collecting the requested information, especially during the current pandemic, and despite the fact that the Cooling Facility includes three COVID-19 emergency projects, including one in a fragile and conflict affected situation.

We regret to note that our comments have not been reflected in the final version of iTAP's assessment and in the conditions iTAP recommends that they are met prior to the execution of the FAA.

COOLING FACILITY

ANNEX 8

GENDER ASSESSMENT AND ACTION PLAN

August 2021

Disclaimer: The gender profiles were prepared, using publicly available sources. The World Bank does not guarantee the accuracy or completeness of the information provided in the present document. To the extent that it reflects opinions, they are the opinions of the individual authors and do not necessarily reflect the views of the World Bank, its staff or Directors. Nothing in this summary should be taken as legal advice.

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1 Introduction

The Cooling Facility is proposed as an innovative, multi-sector and multi-country programmatic financing mechanism with the goal of transforming the cooling sector. The program will increase energy efficiency, reduce or avoid GHG emissions, and accelerate access to cooling for economic growth and human capital development.

To achieve this, the Facility will channel concessional climate finance from the GCF to co-finance IBRD and IDA cooling-related operations that: (i) support investments in efficient, affordable, and sustainable cooling; (ii) catalyze technology and business model innovations across cooling value chains and sectors to reduce barriers for private sector investment; and (iii) help build an enabling environment for sustained, long-term systemic changes for example, for productive uses of energy, in buildings energy efficiency, nutrition, health, and agriculture; as well as (iv) support the integration of sustainable cooling into climate change and SDG plans

The present Gender Assessment and Action Plan provides an overview of issues affecting men and women related to the Facility's implementation in the nine countries: Bangladesh, El Salvador, Kenya, Malawi, North Macedonia, Sao Tome e Principe, Somalia, Panama and Sri Lanka. It details the strategic approach that will be adopted under the Cooling Facility to effectively address these challenges and contribute to promoting female development at different dimensions, for instance through employment in the cooling-related sectors, while enhancing women and men's livelihoods in affected communities, and supporting COVID-19 responses. Planned interventions under the present Gender Action Plan (GAP) are aligned with the GCF third policy objective to "contribute to reducing the gender gap of climate change-exacerbated social, economic, environmental vulnerabilities and exclusions through GCF climate investments that mainstream gender equality issues."

The Cooling Facility's gender activities will focus on strengthening women's roles as employees, entrepreneurs and consumer in energy and cooling-related sectors. The work under the Facility will tangibly assist relevant stakeholders (such as ministries, utilities and businesses) across various sectors such as energy access, energy efficiency and renewable energy, agriculture, health etc. to support the designing and implementation of intervention in cooling-informed operations on the ground. The Facility will work with countries to collect data and generate crucial knowledge of actions and design interventions needed to help close gender gaps in areas that are aligned with the cooling-related sectors and directly support the World Bank Group 2016-2023 Gender Strategy and its four pillars of action¹. The broad experiences and track-record of the ESMAP Gender and Energy Program – which is funding 6 regional energy programs consisting of activities that directly support the 4 pillars of the Strategy - combined with the ESMAP Efficient Clean Cooling Program – which is providing TA support to cooling-informed activities across sectors (working with relevant World Bank Global Practices) - will be leveraged to support activities aimed at helping achieve progress on gender equality through the various cooling projects included in the Facility. The Bank's ecosystem of gender focal points for relevant Global Practices will need to be engaged.

Appropriate gender entry points for the Facility's cooling-informed operations will be case specific, span over a number of sectors (e.g. agriculture, buildings, health, energy, etc), and different geographies (both in terms of countries and in terms of urban versus rural contexts). The Facility's activities will also offer opportunities to tackle the World Bank Group 2016-2023 Gender Strategy and its four pillars of action. For example, ensuring key public services such as clinics or hospitals have access to cooling (in medical facilities and for

¹ The WBG Gender Strategy has 4 objectives: i) improving Human Endowments ii) Removing Constraints for More and Better jobs, iii) Removing Barriers to Women's Ownership and Control of Assets and iv) Enhancing Women's Voice and Agency and Engaging Men and Boys

storage of vaccines and medicine) to reduce infant mortality rates (pillar 1 of the WBG Gender Strategy); and tackling barriers to women's leadership and participation in the e.g. energy and agriculture sector through implementing interventions that e.g. address child care service gaps and increase girls and young women's exposure to jobs in the energy efficiency/cooling sectors (pillar 2). Other areas of support may include, applying new approaches with banks and micro-finance institutions to design financial products and services that meet the needs of female enterprises and consumers in cooling technologies (pillar 3); and enhancing women's voice through inclusion in policy development and participation in project design, including engaging men and boys in gender-based violence (GBV) prevention and response in energy project sites (pillar 4). In addition, the Facility, through the projects it supports, is expected to contribute to meeting the World Bank corporate gender target, measured through the "Gender Tag"². Also, some of the interventions will help countries address the impacts of the COVID-19 global pandemic.

In addition, as described in Annex 6 (Environment and Social Sustainability Framework), environmental and social risks and impact of all projects included in the Cooling Facility will be assessed and managed in accordance with the Bank's Environmental and Social Framework (ESF). At the project-level, provisions on gender in the ESF are focused on gender equality and inclusion, particularly in the context of addressing disadvantaged or vulnerable groups, including the risks of gender-based violence. As per Bank policy, each project also needs to set-up a Grievance Redress Mechanism to ensure any grievances, including regarding sexual exploitation and abuse, are reported and addressed.

The Facility's gender approach will be adapted to each projects' specific design, local barriers and circumstances, but in general the entry points for gender are expected to involve one of more of the following: baseline establishments and data collection; diagnostics barriers; stakeholder engagement; identification of specific actions that are contributing to closing gender gaps; knowledge management (see below). The Cooling Facility will seek to contribute to enhancing the knowledge base on gender issues associated with access to cooling and will seek to contribute to reducing gender gaps in cooling-related sectors (including agriculture, buildings, health and energy) by (i) identifying gaps relevant to the four pillars of the World Bank Group (WBG) Gender Strategy, and (ii) aiming to address these gaps through specific actions supported by the project.

2 Gender Gaps that Matter

While more data and analysis are needed to better understand the gender aspect of access to sustainable cooling and energy efficiency, gender risks are evident (SEforALL 2019). Women and girls are often disproportionately impacted by limited access to sustainable cooling and enhanced access could help make gains in terms of entrepreneurship opportunities and enhanced livelihoods. There is very little evidence that shows what works to close gender gaps in interventions that are dedicated to cooling, however this proposal will extrapolate approaches that are proved to work from other sectors in particular in regards to identify entry points (after a gender assessment).

Gaps that may impact projects under the Facility are outlined briefly below:

- **Jobs and Skills Development**
- **Entrepreneurship**

² Since 2017, the WBG strengthened its monitoring system with regards to gender, focusing on results and outcomes. A "gender tag" was launched to identify projects that effectively promote equality between men and women and are aligned with the WBG gender strategy.

- **Consumer Engagement**
- **Financial services**
- **COVID-19 response and health measures**
- **Data and information**

Alignment with the WBG's Gender Strategy and achieving gender tag: the Bank implements systematically a mechanism to identify operations that aim to close gender gaps in human endowments, more and better jobs, ownership and control of assets, and women's voice and agency. This mechanism is called *Gender Tag*. For a WB lending project to be "gender tagged", it must meet two criteria: (1) explicitly address one or more of the dimensions of the Gender Strategy, and (2) the include a vertical logic with (i) an analysis of the specific gender issues; (ii) actions that seek to address the issues; and (iii) indicators that show the advancement towards closing the gender gaps.

3 Gender Gap Assessment

3.1 Overarching Gender Gaps

When designing cooling projects and programs with a gender equality focus, there are many opportunities to consider issues around access to finance, entrepreneurship, employment and consumer engagement. These may include, for example, working toward government targets for closing gaps between men and women that might be addressed through increasing access to climate-friendly cooling; or including sex-disaggregated information in data collection or survey processes; or making sure that both men and women take part in related consultation processes.

Growing evidence points to significant differences in the needs and wants of women versus men as consumers and entrepreneurs. For example, differences can be observed in how they use energy on a daily basis, how much they are willing and able to adopt cooling technology and measures e.g. in agriculture sector, and how they respond to campaigns marketing products and services. It is therefore essential to tailor any efforts toward consumer-specific behaviors of men and women. For example, women may have lower literacy rates or receive communication through more informal channels than men to e.g. word of mouth versus town meetings etc. This points to a need for communications strategies in these countries to look beyond written materials as they transmit information related to benefits of cooling, consumption patterns, and so forth.

The agriculture sector in a number of developing countries employs a larger proportion of economically active women than is the case for men. Despite their critical role in ensuring food security, women (in particular rural) still face constraints to enter fully and in the same conditions as men in the agriculture sector. Women, as both producers and consumers, have less access to productive resources, technologies (for harvesting and preservation), and markets. These barriers inhibit their potential across the sector, and also limit their economic empowerment and decrease their health, resulting in significant gender gaps that trickle down to families and communities.

Other gender gaps, such as in access to finance or affordability constraints (e.g., women have equal access to financial services in only 60% of the world's countries³), may make it relatively difficult for female-headed households or female business owners or farmers to adopt energy-efficient appliances. To reach policy objectives, it is important to consider these dynamics in the design of relevant programs.

Men and women tend to have different metabolisms, and thus experience air temperature differently, with ramifications for the world's increasing use of air conditioning. In a field study of thermal comfort in offices in India, females were found to accept a wider range of temperatures than males, and require less mechanical cooling (Indraganti, Ooka, and Rijal 2014). Beyond India, a 2019 study conducted in Germany, found that women performed better on certain cognitive tests at higher temperatures than at lower ones, and that overcooled offices had a negative impact on women's productivity (Chang and Kajackaite 2019).

Research has demonstrated that men and women respond differently to information and marketing campaigns. Surveys across countries and different economic conditions show that men tend to display higher levels of awareness and knowledge of environmental issues relative to women, though women exhibit a more positive attitude and behavior toward saving energy than men, suggesting that women are more willing to practice energy-saving behaviors (Huang and Chao 2017). It is likely that there are opportunities to improve the outcomes of efficient cooling programs through information campaigns that are shaped by a better understanding these differences, and through the targeting of approaches to achieve desired results.

An "energy clinic" conducted by the Energy Management Centre in Kerala, India, demonstrated that women can be key agents of change in utilizing more efficient appliances. Women organized demonstration classes on energy conservation in their own villages, and attracted large numbers of attendees, including housewives, teachers, and local administrators (Unnithan 2018). Some energy companies e.g. Tata Power Delhi Distribution Ltd. has documented similar experiences by successfully appointing women as "brand ambassadors" to raise awareness about the benefits of legal connections, help facilitate new connections, and educate consumers on consumption patterns and bill payments, with each brand ambassador covering an area of about 300 households (Shrivastava 2017; World Bank 2017).

Other opportunities to tackle gender equality issues include looking at how cooling institutions can tackle gender gaps apparent in labor force participation rates and during women's recruitment, retention, and promotion. Issues to be addressed include legal barriers to women entering an industry or advancing to certain roles; limited transport and insufficient workplace infrastructure; inadequate institutional sexual harassment policies and reporting mechanisms; a regressive organizational culture in male-dominated fields; gender pay gaps resulting from occupational segregation and unequal pay for equal work; and a lack of care services tailored to women. Such issues are being tackled by India's Energy Efficiency Services Ltd. (EESL) with support from the World Bank's India Energy Efficiency Scale-up Program (P162849) and the ESMAP-funded Women in Power Sector Network in South Asia (WePOWER).

School-based education programs promoting energy efficiency can be an effective way to drive behavior change among households, as children can learn to integrate energy and sustainability in their own habits. They can also be an important entry point to engage young women in high schools to increase their interest in the science and engineering fields. An example is the Girls of Energy initiative conducted by the U.S. Department of Energy's Office

³ The *Global Gender Gap Report*, published by the World Economic Forum, benchmarks 144 countries in their progress toward gender parity across four thematic dimensions: economic participation and opportunity, educational attainment, health and survival, and political empowerment. It has been published annually since 2007.

of Economic Impact and Diversity, whose activities include creating opportunities for more girls to explore the research and technology being developed at the department's National Laboratories. The program also covers energy efficiency, among other topics, in an interactive e-learning book.⁴ It is likely that there are further opportunities to include energy efficiency in curricula focused on promoting women's participation in research and technological innovation.

An enabling environment is crucial to improve women's empowerment and reduce gender gaps. An enabling environment is critical to improve women's empowerment and reduce gender gaps. Despite the fact that many countries are improving gender disparities, such as the inclusion of more women in the labor forces, those efforts are not being translated into decent jobs as women are still overrepresented in low productivity and informal jobs. A lot of the constraints women face are a result of behavioral and social norms in each country: labor segregation, time poverty, violence and gender roles. The Cooling Facility will support fostering the participation of women by integrating gender-specific activities to be carried out.

A robust legal framework and regulations protecting women rights and their access to services are also key for the implementation of activities that aim to close gender gaps. The following section includes a summary of laws and regulations for each country based on the publication *Women, Business and the Law's* (World Bank, 2021)⁵ assessment. This assessment is an annual study that measures the laws and regulations that affect women's economic opportunities. For instance, the COVID-19 has affected women's social and economic capabilities, and the gap between men and women widened due to its impacts. The importance of legal equality is not limited to disaster preparedness or mitigation of the effects, but it is associated with better development outcomes such as lower rates of vulnerable employment and extreme poverty among female workers.

The gender activities will be developed with each Executing Entity to support local institutions having the tools and skills to achieve the proposed objectives. Activities will be in line with each country's respective policy and regulatory context and promote the implementation of assessments to identify the main barriers and constraints that inhibit gender equality. It is important to create awareness among the institutions' personnel to implement all activities effectively. A gender action plan per country is not proposed, but rather the integration of gender consideration in key activities, including ensuring that any data collected will be gender-disaggregated and, when relevant, that vaccine and health outreach campaigns also be targeted at women. In all cases, the EE and the Bank should make sure that consultations consider all relevant stakeholders and include their concerns and suggestions. Finally, during the implementation, tools for monitoring and evaluation should be in place to enable capturing progress, make adjustments as needed and learn from the implementation of gender activities.

Linkages to climate change: gender inequality is one of the inhibitors for development, and the impacts of climate change perpetuate it. As previously mentioned, women have less economic, political, and legal power than men resulting in a more considerable inability to cope with the adverse effects of climate change. The differences according to vulnerabilities such

⁴ For more information on the U.S. Department of Energy's Girls of Energy initiative, see <https://www.energy.gov/diversity/girls-energy>; for the e-book "Girls of Energy," see https://www.energy.gov/sites/prod/files/2016/10/f33/GOE_eBook.pdf.

⁵ "Women, Business and the Law 2021 presents an index covering 190 economies and structured around the life cycle of a working woman. In total, 35 questions are scored across the eight indicators. Overall scores are then calculated by taking the average of each indicator, with 100 representing the highest possible score. Data refer to the laws and regulations that are applicable to the main business city" (World Bank. 2021. *Women, Business and the Law 2021*. Washington, DC: World Bank. doi:10.1596/978-1-4648-1652-9. License: Creative Commons Attribution CC BY 3.0 IGO)

as poverty, jobs' quality, quality of housing, among others, make women more susceptible to climate change, causing them forcibly being displaced, stop working in productive activities, and spend more time in non-paid tasks. Women's empowerment is key to addressing climate change because women play a significant role in natural resources management (e.g., for productive uses such as in farming or cooking activities such as fuel collection). Where pertinent, the Facility will support women in improving their farming productivity (such as offering financial services that allow women to be more efficient, access better technology, and reduce food waste), provide sustainable cooling solutions for buildings that provide services, employment or housing to women.

3.2 General Country-Specific Gender Gaps

A preliminary gender gap assessment was conducted across the countries to be supported by the Facility. An overview of gender gaps identified is outlined below, along with the preliminary identification of potential gender entry points for intervention (at the project level), which will be explored during project design:

3.2.1 Bangladesh

Bangladesh obtained a score of 0.726 in the 2020 WEF Global Gender Report Index, which puts the country in the 50th position. This suggests a substantial improvement in comparison to the 2006 edition of the report, where the country obtained a score of 0.627 and ranked 91st. The country has the highest score in its region, South Asia.

The Constitution of Bangladesh states that women and men have equal rights in all spheres of public life. However, the same legal framework recognizes religious personal laws, which position women in an unequal situation and worse than men. The country has been investing and working in gender equality issues and regulations, including designing, and implementing the National Women Development Policy led by the Ministry of Women and Children Affairs (created in 1972 and acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

- **Legal framework:** Bangladesh has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication Women, Business and the Law 2021 (World Bank, 2021) indicates that Bangladesh scores 49.4 out of 100. It means the country is lower than the regional average achieved in South Asia (63.7). **Employment and payment:** The country does not have a law that restricts women to work in the same fields as men (and can work at night in the same way as men). Also, women at workplaces are protected by a legislation that prevents sexual harassment in employment (Supreme Court of Bangladesh, BNWLA v. Government of Bangladesh, 14 May 2009, Petition No. 5916 of 2008). Women are not able to work in some industrial jobs in the same way as men (including mining, water, and factoring (Labor Act, Secs. 42 and 87)), but some sectors do not have restrictions for women (construction, agriculture, energy, transportation, and others). Women are also prevented from working in jobs that are considered hazardous. In regards of payment, Bangladeshi women does not have a law that mandates equal remuneration for equal value.
- **Maternity Rights:** laws regarding maternity rights are broadly absent. The country lacks parental leave and fathers do not have the option or right to enjoy a paid leave. Also, the government does not administer 100% of the maternity leave benefits (Labor Act, Sec. 47).

- **Assets:** the Muslim Personal Law (Shariat) Application Act, Art. 2 concedes women to have equal ownership rights to immovable property and grants spouses equal administrative authority over assets during marriage. However, the same law states that sons and daughters do not have equal rights to inherit assets from their parents; and widowed women also do not have the same rights to inherit their spouses assets as men.
- **Access to credit:** Bangladesh does not have a law that prohibits discrimination in access to credit based on gender.
- **Marriage:** women cannot obtain a judgment of divorce in the same way as men and also women do not have the same rights to remarry as men.

Income. According to the UNDP's estimates, the female Gross National Income (GNI) per capita in Bangladesh amounts to 2,373 in 2011 PPP \$, which corresponds to 42% of the male GNI per capita (5,700 2011 PPP \$).

Table [1]. Female and male gross national income per capita (2011 ppp terms)

Country	Estimated gross national income per capita (2011 ppp terms) ¹		
	Female	Male	Female as % of male
Bangladesh	2,373	5,700	42

Source: UNDP <http://hdr.undp.org/en/indicators/123506#a>. 1/ Definition: Derived from the ratio of female to male wages, female and male shares of economically active population and gross national income (in 2011 purchasing power parity terms).

Labor Markets. Labor participation in Bangladesh is much lower for women than men; From the whole labor force, women represent only 30.7%. The share is even lower when looking at the share of female employees in the industry (25.3%), manufacturing (32.7%), electricity gas, steam, and air conditioning, supply (8.5%) or professional, scientific and technical activities 8.9%. These gaps are partly explained by labor regulations in the country that forbid women from participating in certain sectors of the economy (Table [2]).

Table [2]. Share of employment by activities and sex (% , 2018 unless stated differently)

Country	Sex	Industry	Manufacturing	Electricity; gas, steam and air conditioning supply	Professional, scientific and technical activities	Total	Women are able to work in the same industries as men
Bangladesh	Female	25.3	32.7	8.5	8.9	30.7	No
	Male	74.7	67.3	91.5	91.1	69.3	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Financial inclusion. Men in Bangladesh present higher levels of financial inclusion than women when comparing the ownership of an account at a financial institution (50 versus 20%), a mobile account (33 versus 10%) or saving money (29 versus 26%) (Table [3]). In addition, the share of women aged 15 or above in the country with either an account at financial institution or a mobile money-service provider is 36%, well below the average in developing countries (58.3%), or South Asia (65.1%).

Table [3]. Indicators of financial inclusion by sex (2017)

Country	Account at a financial institution (% age 15+)		Mobile account (% age 15+)		Saved any money last year (% age 15+)	
	Female	Male	Female	Male	Female	Male
Bangladesh	20%	50%	10%	33%	26%	29%

Source: World Bank Little Book of Gender 2019

Education. Women in Bangladesh are more likely to be enrolled in secondary education than men (68.9 of women versus 58.8% of men). However, this gap is reverted when looking at tertiary education (17 versus 24%). Consequently, men report a higher average number of schooling years (6.8 versus 5.3). In addition, women amount to only one fifth of STEM graduates (Table [4]).

Table [4]. Indicators of education by sex

Country	Sex	Enrolment in primary education (%)	Enrolment in secondary education (%)	Enrolment in tertiary education (%)	Mean years of schooling (years)	Female share of tertiary graduates from STEM (%)
Bangladesh	Female	91.5	68.9	17	5.3	20
	Male	93.7	58.8	24	6.8	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilstat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Health. The maternal mortality ratio in Bangladesh is 176 per 100,000 live births, which ranks 128th among 182 countries included in the Human Development Report. This figure is below the average in developing countries (231) and in line with its region, South Asia (176). However, more than 10 times the maternal mortality ratio in OCDE (14) or Very high human development countries (15) according to the UNDP Human Development Report. In addition, female life expectancy at birth in Bangladesh is almost 4 years higher than the same figure for males (74.3 versus 70.6 years) (Table [5]).

Table [5]. Indicators of health by sex

Country	Maternal mortality ratio (deaths per 100,000 live births)	Maternal mortality ratio (ranking across countries)	Female life expectancy at birth (years)	Male life expectancy at birth (years)
Bangladesh	176	128	74.3	70.6

Source: UNDP. <http://hdr.undp.org/en/data>

Note: rankings among 182 countries with data in UNDP report

Agency. Only one out of five members in Bangladesh's parliament are women (20.6%). The share of women in ministerial positions is even lower, below one in 10 (8%).⁶

Gender Based Violence. According to the United Nations Development Report, more than half of Bangladeshi women aged 15 or older have experience gender-based violence by their partners.⁷

Gender gaps associated with the current project in Bangladesh

Cooling-related activities or sectors targeted: efficient use of energy and resources in buildings and appliances.

Preliminary overview of relevant gender gaps:

- **Women are underrepresented in the energy sector.** A recent study by the World Bank found that in 13 surveyed power sector organizations, only 16 out of 276 (6%) total board members are female. Bangladesh Power Development Board (BPDB) has the highest representation of women with 1,500 out of 16,040 total staff (9%) and 217 out of 2,455 technical staff (9%). Data on the energy efficiency sector are not readily available but are assumed to roughly align with the labor participation observed in the power sector.
- **Women in Bangladesh usually do not have adequate information on technologies available and access to financing or collateral.** The gendered power relations in the household can influence decision-making, behavior change, and/or adoption of improved energy services like implementing renewable technology and/or energy-efficient appliances.
- **The private sector plays an important role to foster women's economic empowerment.** It may thus be important to partner with national and local nongovernment organizations (NGOs) and private companies to implement pilot livelihood programs. Activities that aim to train on use of labor and energy-saving appliances, and to offer financial and business management, credit, market access, marketing strategies.
- **Women face constraints in decision-making at home.** Both women and men would be benefited from efficiency improvements in energy use and buildings and appliances. However, women are less likely to be legal owners or occupiers of their homes, and housing titles tend to be registered in the name of male household members. This limits agency in terms of decision making and also ability to gain access to finance due to collateral constraints for the purchase of energy technologies.

Suggested gender entry points and activities include: (i) develop (or review/enhance/) gender assessment; (ii) carry out consultations, workshops, gender assessment's presentations; (iii) engage policy dialogues to raise awareness about the linkages between cooling and gender; (iv) capacity building/awareness about gender equality; (v) develop gender action plan that includes responsibilities, activities and budgets; (vi) develop and implement a communication/outreach campaign.-

3.2.2 El Salvador

El Salvador obtained a score of 0.706 in the 2020 WEF Global Gender Report Index, which puts the country in the 80th position. This suggests a substantial improvement in comparison to the 2006 edition of the report, where the country obtained a score of 0.684 and ranked 39th. The country has the sixth lowest score in its region, Latin America and the Caribbean.

The Constitution of El Salvador states that women and men have equal rights at all levels. The country has been investing and working in gender equality issues and regulations, including designing, and implementing the National Policy for Women led by The Salvadoran Institute for the Development of Women (created in 1996 and acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

Legal framework: El Salvador has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication *Women, Business and the Law 2021* (World Bank, 2021) indicates that El Salvador scores 88.8 out of 100. It means the country is higher than the regional average achieved in Latin America (80.1).

- **Employment and payment:** The country does not have a law that restricts women to work in the same fields as men (and can work at night in the same way as men). Also, women at workplaces are protected by a legislation that prevents sexual harassment in employment (Código Penal, Art. 165; Ley General de Prevención de Riesgos en Lugares de Trabajo, Art. 7) and sexual harassment at work is considered a crime (Código Penal, Art. 165). A Salvadorian law prohibits discrimination in employment based on gender (Código del Trabajo, Art. 30(12); Código Penal, Art. 246). Women are able to work in all industrial jobs in the same way as men (including mining, water and factoring, construction, agriculture, energy, transportation, and others). Women are not prevented from working in jobs that are considered hazardous. In regards of payment, Salvadorian women do not have a law that mandates equal remuneration for equal value.
- **Maternity Rights:** Mothers in El Salvador have 112 days of paid time during a period between pregnancy and after labor (Código del Trabajo, Art. 309). The country lacks parental leave, but fathers enjoy a 3-day paid leave after labor. Also, the government administers 100% of the maternity leave benefits (Ley del Seguro Social, Arts. 1 y 12). It is also prohibited the dismissal of pregnant worker (Código del Trabajo, Art. 113).
- **Assets:** the Código de Familia, Art. 70 concedes women to have equal ownership rights to immovable property and grants spouses equal administrative authority over assets during marriage. Código Civil, Arts. 983 y 988(1) states that sons and daughters have equal rights to inherit assets from their parents; and widowed women also have the same rights to inherit their spouses assets as men.
- **Access to credit and entrepreneurship:** El Salvador does not have a law that prohibits discrimination in access to credit based on gender. Women also have no restrictions when registering a new business, open a bank account or signing a contract.
- **Marriage:** El Salvador does not have a legal provision that requires a married woman to obey her husband (Código de Familia, Art. 36(1)); women can be called the head of household in the same way as men. The country has specific legislation for addressing domestic violence (Ley Contra la Violencia Intrafamiliar; Ley Especial Integral para una Vida Libre de Violencia para las Mujeres, Art. 8(k)); and women can obtain a judgment of divorce in the same way as men, but they do not have the same rights to remarry as men (Código de Familia, Arts. 17, 23(5) y 115(1)).

Income. According to the UNDP's estimates, the female Gross National Income (GNI) per capita in El Salvador amounts to 6,471 in 2017 PPP \$, which corresponds to 62% of the male GNI per capita (10,501 2017 PPP \$).

Table [6]. Female and male gross national income per capita (2017 ppp terms)

Country	Estimated gross national income per capita (2011 ppp terms) ¹
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	Female	Male	Female as % of male
El Salvador	6,471	10,501	62

Source: UNDP <http://hdr.undp.org/en/indicators/123506#a>. 1/ Definition: Derived from the ratio of female to male wages, female and male shares of economically active population and gross national income (in 2011 purchasing power parity terms).

Labor Markets. Labor participation in El Salvador is much lower for women than men; women represent about the 60% of the labor force. The WEF (2019) indicates that women's rate in the labor force participation is 50% while men's is 83%. Women are less represented as professionals and technical workers (47.6% compared to 52.4% of men). Women are overrepresented in sectors such as services (commerce, hotels, restaurants – 45.5% or women vs. 17.8% of men), manufactures (17.5% vs 14.3% of men), domestic paid work (10.9% vs 0.8% of men) and education (4.3% vs. 1.6% of men) (Table [7]).

Women tend to be included more often in temporary jobs as only 37.3% of them are considered full time formal workers, a number much lower than the 45.1% registered for men.

Table [7]. Share of employment by activities and sex (% , 2017 unless stated differently)

Country	Sex	Services	Manufacturing	Women are able to work in the same industries as men
El Salvador	Female	45.4	17.5	yes
	Male	21.4	14.3	

Source: labor for shares from MTPS <https://www.mtps.gob.sv/wp-content/uploads/descargas/BoletinesEstadisticos/mtps-boletin-laboral-mujeres-2019.pdf> and labor restrictions from World Bank Little Book of Gender 2019

Financial inclusion. Men in El Salvador present higher levels of financial inclusion than women when comparing the ownership of an account at a financial institution (36 versus 24%), a mobile account (2.2 versus 5.2%) or saving money (36 versus 40%) (Table [8]). In addition, the share of women aged 15 or above in the country with either an account at financial institution or a mobile money-service provider is 24%, well below the average in developing countries (58.3%), or Latin America and the Caribbean (51%).

Table [8]. Indicators of financial inclusion by sex (2017)

Country	Account at a financial institution (% age 15+)		Mobile account (% age 15+)		Saved any money last year (% age 15+)	
	Female	Male	Female	Male	Female	Male
El Salvador	24%	36%	2.2%	5.2%	36%	40%

Source: World Bank Little Book of Gender 2019

Education. Women in El Salvador are more likely to be enrolled in secondary education than men at all level (80.6% vs. 79.9% for primary education; 60.8% vs. 59.5% for secondary education; 30.2% vs. 27.0% for tertiary education, for women and men respectively). However, women present a lower literacy rate than men (86.7% vs. 90.6%). In addition, women amount to only one fourth of STEM graduates (Table [9]).

Table [9]. Indicators of education by sex

Country	Sex	Enrolment in primary education (%)	Enrolment in secondary education (%)	Enrolment in tertiary education (%)	Female share of tertiary graduates from STEM (%)

El Salvador	Female	80.6	60.8	30.2	24
	Male	79.9	59.5	27	

Source: enrolment from WEF http://www3.weforum.org/docs/WEF_GGGR_2020.pdf and STEM graduates from World Bank Little Book of Gender 2019

Health. The maternal mortality ratio in El Salvador is 46 per 100,000 live births, which ranks 91st among 182 countries included in the Human Development Report. This figure is below the average in developing countries (231) and below with its region, Latin America and the Caribbean (72.7). However, more than 3 times the maternal mortality ratio in OCDE (14) or Very high human development countries (15) according to the UNDP Human Development Report. In addition, female life expectancy at birth in El Salvador is more than 9 years higher than the same figure for males (77.8 versus 68.5 years) (Table [10]).

Table [10]. Indicators of health by sex

Country	Maternal mortality ratio (deaths per 100,000 live births)	Maternal mortality ratio (ranking across countries)	Female life expectancy at birth (years)	Male life expectancy at birth (years)
El Salvador	46	91	77.8	68.5

Source: UNDP. <http://hdr.undp.org/en/data>

Note: rankings among 182 countries with data in UNDP report

Agency. Only 31% members in Salvadorian's parliament are women. The share of women in ministerial positions is slightly higher (33.3%).⁸

Gender Based Violence. According to the United Nations Development Report, 14.3% of women aged 15 and older have experience intimate partner violence⁹. Other sources¹⁰ suggests that 24.2% of women have suffered of physical partner violence at some point of their lives and 11.5% have experienced sexual violence by a partner at some point. Also, El Salvador have one of the highest femicide rates in region, per year 3.3¹¹ per 100.000 die by gender-based violence.

Gender gaps associated with the current project in El Salvador

Cooling-related activities or sectors targeted: health-oriented projects

Preliminary overview of relevant gender gaps:

- **Women in El Salvador face some constraints when accessing maternal health services:** At early stages of pregnancy or after giving birth, women do not visit health professionals. Complications and risks are not being identified on time resulting in the reduction of women's health or deaths.¹² Also, global data show that about 30% of women who are pregnant are likely to be infected with SARS-CoV-2. Most of women who test positive for COVID-19, will not develop serious complications, however they can infect their newborns.¹³ Health authorities recommend that pregnant women to receive the COVID-19 vaccine.
- **Women live longer than men:** women live longer than men, also they represent a largest proportion of vulnerable population over the age of 65. While women over the age of 65 add to 4.64%, men only account for 3.37%.

⁸ Source: WEF Global Gender Report 2020. http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

⁹ Source: UNDP. <http://hdr.undp.org/en/data>

¹⁰ Source: PAHO <https://iris.paho.org/handle/10665.2/3471>

¹¹ Source: ECLAC: <https://oig.cepal.org/en/indicators/femicide-or-feminicide>

¹² Ministerio de Salud. Línea de base de mortalidad materna en El Salvador. Junio 2005 - mayo 2006. El Salvador. En: https://www.salud.gob.sv/archivos/pdf/documento_LBMM/CONTENIDO_PARTE4.pdf

¹³ <https://www.covid19treatmentguidelines.nih.gov/special-populations/pregnancy/>

- **Women face high levels of gender-based violence:** On average, 26.3% of women report physical and/or sexual abuse by partners at some point in their lives. Victims are usually unemployed, who live in urban areas and also mothers of multiple children. The country also has limited capacity to address gender-based violence cases. Most of the health facilities that are dedicated to examining and serving victims of gender-based violence are not adequate or insufficient. Women who work in domestic task face more often gender-based violence than women who work outside their homes. When entering an emergency service at clinics or hospitals, victims of gender-based violence. Also lockdown measurements increase the incidence of GBV at home.
- **Reverse gender gap is found in COVID-19 cases in the country:** By January 2020, 49539 COVID-19 cases have been recorded in El Salvador. While 51% of the cases correspond to male patients, 49% of the cases are being detected in women.

Suggested gender entry points and activities include: (i) develop a gender assessment (identify priority areas such as maternal health facilities and to capture sex-disaggregated data and information about better access to vaccine); (ii) stakeholder consultations/presentation; (iii) develop and implement a communication/outreach campaign.

3.2.3 Kenya

Kenya obtained a score of 0.671 in the 2020 WEF Global Gender Report Index, which puts the country in the 109th position. This suggests a minor improvement in comparison to the 2006 edition of the report, where the country obtained a score of 0.649 and ranked 73rd.¹⁴ The country ranks 20th among 34 countries in its region, Sub-Saharan Africa.

The Constitution of Kenya states that women and men have the right to equal treatment, including the right to equal opportunities in political, economic, cultural and social spheres. The country has been investing and working in gender equality issues and regulations, including the creation and implementation of the National Policy on Gender and Development led by the Ministry of Public Service and Gender. (created in 2015 and acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

Legal framework: Kenya has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication Women, Business and the Law 2021 (World Bank, 2021) indicates that Kenya scores 80.6 out of 100. It means the country is higher than the regional average achieved in Sub-Saharan Africa (71).

- **Employment and payment:** The country does not have a law that restricts women to work in the same fields as men (and can work at night in the same way as men). Also, women at workplaces are protected by a legislation that prevents sexual harassment in employment (Sexual Offences Act, 2006, Sec. 23; Employment Act, 2007, Sec. 6) and sexual harassment at work is considered a crime (Sexual Offences Act, 2006, Sec. 23). A Kenyan law prohibits discrimination in employment based on gender (Employment Act, Art. 5(3)(a)). Women are able to work in all industrial jobs in the same way as men (including mining, water and factoring, construction, agriculture, energy, transportation, and others). Women are not prevented from working in jobs that are considered hazardous. In regards of payment, Kenyan women are protected by a law that mandates equal remuneration for equal value (Employment Act, Art. 5(5)).

¹⁴ Many countries were added between the two vintages; thus, the drop in the ranking does not necessarily signal the performance of the country.

- **Maternity Rights:** Mothers in Kenya have 90 days of paid time during a period between pregnancy and after labor (Employment Act, Sec. 29(1)). The country lacks parental leave, but fathers enjoy a 14-day paid leave after labor (Employment Act, Art. 29(8)). The government does not administer 100% of the maternity leave benefits (Employment Act, Art. 29). It is also prohibited the dismissal of pregnant worker (Employment Act, Art. 29)
- **Assets:** the Matrimonial Property Act, Sec. 4(a) and 12 respectively concede women to have equal ownership rights to immovable property and grants spouses equal administrative authority over assets during marriage. Law of Succession Act, Art. 35(5) states that sons and daughters have equal rights to inherit assets from their parents; but widowed women do not have the same rights to inherit their spouses assets as men (Law of Succession Act, Art. 36(1)).
- **Access to credit and entrepreneurship:** Kenya does not have a law that prohibits discrimination in access to credit based on gender. Women also have no restrictions open a bank account or signing a contract (Matrimonial Property Act, Art. 4(b)). However, women experience barriers to register a business as they cannot do it in the same way as men (Registration of Business Names Act, Art. 6(1)(h))
- **Marriage:** Kenya does not have a legal provision that requires a married woman to obey her husband; women can be called the head of household in the same way as men. The country has specific legislation for addressing domestic violence (Protection Against Domestic Violence Act, 2015); and women can obtain a judgment of divorce in the same way as men, but they do not have the same rights to remarry as men.

Income. According to the UNDP's estimates, the female Gross National Income (GNI) per capita in Kenya amounts to 2,619 in 2011 PPP \$, which corresponds to 75% of the male GNI per capita (3,490 2011 PPP \$).

Table [11]. Female and male gross national income per capita (2011 ppp terms)

Country	Estimated gross national income per capita (2011 ppp terms) ¹		
	Female	Male	Female as % of male
Kenya	2,619	3,490	75

Source: UNDP <http://hdr.undp.org/en/indicators/123506#a>. 1/ Definition: Derived from the ratio of female to male wages, female and male shares of economically active population and gross national income (in 2011 purchasing power parity terms).

Labor Markets. Labor participation in Kenya is slightly lower for women than men; From the whole labor force, women represent 49.0%. However, the share is much lower when looking at the share of female employees in the industry (13.4%), manufacturing (26.8%), or electricity gas, steam, and air conditioning, supply (19.8%) (Table [12]).

Table [12]. Share of employment by activities and sex (% , 2018 unless stated differently)

Country	Sex	Industry	Manufacturing	Electricity; gas, steam and air conditioning supply	Professional, scientific and technical activities	Total	Women are able to work in the same industries as men
Kenya	Female	13.4	26.8	19.8	NA	49.0	Yes
	Male	86.6	73.2	80.2	NA	51.0	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019. Note: Kenya's numbers are from ILO Modelled Estimates.

Financial inclusion. Men in Kenya present higher levels of financial inclusion than women when comparing the ownership of an account at a financial institution (47 versus 65%), a mobile account (69 versus 77%) or saving money (65 versus 77%) (Table [13]). In addition, the share of women aged 15 or above in the country with either an account at financial institution or a mobile money-service provider is 78%, well above the average in developing countries (58.3%), and more than double the share in South-Saharan Africa (36%).

Table [13]. Indicators of financial inclusion by sex (2017)

Country	Account at a financial institution (% age 15+)		Mobile account (% age 15+)		Saved any money last year (% age 15+)		Women with account at financial inst. or with mobile money-service provider (% age 15+)
	Female	Male	Female	Male	Female	Male	
Kenya	47%	65%	69%	77%	65%	77%	78%

Source: World Bank Little Book of Gender 2019

Education. Women in Kenya are more likely to be enrolled in primary education than men (81.7 of women versus 78.3% of men). However, this gap is reverted when looking at secondary or tertiary education. Consequently, men report a higher average number of schooling years (7.2 versus 6). In addition, women amount to less than one third of STEM graduates (31%) (Table [4]).

Table [14]. Indicators of education by sex

Country	Sex	Enrolment in primary education (%)	Enrolment in secondary education (%)	Enrolment in tertiary education (%)	Mean years of schooling (years)	Female share of tertiary graduates from STEM (%)
Kenya	Female	81.7	45.8	9.7	6	31
	Male	78.3	49	13.2	7.2	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Health. The maternal mortality ratio in Kenya is 510 per 100,000 live births, one of the highest in the world, ranking 164th among 182 countries included in the Human Development Report. This figure is more than twice the average in developing countries (231) but close with its region, South-Saharan Africa (550). In addition, female life expectancy at birth in Kenya is more than 4 years higher than the same figure for males (68.7 versus 64.0 years) (Table [15]).

Table [15]. Indicators of health by sex

Country	Maternal mortality ratio (deaths per 100,000 live births)	Maternal mortality ratio (ranking across countries)	Female life expectancy at birth (years)	Male life expectancy at birth (years)
Kenya	510	164	68.7	64.0

Source: UNDP. <http://hdr.undp.org/en/data>

Note: rankings among 182 countries with data in UNDP report

Agency. Only one out of five members in Kenya's parliament are women (21.8%). The share of women in ministerial positions is slightly higher, one in four (25%).¹⁵

Gender Based Violence. According to the United Nations Development Report, 41% of women in Kenya aged 15 or older have experience gender-based violence by their partners.¹⁶

Gender gaps associated with the current project in Kenya

Cooling-related activities or sectors targeted: efficient and sustainable cooling appliances to both off-grid and on-grid consumers, targeting household end-users, small businesses and manufacturing supply chains. Expansion of credit facility for providers of energy efficient cooling systems.

Preliminary overview of relevant gender gaps:

- **Female-headed households are less likely to have access to energy, for all, grid and off-grid users, in urban and rural areas**¹⁷. Female-headed households are more deprived from electricity than their male counterparts. 16.3% and 50.4% of female headed households in urban and rural areas do not have any source of electricity compared to 9.3% and 45.3% of male-headed households in urban and rural areas, respectively.
- **Financing options may be important to close the access gap for rural female-headed households.** Across Kenya, households where women are primary decision makers are generally less willing (or able) to pay for off-grid solar solutions up front but were more willing to purchase a solar solution if the repayment period was over a longer period of time (6–12 months).
- **Women's income is limited.** For instance, when offered an off-grid solar home system, only 17% of female-headed households were willing (or able) to pay its full price up front, compared to 31% of male-headed households. However, when a payment plan of 12 months was offered, more female-headed households (26.8%) than male-headed households (16.3%) were willing to pay for this solar-solution.
- **Regardless of their willingness to pay of energy access, women in Kenya have less access to financial options than men.**¹⁸ For example, when looking at the shares owning an account (77.7% of females versus 85.8% of males), saving money (64.7% versus 76.6%) or borrowing money (61.7% versus 67.4%). Moreover, women tend to rely more on informal sources of credit. 13.8% of adult women borrowed from a financial institution or used a credit card, compared to 25.3% of adult males. Meanwhile, 24.2% of adult women borrowed through a savings club compared to 14.6% of adult men.

Suggested ender entry points and activities include: (i) develop (or review/enhance/) gender assessment; (ii) carry out consultations, workshops, gender assessment's presentations; (iii) engage in policy dialogues to raise awareness about the linkages between cooling and gender; (iv) capacity building/awareness about gender equality; (v)) develop agender action plan that includes responsibilities, activities and budgets (vi): impart informational sessions for female-led firms; (vii) assessment on financial barriers female-

¹⁵ Source: WEF Global Gender Report 2020. http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

¹⁶ Source: UNDP. <http://hdr.undp.org/en/data>

¹⁷ Source: World Bank (2020) Kenya - Multi-Tier Framework (MTF) Survey <https://energydata.info/dataset/kenya-multi-tier-framework-mtf-survey>

¹⁸ World Bank (2017) Findex Survey <https://globalfindex.worldbank.org/>

enterprises face; (viii) develop training/capacity/skill building activities for women, (ix) develop and implement a communication/outreach campaign.

3.2.4 Malawi

Malawi obtained a score of 0.664 in the 2020 WEF Global Gender Report Index, which puts the country in the 116th position. This is the same score the country received in the 2006 edition of the report; however, a lower ranking. Malawi ranked 91st in the 2006 vintage.¹⁹ The country ranks 23rd in its region, South-Saharan Africa.

The Constitution of Malawi states that women have the right to fully participate in all spheres of Malawian society on the basis of equal opportunities with men; and that women are protected by the implementation of the principles of nondiscrimination. The country has been investing and working in gender equality issues and regulations, including the creation for gender equality coordination of the Ministry of Gender, Children, Disability and Social Welfare (created in 1996 and acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

Legal framework: Malawi has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication Women, Business and the Law 2021 (World Bank, 2021) indicates that Malawi scores 77.5 out of 100. It means the country is higher than the regional average achieved in Sub-Saharan Africa (71).

- **Employment and payment:** The country does not have a law that restricts women to work in the same fields as men (and can work at night in the same way as men). Also, women at workplaces are protected by a legislation that prevents sexual harassment in employment and sexual harassment at work is considered a crime (Gender Equality Act, 2013, Secs. 6(2) and 7(2)). A Malawian law prohibits discrimination in employment based on gender (Employment Act, Sec. 5). Women are able to work in all industrial jobs in the same way as men (including mining, water and factoring, construction, agriculture, energy, transportation, and others). Women are not prevented from working in jobs that are considered hazardous. In regards of payment, Malawian women are protected by a law that mandates equal remuneration for equal value (Employment Act, Sec. 6).
- **Maternity Rights:** Mothers in Malawi have 56 days of paid time during a period between pregnancy and after labor (Employment Act, Sec. 47(1)). The country lacks parental leave, and fathers cannot enjoy a paid leave after labor. The government does not administer 100% of the maternity leave benefits (Employment Act, Sec. 47). It is also prohibited the dismissal of pregnant worker (Employment Act, Sec. 49)
- **Assets:** The Married Women's Property Act, Secs. 1, 2 and 5 respectively concede women to have equal ownership rights to immovable property and grants spouses equal administrative authority over assets during marriage. Deceased Estates (Wills, Inheritance and Protection) Act, Sec. 17 states that sons and daughters have equal rights to inherit assets from their parents; and widowed women have the same rights to inherit their spouses' assets as men.
- **Access to credit and entrepreneurship:** Malawi does not have a law that prohibits discrimination in access to credit based on gender. Women also have no restrictions open a bank account or signing a contract (Constitution of Malawi, Art. 24(1)(a)(i).

¹⁹ Many countries were added between the two vintages; thus, the drop in the ranking does not necessarily signal the performance of the country.

- **Marriage:** Malawi does not have a legal provision that requires a married woman to obey her husband; women can be called the head of household in the same way as men (Marriage, Divorce and Family Relations Act, Arts. 48-50). The country has specific legislation for addressing domestic violence (Prevention of Domestic Violence Act); and women can obtain a judgment of divorce in the same way as men, but they do not have the same rights to remarry as men.

Income. According to the UNDP's estimates, the female Gross National Income (GNI) per capita in Malawi amounts to 925 in 2011 PPP \$, which corresponds to 6% of the male GNI per capita (1,400 2011 PPP \$).

Table [16]. Female and male gross national income per capita (2011 ppp terms)

Country	Estimated gross national income per capita (2011 ppp terms) ¹		
	Female	Male	Female as % of male
Malawi	925	1,400	66

Source: UNDP <http://hdr.undp.org/en/indicators/123506#a>. 1/ Definition: Derived from the ratio of female to male wages, female and male shares of economically active population and gross national income (in 2011 purchasing power parity terms).

Labor Markets. The labor force participation rate for women in Malawi is slightly below the participation rate for men; from the whole labor force, women represent only 48.1%. However, the share is significantly even lower when looking at the share of female employees in the industry (36.0%), manufacturing (40.7%), or electricity gas, steam, and air conditioning, supply (9.2%) (Table [17]).

Table [17]. Share of employment by activities and sex (% , 2018 unless stated differently)

Country	Sex	Industry	Manufacturing	Electricity; gas, steam and air conditioning supply	Professional, scientific and technical activities	Total	Women are able to work in the same industries as men
Malawi	Female	36.0	40.7	9.2	NA	48.1	Yes
	Male	64.0	59.3	90.8	NA	51.9	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Financial inclusion. Men in Malawi present higher levels of financial inclusion than women when comparing the ownership of an account at a financial institution (29 versus 19%), a mobile account (23 versus 18%) or saving money (53 versus 51%) (Table [18]). In addition, the share of women aged 15 or above in the country with either an account at financial institution or a mobile money-service provider is 30%, well below the average in developing countries (58.3%), or South-Saharan Africa (36%).

Table [18]. Indicators of financial inclusion by sex (2017)

Country	Account at a financial institution (% age 15+)		Mobile account (% age 15+)		Saved any money last year (% age 15+)		Women with account at financial inst. or with mobile money-service provider (% age 15+)
	Female	Male	Female	Male	Female	Male	
Malawi	18%	29%	18%	23%	51%	53%	30%

Source: World Bank Little Book of Gender 2019

Education. Women in Malawi are more likely to be enrolled in primary or secondary education than men. However, this gap is reverted when looking at tertiary education (0.6 versus 1%). Due to these difference and gaps in education attainment, men report a higher average number of schooling years (5.1 versus 4.1).

Table [19]. Indicators of education by sex

Country	Sex	Enrolment in primary education (%)	Enrolment in secondary education (%)	Enrolment in tertiary education (%)	Mean years of schooling (years)	Female share of tertiary graduates from STEM (%)
Malawi	Female	97.1	34.8	0.6	4.1	NA
	Male	92	33.7	1	5.1	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Health. The maternal mortality ratio in Malawi is 634 per 100,000 live births, which is one of the highest in the world, 170th among 182 countries included in the Human Development Report. This figure is above the average in developing countries (231) or Malawi's region, South-Saharan Africa (550). In addition, female life expectancy at birth in Malawi is almost 6 years higher than the same figure for males (66.9 versus 60.7 years) (Table [20]).

Table [20]. Indicators of health by sex

Country	Maternal mortality ratio (deaths per 100,000 live births)	Maternal mortality ratio (ranking across countries)	Female life expectancy at birth (years)	Male life expectancy at birth (years)
Malawi	634	170	66.9	60.7

Source: UNDP. <http://hdr.undp.org/en/data>

Note: rankings among 182 countries with data in UNDP report

Agency. Only around one out of five parliament members in Malawi are women (22.9%). The share of women in ministerial positions is even lower, (16.7%).²⁰

Gender Based Violence. According to the United Nations Development Report, more 38% of women in Malawi aged 15 or older have experience gender-based violence by their partners.²¹

Gender gaps associated with the current project in Malawi

Cooling-related activities or sectors targeted: agriculture value chain products; MSMEs in which cooling activities are relevant to their business, such as dairy, horticulture (fruits and vegetables)

Preliminary overview of relevant gender gaps:

- **Women's entrepreneurship is booming.** At the entrepreneurship level, despite being less wealthy, less developed and lagging in terms of knowledge assets, financial access and supporting entrepreneurial conditions, female-led businesses in Malawi, representing 30.3% of the country's businesses, were ranked no 7 globally according to the Mastercard Index of Women Entrepreneurs index.
- **In Malawi, women farmers have lower levels of education compared to male farmers.** Women have less access to agricultural productive endowments and financial resources, which severely affects productivity.

²⁰ Source: WEF Global Gender Report 2020. http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

²¹ Source: UNDP. <http://hdr.undp.org/en/data>

- **Women's productivity is lower than men.** Agricultural productivity on female managed plots has been estimated to be 25% lower than on male-managed plots due to limited access to productive endowments. Also, data from the UN shows that women have lower access to agricultural machinery and production technologies.

Suggested ender entry points and activities include: (i) develop (or review/enhance/) gender assessment; (ii) carry out consultations, workshops, gender assessment's presentations; (iii) engage in policy dialogues to raise awareness about the linkages between cooling and gender; (iv) capacity building/awareness about gender equality; (v)) develop agender action plan that includes responsibilities, activities and budgets (vi): impart informational sessions for female-led firms; (vii) assessment on financial barriers female-enterprises face; (viii) develop training/capacity/skill building activities for women, (ix) develop and implement a communication/outreach campaign.

3.2.5 North Macedonia

North Macedonia obtained a score of 0.711 in the 2020 WEF Global Gender Report Index, which puts the country in the 70th position. This suggests a small improvement in comparison to the 2006 edition of the report, where the country obtained a score of 0.698 and ranked 28th.²² The country ranks 16th among 26 countries in its region, Eastern Europe and Central Asia.

The Law on Equal Opportunities for Women and Men establishes that in North Macedonia women and men have equal rights at all levels. The country has been investing and working in gender equality issues and regulations, including designing, and implementing the National Strategy for Gender Equality led by Ministry of Labour and Social Policy (acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

Legal framework: North Macedonia has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication Women, Business and the Law 2021 (World Bank, 2021) indicates that North Macedonia scores 85 out of 100. It means the country is higher than the regional average achieved in Europe and Central Asia (83.4).

- **Employment and payment:** The country does not have a law that restricts women to work in the same fields as men (and can work at night in the same way as men). Also, women at workplaces are protected by a legislation that prevents sexual harassment in employment and sexual harassment at work is consider a crime (Law on Protection Against Harassment at the Workplace, Arts. 1 and 7; Law on Equal Opportunities for Women and Men, Art. 3(3); Law on Prevention and Protection against Discrimination, Art. 41; Law on Protection Against Harassment at the Workplace, Art. 36, 36a and 37). A N. Macedonian law prohibits discrimination in employment based on gender (Labor Relations Act, Arts. 6 and 7). However, women are not able to work in all industrial jobs in the same way as men (including mining, construction and factories are not allowed (Labor Relations Act, Art. 160 and 131) while water, agriculture, energy, transportation, and others are allowed). Women are not prevented from working in jobs that are considered hazardous. In regards of payment, N.

²² Many countries were added between the two vintages; thus, the drop in the ranking does not necessarily signals the performance of the country.

Macedonian women are not protected by a law that mandates equal remuneration for equal value.

- **Maternity Rights:** Mothers in North Macedonia have 270 days of paid time during a period between pregnancy and after labor (Labor Relations Act, Art. 165). The country lacks parental leave, but fathers can enjoy a paid leave after labor (2 days) (Labor Relations Act, Art. 146; General Collective Bargaining Agreement for the Private Sector in the Field of Economy, Article 42). The government administers 100% of the maternity leave benefits (Law on Health Insurance, Arts. 14 and 17). It is also prohibited the dismissal of pregnant worker (Labor Relations Act, Art. 101)
- **Assets:** Constitution of the Republic of North Macedonia, Art. 9 and the Law on Ownership and Other Real Rights, Arts. 70-72 concede women to have equal ownership rights to immovable property and grants spouses equal administrative authority over assets during marriage respectively. The Law on Inheritance of 1996, Arts. 3 and 13 states that sons and daughters have equal rights to inherit assets from their parents; and widowed women have the same rights to inherit their spouses' assets as men.
- **Access to credit and entrepreneurship:** North Macedonia prohibits discrimination in access to credit based on gender (Law on Prevention of and Protection Against Discrimination, Art. 11). Women also have no restrictions open a bank account, signing a contract, and register a business in the same way as men.
- **Marriage:** North Macedonia does not have a legal provision that requires a married woman to obey her husband; women can be called the head of household in the same way as men. The country has specific legislation for addressing domestic violence (Law on Prevention, Suppression and Protection from Domestic Violence; Criminal Code, Art. 122(21)); and women can obtain a judgment of divorce and have the same rights to remarry as men.

Income. According to the UNDP's estimates, the female Gross National Income (GNI) per capita in North Macedonia amounts to 9,464 in 2011 PPP \$, which corresponds to 58% of the male GNI per capita (16,279 2011 PPP \$).

Table [21]. Female and male gross national income per capita (2011 ppp terms)

Country	Estimated gross national income per capita (2011 ppp terms) ¹		
	Female	Male	Female as % of male
North Macedonia	9,464	16,279	58

Source: UNDP <http://hdr.undp.org/en/indicators/123506#a>. 1/ Definition: Derived from the ratio of female to male wages, female and male shares of economically active population and gross national income (in 2011 purchasing power parity terms).

Labor Markets. Labor participation in North Macedonia is much lower for women than men; From the whole labor force, women represent only 39.6%. The share is even lower when looking at the share of female employees in the industry (33.8%), electricity gas, steam, and air conditioning, supply (16.7%). However, closer to parity when looking at manufacturing (47.6%) or professional, scientific and technical activities (54.8%). These gaps are partly explained by labor regulations in the country that forbid women from participating in certain sectors of the economy (Table [22]).

Table [22]. Share of employment by activities and sex (% , 2018 unless stated differently)

Country	Sex	Industry	Manufacturing	Electricity; gas, steam and air conditioning supply	Professional, scientific and technical activities	Total	Women are able to work in the same industries as men
North Macedonia	Female	33.8	47.6	16.7	54.8	39.6	No
	Male	66.2	52.4	83.3	45.2	60.4	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Financial inclusion. Men in North Macedonia present higher levels of financial inclusion than women when comparing the ownership of an account at a financial institution (73 versus 80%), a mobile account (11 versus 13%) or saving money (33 versus 38%) (Table [23]). In addition, the share of women aged 15 or above in the country with either an account at financial institution or a mobile money-service provider is 73%, well above the average in developing countries (58.3%), or Europe and Central Asia (54%).

Table [23]. Indicators of financial inclusion by sex (2017)

Country	Account at a financial institution (% age 15+)		Mobile account (% age 15+)		Saved any money last year (% age 15+)		Women with account at financial inst. or with mobile money-service provider (% age 15+)
	Female	Male	Female	Male	Female	Male	
North Macedonia	73%	80%	11%*	13%*	33%	38%	73%

Source: World Bank Little Book of Gender 2019.

*For North Macedonia, this indicator reflects usage of mobile phone or the internet to access an account in 2017 (% age 15+)

Education. Women and men in North Macedonia are almost equally likely to be enrolled in primary education, however, a gap favoring men appear when looking at secondary and tertiary education. Consequently, men report a higher average number of schooling years (10.2 versus 9.2). Women encompass a lower share of STEM graduates (45%) (Table [24]).

Table [24]. Indicators of education by sex

Country	Sex	Enrolment in primary education (%)	Enrolment in secondary education (%)	Enrolment in tertiary education (%)	Mean years of schooling (years)	Female share of tertiary graduates from STEM (%)
North Macedonia	Female	91.9	77.6	45.9	9.2	45
	Male	91.5	80.2	36.7	10.2	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Health. The maternal mortality ratio in North Macedonia is 8 per 100,000 live births, which ranks 29th among 182 countries included in the Human Development Report. This figure is well below the average in developing countries (231) or its region, Europe and Central Asia (25). The ratio is even lower than the maternal mortality ratio in OCDE (14) or Very high human development countries (15) according to the UNDP Human Development Report. In addition, female life expectancy at birth in North Macedonia is 4 years higher than the same figure for males (77.7 versus 73.7 years) (Table [25]).

Table [25]. Indicators of health by sex

Country	Maternal mortality ratio (deaths per	Maternal mortality ratio (ranking across countries)	Female life expectancy at birth (years)	Male life expectancy at birth (years)
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	100,000 live births)			
North Macedonia	8	29	77.7	73.7

Source: UNDP. <http://hdr.undp.org/en/data>

Note: rankings among 182 countries with data in UNDP report

Agency. Close to 40% of parliament members in North Macedonia are women. However, the share of women in ministerial positions is only 16%).²³

Gender Based Violence. According to the United Nations Development Report, 10% of women in North Macedonia aged 15 or older have experience gender-based violence by their partners.²⁴

Gender gaps associated with the current project in North Macedonia

Cooling activities or sectors targeted: energy efficiency investments in public buildings (central government and municipal schools, health care facilities, universities, etc) to lower and avoid cooling loads through the development and implementation of a sustainable financing mechanism.

Preliminary overview of relevant gender gaps

- **Gender-based violence represents a priority public health issue:** the country has established dedicated facilities administrated by the Ministry of Health that address and prevent victimization of women. Gender-based violence results on women's physical and mental injuries and traumas. Also, evidence shows that women that experience physical or sexual violence report higher rates of depression, abortion and acquiring HIV compared to women have do not experience this type of violence.
- **Vocational differences can be spotted between male and female students' secondary qualifications.** From all female students, only 1% choose mechanical engineering and 5% chooses electrical engineering. Meanwhile 16% of male students choose mechanical engineering and 20% choose electrical engineering.
- **Women are underrepresented in the STEM careers.** Overall, 10% of all female students choose an engineering qualification compared to 35% of male students. Women in North Macedonia account to only 17.5% of all researchers in science, technology, and innovation working on the field of engineering and technology.
- **The labor market is also clustered by sex in North Macedonia.** From all female workers, only 0.4% are in the electricity; gas, steam and air conditioning supply, compared to 2.8% of male workers. Moreover, the male labor force is bigger due to higher male labor participation. Consequently, male workers account for 91% of the employees in electricity; gas, steam and air conditioning supply.

Suggested gender entry points and activities include: (i) develop (or review/enhance/) gender assessment; (ii) carry out consultations, workshops, gender assessment's presentations; (iii) engage policy dialogues to raise awareness about the linkages between cooling and gender; (iv) capacity building/awareness about gender equality; (v) develop agender action plan that includes responsibilities, activities and budgets; (vi) develop and implement a communication campaign.

²³ Source: WEF Global Gender Report 2020. http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

²⁴ Source: UNDP. <http://hdr.undp.org/en/data>

3.2.6 Panama

Panama obtained a score of 0.730 in the 2020 WEF Global Gender Report Index, which puts the country in the 46th position. This suggests a substantial improvement in comparison to the 2006 edition of the report, where the country obtained a score of 0.693 and ranked 31st.²⁵ The country ranks 12th among 25 countries in its region, Latin America and the Caribbean.

The Constitution of Panama establishes that Panamanian women and men have equal rights at all levels. The country has been investing and working in gender equality issues and regulations, including the creation of the National Women's Institute (created in 2008 and acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

Legal framework: Panama has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication Women, Business and the Law 2021 (World Bank, 2021) indicates that Panama scores 79.4 out of 100. It means the country is lower than the regional average achieved in Latin America and the Caribbean (80.1).

- **Employment and payment:** The country does not have a law that restricts women to work in the same fields as men (and can work at night in the same way as men). Also, women at workplaces are protected by a legislation that prevents sexual harassment in employment and sexual harassment at work is considered a crime (Código de Trabajo, Arts. 127(12) y 138(15); Ley Núm. 82 de 2013, Arts. 4(20) y 45; and Ley Núm. 82 de 2013, Arts. 4(20) y 45). A Panamanian law prohibits discrimination in employment based on gender (Ley Núm. 82, Arts. 4(20) y 45). and women are able to work in all industrial jobs in the same way as men (including mining, construction, factories, water, agriculture, energy, transportation, and others). Women are not prevented from working in jobs that are considered hazardous (Código de Trabajo, Art. 104). In regards of payment, Panamanian women are not protected by a law that mandates equal remuneration for equal value.
- **Maternity Rights:** Mothers in Panama have 98 days of paid time during a period between pregnancy and after labor (Código de Trabajo, Art. 107). The country lacks parental leave, but fathers can enjoy a paid leave after labor (3 days) (Ley Núm. 27, Arts. 1 y 3). The government administers 100% of the maternity leave benefits (Código de Trabajo, Art. 107). It is also prohibited the dismissal of pregnant worker (Código de Trabajo, Art. 106)
- **Assets:** Código de Familia, Arts. 103 y 104 concedes women to have equal ownership rights to immovable property and grants spouses equal administrative authority over assets during marriage respectively. The Código Civil, Arts. 661-663 states that sons and daughters have equal rights to inherit assets from their parents; and widowed women have the same rights to inherit their spouses' assets as men.
- **Access to credit and entrepreneurship:** Panama does not have a law that prohibits discrimination in access to credit based on gender, but women do not have no restrictions open a bank account, signing a contract, and register a business in the same way as men.
- **Marriage:** Panama does not have a legal provision that requires a married woman to obey her husband; women can be called the head of household in the same way as men. The country has specific legislation for addressing domestic violence (Ley Núm.

²⁵ Many countries were added between the two vintages; thus, the drop in the ranking does not necessarily signal the performance of the country.

38 de 2001; Ley Núm. 82 de 2013, Arts. 2, 4(3) y (10)); and women can obtain a judgment of divorce and have the same rights to remarry (Código de Familia, Arts. 35(2) y 219) as men.

Income. According to the UNDP's estimates, the female Gross National Income (GNI) per capita in Panama amounts to 16,107 in 2011 PPP \$, which corresponds to 65% of the male GNI per capita (24,787 2011 PPP \$).

Table [26]. Female and male gross national income per capita (2011 ppp terms)

Country	Estimated gross national income per capita (2011 ppp terms) ¹		
	Female	Male	Female as % of male
Panama	16,107	24,787	65

Source: UNDP <http://hdr.undp.org/en/indicators/123506#a>. 1/ Definition: Derived from the ratio of female to male wages, female and male shares of economically active population and gross national income (in 2011 purchasing power parity terms).

Labor Markets. Labor participation in Panama is lower for women than men; from the whole labor force, women represent only 41.0%. The share is even lower when looking at the share of female employees in the industry (21.6%) and electricity gas, steam, and air conditioning, supply (17.0%)²⁶. In addition, the country has labor regulations that forbid women from participating in certain sectors of the economy (Table [27]).

Table [27]. Share of employment by activities and sex (% , 2018 unless stated differently)

Country	Sex	Industry	Manufacturing	Electricity; gas, steam and air conditioning supply	Professional, scientific and technical activities	Total	Women are able to work in the same industries as men
Panama	Female	21.6	43.0	17*	48.6	41.0	Yes
	Male	78.4	57.0	83.0	51.4	59.0	

Source: labor force shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019.

Note: Numbers with an asterisk are considered unreliable by ILO due to small samples.

Education. Women in Panama are more likely to be enrolled in secondary or tertiary education than men. Of all women in the corresponding education ages, 65.9% are enrolled in secondary education and 58.6% in tertiary education. In contrast, from all men in the same age groups, only 61.7% are enrolled in secondary education, and 37.3% are enrolled in tertiary education. Consequently, women report a higher average number of schooling years (10.4 versus 9.9). In addition, women amount to almost half of STEM graduates (Table [28]).

Table [28]. Indicators of education by sex

Country	Sex	Enrolment in primary education (%)	Enrolment in secondary education (%)	Enrolment in tertiary education (%)	Mean years of schooling (years)	Female share of tertiary graduates from STEM (%)
Panama	Female	85.9	65.9	58.6	10.4	49
	Male	86.6	61.7	37.3	9.9	

²⁶ The estimate of the share of employees by sex in the Electricity; gas, steam and air conditioning supply sector in Panama is considered unreliable by ILO due to small samples.

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Health. The maternal mortality ratio in Panama is 94 per 100,000 live births, which ranks 109th among 182 countries included in the Human Development Report. This figure is below the average in developing countries (231) but above its region, Latin America and the Caribbean (68). In addition, female life expectancy at birth in Panama is more than 6 years higher than the same figure for males (81.6 versus 75.2 years) (Table [29]).

Table [29]. Indicators of health by sex

Country	Maternal mortality ratio (deaths per 100,000 live births)	Maternal mortality ratio (ranking across countries)	Female life expectancy at birth (years)	Male life expectancy at birth (years)
Panama	94	109	81.6	75.2

Source: UNDP. <http://hdr.undp.org/en/data>

Note: rankings among 182 countries with data in UNDP report

Agency. Only one out of five members in Panama's parliament are women (21.1%). Additionally, the share of women in ministerial positions close to one in four (26.7%).²⁷

Gender Based Violence. According to the United Nations Development Report, almost one in seven women in Panama aged 15 or older has experience gender-based violence by their partners.²⁸

Gender gaps associated with the current project in Panama

Cooling-related activities or sectors targeted: implementation of innovative energy efficiency fund, working through the local banking sector, to channel financing towards energy efficient cooling appliances; green buildings. Policy and communications activities are also planned.

Preliminary overview of relevant gender gaps

- **Financial inclusion.** Men in Panama present higher levels of financial inclusion than women when comparing the ownership of an account at a financial institution (50 versus 42%), a mobile account (5 versus 2%) or saving money (55 versus 42%) (Table [30]). In addition, the share of women aged 15 or above in the country with either an account at financial institution or a mobile money-service provider is 42%, well below the average in developing countries (58.3%), or Latin America and the Caribbean (52%).

Table [30]. Indicators of financial inclusion by sex (2017)

Country	Account at a financial institution (% age 15+)		Mobile account (% age 15+)		Saved any money last year (% age 15+)		Women with account at financial inst. or with mobile money-service provider (% age 15+)
	Female	Male	Female	Male	Female	Male	
Panama	42%	50%	2%	5%	42%	55%	42%

Source: World Bank Little Book of Gender 2019

- **Women are underrepresented in the STEM field.**

²⁷ Source: WEF Global Gender Report 2020. http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

²⁸ Source: UNDP. <http://hdr.undp.org/en/data>

- In 2013, 32% of all researchers in science, technology, and innovation in Panama were female. In the same year, 19% of researchers in engineering were female.
- The percent of share of tertiary-level STEM students in the country is 52%. However, the share drops to 44% by the time of graduation.
- Male workers account for 83% of the electricity; gas, steam and air conditioning supply employees.
- **Suggested gender entry points and activities** include: (i) develop (or review/enhance/) gender assessment; (ii) carry out consultations, workshops, gender assessment's presentations; (iii) engage policy dialogues to raise awareness about the linkages between cooling and gender; (iv) capacity building/awareness about gender equality; (v) develop agender action plan that includes responsibilities, activities and budgets; (vi): impart informational sessions for female-led firms; (vii) assessment on financial barriers female-enterprises face; (viii) develop training/capacity/skill building activities for women, (ix) develop and implement a communication campaign.

3.2.7 Sao Tome and Principe

The Human Development Report 2020 and the Gender Development Index (GDI) define a ratio of the female to male Human Development Index (HDI). The GDI shows gender inequalities in achievement health, education and economic resources. For 2019, the total GDI for Sao Tome e Principe was 0.906, identifying gender gaps against women.²⁹

The Constitution of Sao Tome and Principe guarantees equal rights and obligations to men and women in the country. The country has been investing and working in gender equality issues and regulations, including the design and implementation of the National Strategy for Gender Equality and Equity led by the National Institute For The Promotion Of Gender Equality And Equity (created in 2008 and acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

Legal framework: Sao Tome and Principe has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication Women, Business and the Law 2021 (World Bank, 2021) indicates that Sao Tome and Principe scores 86.3 out of 100. It means the country is higher than the regional average achieved in Sub-Saharan Africa (71).

- **Employment and payment:** The country has a law allows women to get a job in the same way as men (Law No. 2/77, Art. 28). Also, women at workplaces are protected by a legislation that prevents sexual harassment in employment and sexual harassment at work is consider a crime (Penal Code, Art. 166(2); Labour Code, Art. 18(2)). A Santomean law prohibits discrimination in employment based on gender (Law No. 6/2019, Arts. 16, 17, 21 and 22). and women are able to work in all industrial jobs in the same way as men (including mining, construction, factories, water, agriculture, energy, transportation, and others). Women are not prevented from working in jobs that are considered hazardous. In regards of payment, Santomean women are not protected by a law that mandates equal remuneration for equal value.
- **Maternity Rights:** Mothers in Sao Tome and Principe have 60 days of paid time during a period between pregnancy and after labor (Labor Law Art. 140(1)). The country lacks parental leave, and fathers cannot enjoy a paid leave after labor. The government administers 100% of the maternity leave benefits (Presidential Decree No. 3/2004, Art.

²⁹ http://hdr.undp.org/sites/all/themes/hdr_theme/country-notes/STP.pdf

4; Decree-Law No. 25/2014, Art. 96(1)(b); Law No. 6/2019, Art. 249). It is also prohibited the dismissal of pregnant worker (Law No. 6/2019, Art. 262)

- **Assets:** Law No. 2/77, Arts. 35 & 36, and 38(2) concedes women to have equal ownership rights to immovable property and grants spouses equal administrative authority over assets during marriage respectively. The Law No. 2/77, Art. 65 states that sons and daughters have equal rights to inherit assets from their parents.
- **Access to credit and entrepreneurship:** Sao Tome and Principe does not have a law that prohibits discrimination in access to credit based on gender, but women do not have no restrictions open a bank account, signing a contract, and register a business in the same way as men.
- **Marriage:** Sao Tome and Principe does not have a legal provision that requires a married woman to obey her husband; women can be called the head of household in the same way as men. The country has specific legislation for addressing domestic violence (Law on Domestic and Family Violence); and women can obtain a judgment of divorce and have the same rights to remarry (Law 19/2018, Art. 25(1)) as men.

Income. Sao Tome e Principe has low Gross National Income (GNI) per capita and it is estimated for US\$2,462 for women and US\$5,439 for men. Also, 37.9% of households are headed by women and 62.1% by men. On average, the monthly household expenditure is 16% lower for female-headed households than male-headed households. Female-headed households are also overrepresented in the lowest expenditure quintiles in the country. Women also

Labor Markets. Labor participation in Sao Tome e Principe is much lower for women than men women's rate in the labor force participation is 43% while men's is 76%.³⁰ Also, only 44% of women in the labor force are waged and salaried workers, as opposed of 56% of men. 55% percent of women who belong to the labor force are considered as vulnerable employment. Only 3% of women work in the industry sector (vs. 25% of men) and women are overrepresented in the service sector (84% vs. 45% of men).

Education. Women in Sao Tome e Principe are about the same to be enrolled in primary education levels (96% for both boys and girls) and are slightly more likely to be enrolled in secondary education levels than men (68% vs. 61%).

Health. The maternal mortality ratio in Sao Tome e Principe is 130 per 100,000 live births. This figure is below the average in developing countries (231) and also below the Sub-Saharan Africa (547) region. Also, it is more than 10 times the maternal mortality ratio in OCDE (14) or very high human development countries (15) according to the UNDP Human Development Report. In addition, female life expectancy at birth in Sao Tome e Principe is 4 years higher than the same figure for males (72.8 versus 68 years).

Agency. Only 18% members in Sao Tome e Principe's parliament are women. The share of women in ministerial positions is about the same (18%).³¹

Gender-Based Violence: About 1 in 3 women have experience some form of GBV at the hands of their husbands or partners.

Cooling-related activities or sectors targeted: in health-related projects

³⁰Source: WBG. <http://hdl.handle.net/10986/31689>

³¹ Source: WBG. <http://hdl.handle.net/10986/31689>

Preliminary overview of relevant gender gaps:

- **COVID-19 global emergency has impacted the country at all levels:** Sao Tome e Principe is currently facing an economic crisis resulted from the effect of the COVID-19 emergency, primary through losses in the tourism industry. COVID-19 cases in the country represent one among the highest infection rates the Sub-Saharan Africa region.
- **Gender-Based Violence:** lockdowns and stay-at-home orders triggers potential increases of GBV; it is important to provide women of tools for economic empowerment (including the most productive sectors including the STEM field)

Suggested gender entry points and activities include: (i) develop a gender assessment (identify priority areas such as maternal health facilities and to capture sex-disaggregated data and information about better access to vaccine); (ii) stakeholder

3.2.8 Somalia

Data and information regarding gender gaps in Somalia are limited. This situation represents a constraint to tackle issues that inhibits gender equality in the country. In the Gender Inequality Index (2012) report, the country has registered a score of 0.776 (with a maximum of 1 denoting complete inequality), placing Somalia at the fourth highest in gender inequality globally. Moreover, women in the country are being affected more by conflict and violence, inhibiting their potential to exit poverty.

The Constitution of Somalia provides that all citizens have equal rights and obligations regardless of sex, and that the State must not discriminate against any person on the basis of gender. The country has been investing and working in gender equality issues and regulations, including the design and implementation of the National Gender Policy of Somalia led by the Ministry of Women and Human Rights Development (created in 2014 and acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

Legal framework: Somalia has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication Women, Business and the Law 2021 (World Bank, 2021) indicates that Somalia scores 46.9 out of 100. It means the country is lower than the regional average achieved in Sub-Saharan Africa (71).

- **Employment and payment:** The country has a law that does not restrict women to work in in the same fields as men (Labor Code, Art. 4). However, women at workplaces are not protected by a legislation that prevents sexual harassment in employment. A Somalian law prohibits discrimination in employment based on gender (Labor Code, Arts. 3 and 70) but women are not able to work in all industrial jobs in the same way as men (including mining, construction, factories, water, agriculture, energy, transportation, and others; Labor Code, Art. 88). Women are not prevented from working in jobs that are considered hazardous. In regards of payment, Somalian women are protected by a law that mandates equal remuneration for equal value (Labor Code, Art. 70).
- **Maternity Rights:** Mothers in Somalia have 98 days of paid time during a period between pregnancy and after labor (Labor Code, Art. 91(2)). The country lacks parental leave, and fathers cannot enjoy a paid leave after labor. The government does not administer 100% of the maternity leave benefits (Labor Code, Art. 91(2)). It is also prohibited the dismissal of pregnant worker (Labor Code, Art. 91)

- **Assets:** There are no restrictions that do not concede women to have equal ownership rights to immovable property and Family Law, Arts. 1 and 29(3) grants spouses equal administrative authority over assets during marriage. The Constitution of Somalia, Art. 2(3) states that sons and daughters do not have equal rights to inherit assets from their parents; and widowed women do not have the same rights to inherit their spouses' assets as men (Constitution of Somalia, Art. 2(3)).
- **Access to credit and entrepreneurship:** Somalia does not have a law that prohibits discrimination in access to credit based on gender, but women do not have restrictions open a bank account, signing a contract, and register a business in the same way as men.
- **Marriage:** Somalia does not have a legal provision that requires a married woman to obey her husband; however, women cannot be called the head of household in the same way as men (Family Law, Art. 4(2)). The country does not have a specific legislation for addressing domestic violence; and women cannot obtain a judgment of divorce and do not have the same rights to remarry.

Income. Somalia has one of the lowest Gross National Income (GNI) per capita in the world and it is estimated at US\$400, however there is not recent data that shows the differences between men and women's income. Women face high levels of poverty and this is translated to 49% of female-headed households are poor, a number slightly lower than the 54% of male-headed households.

Labor Markets. Labor participation in Somalia is much lower for women than men; the Gender Inequality Index (2012) indicates that women's rate in the labor force participation is 19.1% while men's is 74.3%. According to ILO 2017 data, 87.9% of women in Somalia are self-employer, a higher number compared to men (70.2%). Also, the share of females employed in agriculture was 91.9% when compared to 84.8% of men. Somalia is a predominantly a pastoralist economy, and livestock is considered family's wealth and has traditionally been the property of men; women, on the other hand, manage the sale and exchange of livestock products such as milk and ghee and spend their earnings on household needs.

Education. Women in Somalia are less likely to be enrolled in any education level than men. For every 10 boys enrolled in primary school there are only 7 girls and for every 10 young men enrolled in secondary education there are only 4 young women. While other educational reports might show parity in Somalia, the gender disparity in rural Somalia is higher than urban areas. It is important to note that the enrollment rates of girls in displaced communities is higher than other rural areas.

Health. The maternal mortality ratio in Somalia is 732 per 100,000 live births, one of the highest in the world. This number is above the average in developing countries (231) and above the Sub-Saharan Africa (547) region. Moreover, the figure is more than 52 times the maternal mortality ratio in OCDE (14) or very high human development countries (15) according to the UNDP Human Development Report. In addition, female life expectancy at birth in Somalia is 3 years higher than the same figure for males (58 versus 55 years) (Table [31]).

Table [31]. Indicators of health by sex

Country	Maternal mortality ratio (deaths per 100,000 live births)	Female life expectancy at birth (years)	Male life expectancy at birth (years)
Somalia	732	58	55

Source: WBG. <http://hdl.handle.net/10986/31689>

Agency. Only 24% members in Somalia's parliament are women. The share of women in ministerial positions is even lower (7%).³²

Preliminary overview of relevant gender gaps:

- **Women in Somalia face some constraints when accessing maternal health services:** Most of the country has limited or no access to medical services and most of the remote communities rely on traditional birth attendants to handle deliveries. This scenario exacerbates women's vulnerabilities. As previously stated, evidence shows that about 30% of women who are pregnant are likely to be infected with SARS-CoV-2. Most of women who test positive for COVID-19, will not develop serious complications, however they can infect their newborns.³³ Health authorities recommend that pregnant women to receive the COVID-19 vaccine.
- **Women's autonomy is limited for accessing health services:** women lack of empowerment and tools to reach health care services. For instance, for emergency C-section, women have indicated that the prevailing norm is for women or providers to seek the consent of a male family member or clan leader, creating delays that can prove fatal to both the mother and the unborn child.³⁴
- **Reverse gender gap is found in COVID-19 cases in the country:** By January 2020, 4,726 COVID-19 cases have been recorded in the Somali territory. While 26% of the cases correspond to female patients, 74% of the cases were detected in men.

Suggested gender entry points and activities include: (i) develop a gender assessment (identify priority areas such as maternal health facilities and to capture sex-disaggregated data and information about better access to vaccine); (ii) stakeholder consultation.

3.2.9 Sri Lanka

Sri Lanka obtained a score of 0.68 in the 2020 WEF Global Gender Report Index, which puts the country in the 102th position. This suggests a substantial drop in comparison to the 2006 edition of the report, where the country obtained a score of 0.72 and ranked 13th.³⁵ The country has the third highest score among 7 countries in its region, South Asia.

The Constitution of Sri Lanka states that all persons are equal before the law and are entitled to the equal protection of the law and no citizen shall be discriminated against on the grounds of race, religion, language, caste, sex, political opinion, place of birth or any such grounds. The country has been investing and working in gender equality issues and regulations, including the design and implementation of the Bill of Women's Rights and developed a National Plan of Action for Women led by the Ministry of Women and Child Affairs (created initially in 1978 and consolidated in 2015 and acting as the women's rights machinery). The following are some of the laws and rules applicable for women in the country:

Legal framework: Sri Lanka has laws and regulations that protect women against discrimination and violence and recognize their rights. The publication Women, Business and

³² Source: WBG. <http://hdl.handle.net/10986/31689>

³³ <https://www.covid19treatmentguidelines.nih.gov/special-populations/pregnancy/>

³⁴ <https://conflictandhealth.biomedcentral.com/articles/10.1186/s13031-019-0241-x>

³⁵ Many countries were added between the two vintages; thus, the drop in the ranking may overrepresent the change in the performance of the country.

the Law 2021 (World Bank, 2021) indicates that Sri Lanka scores 65.6 out of 100. It means the country is higher than the regional average achieved in South Asia (63.7).

- **Employment and payment:** The country has a law that does not restrict women to work in in the same fields as men. Women at workplaces are protected by a legislation that prevents sexual harassment in employment (Penal Code, Sec. 345(1)) and sexual harassment at work is considered a crime (Penal Code, Sec. 345). Sri Lank lacks a law that prohibits discrimination in employment based on gender. and women are not able to work in some industrial jobs in the same way as men (including mining (Mines and Mineral Act No. 33, Sec. 55; Employment of Females in Mines Ordinance No.13, Sec. 2) and factories (Factories Ordinance No. 45 of 1942, Secs. 25, 67 and 67A (2)) are not allowed, while construction. Agriculture, energy, water, transportation, and other fields are allowed). Women are not prevented from working in jobs that are considered hazardous. In regards of payment, Sri Lankan women are not protected by a law that mandates equal remuneration for equal value.
- **Maternity Rights:** Mothers in Somalia have 84 days of paid time during a period between pregnancy and after labor (Shop and Office Employees Act, Sec. 18B; Shop and Office Employees (Regulation of Employment and Remuneration) (Amendment) Act, No. 14 of 2018). The country lacks parental leave, and fathers cannot enjoy a paid leave after labor. The government does not administer 100% of the maternity leave benefits (Shop and Office Employees Act, Sec. 18B; Shop and Office Employees (Regulation of Employment and Remuneration) (Amendment) Act, No. 14 of 2018)
- **Assets:** The Married Women's Property Ordinance, Sec. 5 concedes women to have equal ownership rights to immovable property and grants spouses equal administrative authority over assets during marriage. The Matrimonial Rights and Inheritance Ordinance, Sec. 24 and Sec 22 states that sons and daughters have equal rights to inherit assets from their parents; and widowed women have the same rights to inherit their spouses' assets as men respectively.
- **Access to credit and entrepreneurship:** Sri Lanka does not have a law that prohibits discrimination in access to credit based on gender, but women do not have restrictions open a bank account, signing a contract, and register a business in the same way as men.
- **Marriage:** Sri Lanka does not have a legal provision that requires a married woman to obey her husband. Women can be called the head of household in the same way as men. The country has a specific legislation for addressing domestic violence (Prevention of Domestic Violence Act, No. 34 of 2005); and women can obtain a judgment of divorce and have the same rights to remarry as men.

Income. According to the UNDP's estimates, the female Gross National Income (GNI) per capita in Sri Lanka amounts to 6,766 in 2011 PPP \$, which corresponds to 40% of the male GNI per capita (16,852 2011 PPP \$).

Table [32]. Female and male gross national income per capita (2011 ppp terms)

Country	Estimated gross national income per capita (2011 ppp terms) ¹		
	Female	Male	Female as % of male
Sri Lanka	6,766	16,852	40

Source: UNDP <http://hdr.undp.org/en/indicators/123506#a>. 1/ Definition: Derived from the ratio of female to male wages, female and male shares of economically active population and gross national income (in 2011 purchasing power parity terms).

Labor Markets. Labor participation in Sri Lanka is much lower for women than men; From the whole labor force, women represent only 35.7%. The share is also lower when looking at the share of female employees in the industry (34.0%), manufacturing (48.0%), electricity gas, steam, and air conditioning, supply (11.4%) or professional, scientific and technical activities 33.7% (Table [33]).

Table [33]. Share of employment by activities and sex (% , 2018 unless stated differently)

Country	Sex	Industry	Manufacturing	Electricity; gas, steam and air conditioning supply	Professional , scientific and technical activities	Total	Women are able to work in the same industries as men
Sri Lanka	Female	34.0	48.0	11.4	33.7	35.7	No
	Male	66.0	52.0	88.6	66.3	64.3	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Financial inclusion. Men in Sri Lanka present slightly higher levels of financial inclusion than women when comparing the ownership of an account at a financial institution (74 versus 73%), a mobile account (1 versus 4%) or saving money (42 versus 45%) (Table [34]). In addition, the share of women aged 15 or above in the country with either an account at financial institution or a mobile money-service provider is 73%, well above the average in developing countries (58.3%), or South Asia (65.1%).

Table [34]. Indicators of financial inclusion by sex (2017)

Country	Account at a financial institution (% age 15+)		Mobile account (% age 15+)		Saved any money last year (% age 15+)		Women with account at financial inst. or with mobile money-service provider (% age 15+)
	Female	Male	Female	Male	Female	Male	
Sri Lanka	73%	74%	1%	4%	42%	45%	73%

Source: World Bank Little Book of Gender 2019

Education. Women in Sri Lanka are more likely to be enrolled in secondary or tertiary education than men. Of all women in the corresponding education ages, 90.4% are enrolled in secondary education and 23.4% in tertiary education. In contrast, from all men in the same age groups, only 87.7% are enrolled in secondary education, and 15.8% are enrolled in tertiary education. However, probably due to differences in attainment, women report a lower average number of schooling years (10.5 versus 11.6). In addition, women amount to 40% of all STEM graduates (Table [35]).

Table [35]. Indicators of education by sex

Country	Sex	Enrolment in primary education (%)	Enrolment in secondary education (%)	Enrolment in tertiary education (%)	Mean years of schooling (years)	Female share of tertiary graduates from STEM (%)
Sri Lanka	Female	96.4	90.4	23.4	10.5	40
	Male	98.1	87.7	15.8	11.6	

Source: labor for shares from ILO https://www.ilo.org/ilostat-files/Documents/Bulk_ilostat_en.html# and labour restrictions from World Bank Little Book of Gender 2019

Health. The maternal mortality ratio in Sri Lanka is 30 per 100,000 live births, which ranks 71st among 182 countries included in the Human Development Report. This figure is below the average in developing countries (231) and well below the average in its region, South Asia

(176). However, more than 2 times the maternal mortality ratio in OCDE (14) or Very high human development countries (15) according to the UNDP Human Development Report. In addition, female life expectancy at birth in Sri Lanka is almost 7 years higher than the same figure for males (80.1 versus 73.4 years) (Table [36]).

Table [36]. Indicators of health by sex

Country	Maternal mortality ratio (deaths per 100,000 live births)	Maternal mortality ratio (ranking across countries)	Female life expectancy at birth (years)	Male life expectancy at birth (years)
Sri Lanka	30	71	80.1	73.4

Source: UNDP. <http://hdr.undp.org/en/data>

Note: rankings among 182 countries with data in UNDP report

Agency. Only one out of almost 20 members in Sri Lanka's parliament are women (5.3%). Similarly, the share of women in ministerial positions is only 6.7%).

Gender gaps associated with the current project in Sri Lanka

Cooling-related activities or sectors targeted: Energy efficient chillers and other energy efficiency and renewable energy measures in the commercial and industrial building sector through Energy Service Companies (ESCOs). Policy and regulatory, as well as capacity buildings activities are also planned.

Preliminary overview of relevant gender gaps

- **Sri Lanka has made good progress in achieving gender equity in health and education, brought about through giving women their right to vote, free education and health services.** Area that still needs attention is women's engagement in labor, their economic empowerment, and their involvement in governance mechanisms, especially at higher levels.
- **Women are accessing fewer basic services than men.** While there is a high level of basic literacy among women, in practice traditional patriarchal attitudes limit the interaction of a vast majority of women in community decision making and in commerce.
- **For women, labor force participation rates are low; they are paid less than men, a difference not explained by their productive capacity and holds true for all sectors.** A related area for improvement is women's empowerment.

Suggested gender entry points and activities include: (i) develop (or review/enhance/) gender assessment; (ii) carry out consultations, workshops, gender assessment's presentations; (iii) engage policy dialogues to raise awareness about the linkages between cooling and gender; (iv) capacity building/awareness about gender equality; (v)) develop agender action plan that includes responsibilities, activities and budgets; (vi) develop and implement a communication campaign.

4 Sector Gender-Specific Gaps Summary

The following is a matrix that summarizes **possible entry points** that could be addressed by country after producing a gender assessment. Countries face multiple dimensions of gender inequalities and its possible to contribute to addressing them through cooling-related sectors.

	Economic Empowerment				Access to services				Voice and Agency	
	Low labor force participation (all relevant sub-sectors)	Low Income Levels	Low productivity	Low entrepreneurship	Less Access to Energy	Low access to financial resources	Low access to Education (basic and STEM fields)	Low access/ ownership to inputs and proper equipment	Less decision-making power	Gender-Based Violence
Countries										
Bangladesh	Yes			Yes		Yes			Yes	
El Salvador	Yes									Yes
Kenya		Yes			Yes	Yes				
Malawi			Yes	Yes			Yes			
North Macedonia	Yes						Yes			
Panama						Yes	Yes			
Sao Tome e Principe	Yes									Yes
Somalia			Yes							Yes
Sri Lanka	Yes	Yes					Yes		Yes	

Potential Gender Entry Points by Theme

The following is a list of **examples of possible interventions** that could be proposed, designed, implemented and monitored to address gender gaps found **in the countries** to be included in individual gender action plans as relevant.

A. Foster Women's Economic Empowerment

- Implement employment initiatives for women in the sector to provide labor opportunities for both men and women. Carrying out consultations on gender employment issues and opportunities could engage with employers.
- Create and implement Human Resources reforms or staff policies; hiring women for consumer or public facing marketing or advocacy aspects.
- Incentivize data collection on male versus female participation in firm ownership and leadership at the MSME level.
- Explore ways to best incorporate the needs of women and youth in project activities as a way of enhancing their livelihood.
- Increased representation of women in the engineering/technical professions with a focus on energy efficiency
- Track data on female staff hired under service contracts for technical and design work under the Project
- Provide targeted training program for women and increasing awareness about employment opportunities under the Project investments.
- Create jobs and develop skills for women and men in the cooling appliance sector.
- Provide opportunities in most productive sector to reduce the incidence of GBV.

B. Increase Women's Access to services

- Design and implement targeted approaches to reach female-headed households or female customers or business to expand affordable energy access to women.
- Offer financial products and services that address women's needs. Approaches might include e.g. consumer finance mechanisms include micro-consignment, revolving loan funds, and flexible repayment plans. Use of more concessional forms of financing or more intensive technical assistance to enhance gender inclusivity in financial services and products
- Foster the financial inclusion of women throughout the support provided to local banks.
- Ensure energy efficient cooling investment also cover health or financial facilities that provide services to women and address their needs.
- Produce assessments on what barriers and opportunities there are to closing gender gaps around access to services, education, financial products and other services for women.
- Produce assessments of needs of women and men around cooling e.g. around transport or storage may differ and how livelihoods and income could be enhanced especially for women who may not be capturing potential catch to market value-additions through cooling.
- Produce assessments of extent to which reduction of waste and loss impacts the national food security, particularly for women and children.
- Gather information for understanding different motivations for adopting cooling technologies and barriers to accessing finance to purchase products.
- Foster the access of women to health facilities and services (including the services to mitigate the effects of the COVID-19 global pandemic and the access to its vaccine).

Improve leadership and decision-making by women

- Include a baseline survey that looks at gender differences in energy user behaviors, perceptions of energy efficiency, and awareness and attitudes on energy efficiency among employees and users of buildings.
- Design and implement activities to address any potential gender gaps in the area of energy efficiency, such as conducting awareness raising campaigns on energy efficiency, considering the engagement of both men and women and tailor messages based on gender-differentiated concerns, priorities, and perceptions toward energy efficiency in buildings.
- Assess studies to understand different motivations for adopting cooling technologies and barriers to accessing finance to purchase products
- Design communication campaign to change consumer patterns and behaviors.

5 Strategic Approach

5.1 Facility Level:

The Facility will support activities that contribute to closing gender gaps across sub-projects e.g. data collection, access to finance and skills development activities etc. In addition, at every step of the project cycle, the Gender and Energy program of the World Bank Energy Sector Management Assistance Program (ESMAP) will be available to provide technical assistance to the project team and the client country.

The Facility will encourage the recruitment of a balanced female-to-male employment ratio within the Project Management Unit (PMU) team to the extent possible. The standard World Bank implementation process will be put in place and include a PMU in each country, responsible for implementing and monitoring each project. If needed, PMU staff members including the Project Manager and the Environment and Social Protection Specialists may receive capacity-building trainings. For each project supported under the Cooling Facility, a social/gender specialist is expected to be included in each PMU team. In addition, the ESMAP Gender and Energy Program will provide complementary support to the Facility projects in the form of technical advice for the Bank's implementation/supervision of Facility projects.

This will provide support to consider identified gaps (at all gender dimensions) throughout the project cycle. **Obtaining stakeholders buy-in will be key in ensuring that interventions to promote gender equality activities are effectively implemented with measurable outcomes.** Institutional and behavioral change are a long-time process that will require sustained support of all relevant stakeholders (public and private) throughout the program cycle. Countries' adherence to international agreements and national legal frameworks on gender -equality, are a good starting point to strengthen a government's support for gender-interventions. The following strategy will be followed to increase stakeholders' commitments, to the extent possible:

- Engage in policy-dialogue with client countries through dedicated meetings, to build awareness and sensitivity about the benefits of promoting women's economic empowerment throughout interventions that foster their employment and their entrepreneurship;
- Closely align proposed activities with each country's national priorities and gender legal framework, such as community development or female employment;

- Identify an institutional gender-focal point to serve as interlocutor during project implementation, if appropriate;
- Offer technical support to help investing in workforce diversification, building on existing internal gender strategies when available. This support may take the form of workshops on gender-related concepts with high-level stakeholders, or a review of training plans to better integrate female employees.
- Promote the access of the COVID-19 vaccine in health facilities that address women's needs.

During projects design, the private sector can be involved in financing interventions to i) create income-generating opportunities for women, ii) implement mechanisms to prevent and respond to associated incidents or risks, iii) increase rising awareness rates among female-headed households, for instance through capacity-building workshop, communication or social media campaigns, etc.

5.2 Project Level:

Considering the diversity of pre-identified projects to be supported by the Facility, as well as the fact that they are at different stages of maturity, there is no single model for all projects.

- **Data collection to document the baseline situation and identify gaps.** Baseline data will be collected to set targets and measure progress during implementation, it will also support existing efforts where baseline sex-disaggregated data are weak. Research will build upon existing country gender diagnostics and action plan when available, to enable the understanding and consideration of national priorities. To support project design and implementation, stakeholders' consultations will involve female participants, to understand men and women differentiated needs. In addition, initial assessments will include an analysis of the differentiated impact on men and women, through the involvement of gender and energy experts (Environmental and Social Impact Assessment, Resettlement Action Plans etc.). Gaps could be identified but not limited to female labor issue or inadequate access to finance for the cooling sector, and design mitigation measures that minimize gender gaps such as enhanced productive uses of energy, enhanced knowledge and skills of small and micro-business, households and farmers.
- **Identifying and implementing relevant actions.** Based on the initial diagnosis and inclusive stakeholders' consultations, targeted activities will be identified to address key gender gaps. These may include (i) actions to promote female employment (including capacity building and programs for women to gain hands-on experience in the sector such as support for school-to-work transition), (ii) financing local communities projects empowering women in the area of implementation of the projects (such as activities generating revenues), (iii) implementation of prevention measures to mitigate associated incidents or risks in the area of implementation of the projects and (iv) facilitating energy access for female-headed households (including capacity-building workshops to maintain and develop energy services).
- **M&E and Knowledge management.** In 2017, the WBG started implementing the "gender-tag" mechanisms to identify projects that effectively promote equality between men and women and are aligned with the WBG gender strategy. The Cooling Facility will adopt a gender perspective and aim to close key gaps in its supported projects. Results will be monitored throughout the project and be used for real-time adjustments during implementation. To expand the knowledge base, projects will build on lessons

learnt from prior projects as well as regional and country base knowledge production and exchange learning opportunities.

- **Engaging stakeholders** with specific needs and risks of gender inequality in meaningful consultation, information disclosure and responsive grievance redress throughout the project life (as set out in the Stakeholder Engagement Plan). The Cooling Facility will support carrying out activities to strengthen women's participation and voice during consultations. Consultations and public meetings with stakeholders will be conducted throughout the project cycle in a participatory manner and with a special focus on engaging women. This will help gaining knowledge about community concerns during project preparation and help monitor impact and undertake real-time correction during implementation. Special attention will be paid to using local dialect, cultural concerns when relevant and including women from minority ethnic groups and other socially vulnerable groups.

5.3 *Expected Gender Co-Benefits Resulting from projects:*

The Facility includes projects in different countries that are at different stages of development. Three of them are at early stages of development (Bangladesh, Panama and Sri Lanka); while the other cooling-related projects supported by the Facility will build on a World Bank-approved parent project (and be processed as "additional financing" to the parent project), which already includes a framework for addressing gender³⁶. They are summarized below (the expected gender co-benefits of the early stage projects will be defined during project preparation, along with gender-related indicators in line with the WB Gender Strategy mentioned above, and will be included in the respective Project Appraisal Document):

Kenya: The parent project intends to contribute to closing the relevant gender gaps in Kenya, particularly in access to electricity, by (a) ensuring women's access to solar products and cleaner and more efficient cookstoves; (b) increasing women's awareness about solar energy use and the productive uses of solar power; (c) ensuring women's participation in decision making around the installation and management of the solar systems; (d) providing women with employment opportunities in marketing and installation of solar products; and (e) mitigating potential risks such as displacement or gender-based violence (GBV) that women might face due to project implementation. In addition, the project M&E system will adopt and integrate several results indicators to monitor and assess both progress in implementing gender-related activities and the project benefits for women and men.

Malawi: The parent project is expected to generate significant benefits for Malawi's economy. The proposed project will contribute to generating positive fiscal returns by increasing the tax and government revenues the government receives from the registration of new businesses and the expansion of existing businesses and by leveraging private sector financing, thereby reducing the fiscal burden associated with government-funded SME finance programs. A critical expected outcome of this project is to encourage the participation of youth and women-owned businesses. Hence, firms will be selected into Stage 1 of the program based on experimental design and stratified by age and gender. In addition, training programs will be customized and will be informed by evidence (firm survey) to meet the needs of the youth and women-led firms, with an emphasis on personal initiative skills training. Some of the other topics that may be covered in the life skills training modules are likely to include a combination of socioemotional skills training and information related to gender-based violence (GBV),

³⁶ The exception is El Salvador, where the inclusion of gender elements into the design of interventions will be done in the context of the Additional Financing project, including meeting the Bank's Gender-Tag corporate commitment (an exception was granted for the Parent project given its emergency nature).

returns to education, job readiness, financial literacy, and cross-cutting topics, such as nutrition and civic engagement.

North Macedonia: The parent project will address an employment gap in the energy sector through providing women a targeted training program and increasing awareness about employment opportunities under the project investments. The project will engage with industry associations and technical universities to increase awareness and information about the project's offerings, in particular employment opportunities created under technical and design work related to project investments. Furthermore, the project will offer skills training to women, which will be based on skills requirements needed for technical and design work under the project. The Project will monitor the impact through the following indicators: (i) number of women attended skills training activities and certifications obtained and (ii) percentage of female staff hired under service contracts for technical and design work

Sao Tome and Principe: In the context of COVID-19, the parent project will continue to support communication campaigns, including messages regarding appropriate care for sick family members, resources available, coping strategies, etc. to minimize the health and psychosocial impacts of COVID-19 on caregivers and to disseminate targeted community messages on GBV. The project will ensure that guidelines for pandemic preparedness, response, and planning consider the unique needs of the most vulnerable populations, as well as gendered roles and responsibilities. As such, the project will ensure capture of vaccine coverage data disaggregated by gender to ensure coverage is equitable by gender. Special attention will be paid to capturing information on the population of female health workers and volunteers vaccinated. Furthermore, the project will ensure that all training opportunities (surveillance, epidemiology and laboratory diagnostics) will be gender sensitive and the outputs of such trainings will be tracked by gender disaggregated indicators being revised.

Somalia: The parent project emphasizes addressing persistent gender challenges and monitoring the progress in the gender aspects of the project. First, given high gender disparities in Somalia, as previously noted, the Project will support the implementation of a package of GBV services (awareness raising, case identification, counselling, and management), with service providers to track the status of GBV service utilization by women and any challenges that arise. As the Project is expected to be gender-tagged, this will be tracked through the Project's intermediate results indicator (IRI) focused on tracking the number of GBV services delivered to facilitate course correction. The Project will also closely monitor another IRI on the number of outpatient department visit, which is disaggregated to track the percentage of female outpatient department visits, to monitor whether the Project contributes to an increase in women's service utilization. The Project Grievance Redress Mechanism (GRM) will also be sensitized to receive and respond to sexual abuse, harassment, exploitation, and disproportionate use of force among females.

6 Gender Action Plan

The Gender Action Plan proposed below is indicative at this stage and may be further amended and adapted according to knowledge gathered during the project inception phase, to reflect local and project context.

This gender action plan will cover all activities within the facility and are intended to close gender gaps women face at different levels. Some of the areas on which the facility may focus (but will not be limit to) are:

- **Entrepreneurship:** Women entrepreneurs often lack information about access to finance options in the country. In addition, development of enterprises requires

technical assistance support tailored to women's needs. Partnering with business networks, such as female exporters or associations under chambers of commerce, is key to creating awareness about credit lines among women entrepreneurs. From a consumer perspective they will also have different needs and wants in terms of cooling technology products and services and the sectors they operate in (e.g. agriculture or transport).

- **Financial services**—not only accounts, but payments, savings, and credit—give women some basic tools they need to enhance their livelihoods and economic status. Under the current ecosystem, bank accounts are key to process payments for cooling technologies, and strategies need to be developed to facilitate women's participation to the financial sector and cooling market. The use of digital payments can further reduce access barriers for women. Affordability is a key part of this focus.
- **Jobs and Skills Development:** Job creation and livelihood opportunities will be mapped out as part as part job creation and skills development aspects of the Facility. For instance, having women sales agents and local suppliers will create opportunities in terms of jobs or looking at labor force participation gaps between women and men in the e.g. cooling sector and implementing entities.
- **Consumer/stakeholder Engagement:** Growing evidence points to significant differences in the needs and wants of women versus men as e.g. consumers and entrepreneurs, as well as caregivers. For example, differences can be observed in how they use energy on a daily basis, how much they are willing and able to adopt cooling technology and measures e.g. in agriculture sector), and how they respond to campaigns marketing products and services. Differences have also been observed in the context of caregiving responsibilities in communities and families. It is therefore essential to tailor any efforts toward consumers specific behaviors of men and women. This points to a need for communications strategies in these countries to look beyond written materials as they transmit information related to benefits of cooling-related services, consumption patterns, and so forth. It could also be advisable to leverage women's groups.
- **COVID-19 response and health measures:** Women often lack empowerment and autonomy to access services, sometimes related to health and decision making. Women must not be excluded from accessing services, as they can act as agents of change. Families and communities can benefit from informed actions taken by women to counter the negative effects of the global pandemic.
- **Data and information:** collecting sex-disaggregated data and gender-specific information help to assess the projects' results and impacts on gender equality and also help to scale-up refine activities for future interventions.

Available Funding: Funding under Component 1 of the Facility will be dedicated to support gender-relevant activities³⁷ in the countries in the Facility, taking into account the particular context and needs of each country/project and in line with the Facility's Gender Action Plan. This includes at least USD200,000 per country (i.e., USD 1.8 million at the Facility-level) to support the development of a gender assessment. This gender assessment will identify relevant gender gaps in each country including barriers women have to access to services associated with cooling such as jobs, finance, assets, etc., and countries will have the opportunity to identify priorities to develop specific activities that will contribute to close gender gaps.

Beneficiaries: Men and women are the Facility's beneficiaries; all activities are gender informed and created in coordination with the IBRD/IDA parent project (where applicable), which will include gender mainstreaming activities. Each of the projects are pending to finalize their design and the details about specific beneficiaries. For five of the Facility's cooling

³⁷ This funding complements WB-funding dedicated to gender-related activities under each project.

projects, the parent projects are already aligned with the gender-tag WB's corporate commitments (see footnote 2). WB's projects that are investing in gender equality issues are focusing in three dimensions: human endowments (health and education); economic opportunities (participation in economic activities and access and to control ok key productive assets); and voice an agency (freedom from violence, political participation etc.). The World Bank Gender Strategy (which applies to the Cooling Facility's projects) focuses on addressing socially determined gender inequalities, which are rooted in behaviors, institutions, and market forces. Women and men are heterogenous groups (that are also stratified by race, ethnicity, disability, sexual orientation, and gender identity and overlap with income level, geographic location, migratory status, etc.). The WB understands that most inequalities based on gender have put females at a disadvantage. Therefore, transformative actions that aim to close gender gaps are planning to target and improve women's status by working with relevant stakeholders to empower women.

Activities	Indicators and Targets	Timeline	Responsibilities
Impact Statement: The Cooling Facility will follow a systematic approach and provide support to individual projects interventions to contribute to closing gender gaps with a focus on data collection, policy dialogue, capacity building and training and skills development activities. The expected impact includes enhanced female economic participation in cooling related sectors, increased women's access to services (energy, cooling, financial), and enhanced productivity and income for female-led businesses.			
Outcome Statement: The Facility will focus on 4 key areas a) generating sector specific country-level data collection that looks at gaps between women and men; b) policy dialogue activities to help recognize and embed actions to close gender gaps; c) project and program support to promote gender equality in the projects and program that are supported by the facility and d) exchange and learning opportunities.			
Implementation of the Facility	Number of direct beneficiaries: Total: 8.1 million Number of women: 4. 1 million (51%)	At the end of the program.	Executing Entities with the support of the WB team. Gender Specialist will be responsible to implement and report on the activities
Output(s) Statement: Output 1: Gender Analysis: Sector specific country-level data collection that looks at gaps between women and men Men and women have different motivations and priorities about the use of energy services and adoption of energy technologies and products and make decisions differently. Collecting baseline information about the sex of prospective beneficiaries and respondents (e.g., by household) in surveys or other information-gathering activities can provide useful input to policy and program design. In addition, a focus on gender gaps with regards to access to finance, entrepreneurship and employment, as well as health services will be key			
To contribute to close a knowledge gender gap for the cooling-related sectors in each country: by conducting a gender-sensitive gender assessment that describes different needs, cultural constraints, and perspectives of women and men in terms of access to cooling-related services/products, access to finance and employment, health services, including those associated directly or indirectly to immunization (including COVID-19) . The findings of the gender assessments will be presented and discussed at country-level multi-stakeholder events (e.g. consultation, workshops) and will help develop recommendations on	Number of gender assessments developed or	On a rolling basis, at the time of project design Gender assessments: (2023: 3; 2024: 3; 2025: 3)	Executing Entities with the support of the WB team. Gender Specialist will be responsible to oversee implementation and report on the activities.

the design of the Facility projects and help generate gender and cooling insights more generally at the Facility level that may be helpful for .	improved. Baseline: 0³⁸. Target: 9³⁹. Number of multi-stakeholder event (e.g. consultation, workshop) held, including awareness raising on gender equality issues. Baseline: 0, Target: 9.	Multistakeholder events: (2023: 3; 2024: 3; 2025: 3)	
Output 2: Policy dialogue activities to recognize and embed actions to close gender gaps In many cases, policy targets for closing gaps between men and women can be addressed through actions in cooling-related projects. For example, national or sector targets for enhancing engagement with women's groups and enhancing the voice and agency of female citizens in service delivery or development programs could be met through relevant approaches in project design. Looking at the interplay between policy and opportunities in the design of projects and supported interventions is therefore key. Activities under this output will include			
To identify barriers, constraints, and opportunities to foster the economic participation of women in the cooling-related sectors at the country level by producing and disseminating insights and opportunities from regulatory and policy framework assessments To train and create capacity within the PMU through workshops and trainings targeted at the PMU staff on gender and cooling-related issues) To create specific gender action plans with relevant gathered information that promote equal access of women and men to opportunities connected to	Number of relevant policy dialogue engaged with client countries to build awareness and sensitivity, on gender equality issues, including benefits of promoting women's economic	By the Facility's end date Policy dialogues: (2025: 3; 2026: 3)	Executing Entities with the support of the WB Team Gender Specialist will be responsible to oversee implementation and report on the activities.

³⁸ Countries have to develop a gender assessment at an individual basis.

³⁹ The extent of assessment may vary by country and may be more limited, especially, in the context of countries where the Cooling Facility is supporting COVID-19 emergency response operations and countries in a FCV (fragile, conflict and violence) situations (Somalia).

operations in terms of direct technical and operations skills and employment or bids/contracts etc.	empowerment, removal of barriers to female employment in cooling-related sectors and enhancing gender considerations in service delivery. Baseline: 0 Target: 6⁴⁰. • Number of capacity-building workshops organized with PMU staff. Baseline: 0 Target: 6⁴¹ • Number of beneficiaries: • Share of women in capacity building activities: Baseline 0; Target: country dependent • Share of men in capacity building activities: Baseline 0; Target: country dependent • Number of gender action plans developed: Baseline: 0, Target: 6⁴²	Workshops (2025: 3; 2026: 3) Gender action plans (2025: 3; 2026: 3)	
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⁴⁰ Bangladesh, Kenya, Malawi, North Macedonia, Panama, and Sri Lanka

⁴¹ IDEM

⁴² IDEM

Output 3: Project support to promote gender equality in the projects that are supported by the Facility. At the project level, gender support will be provided (in coordination with/led by the World Bank staff and/or carried out by consultants) at the design and implementation phases. The proposed general approach that will be followed is outlined below.			
To assist projects to meet the WB's corporate commitments of the gender tag by including an analysis, actions and indicators that aim to close gender gaps in the cooling-related sectors and are aligned with vertical logic of the project: To assess barriers and constraints: by identifying barriers that need to be addressed to promote the participation of women e.g. from national policies and programs that directly address female labour issue in the cooling-related sectors, down to main entity level (e.g., ministry, utility, regional bank) HR policies, recruitment, awareness raising and promotion processes. To engage with key stakeholders: by organizing events/consultation activities. Relevant stakeholders (e.g. policy makers, private/government, institutions, NGOs, schools, and universities) aim to closing gender gaps (including in the COVID-19 vaccination process and in reduction of GBV). To identify and prioritize actions that contribute to close gender gaps: Possible actions include e.g. enhance the productive uses of energy in agricultural, industrial and service sectors by enhancing the knowledge and skills of small and micro-business, households and farmers on how to use newfound access for profitable enterprise or addressing the barriers that need to be addressed e.g. from national policies and programs that directly address female labor participation issues in the cooling and related sector sectors (including the health sector and its cold chain for vaccines and medicine).	- Number of projects that will meet the WB Gender Tag (Baseline: 44% Target 60% of total number of projects)⁴³ - Number of information sessions held with female-led firms⁴⁴ to share information on opportunities in contracts/tenders in projects. Baseline: 0 Target: 2.⁴⁵ - Number of assessments conducted to determine access to finance barriers for female enterprises and for product adoption at the	On a rolling basis, as projects and programs come through Gender tag: at the end of the program (monitored on yearly basis) Informational sessions (2025: 1; 2026: 1) Assessments (2025: 1; 2026: 2) Campaigns: (2024: 3; 2025: 3; 2026: 3)	Executing Entities with the support of the WB Team Gender Specialist will be responsible to oversee implementation and report on the activities.

⁴³ Gender Tag (see page 5 for an explanation) is a WBG's corporate requirement as a percentage of lending operations. Baseline is set as the number of gender tagged Energy projects during the fiscal year 18 (July 2017 to June 2018)

⁴⁴ A business qualifies as a woman-owned business if it meets the following criteria: (A) $\geq 51\%$ owned by woman/women; OR (B) $\geq 20\%$ owned by woman/women; AND (i) has ≥ 1 woman as CEO/COO/President/Vice President; AND (ii) has $\geq 30\%$ of the board of directors composed of women, where a board exists (IFC)

⁴⁵ Kenya, Malawi

	consumer level Baseline: 0, Target: 3. - Number of companies/institutions benefiting from capacity building and training opportunities provided by the Facility on women's employment issues. Baseline: 0 Target: 6⁴⁶. - Number of information campaigns, outreach or behavior change initiatives conducted that tailor messages that target both men and women. Baseline: 0, Target: 9⁴⁷.		
Output 4: Exchange and learning opportunities through the Cooling Facility Enhance knowledge and learning of stakeholders by providing knowledge building opportunities to integrate a focus on gender equality in cooling and investment, through:			
To raise awareness and leverage the gender equality dialogue at the country level by the organization of fora where gender will be the focus of panels, workshops and discussions. Events will be organized that will invite beneficiaries,	Number of panels and discussions focusing on gender held through Facility	At the time of the delivery of workshops, on a rolling basis	Executing Entity with the support of the WB Team Gender Specialist will be

⁴⁶ Bangladesh, Kenya, Malawi, North Macedonia, Panama and Sri Lanka

⁴⁷ The extent and type of information campaigns, outreach or behavior change initiative may vary by country and may be more limited, especially, in the context of countries where the Cooling Facility is supporting COVID-19 emergency response operations and countries in a FCV (fragile, conflict and violence) situations (Somalia).

potential future beneficiaries, key country representatives and partners under the Facility to share best-practices and agreed commitments on gender under projects. To strengthen capacity and technical skills at the country level by the provision of gender focused capacity building workshops, To contribute to create global knowledge on gender equality in context of cooling-related projects by developing case studies from insights from Facility projects.	support. Baseline: 0, Target: 4. - Number capacity building training that includes gender and cooling/EE entry points. Baseline: 0, Target: 6. Number of beneficiaries: - Share of women in capacity building activities: Baseline 0; Target: country dependent - Share of men in capacity building activities: Baseline 0; Target: country dependent - Number of case studies produced on gender by the Facility. Baseline: 0, Target: 3.	Panels: (2025: 2; 2026: 2) Trainings: (2025: 3; 2026: 3) Case studies: (2026: 3)	responsible to oversee implementation and report on the activities.
Output 5: Facilitate benefit access of cooling, refrigeration, cold chains, logistics, and green solutions: At the project level, each team will work on activities that aim to close gender gaps with a mainstreaming approach. The country-specific gender gaps have been or will be assessed and incorporated within each proposal's vertical logic considering their context and resources. Each project will benefit men and women, and each country/project will be responsible for monitoring specific indicators and activities. The following approach would be adopted and tailored according to each project context. Selection of specific activities/interventions will be made during project preparation .			
Improve access for men and women to new cooling, refrigeration, cold chains, logistics and green solutions by designing interventions that aim to	Number of initiatives/programs	At the time of the delivery of	Executing Entities with the support of the WB Team Gender

reduce a gender gap in employment and entrepreneurship and/or access to cooling-related service.	that facilitate women's access to new solutions. Baseline: 0; Target: 6⁴⁸ • Number of beneficiaries: • Number and percentage of female beneficiaries. Baseline: 0; Target: country dependent. • Number and percentage of male beneficiaries. Baseline: 0; Target: country dependent. • Number of new jobs for women: Baseline 0; target: country dependent. • Number of women-owned businesses benefited. Baseline: Target: country-dependent	workshops, on a rolling basis	Specialist will be responsible to oversee implementation and report on the activities.
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Additional Information:

Knowledge- Links between Gender and Energy and other Sectors

⁴⁸ Bangladesh, Kenya, Malawi, North Macedonia, Panama and Sri Lanka

The ESMAP Gender and Energy Program has produced various reports and products over the years on issues of gender and energy across sectors based on case studies and best practices. A brief summary of reports and guidance is summarized below. In addition, the Bank has also produced reports and guidance related to gender in the context of different sectors. A number of these are listed below.

Resource	Summary
Cooling for All and Gender – Towards Inclusive, Sustainable Cooling Solutions (SEforALL, 2021)	This knowledge brief is the first such analysis of the gender-based impacts of a lack of access to cooling. It examines the gender-related challenges that should be considered, analyzed and addressed to ensure cooling interventions and finance acknowledge gender-differentiated impacts and adapt to maximize equitable access. This brief offers a series of recommended steps to address these challenges while pursuing universal sustainable cooling and gender equality. Governments, development finance institutions and nongovernmental organizations should raise awareness regarding the gender-based impacts of a lack of access to cooling, bolster policies for protections, drive research to understand gender disparities and increase investments to gender-transformative solutions that also deliver sustainable cooling.
Integrating Behavior Change in Energy Efficiency Programs in Developing Countries: A Practitioner’s Guide (ESMAP 2020)	The purpose of this guide is to provide practitioners in the field of energy efficiency with an overview of the opportunities available to integrate behavior change strategies in demand-side energy efficiency programs. The guide offers relevant suggestions, references, and examples, with a focus on communication and gender equality considerations. While behavior change principles can be applied to energy efficiency programs in various sectors (public, commercial, industrial, residential, and transportation), this guide centers particularly on the residential sector.
Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers (ESMAP 2019)	Approximately 1.2 billion people will need to gain access to electricity to achieve universal access by 2030. The only way to achieve this is through a combination of the main grid extension, mini grids, and off-grid solar. This report shows that about half a billion people can be cost-effectively provided with electricity through mini grids. The combination of falling costs, dramatic increase in the quality of service, and favorable enabling environments have made modern mini grids a scalable option to complement grid extension and solar home systems. The report is the most comprehensive study on mini grids to date. It takes stock of the global mini grid market and industry; analyzes in detail the solar-hybrid mini grid costs and technological innovations; shows the importance of introducing income-generating machinery and micro-finance early in the planning process; and provides policy makers, investors and developers with insights on how mini grids can be scaled up.

Gender Equality in the Geothermal Energy Sector-Road to Sustainability (ESMAP 2019)	The report is a primer on advancing gender equality in the geothermal energy sector. Based on good practices and lessons learned, it introduces ways that geothermal projects can mitigate risks and pursue opportunities to address gender gaps within the project cycle. It outlines the risks and opportunities associated with (i) changes in land and natural resource use, (ii) changes to employment and economic patterns, and (iii) changes to the environment and health. Beyond mapping risks and opportunities in another energy sector, the report makes the case for focusing on the gaps between men and women from a project's outset.
Key Entry Points for Gender Equity in Energy Access, Energy Policy, Renewable Energy, and Transmission and Distribution (World Bank 2019c)	Short one-pagers provide hands-on, practical guidance to help task teams close gender gaps in energy operations. They address four energy subsectors—energy access, energy policy, renewable energy, and transmission and distribution—and provide entry points for achieving objectives of the Gender Strategy.
Getting to Gender Equality in Energy Infrastructure: Lessons from Electricity Generation, Transmission, and Distribution Projects (ESMAP 2018)	The report examines the social and gender footprint of large-scale electricity generation, transmission, and distribution projects to establish a foundation on which further research and replication of good practices can be built. The main impact pathways analyzed are access to land and labor markets, along with social sustainability considerations, including health and education.
Integrating Gender Considerations into Energy Operations (World Bank 2013)	Provides project teams with basic tools for integrating gender considerations into energy sector activities with country examples and a focus on monitoring and evaluation.
Gender in Agriculture Sourcebook (World Bank, FAO, IFAD 2009)	The document combines descriptive accounts of national and international experience in investing in agriculture with practical operational guidance on to how to design agriculture-for development strategies that capitalize effectively on the unique properties of agricultural growth and rural development involving women and men as a high-impact source of poverty reduction. It looks at gender equality and women's empowerment, and the associated principles have the potential to make a difference in the lives of hundreds of millions of rural poor.
The Cost of the Gender Gap in Agricultural Productivity – in Malawi, Tanzania and Uganda (World Bank, UN Women, UNEP, UNDP 2015)	The report provides a unique quantification of the costs in terms of lost growth opportunities and an estimate of what societies, economies, and communities would gain if the gender gap in agriculture is addressed. The findings of this report are striking and send a strong signal to policy makers in Africa as well as development partners that closing the gender gap is smart economics.

Gender in fisheries and aquaculture – Module 13 (World Bank - http://siteresources.worldbank.org/INT/GENAGRLIVSOUBOOK/Resources/Module13.pdf)	This Module details investments that address livelihood problems arising from the ongoing changes in production systems, marketing, and technology in the fisheries and aquaculture sector and examines investments that reflect gender inequities that exist in many societies. These gender inequities include the comparatively low value attached to work done by women, and women's limited access to essential resources such as ponds, new technology, education, and information and skills. These inequities reflect societal norms of masculinity and femininity that determine who can and should do what and are visible in local communities, in institutions serving these communities, and in the way many national and international organizations operate
Gender Data Portal (World Bank - https://datatopics.worldbank.org/gender/)	The Gender Data Portal is the World Bank Group's comprehensive source for the latest sex-disaggregated data and gender statistics covering demography, education, health, access to economic opportunities, public life and decision-making, and agency.